FRM Part I Exam

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Questions with Answers - Quantitative Analysis

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Reading 12: Fundamentals of Probability

Q.315 The probability of an increase in the annual dividend paid out to shareholders of ABC Limited is 0.4. The probability of an increase in share price given an increase in dividends is 0.7. Determine the joint probability of an increase in dividends and an increase in the share price.

A. 0.28

B. 0.14

C. 0.72

D. 0.3

The correct answer is **A.**

Let:
A be the event that the dividend is increased and,
B be the event that the share price increases
Therefore, $P(A) = 0.4$ and $P(B \mid A) = 0.7$
The joint probability of an increase in dividends and an increase in share price is $P(B \cap A)$
The multiplication rule of probability states that:

$$P(B|A) = \frac{(B \cap A)}{P(A)}$$

Hence

$$P(B \cap A) = P(B|A) * P(A) = 0.7 * 0.4 = 0.28 \text{ or } 28\%$$

(Note that $P(A \cap B) = P(B \cap A)$).

Q.317 A financial risk manager exam candidate is asked two questions. The probability that she gets the first question correct is 0.4 and the probability that she gets the second question correct is 0.5. Given that the probability that she gets both questions correct is 0.2, determine the probability that she gets either the first or the second question correct.

A. 0.9

B. 0.7

C. 0.1

D. 0.4

The correct answer is **B.**

Let A = {gets first question right} and B = {gets second question right}

Therefore, $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$

We want to determine $P(A \cup B)$;

The addition rule states that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Hence,

$$P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$$

Q.318 An empirical study of ABC stock listed on the New York Exchange reveals that the stock has closed higher on one-third of all days in the past few months. Given that up and down days are independent, determine the probability of ABC stock closing higher for six consecutive days.

A. 0.17

B. 0.0137

C. 0.03704

D. 0.00137

The correct answer is **D.**

From the information above, we can establish that the probability of closing higher = $\frac{1}{3}$

Using independence, the probability of 6 consecutive "highs" = $\frac{1}{3} * \frac{1}{3} * \frac{1}{3} * \frac{1}{3} * \frac{1}{3} * \frac{1}{3} = \frac{1}{729}$

(The calculation above follows from the fact that if A and B are independent events, then $P(A \cap B) = P(A) * P(B)$.)

Things to Remember

- Independence of events in probability theory
- Probability of independent events occuring consecutively
- Calculation of probabilities using the multiplication rule for independent events
- Understanding of basic probability concepts

Q.319 A fruit juice shop allows customers to choose apple juice, mango juice or passion juice. The probability of a customer ordering passion juice is 0.45, mango juice and apple juice 0.19, passion juice and mango juice 0.15, passion juice and apple juice 0.25, passion juice or mango juice 0.6, passion juice or apple juice 0.84, and 0.9 for at least one of them.
Find the probability that a customer orders all the three juices.

A. 0.64

B. 0.1

C. 0.3

D. 0.25

The correct answer is **B.**

Let:

A be the event that a customer chooses/orders apple juice

M be the event that a customer chooses mango juice

S be the event that a customer chooses passion fruit

We can easily establish that:

$P(S) = 0.45, P(M \cap A) = 0.19, P(M \cap S) = 0.15,$
$P(A \cap S) = 0.25, P(M \cup S) = 0.6, P(A \cup S) = 0.84, P(A \cup M \cup S) = 0.9$

We need to determine $P(A \cap M \cap S)$:

Borrowing from the addition rule with three sets,

$P(A \cup M \cup S) = P(A) + P(M) + P(S) - P(M \cap A) - P(M \cap S) - P(A \cap S) + P(A \cap M \cap S) \ldots\ldots \text{equatio}$

$$P(M \cup S) = P(M) + P(S) - P(M \cap S),$$
$$P(M) = 0.6 + 0.15 - 0.45 = 0.3$$

Similarly,

$$P(A \cup S) = P(A) + P(S) - P(A \cap S)$$
$$P(A) = 0.84 - 0.45 + 0.25 = 0.64$$

Therefore applying equation (I),

$$0.9 = 0.64 + 0.3 + 0.45 - 0.19 - 0.15 - 0.25 + P(A \cap M \cap S)$$

Which gives us $P(A \cap M \cap S) = 0.1$

Q.321 During a lottery, 400 names are fed into a computer program. Five of the names are identical. If a name is drawn from the program at random, what is the probability that one of these 5 names will be drawn?

A. 0.0125

B. 0.25

C. 0.0025

D. 0.0625

The correct answer is **A.**

$$P(\text{name } 1 \cup \text{name } 2 \cup \text{name } 3 \cup \text{name } 4 \cup \text{name } 5)$$
$$= \frac{1}{400} + \frac{1}{400} + \frac{1}{400} + \frac{1}{400} + \frac{1}{400} = 0.0125$$

Q.356 A financial risk manager has three routes to get to the office. The probability that she gets to the office on time using routes X, Y, and Z are 60%, 65%, and 70%. She does not have a preferred route and is therefore equally likely to choose any of the three routes. Calculate the probability that she chose route Z given that she arrives to work on time.

A. 0.359

B. 0.233

C. 0.216

D. 0.2

The correct answer is **A.**

Define X to be the event "chooses route X." Let Y and Z have similar definitions.
Define O to be the event that she arrives on time

We wish to determine $P(Z|O)$. Then:

$$\begin{aligned} P(Z|O) &= \frac{P(Z) * P(O|Z)}{[P(Z) * P(O|Z) + P(Y) * P(O|Y) + P(X) * P(O|X)]} \\ &= \frac{(\frac{1}{3} * 0.7)}{[(\frac{1}{3} * 0.7) + (\frac{1}{3} * 0.65) + (\frac{1}{3} * 0.6)]} \\ &= \frac{0.2333}{(0.2333 + 0.2167 + 0.2)} \\ &= 0.3589 \end{aligned}$$

Q.357
A life assurance company insures individuals of all ages. A manager compiled the following statistics of the company's insured persons:

Age of insured	Mortality (Probability of death) [arbitrary]	Portion of company's insured persons
16-20	0.04	0.10
21-30	0.05	0.29
31-65	0.10	0.49
66-99	0.14	0.12

If a randomly selected individual insured by the company dies, calculate the probability that the dead client was age 16-20.

A. 0.9525

B. 0.0593

C. 0.063

D. 0.0475

The correct answer is **D.**

Define the following events:

B = Event of death

B_1 = Event the insured's age is in the range 16-20

B_2 = Event the insured's age is in the range 21-30

B_3 = Event the insured's age is in the range 31-65

B_4 = Event the insured's age is in the range 66-99

We wish to determine $P(B_1|B)$

$$\begin{aligned}
P(B_1|B) &= \frac{(P(B_1) * P(B|B_1))}{[P(B_1) * P(B|B_1) + (P(B_2) * P(B|B_2) + (P(B_3) * P(B|B_3) + (P(B_4) * P(B|B_4)]} \\
&= \frac{(0.1 * 0.04)}{[(0.1 * 0.04) + (0.29 * 0.05) + (0.49 * 0.1) + (0.12 * 0.14)} \\
&= \frac{0.004}{(0.004 + 0.0145 + 0.049 + 0.0168)]} \\
&= 0.04745 \text{ or } 4.7\%
\end{aligned}$$

Q.358

A life assurance company insures individuals of all ages. A manager compiled the following statistics of the company's insured persons:

Age of insured	Mortality (Probability of death) [arbitrary]	Portion of company's insured persons
$16-20$	0.04	0.10
$21-30$	0.05	0.29
$31-65$	0.10	0.49
$66-99$	0.14	0.12

If a randomly selected individual insured by the company dies, calculate the probability that the dead client was in age range 21-30.

A. 0.172

B. 0.04

C. 0.168

D. 0.145

The correct answer is **A.**

We wish to determine $P(B_2|B)$

$$\begin{aligned}
P(B_2|B) &= \frac{(P(B_2)*P(B|B_2))}{[P(B_2)*P(B|B_2)+(P(B_1)*P(B|B_1)+(P(B_3)*P(B|B_3)+(P(B_4)*P(B|B_4)]} \\
&= \frac{(0.29*0.05)}{[(0.29*0.05)+(0.1*0.04)+(0.49*0.1)+(0.12*0.14)} \\
&= \frac{0.0145}{(0.0145+0.004+0.049+0.0168)} \\
&= 17.2\%
\end{aligned}$$

Q.360 A life assurance company insures individuals of all ages. A manager compiled the following statistics of the company's insured persons:

Age of insured	Mortality (Probability of death) [arbitrary]	Portion of company's insured persons
16-20	0.04	0.10
21-30	0.05	0.29
31-65	0.10	0.49
66-99	0.14	0.12

Calculate the probability that the dead client was between 66 and 99 years.

A. 0.047

B. 0.199

C. 0.12

D. 0.201

The correct answer is **B.**

To calculate the probability that the dead client was between 66 and 99 years, we need to use Bayes' theorem, which states:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

In this case, we want to calculate the probability that the dead client was between 66 and 99 years old (event A) given that they have died (event B). We know the mortality rates and the proportion of insured persons for each age group, so we can calculate the probabilities of dying and the marginal probability of dying:

$$P(\text{Dying}) = 0.04 \times 0.10 + 0.05 \times 0.29 + 0.10 \times 0.49 + 0.14 \times 0.12 = 0.0843$$

Therefore,

$$P(66\text{-}99|\text{Dying}) = \frac{P(\text{Dying}|66\text{-}99) \times P(66\text{-}99)}{P(\text{Dying})}$$
$$= \frac{0.14 \times 0.12}{0.0843} = 0.1993$$

Q.361 An investment firm classifies capital projects into three different categories, depending on risk level: Standard, Preferred, and Ultra-preferred. Of the firm's projects, 60% are standard, 30% are preferred, and 10% are ultra-preferred. The probabilities of a project making a loss are 0.01, 0.005, and 0.001 for categories standard, preferred, and ultra-preferred respectively.
If a capital project makes a loss in the next year, then what is the probability that the project was standard (correct to 2 decimal places)?

A. 0.79

B. 0.73

C. 0.22

D. 0.15

The correct answer is **A.**

Let:

L = Event a project makes a loss

S = Event of a standard project

P_1 = Event of a preferred project

U = Event of a ultra-preferred project

We wish to determine $P(S|L)$

$$
\begin{aligned}
P(S|L) &= \frac{(P(S) * P(L|S))}{[P(P_1) * P(L|P_1) + P(U) * P(L|U)]} \\
&= \frac{(0.6 * 0.01)}{[(0.6 * 0.01) + (0.3 * 0.005) + (0.1 * 0.001)]} \\
&= \frac{0.006}{[0.006 + 0.0015 + 0.0001]} \\
&= 0.7895 \text{ or } 79\%
\end{aligned}
$$

Q.363 Upon arrival at a cancer treatment center, patients are categorized into one of four stages, namely: stage 1, stage 2, stage 3, and stage 4. In the past year,
i. 10% of patients arriving were in stage 1
ii. 40% of patients arriving were in stage 2
iii. 30% of patients arriving were in stage 3
iv. The rest of the patients were in stage 4
v. 10% of stage 1 patients died
vi. 20% of stage 2 patients died
vii. 30% of stage 3 patients died
viii. 50% of stage 4 patient died

Given that the patient died, what is the probability that the patient was in stage 4 cancer?

A. 0.86

B. 0.1

C. 0.36

D. 0.5

The correct answer is **C.**

We wish to determine $P(C_4|D)$

$$\begin{aligned} P(C_4|D) &= \frac{(P(C_4) * P(D|C_4))}{[(P(C_4) * P(D|C_4) + ((P(C_1) * P(D|C_1) + (P(C_2) * P(D|C_2) + (P(C_3) * P(D|C_3)]} \\ &= \frac{(0.2 * 0.5)}{[(0.2 * 0.5) + (0.1 * 0.1) + (0.4 * 0.2) + (0.3 * 0.3)]} \\ &= \frac{0.1}{(0.1 + 0.01 + 0.08 + 0.09)} \\ &= 36\% \end{aligned}$$

Q.364 You are an analyst at a large mutual fund. After examining historical data, you establish that all fund managers fall into two categories: superstars (S) and ordinaries (O).
Superstars are by far the best managers. The probability that a superstar will beat the market in any given year stands at 70%. Ordinaries, on the other hand, are just as likely to beat the market as they are to underperform it. Regardless of the category in which a manager falls, the probability of beating the market is independent from year to year. Superstars are rare diamonds because only a meager 16% of all recruits turn out to be superstars.

During the analysis, you stumble upon the profile of a manager recruited three years ago, who has since gone on to beat the market every year.

Determine the probability that the manager is a superstar today, given that he has beaten the market for three consecutive years.

A. 0.3433

B. 0.15988

C. 0.6567

D. 0.16

The correct answer is **A.**

Let's define the following events:

- S: the manager is a superstar
- O: the manager is ordinary
- B: the manager beats the market in a given year
- U: the manager underperforms the market in a given year

From the information given in the problem, we have:

- $P(S) = 0.16$ (the prior probability of being a superstar)
- $P(O) = 0.84$ (the prior probability of being ordinary)
- $P(B|S) = 0.7$ (the probability of beating the market given the manager is a superstar)
- $P(B|O) = 0.5$ (the probability of beating the market given the manager is ordinary)

We want to find $P(S|B^3)$, the probability that the manager is a superstar today, given that he has beaten the market for three consecutive years.

Using Bayes' theorem, we have:

$$P(S|B^3) = \frac{P(B^3|S) \cdot P(S)}{P(B^3)}$$

We can calculate each of the terms separately first:

$P(B^3|S) = P(B|S)^3 = 0.7^3 = 0.343$ (the probability of beating the market for three consecutive years given the manager is a superstar)

$$P(B^3) = P(B^3|S) \cdot P(S) + P(B^3|O) \cdot P(O) = 0.343 \cdot 0.16 + 0.5^3 \cdot 0.84 = 0.15988$$

(the probability of beating the market for three consecutive years regardless of whether the manager is a superstar or ordinary)

$$P(S|B^3) = \frac{0.343 \cdot 0.16}{0.15988} \approx 0.3433$$

Therefore, the probability that the manager is a superstar today, given that he has beaten the market for three consecutive years, is approximately 0.3433.

Things to Remember

- Bayes' Theorem is a fundamental concept in probability theory that describes the probability of an event, based on prior knowledge of conditions that might be related to the event.
- Conditional probability is the probability of an event given that another event has already occurred.
- Independence of events means that the occurrence of one event does not affect the probability of the occurrence of another event.
- Prior probability is the initial probability of an event before new evidence is taken into account.
- Posterior probability is the revised probability of an event after taking into account new evidence.

Q.365 You are an analyst at a large mutual fund. After examining historical data, you establish that all fund managers fall into 2 categories: superstars (S) and ordinaries (O).
Superstars are by far the best managers. The probability that a superstar will beat the market in any given year stands at 70%. Ordinaries, on the other hand, are just as likely to beat the market as they are to underperform it. Regardless of the category in which a manager falls, the probability of beating the market is independent of year to year. Superstars are rare diamonds because only a meager 16% of all recruits turn out to be superstars.

During the analysis, you stumble upon the profile of a manager recruited 3 years ago, who has since gone on to beat the market every year.

What is the probability that the manager is a superstar as at present?

A. 0.46

B. 0.34

C. 0.84

D. 0.16

The correct answer is **B.**

We need to determine P(S | 3B): The probability that the manager is a superstar given that they have managed to beat the market in three consecutive years. As such, we need to apply Bayes' theorem.

$$P(S|3B) = P(S) * \frac{P(3B|S)}{P(3B)}$$

Now, we already have $P(S) = 16\% = \frac{4}{25}$

$$\begin{aligned} P(3B|S) &= (\frac{7}{10})^3 \text{ since performance is independent from one year to the next} \\ &= \frac{343}{1000} \end{aligned}$$

$$\begin{aligned} P(3B) &= \text{unconditional probability of beating the market in 3 consecutive years} \\ &= \text{weighted average probability of 3 marketing-beating years over both superstars and ord} \\ &= P(3B|S) * P(S) + P(3B|O) * P(O) \\ &= [(\frac{7}{10})^3 * \frac{4}{25}] + [(\frac{1}{2})^3 * \frac{21}{25}] \\ &= (\frac{343}{1000} * \frac{4}{25}) + (\frac{1}{8} * \frac{21}{25}) \\ &= \frac{1372}{25000} + \frac{21}{200} \\ &= 16\% \end{aligned}$$

Therefore,

$$16\% * \frac{34.3\%}{16\%} = 34.3\% \text{ or } 0.343$$

Q.366 You are an analyst at a large mutual fund. After examining historical data, you establish that all fund managers fall into two categories: superstars (S) and ordinaries (O).
Superstars are by far the best managers. The probability that a superstar will beat the market in any given year stands at 70%. Ordinaries, on the other hand, are just as likely to beat the market as they are to underperform it. Regardless of the category in which a manager falls, the probability of beating the market is independent of year to year. Superstars are rare diamonds because only a meager 16% of all recruits turn out to be superstars.

During the analysis, you stumble upon the profile of a manager recruited three years ago, who has since gone on to beat the market every year.

What is the conditional probability for a non-superstar manager given that he/she has beaten the market for 3 years?

A. 0.66

B. 0.7

C. 0.45

D. 0.64

The correct answer is **A.**

First, we need to determine P(S | 3B): The probability that the manager is a superstar given that they have managed to beat the market in three consecutive years. As such, we need to apply Bayes' theorem.

$$P(S|3B) = P(S) * \frac{P(3B|S)}{P(3B)}$$

Now, we already have $P(S) = 16\% = \frac{4}{25}$

$$P(3B|S) = \left(\frac{7}{10}\right)^3 \text{ since performance is independent from one year to the next}$$
$$= \frac{343}{1000}$$

$$
\begin{aligned}
P(3B) &= \text{unconditional probability of beating the market in 3 consecutive years} \\
&= \text{weighted average probability of 3 marketing-beating years over both superstars and ord} \\
&= P(3B|S) * P(S) + P(3B|O) * P(O) \\
&= [(\frac{7}{10})^3 * \frac{4}{25}] + [(\frac{1}{2})^3 * \frac{21}{25}] \\
&= (\frac{343}{1000} * \frac{4}{25}) + (\frac{1}{8} * \frac{21}{25}) \\
&= \frac{1372}{25000} + \frac{21}{200} \\
&= 16\%
\end{aligned}
$$

Therefore,

$$16\% * \frac{34.3\%}{16\%} = 0.343$$

Finally, $P(O|3B) = 1 - P(S|3B) = 1 - 0.343 = 0.66$

Q.367 A human health organization tracked a group of individuals for five years. At the commencement of the study, 25% were categorized as heavy smokers, 40% as light smokers, and the remaining as nonsmokers. Results revealed that light smokers were twice as likely as nonsmokers to die during the half-decade study, but only half as likely as heavy smokers. During the period, a randomly selected group member passed on.
Compute the probability that the individual who died was a heavy smoker.

A. 19%

B. 53%

C. 47%

D. 18%

The correct answer is **C.**

To solve this problem, we use Bayes' Theorem to find the probability that an individual who died was a heavy smoker, given the probabilities of dying for each group of individuals. Define:

$$
\begin{aligned}
H &: \text{The event that an individual is a heavy smoker.} \\
L &: \text{The event that an individual is a light smoker.} \\
N &: \text{The event that an individual is a nonsmoker.} \\
D &: \text{The event that an individual dies.}
\end{aligned}
$$

Given probabilities:

$$
\begin{aligned}
P(H) &= 0.25 \quad \text{(25\% are heavy smokers)}\\
P(L) &= 0.40 \quad \text{(40\% are light smokers)}\\
P(N) &= 0.35 \quad \text{(remaining 35\% are nonsmokers)}
\end{aligned}
$$

Let $P(D|N) = x$ be the probability that a nonsmoker dies.

Thus:

- Probability that light smokers are twice as likely to die compared to nonsmokers: $P(D|L) = 2x$

- Probability that heavy smokers are twice as likely as light smokers to die, hence four times as likely as nonsmokers: $P(D|H) = 4x$

Using the law of total probability for $P(D)$:

$$
\begin{aligned}
P(D) &= P(D|H)P(H) + P(D|L)P(L) + P(D|N)P(N)\\
&= 4x \cdot 0.25 + 2x \cdot 0.40 + x \cdot 0.35\\
&= 1.0x + 0.8x + 0.35x\\
&= 2.15x
\end{aligned}
$$

Calculate $P(H|D)$:

$$
P(H|D) = \frac{P(D|H)P(H)}{P(D)} = \frac{4x \cdot 0.25}{2.15x} = \frac{1.0}{2.15} \approx 0.4651
$$

Therefore, the probability that the individual who died was a heavy smoker is approximately 47%
.

Q.369 Peter selects a coin from a pair of coins and tosses it. While coin 1 is double-headed, coin 2 is a normal unbiased coin. After the toss, the result is a head. Calculate the probability that it was coin 1 which was tossed.

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. 0.5

D. 0.75

The correct answer is **B.**

We need to determine $P(\text{coin 1}|\text{head})$. By applying Bayes' theorem,

$$P(\text{coin 1}|\text{head}) = \frac{P(\text{coin 1}) \times P(\text{head}|\text{coin 1})}{(P(\text{coin 1}) \times P(\text{head}|\text{coin 1}) + P(\text{coin 2}) \times P(\text{head}|\text{coin 2}))} = \frac{(1 \times \frac{1}{2})}{1 \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} = \frac{2}{3}$$

Note:

- $P(\text{coin 1}) = P(\text{coin 2}) = \frac{1}{2}$ since we have two coins (pair) of coins that are equally likely to occur
- $P(\text{head }|\text{coin 1}) = 1$ since coin 1 is double headed
- $P(\text{head }|\text{coin 2}) = \frac{1}{2}$ since coin 2 is unbiased since both the head and the tail are equally likely to occur

Q.3251 The probability that the Eurozone economy will grow this year is 18%, and the probability that the European Central Bank (ECB) will loosen its monetary policy is 26%. Assuming that the joint probability that the Eurozone economy will grow and the ECB will loosen its monetary policy is 12%, then the probability that either the Eurozone economy will grow or the ECB will loosen its the monetary policy is *closest to*:

A. 14%

B. 32%

C. 44%

D. 6%

The correct answer is **B.**

The addition rule of probability is used to solve this question:

$P(E) = 0.18$ (the probability that the Eurozone economy will grow is 18%)

$p(M) = 0.26$ (the probability that the ECB will loosen the monetary policy is 26%)

$p(EM) = 0.12$ (the joint probability that Eurozone economy will grow and the ECB will loosen its monetary policy is 12%)

The probability that either the Eurozone economy will grow or the central bank will loosen its the monetary policy:

$$p(E \text{ or } M) = p(E) + p(M) - p(EM) = 0.18 + 0.26 - 0.12 = 0.32$$

Q.3252 A mathematician has given you the following conditional probabilities:

$p(O\|T) = 0.62$	Conditional probability of reaching the office if the train arrives on time
$p(O\|Tc) = 0.47$	Conditional probability of reaching the office if the train does not arrive on time
$p(T) = 0.65$	Unconditional probability of the train arriving on time
$p(O) = ?$	Unconditional probability of reaching the office

What is the unconditional probability of reaching the office, p(O)?

A. 0.5675

B. 0.4325

C. 0.3265

D. 0.3333

The correct answer is **A.**

This question can be solved using the total probability rule.

If $p(T) = 0.65$ (Unconditional probability of train arriving at time is 0.65), then the unconditional probability of the train not arriving on time $p(Tc) = 1 - p(T) = 1 - 0.65 = 0.35$.

Now, we can solve for $p(O) = p(O|T) * p(T) + p(O|Tc) * p(Tc) = 0.62 * 0.65 + 0.47 * 0.35 = 0.5675$

Q.3254 An investor owns shares of both Apple and Microsoft. The two companies operate independently. The investor assumes that the probability of Apple's share price declining by more than 5% this year is 0.4 while the probability of Microsoft's share price declining by more than 5% is 0.3. What is the probability that either Apple or Microsoft share prices will decline in price by more than 5% this year?

A. 0.58

B. 0.12

C. 0.7

D. 0.67

The correct answer is **A.**

We know that,

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) * P(B) \\ &= 0.4 + 0.3 - 0.4 * 0.3 = 0.58 \end{aligned}$$

Q.3255 The probabilities that Bond A and Bond X will default in the next two years are 10% and 8%, respectively. The probability that both bonds will default simultaneously in the next two years is 5%. The probability that Bond A will default given that Bond X has already defaulted is *closest to*:

A. 62.50%.

B. 50%.

C. 80%.

D. 37.50%

The correct answer is **A.**

$P(X) = 8\%$

$P(A) = 10\%$

$P(X \cap A) = 5\%$

As per the conditional probability:

$$P(A|X) = \frac{P(X \cap A)}{P(X)} = \frac{5\%}{8\%} = 62.5\%$$

Q.3256 There is a 40% chance that ABX will announce negative quarterly results tomorrow. On any given day, there is a 55% chance that the company's stock price will decrease. If negative quarterly results are announced, the probability that the stock price will decline is 85%. Tomorrow, the probability that ABX will announce negative quarterly results or that the stock will decrease in price is *closest to*:

A. 0.72

B. 0.67

C. 0.85

D. 0.61

The correct answer is **D.**

To calculate the probability that ABX will announce negative quarterly results or that the stock will decrease in price, we can use the addition rule of probability. The probability of event A

occurring is given as 0.40, and the probability of event B occurring is given as 0.55.
The probability of event B given event A (stock price decreasing given negative quarterly results) is given as 0.85.
Using the addition rule, the probability of either event A or event B occurring is calculated as:
P(A or B) = P(A) + P(B) - P(A and B)

Since the events are independent, P(A and B) = P(A) * P(B|A)
Substituting the given probabilities:
P(A or B) = 0.40 + 0.55 - (0.40 * 0.85)
P(A or B) = 0.95 - 0.34
P(A or B) ≈ 0.61

Things to Remember

- The addition rule of probability states that the probability of either of two mutually exclusive events occurring is the sum of their individual probabilities.
- The multiplication rule of probability is used when events are not independent, and it calculates the probability of both events occurring.
- Independence of events means that the occurrence of one event does not affect the probability of the other event occurring.
- Conditional probability is the probability of one event happening given that another event has already occurred.
- Bayes' Theorem is a formula that describes how to update the probabilities of hypotheses when given evidence.

Q.3257 The probability that a portfolio manager reads Business News weekly is 0.50, while the probability that a portfolio manager reads BloomField News is 0.40. If the probability that a portfolio manager reads both Business News and BloomField News is 0.30, then the probability that a portfolio manager does not read any of the two newspapers is *closest* to:

A. 0.30.

B. 0.40.

C. 0.50.

D. 0.6

The correct answer is **B.**

For simplicity, let

A = Business News; and

B = BloomField News.

$p(A) = 0.50$

$p(B) = 0.40$

$P(A \cap B) = 0.30$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$P(A \cup B) = 0.50 + 0.40 - 0.30 = 0.60$$

The probability that someone reads A or B is 0.60. Therefore, the probability that a person does not read any of the two:

$$p(\text{person does not read any of the two}) = 1 - 0.60 = 0.40$$

Q.3590 An athlete takes part in two different events. The probability that she wins the first event is 0.3 and the probability that she wins the second event is 0.4. Given that the probability that she wins the first and the second event is 0.1, calculate the probability that she wins either the first or the second event.

A. 0.2

B. 0.5

C. 0.6

D. 0.1

The correct answer is **C.**

If A = P(wins first event) and B = P(wins second event), then we are told:

$P(A) = 0.3$; $P(B) = 0.4$; and $P(A \cap B) = 0.1$

We want,

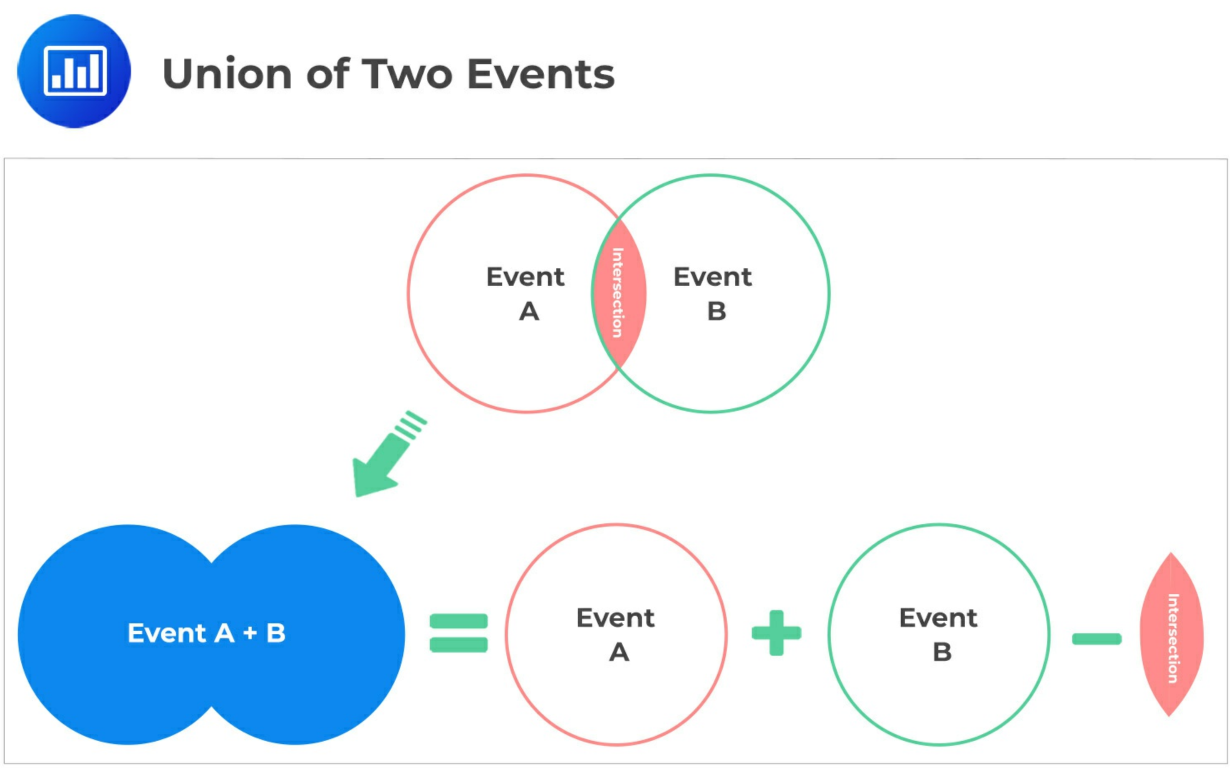
$$P(A \cup B)$$

Recall that:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Thus,

$$P(A \cup B) = 0.3 + 0.4 - 0.1 = 0.6$$



Q.3591 A homeowners insurer offers a discount for homeowners that either have a sprinkler system or live within 5 miles of a fire station. 60% of homeowners qualify for the discount. Only 15% of homeowners have a sprinkler system, and none of those homeowners live within 5 miles of a fire station. If the events are mutually exclusive, what is the probability a randomly selected homeowner lives within 5 miles of a fire station?

A. 15%

B. 25%

C. 30%

D. 45%

The correct answer is **D.**

We use the formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

However, since the events are mutually exclusive, then $P(A \cap B) = 0$,

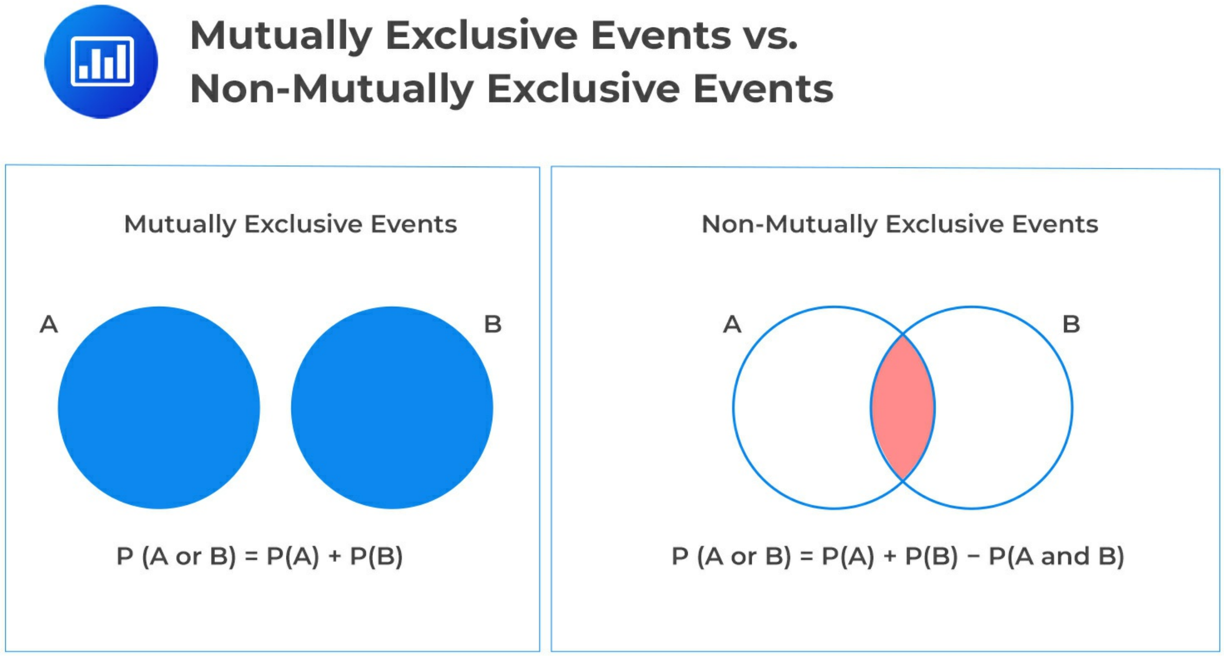
Thus,

$$.60 = .15 + x - 0$$

$$X = .45 = 45\%$$

Things to Remember

- Two events, A and B, are said to be mutually exclusive if the occurrence of A rules out the occurrence of B, and vice versa. For example, a car cannot turn left and turn right at the same time.
- For two mutually exclusive events, A and B, there's **no intersection**, and therefore P(A or B) = P(A) + P(B)
- For non-mutually exclusive events, however, the intersection **contains some outcomes** and has to be subtracted to get P(A or B)



Q.3592 A patient is considered high risk for a heart attack if they either have high cholesterol or high blood pressure. What percentage of patients are NOT considered high risk for a heart attack if 25% have high cholesterol, 30% have high blood pressure and 10% have both high cholesterol and high blood pressure?

A. 45%

B. 50%

C. 55%

D. 60%

The correct answer is **C.**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$.25 + .30 - .10 = .45$$

Therefore, 45% of patients are considered high risk. We now need to find the percentage of patients NOT considered high risk:

$$P(A \cup B)' = 1 - P(A \cup B) = 1 - .45 = .55 \quad \text{or} \quad 55\%$$

Q.3596 55% of an insurer's policyholders are male and 45% are female. The chances of a male having a claim stand at 10% while the chances of a female having a claim stand at 7%. What is the probability that NO ONE will have a claim?

A. 83%

B. 90%

C. 91%

D. 93%

The correct answer is **C.**

We know that:
P(Male) = 55%
P(Female) = 45%
P(Claim|Male)=P(Claim given it's a Male)=10%
P(Claim|Female)=P(Claim given it's a Female)=7%

This question makes use of the conditional probability rule:

$$P(\text{Claim}|\text{Male}) = \frac{P(\text{Claim} \cap \text{Male})}{P(\text{Male})}$$

$$\Rightarrow P(\text{Claim} \cap \text{Male}) = P(\text{Claim}|\text{Male}) \times P(\text{Male}) = .55 \times .10$$

We now have to also add together male and female drivers:

$$.55 \times .10 + .45 \times .07 = .0865$$

And since we're interested in no one having a claim:

$$1 - .0865 = .9135 \quad \text{or} \quad 91\%$$

Q.3601 A company insures both male and female drivers. At the moment, the company has insured an equal number of male and female drivers. Males have a 0.15 chance of having a claim during a policy period while females have a 0.10 chance of having a claim. If a driver is randomly selected from the population, what is the probability that the driver has no claim during the policy period?

A. 73.5%

B. 76.5%

C. 77.5%

D. 87.5%

The correct answer is **D.**

We know that:
P(Male) = 50%
P(Female) =50%
P(Claim|Male)=P(Claim given it's a Male)=0.15
P(Claim|Female)=P(Claim given it's a Female)=0.10

This question makes use of the conditional probability rule:

$$P(\text{Claim}|\text{Male}) = \frac{P(\text{Claim} \cap \text{Male})}{P(\text{Male})}$$

$$\Rightarrow P(\text{Claim} \cap \text{Male}) = P(\text{Claim}|\text{Male}) \times P(\text{Male}) = 0.15 \times 0.50$$

$$\Rightarrow P(\text{Claim} \cap \text{Female}) = P(\text{Claim}|\text{Female}) \times P(\text{Female}) = 0.10 \times 0.50$$

We now have to also add together male and female drivers:

$$0.15 \times 0.50 + 0.10 \times 0.50 = 0.125$$

And since we're interested in the probability that the selected driver has no claim:

$$1 - 0.125 = 0.875 \text{ or } 87.5\%$$

Q.3603 For a certain insured, the probability of making no claim during a policy period is 0.60. The probability of making only 1 claim is 0.25. What is the probability that this insured makes more than 1 claim during the policy period?

A. 0.15

B. 0.25

C. 0.35

D. 0.6

The correct answer is **A.**

"More than 1" is the complement of "no more than 1" which is 0 or 1

$$P(A \cup B) = P(A) + P(B) = .60 + .25 = .85$$

$$P(A \cup B)' = 1 - P(A \cup B) = 1 - .85 = .15$$

Q.3606 60% of an insurer's policyholders are male and 40% are female. The chance of a female having a claim is twice the chance of a male having a claim. Given a randomly selected policyholder has a claim, what's the probability that the policyholder is a male?

A. 35%

B. 65%

C. 57%

D. 43%

The correct answer is **D.**

Let the probability of a male having a claim be P(C|M) and that of a female having a claim be P(C|F).

The chance of a female having a claim is twice the chance of a male having a claim, that is, 2P(C|M) = P(C|F)

Thus,

$$\begin{aligned} P(M|C) &= \frac{[P\ (C|M)^*\ P(M)]}{[P(M)^*P(C|M)+P(F)^*P(C|F)]} \\ &= \frac{[P\ (C|M) * 0.60)]}{[0.60 * P(C|M) + 0.40 * P(C|F)]} \end{aligned}$$

Replacing 2P(C|M) = P (C|F)

$$\begin{aligned} P(M|C) &= \frac{[P\ (C|M) * 0.60)]}{[0.60 * P(C|M) + 0.40 * 2 * P(C|M)]} \\ &= \frac{0.60}{(0.60 + 0.80)} \\ &= 0.4286 = 42.86\% \end{aligned}$$

Q.3607 55% of an insurer's policyholders are male and 45% are female. The chance of a male having a claim is 10% and the chance of a female having a claim is 7%. Given a randomly selected policyholder has a claim, what's the probability she is a female?

A. 25%

B. 33%

C. 36%

D. 38%

The correct answer is **C.**

This question makes use of Bayes' theorem. We know that: $P(M) = 0.55$ and $P(F) = 0.45$
Also, $P(C|M) = 0.10$ and
$P(C|F) = 0.07$
Now, we need $P(F|C)$. Using Bayes' Theorem,

$$\begin{aligned} P(F|C) &= \frac{P(C|F).P(F)}{P(C|F).P(F) + P(C|M).P(M)} \\ &= \frac{0.07 \times 0.45}{0.07 \times 0.45 + 0.10 \times 0.55} \\ &= \frac{0.0315}{0.0865} \\ &= 0.3642 \approx 36\% \end{aligned}$$

Q.3609 A company insures red and black cars, male and female drivers and writes policies in 2 territories (A and B). There are 300 male drivers and 200 female drivers in total. There are 150 males who drive red cars and 100 females who drive red cars. 100 male and 100 female drivers live in territory A and 50 of each, males and females, drive red cars in territory A. Given that a randomly selected policyholder drives a black car, what is the probability that they are female and live in territory B?

A. 20%

B. 25%

C. 33%

D. 40%

The correct answer is **A.**

		Territory A	Territory B
Male	Red Car	50	100
Male	Black Car	50	100
Female	Red Car	50	50
Female	Black Car	50	50

$$P\text{ (Female and Territory B} \mid \text{Black Car)} = \frac{\text{P (Female, Territory B and Black Car)}}{\text{P (Black Car)}}$$

$$= \frac{50}{250} = .20 \text{ or } 20\%$$

Q.3610 A patient is considered high risk for a heart attack if they either have high cholesterol or high blood pressure and the two events are independent. In a given population, 25% have high cholesterol and 30% have high blood pressure. If a randomly selected person has high blood pressure, what is the probability they also have high cholesterol?z

A. 15%

B. 20%

C. 25%

D. 33%

The correct answer is **C.**

Let's denote the following probabilities:

- **P(C)** = Probability of having high cholesterol = 0.25
- **P(B)** = Probability of having high blood pressure = 0.30
- **P(C ∩ B)** = Probability of having both high cholesterol and high blood pressure

Since the events are independent:

$P(C \cap B) = P(C) \times P(B)$

Now, substitute the given probabilities:

$P(C \cap B) = 0.25 \times 0.30 = 0.075$

Next, we need to find the conditional probability that a person has high cholesterol given that they have high blood pressure, denoted as **P(C|B)**.

Using the definition of conditional probability:

$P(C|B) = P(C \cap B) / P(B)$

Substitute the known values:

$P(C|B) = 0.075 / 0.30 = 0.25$

Therefore, the probability that a person has high cholesterol given that they have high blood pressure is **0.25**.

Things to Remember

- Conditional Probability: Conditional probability is the probability of an event occurring given that another event has already occurred. It is denoted as P(A|B) where A and B are events.
- Independence of Events: Two events are considered independent if the occurrence of one event does not affect the occurrence of the other event. Mathematically, $P(A \cap B) = P(A) * P(B)$ if A and B are independent events.
- Marginal Probability: Marginal probability is the probability of a single event occurring without any reference to any other event. It is denoted as P(A) or P(B).
- Joint Probability: Joint probability is the probability of two events occurring together. It is denoted as $P(A \cap B)$.

- Bayes' Theorem: Bayes' Theorem is a mathematical formula used to determine the conditional probability of an event based on prior knowledge of conditions that might be related to the event.

Q.3611 A patient is considered high risk for a heart attack if they either have high cholesterol or high blood pressure. In a given population, 45% of people are considered high risk for a heart attack, (25% have high cholesterol, 30% have high blood pressure). If a randomly selected person has high blood pressure, what is the probability they also have high cholesterol?

A. 15%

B. 20%

C. 25%

D. 33%

The correct answer is **D.**

$$P\,(\text{High Cholesterol} \mid \text{High Blood Pressure}) = \frac{P\,(\text{Both})}{P\,(\text{High BP})}$$

$$P\,(A \cup B) = P\,(A) + P\,(B) - P\,(A \cap B)$$

$$.45 = .25 + .30 - x$$

$$P\,(A \cap B) = .10$$

$$\frac{P\,(\text{Both})}{P\,(\text{High BP})} = \frac{.10}{.30} = \frac{1}{3} \text{ or } 33\%$$

Q.3613 Given the following chart describing the claims of an auto insurer during a policy period, calculate $P(C|M)$. (Assume that the number of male and female claims are independent of each other.)

	Male (M)	Female (F)	Total
Claim (C)	100	200	300
No Claim (X)	400	600	1000
Total	500	800	1300

A. 8%

B. 10%

C. 20%

D. 23%

The correct answer is **C.**

To find $P(C \mid M)$, the probability of a claim given that the policyholder is male, we use the formula for conditional probability:

$$P(C \mid M) = \frac{P(C \cap M)}{P(M)}$$

Where:

- $P(C \cap M)$ is the joint probability of both a claim and the policyholder being male.
- $P(M)$ is the marginal probability of the policyholder being male.

From the table, we can find these probabilities:

- $P(C \cap M) = \frac{100}{1300}$, as there are 100 claims by males out of 1300 total observations.
- $P(M) = \frac{500}{1300}$, as there are 500 males out of 1300 total observations.

Now we can calculate the conditional probability:

$$P(C \mid M) = \frac{P(C \cap M)}{P(M)} = \frac{\frac{100}{1300}}{\frac{500}{1300}} = \frac{100}{500} = 20\%$$

Therefore, the probability that a claim is made given that the policyholder is male is 20%.

Q.3615 An insurance company classifies its policyholders into three tiers – standard, preferred, and ultra preferred. 40% of standard tier policyholders are male, 50% of preferred tier policyholders are male and 75% of ultra preferred tier policyholders are male. There is an equal number of policyholders in each tier. If a policyholder is selected at random, what is the chance she is female?

A. 25%

B. 30%

C. 33%

D. 45%

The correct answer is **D.**

Let event F be “female”, event S be “standard”, event P be “preferred” and event U be “ultrapreferred”

$$P(F) = P(F/S) * P(S) + P(F/P) * P(P) + P(F/U) * P(U)$$

$$P(F) = .60 * (1/3) + .50 * (1/3) + .25 * (1/3)$$

$$P(F) = .45 \quad \text{or} \quad 45\%$$

Things to Remember

- Probability theory is a branch of mathematics that deals with calculating the likelihood of a given event's occurrence.
- Conditional probability is the probability of an event occurring given that another event has already occurred.
- Bayes' Theorem is a fundamental theorem in probability theory that describes the probability of an event based on prior knowledge of conditions that might be related to the event.
- Gender distribution in a population is an important factor to consider when calculating probabilities related to gender-specific events.

Q.3616 An insurance company classifies its policyholders into three tiers – standard, preferred, and ultra preferred. 40% of standard tier policyholders are male, 50% of preferred tier policyholders are male and 75% of ultra preferred tier policyholders are male. There is an equal number of policyholders in each tier. If a male policyholder is selected at random, what is the chance he is classified as a standard tier?

A. 15%

B. 24%

C. 30%

D. 33%

The correct answer is **B.**

Using Bayes Theorem,

$$P\,(S/M) = \frac{P\,(M/S) \times P\,(S)}{P\,(M)}$$

Where

$$\begin{aligned} P\,(M) &= P\,(M/S) \times P\,(S) + P\,(M/P) \times P\,(P) + P\,(M/U) \times P\,(U) \\ &= 0.40 \times \left(\frac{1}{3}\right) + 0.50 \times \left(\frac{1}{3}\right) + 0.75 \times \left(\frac{1}{3}\right) \\ &= 0.55 \end{aligned}$$

Thus,

$$P\,(S/M) = \frac{(0.40 \times (1/3))}{(0.55)} = 0.24 \quad \text{or} \quad 24\%$$

Q.3617 An insurance company classifies its policyholders into three tiers – standard, preferred and ultra preferred with a 25%/50%/25% distribution. The chance of a policyholder in the standard tier having a claim is 10%, in the preferred tier it is 5% and in the ultra preferred tier it is 2%. Given a policyholder has a claim, what is the probability they came from the ultra preferred tier?

A. 5%

B. 7%

C. 9%

D. 11%

The correct answer is **C.**

To find this probability, we used Bayes' Theorem:

$$P(U|C) = \frac{P(C|U)P(U)}{P(C)}$$

Where:

$$P(C) = P(C|S)P(S) + P(C|P)P(P) + P(C|U)P(U)$$

Given the information:

- The distribution of policyholders is 25% standard, 50% preferred, and 25% ultra preferred.
- The probability of a claim for a policyholder in the standard tier is 10%, in the preferred tier is 5%, and in the ultra preferred tier is 2%.

We calculated the overall probability of a claim, P(C), as:

$$P(C) = 0.10 \times 0.25 + 0.05 \times 0.50 + 0.02 \times 0.25 = 0.055$$

Finally, we calculated the probability that a policyholder came from the ultra preferred tier given that they have a claim:

$$P(U|C) = \frac{0.02 \times 0.25}{0.055} = 0.09 = 9\%$$

Q.3618 There are three different bags. The first bag contains 3 square blocks and 2 round blocks. The second bag contains 2 square blocks and 3 round blocks. The third bag contains 5 round blocks. In an experiment, a bag is randomly chosen, and then a block is chosen from the bag. What is the probability that a round block is chosen?

A. 1/5

B. 1/3

C. 2/5

D. 2/3

The correct answer is **D.**

Let:

$P(R)$=Probability that the block is round

$P(R/1)$= Probability that the block is round given that it is from the first bag = $\frac{2}{5}$

$P(R/2)$= Probability that the block is round given that it is from the second bag = $\frac{3}{5}$

$P(R/3)$= Probability that the block is round given that it is from the third bag = $\frac{5}{5} = 1$

$P(R) = P(R/1) * P(1) + P(R/2) * P(2) + P(R/3) * P(3)$
$P(R) = (2/5) * (1/3) + (3/5) * (1/3) + 1 * (1/3) = 2/3$

Things to Remember

- Conditional Probability: In this question, conditional probability is used to calculate the probability of choosing a round block given the bag from which it is chosen.

- Total Probability Rule: The total probability rule is applied to find the overall probability of choosing a round block by considering the probabilities from each bag.

- Probability of an Event: Probability is a measure of the likelihood that an event will occur. It ranges from 0 to 1, where 0 indicates impossibility and 1 indicates certainty.

- Random Experiment: A random experiment is a process or course of action whose outcome cannot be predicted with certainty.

Q.3619 There are three different bags. The first bag contains 3 square blocks and 2 round blocks. The second bag contains 2 square blocks and 3 round blocks. The third bag contains 5 round blocks. In an experiment, a bag is randomly chosen and then a block picked. Given a round block was selected, what is the probability it came from the second bag?

A. 1/5

B. 3/10

C. 1/3

D. 2/5

The correct answer is **B.**

Let "1" be the event that the first bag is picked,
"2" be the event that the second bag is picked,
"3" be the event that the third bag is picked,
and "R" be the event that a round block is drawn
$P(2/R) = P(R/2) * P(2)/P(R)$

$P(R) = P(R/1) * P(1) + P(R/2) * P(2) + P(R/3) * P(3)$

$P(R) = (2/5) * (1/3) + (3/5) * (1/3) + 1 * (1/3) = 2/3$

$P(2/R) = (3/5) * (1/3)/(2/3) = 3/10$

Things to Remember

- Conditional Probability: This concept is used to calculate the probability of an event given that another event has already occurred.
- Bayes' Theorem: A mathematical formula used to determine the conditional probability of an event based on prior knowledge of conditions that might be related to the event.
- Probability of an Event: The likelihood of an event occurring, usually represented as a number between 0 and 1.
- Random Experiment: A process that leads to one of several possible outcomes with uncertain results.

Q.3620 There are two bags with red and white balls. The first bag has 5 red and 5 white balls.

The second bag has 3 red and 2 white balls. A bag is randomly selected, and a ball is drawn without replacement. If the ball is red, another ball is selected from the same bag. If the ball is white, another ball is selected from the other bag. What is the probability that the second ball drawn is red?

A. 40%

B. 50%

C. 51%

D. 53%

The correct answer is **C.**

$P(2^{nd}\ \ red) = P(2^{nd}\ \ red/1^{st}\ \ red) * P(1^{st}\ \ red) + P(2^{nd}\ \ red/1^{st}\ \ white) * P(1^{st}\ \ white)$

$P(1^{st}\ \ red) = \frac{1}{2} * 1/2 + 1/2 * 3/5 = .55$

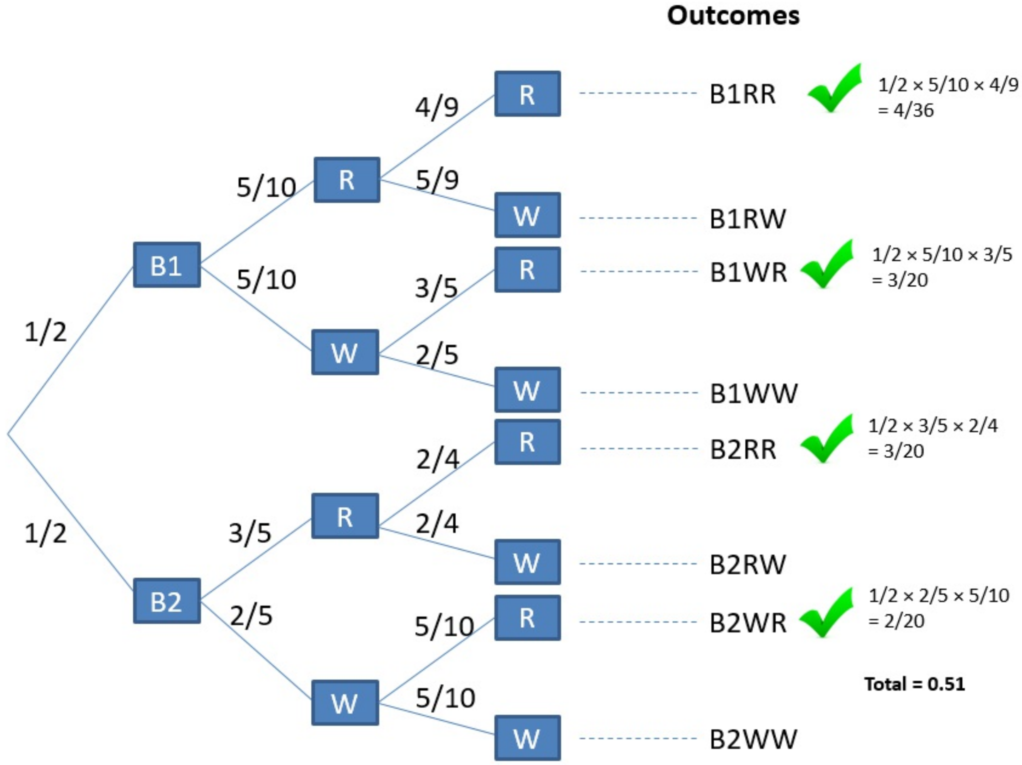
$P(2^{nd}\ \ red/1^{st}\ \ red) = (1/2) * (4/9) + (3/5) * (2/4) = .52$

$P(2^{nd}\ \ red/1^{st}\ \ white) = (1/2) * (3/5) + (2/5) * (1/2) = .5$

$P(1^{st}\ \ white) = \frac{1}{2} * 5/10 + 1/2 * 2/5 = .45$

$P(2^{nd}\ \ red) = .55 * .52 + .45 * .50 = .51 \quad or \quad 51\%$

Alternative approach using a probability tree



Things to Remember

- Conditional probability is the likelihood of an event occurring given that another event has already occurred.
- Bayes' Theorem is a formula that describes how to update the probabilities of hypotheses when given evidence.
- Total probability rule states that the sum of the probabilities of all possible outcomes in a sample space is equal to 1.
- Independence of events means that the occurrence of one event does not affect the probability of the occurrence of another event.
- Random variables are variables whose values depend on outcomes of a random phenomenon.
- Expected value is the long-run average value of repetitions of the experiment it represents.
- Probability tree diagrams are graphical representations of possible outcomes and their probabilities in sequential events.

Q.3621 In a game, a coin is flipped. If the coin is heads, the player rolls one die. If the coin turns up tails, the player rolls two dice and the player moves their playing piece the number of spots shown on the die or dice. Given that on a player's turn, he moves 5 spaces, what is the probability he flipped tails on the coin?

A. 1/10

B. 1/5

C. 2/5

D. 1/3

The correct answer is **C.**

$P(5) = P(5/T) * P(T) + P(5/H) * P(H)$
$P(5) = (1/9) * (1/2) + (1/6) * (1/2) = 5/36$

$P(T/5) = P(5/T) * P(T)/P(5)$

$P(T/5) = (1/9) * (1/2)/(5/36) = 2/5$

Further explanation:
The probability P(5/H) is a conditional probability, indicating that the player has already flipped the coin and obtained a head (H). Given this information, the player only rolls the dice once, as the coin is showing a head. It's important to note that rolling a dice has six possible outcomes (1, 2, 3, 4, 5, 6), each with an equal probability of 1/6.

Therefore, the probability of the player rolling the dice and obtaining a 5 is 1/6.

The question states that if the coin turns up tails, the player will roll two dice. In this case, the player will have a sample size of 36 (6 outcomes for the first dice multiplied by 6 outcomes for the second dice). The question further specifies that the player moves 5 times. If the coin turns out to be tails, the possible outcomes are (1,4), (2,3), (3,2), and (4,1) out of the total sample space of 36 outcomes. Therefore, the probability P(5|T) is calculated as 4/36, which simplifies to 1/9.

Things to Remember

- Conditional probability is the likelihood of an event occurring given that another event has already occurred.
- Sample space is the set of all possible outcomes in a probability experiment.
- Probability of an event is calculated as the number of favorable outcomes divided by the total number of outcomes in the sample space.
- Independence of events means that the occurrence of one event does not affect the probability of another event occurring.
- Bayes' Theorem is a formula that describes how to update the probability of a

hypothesis as more evidence or information becomes available.

Q.3624 An insurance company writes business in three territories: A, B and C. They have 150 policyholders in territory A, 250 in territory B and 300 in territory C. A person is twice as likely to have a claim in territory B than territory A and 3 times as likely to have a claim in territory C than territory A. On average, 50 people have a claim every policy period. Given a claim occurs, what is the probability it was a policyholder in territory C?

A. 21%

B. 33%

C. 43%

D. 58%

The correct answer is **D.**

$$P(C|\text{Claim}) = \frac{P(\text{Claim}|C) * P(C)}{P(\text{Claim})}$$

$$\begin{aligned} P(\text{Claim}) &= P(\text{Claim}|A) * P(A) + P(\text{Claim}|B) * P(B) + P(\text{Claim}|C) * P(C) \\ &= x * \left(\frac{150}{700}\right) + 2x * \left(\frac{250}{700}\right) + 3x * \left(\frac{300}{700}\right) \\ &= x * \left(\frac{1550}{700}\right) \end{aligned}$$

But we also know that the probability of a claim from all territories combined is 50/700. Thus, $\frac{1550x}{700} = \frac{50}{700} \Rightarrow x = 0.032$

But, $P(C) = \frac{300}{700}$

Therefore,

$$\begin{aligned} P(C|\text{Claim}) &= \frac{3 * 0.032 * \frac{300}{700}}{\frac{50}{700}} \\ &= 0.576 \approx 58\% \end{aligned}$$

Q.3625 A test for heart disease results in a correct positive diagnosis 95% of the time and a correct negative diagnosis 99% of the time. 25% of the population has heart disease. What is the probability of a positive test?

A. 0.275

B. 0.15

C. 0.245

D. 0.95

The correct answer is **C.**

Let's define the following events:
+ = positive diagnosis
- = negative diagnosis
H = heart disease
H' = no heart disease
$P(+|H) = .95$
$P(-|H) = .99$
$P(+) = P(+|H) * P(H) + P(+|H^{'}) * P(H^{'})$
$= .95 * .25 + .01 * .75 = .245$

Alternative Approach

The question asks for the probability of a positive test result, whether or not the person being tested actually has heart disease. This can be obtained using the total probability rule, taking into account both the probability of a true positive result (the person has heart disease and the test is positive) and a false positive result (the person does not have heart disease but the test is positive).

The given data are:

- The prevalence of heart disease in the population (P(D)), which is 25%, or 0.25.
- The probability of a correct positive diagnosis (P(T|D)), which is 95%, or 0.95.
- The probability of a correct negative diagnosis (P(~T|~D)), which is 99%, or 0.99.

The probability of not having the disease (P(~D)) is 1 - P(D), or 0.75. The probability of a positive test given that the person does not have the disease (P(T|~D)) is 1 - P(~T|~D), or 0.01

Now, using the total probability rule, we can calculate the total probability of a positive test result (P(T)), whether the person being tested has the disease or not:

$P(T) = P(D) \times P(T|D) + P(\sim D) \times P(T|\sim D)$
Substituting the known values:
$P(T) = 0.25 \times 0.95 + 0.75 \times 0.01 = 0.2375 + 0.0075 = 0.245$
So, the probability of a positive test result, whether the person has heart disease or not, is 0.245

(or 24.5% when expressed as a percentage)

Things to Remember

- Prevalence rate of a disease in a population affects the interpretation of diagnostic test results.
- True positive rate (Sensitivity) and true negative rate (Specificity) are important measures of a diagnostic test's accuracy.
- False positive rate and false negative rate are also crucial in evaluating the performance of a diagnostic test.
- Total Probability Rule is a fundamental concept in probability theory used to calculate the overall probability of an event by considering all possible outcomes.
- Bayes' Theorem is another important concept in probability theory that describes the probability of an event based on prior knowledge of conditions that might be related to the event.

Q.3643 A company insures red and black cars, male and female drivers and writes policies in 2 territories A and B. There are 300 male drivers and 200 female drivers in total. There are 150 males who drive red cars and 100 females who drive red cars. 100 male and 100 female drivers live in territory A and 50 of each, males and females, drive red cars in territory A. What is the probability that a randomly selected driver is either female or lives in territory B?

A. $\frac{2}{5}$

B. $\frac{7}{10}$

C. $\frac{4}{5}$

D. $\frac{9}{10}$

The correct answer is **C.**

Using:

$$P\,(A \cup B) = P\,(A) + P\,(B) - P\,(A \cap B)$$

We need,

$$P\,(\text{female or living in territory B}) = P\,(\text{Female}) + P\,(\text{ Living in territory B}) - P\,(\text{female and living i}$$

$$= \frac{200}{500} + \frac{300}{500} - \frac{100}{500} = \frac{400}{500} = \frac{4}{5}$$

Q.5302 In a financial institution following the Three Lines of Defense Model, the following information is given:

- The First Line of Defense (Business Units) detects 70% of all operational risk incidents.
- The Second Line of Defense (Risk Management) detects 60% of the remaining undetected incidents.
- The Third Line of Defense (Internal Audit) detects 50% of the remaining undetected incidents after the Second Line of Defense.

What is the probability that an operational risk incident is detected by at least one of the three lines of defense?

A. 91%

B. 94%

C. 87%

D. 89%

The correct answer is **B.**

To calculate the overall probability of an operational risk incident being detected, we can use the formula for the probability of the complement event:

$P(A \cup B \cup C) = 1 - P(A' \cap B' \cap C')$

Where A, B, and C represent the detection probabilities of the three lines of defense, and A', B', and C' represent their complements (i.e., the probabilities of not detecting an incident).

First, we need to find the probabilities of not detecting an incident for each line of defense:

- $A' = 1 - A = 1 - 0.7 = 0.3$
- $B' = 1 - B = 1 - 0.6 = 0.4$
- $C' = 1 - C = 1 - 0.5 = 0.5$

Next, we need to find the probability of none of the lines of defense detecting an incident, which is given by the product of their respective probabilities of not detecting an incident:

$P(A' \cap B' \cap C') = A' \times B' \times C' = 0.3 \times 0.4 \times 0.5 = 0.06$

Now we can find the probability of at least one line of defense detecting an incident:

$P(A \cup B \cup C) = 1 - P(A' \cap B' \cap C') = 1 - 0.06 = 0.94$

Things to Remember

- The Three Lines of Defense Model is a risk management framework used in the financial industry to ensure robust risk management practices.
- The First Line of Defense consists of business units that own and manage risks on a day-to-day basis.
- The Second Line of Defense involves risk management functions that oversee and monitor the risks taken by the business units.

- The Third Line of Defense comprises internal audit functions that provide independent assurance on the effectiveness of risk management processes.
- Probability calculations often involve understanding complements and using them to find the probability of the desired event.
- The concept of detecting operational risk incidents by multiple lines of defense helps in creating a comprehensive risk management framework.

Q.5325 In a financial institution, the risk management department is monitoring the probability of a cyber-attack on their systems. Based on historical data, they have determined that the probability of a cyber-attack occurring in any given year is 5% ($P(A)$). They have also learned that a specific type of security breach (B) has been observed in 75% of past cyber-attacks ($P(B|A)$). Additionally, the probability of observing this specific type of security breach without a cyber-attack is 1% ($P(B|A')$). Using Bayes' theorem, what is the probability that a cyber-attack is occurring given that the specific type of security breach has been observed ($P(A|B)$)?

A. 16.13%

B. 31.91%

C. 79.79%

D. 58.51%

The correct answer is **C.**

To solve this problem, we can use Bayes' theorem, which states:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

We are given $P(B|A)$, $P(A)$, and $P(B|A')$, but we need to find $P(B)$. We can do that using the Law of Total Probability:

$$P(B) = P(B|A) \times P(A) + P(B|A') \times P(A')$$

We know that $P(A) = 0.05$ and $P(A') = 1 - P(A) = 0.95$.

So, $P(B) = 0.75 \times 0.05 + 0.01 \times 0.95 = 0.037 + 0.0095 = 0.04700$.

Now we can apply Bayes' theorem:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} = \frac{0.75 \times 0.05}{0.04700} \approx 0.79787$$

Thus, the probability that a cyber-attack is occurring given that the specific type of security breach has been observed is approximately 79.79%.

Things to Remember

- Bayes' Theorem is a fundamental concept in probability theory that describes the probability of an event, based on prior knowledge of conditions that might be related to the event.
- The Law of Total Probability is a fundamental rule relating marginal probabilities to conditional probabilities.
- Conditional probability is the probability of an event given that another event has already occurred.
- Understanding the relationship between conditional probabilities and marginal probabilities is crucial in risk management and decision-making processes.

Reading 13: Random Variables

Q.314 The following is the probability mass function for a discrete random variable X,

$$P(x) = \frac{x}{100}; X = 10, 20, 30, 40$$

Determine the CDF at X =30, i.e., F(30)

A. 0.2

B. 0.3

C. 0.7

D. 0.6

The correct answer is **D.**

The cumulative distribution function (cdf) F(x) defines the probability of a random variable X, and assumes a value equal to or less than a specified value, x. As such:

$$\begin{aligned} F(30) &= P(X \leq 30) \\ &= P(X = 10) + P(X = 20) + P(X = 30) \\ &= \frac{10}{100} + \frac{20}{100} + \frac{30}{100} \\ &= 0.6 \text{ or } 60\% \end{aligned}$$

Q.330 A discrete random variable Y has probability function given by:

Y	0	1	2
P(Y = y)	0.3	0.6	0.1

Calculate Var(Y).

A. 0.2

B. 0.36

C. 0.8

D. 1

The correct answer is **B.**

The variance of any given random variable is given by:

$$Var(Y) = E(Y^2) - E^2(Y)$$

$$E(Y) = \sum Y P(Y = y) = 0 * 0.3 + 1 * 0.6 + 2 * 0.1 = 0.8$$

$$E(Y^2) = \sum Y^2 P(Y = y) = 0^2 * 0.3 + 1^2 * 0.6 + 2^2 * 0.1 = 1$$

Therefore,

$$Var(Y) = 1 - 0.8^2 = 0.36$$

Q.334 Which of the following best describes the concept of skewness in statistics?

A. The degree to which a distribution is symmetric about its mean.

B. The degree to which a distribution is nonsymmetric about its median.

C. The degree to which a distribution is nonsymmetric about its mean.

D. The degree to which a random variable spreads around its mean.

The correct answer is **C.**

Skewness in statistics is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. In other words, skewness tells us to what degree the distribution is non-symmetric about its mean. If the distribution of data is perfectly symmetric, the skewness will be zero. However, if the distribution is not symmetric, then the skewness will not be zero. For instance, if the distribution has a longer or fatter tail on the right side than on the left side, it is said to be positively skewed. Conversely, if the distribution has a longer or fatter tail on the left side than on the right side, it is said to be negatively skewed. Skewness is an important concept in statistics as it can provide insights into the underlying data generating process and potential outliers or anomalies in the data.

Choice A is incorrect. Skewness does not measure the degree of symmetry about the mean. Instead, it measures the degree of asymmetry or nonsymmetry about the mean.

Choice B is incorrect. Skewness does not measure nonsymmetry about the median. It specifically refers to nonsymmetry about the mean of a distribution.

Choice D is incorrect. The spread of a random variable around its mean is measured by variance or standard deviation, not skewness. Skewness measures asymmetry in a distribution's shape.

Things to Remember

- Skewness is a measure of the asymmetry of a probability distribution.
- A distribution with a longer or fatter tail on the right side is positively skewed.
- A distribution with a longer or fatter tail on the left side is negatively skewed.
- Skewness provides insights into the underlying data generating process and potential outliers.
- Skewness is not a measure of symmetry about the mean, but rather of asymmetry or nonsymmetry.
- Variance and standard deviation measure the spread of a random variable around its mean, not skewness.

Q.336 Mary Noel, FRM, is tasked with analyzing the returns of two different assets – A and B. She finds that the two assets have the same mean, variance, and skewness, but A has a higher kurtosis than B. Which of the following statements is most likely true?

A. Asset A is riskier than asset B.

B. Asset B is riskier than asset A.

C. Both assets are highly profitable.

D. Assets A and B will earn negative returns in the long term.

The correct answer is **A.**

Kurtosis, in the realm of finance, is a statistical measure that is used to describe the distribution of observed data around the mean. It is particularly useful in identifying the presence of outliers in a data set. A higher kurtosis value indicates a higher probability of extreme values or outliers, which in turn implies a higher level of risk. In this context, since Asset A has a higher kurtosis than Asset B, it is indicative of a higher likelihood of extreme returns, thereby making Asset A riskier than Asset B. This conclusion holds true despite the fact that both assets have the same mean, variance, and skewness. It is important to note that while mean, variance, and skewness are crucial in understanding the distribution of returns, kurtosis provides additional insight into the tail risk or the risk of extreme outcomes, which is a critical aspect of financial risk management.

Choice B is incorrect. The riskiness of an asset is not solely determined by its kurtosis. While

Asset A has a higher kurtosis, indicating more extreme values or outliers, this does not automatically make Asset B riskier. Both assets have the same mean and variance, which are key measures of risk.

Choice C is incorrect. The profitability of an asset cannot be inferred from its mean, variance, skewness or kurtosis alone. These statistical measures provide information about the distribution of returns but do not directly indicate profitability.

Choice D is incorrect. The assertion that both assets will earn negative returns in the long term cannot be made based on the given information about their mean, variance, skewness and kurtosis. These measures do not predict future performance or direction of returns.

Things to Remember

- Mean, variance, skewness, and kurtosis are important statistical measures used in analyzing the distribution of returns of assets.
- Kurtosis helps in identifying the presence of outliers in a data set and provides insights into the tail risk or the risk of extreme outcomes.
- A higher kurtosis value indicates a higher probability of extreme values or outliers, implying a higher level of risk.
- Riskiness of an asset is not solely determined by its kurtosis; other factors such as mean and variance also play a crucial role.
- Profitability of an asset cannot be directly inferred from statistical measures like mean, variance, skewness, or kurtosis.
- Mean, variance, skewness, and kurtosis do not predict future performance or direction of returns; they provide insights into the distribution of historical returns.

Q.3262 Assume you're a financial risk manager at an investment management firm where you're given the task to estimate the dispersion of a specific equity price around its forecasted value. As a financial risk manager, calculate the variance of equity value using the data provided in the following table.

Probability	Equity Value
0.33	$62.15
0.39	$60.75
0.28	$63

A. 0.87

B. 0.93

C. 0.75

D. 0.78

The correct answer is **A.**

$$
\begin{aligned}
Var(X) &= E(X^2) - [E(X)]^2 \\
E(X) &= 0.33 \times 62.15 + 0.39 \times 60.75 + 0.28 \times 63.0 = 61.842 \\
E(X^2) &= 0.33 \times 62.15^2 + 0.39 \times 60.75^2 + 0.28 \times 63.0^2 = 3,825.3048 \\
Var(X) &= 3,825.3048 - 61.842^2 = 0.8718
\end{aligned}
$$

Q.3268 An equity research analyst forecasts the share price of Equidor Inc.'s stock and the probability of achieving the price target. The forecast made by the analyst is given in the following exhibit.

Exhibit 1: Share Price Forecast

Probability	Share Price
20%	$32.00
25%	$28.00
40%	$34.00
15%	$38.00

What is the variance of Equidor's stock price?

A. 12.45

B. 10.51

C. 16.324

D. 12.213

The correct answer is **B.**

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ E(X) &= 0.20(32) + 0.25(28) + 0.40(34) + 0.15(38) = 32.7 \\ [E(X^2)] &= 0.20 \times 32^2 + 0.25 \times 28^2 + 0.40 \times 34^2 + 0.15 \times 38^2 = 1{,}079.8 \\ \text{Var}(X) &= 1{,}079.8 - 32.7^2 = 10.51 \end{aligned}$$

Q.3739 Let X have the following probability density function:

$$f_X(x) = \begin{cases} 0.15 & x = 1 \\ 0.25 & x = 2 \\ 0.35 & x = 3 \\ C & x = 4 \end{cases}$$

Calculate the mode of the distribution.

A. 1.5

B. 2.0

C. 2.5

D. 3.0

The correct answer is **D.**

First, find C, to see the weight attached to this variable:

$$C = 1 - 0.15 - 0.25 - 0.35 = .25$$

The mode is the most likely value of the distribution (with the highest probability), so in this case it is x=3.

Q.3836 The average salary for an employee at Capital Asset Managers is $50,000 per year. This year, the management has decided to award bonuses to every employee:

- A Christmas bonus of $1,000
- An incentive bonus equal to 15% of the employee’s salary

Determine the mean bonus received by employees.

A. $8,500

B. $4,250

C. $500

D. $10,500

The correct answer is **A.**

To compute the bonus, we apply the following linear transformation to each employee's salary:

$$Y = \alpha + \beta x$$
$$Y = 1,000 + 0.15x$$

where Y is the transformed variable (bonus), X is the original variable (salary), β is the scale constant, and α is the shift constant.

The mean bonus, i.e., mean of Y, is given by:

$$E(Y) = E(\alpha + \beta x) = \alpha + \beta E(X)$$

We know that the mean salary is $50,000. Thus,

$$E(Y) = \$1,000 + 0.15(\$50,000) = \$8,500$$

Q.3837 At Capital Bank, the average salary among sales employees is \$30,000 per year, and they are also entitled to a bonus of \$0.05 for every dollar of sales brought in. Average sales amount to \$300,000 per year. Determine the mean compensation received by employees.

A. \$165,000

B. \$45,000

C. \$22,500

D. \$330,000

The correct answer is **B.**

To compute the mean compensation, we apply the following linear transformation to sales:

$$Y = \alpha + \beta x$$
$$Y = 30,000 + 0.05x$$

where Y is the transformed variable (mean compensation), X is the sales variable, β is the scale constant (\$0.05/\$1)s, and α is the shift constant.

The mean compensation, i.e., mean of Y, is given by:

$$E(Y) = E(\alpha + \beta x) = \alpha + \beta E(X)$$

We know that average sales, E(X) = \$300,000. Thus,

$$E(Y) = \$30,000 + 0.05(\$300,000) = \$45,000$$

Q.3838 The average salary for an employee at Capital Asset Managers is $50,000 per year, with a variance of 6,000,000. This year, the management has decided to award bonuses to every employee:

- A Christmas bonus of $1,000
- An incentive bonus equal to 15% of the employee's salary

Calculate the standard deviation of employee bonuses.

A. $8,500

B. $250

C. $367

D. $10,500

The correct answer is **C.**

To compute the bonus, we apply the following linear transformation to each employee's salary:

$$Y = \alpha + \beta x$$
$$Y = 1,000 + 0.15x$$

where Y is the transformed variable (bonus), X is the original variable (salary), β is the scale constant, and α is the shift constant.

The variance of Y, i.e., variance of bonuses, is given by:

$$\text{Var}(Y) = \text{Var}(\alpha + \beta x) = \beta^2 \text{Var}(X) = \beta^2 \sigma^2$$

The variance of salary [Var(X) = σ], is 6,000,000. Thus,

$$\text{Var}(Y) = 0.15^2 \times 6,000,000 = 135,000$$

Standard deviation is equal to the square root of the variance, and therefore the standard deviation of employee bonuses is equal to the square root of 135,000 or $367.42

Q.3839 At Capital Bank, the compensation framework is made up of a basic salary plus bonuses. The average salary among sales employees is $30,000 per year, and they are also entitled to a bonus of $0.05 for every dollar of sales brought in. Average sales amount to $300,000 per year with a variance of 5,000,000. Determine the standard deviation of compensation received by employees.

A. $165

B. $450

C. $222

D. $112

The correct answer is **D.**

To compute the mean compensation, we apply the following linear transformation to sales:

$$Y = \alpha + \beta x$$
$$Y = 30,000 + 0.05x$$

where Y is the transformed variable (mean compensation), X is the sales variable, β is the scale constant ($0.05/$1), and α is the shift constant

The variance of Y, i.e., variance of compensation, is given by:

$$\text{Var}(Y) = \text{Var}(\alpha + \beta x) = \beta^2 \text{Var}(X) = \beta^2 \sigma^2$$

The variance of sales [Var(X) = σ^2], is 5,000,000. Thus,

$$\text{Var}(Y) = 0.05^2 \times 5,000,000 = 12,500$$

Standard deviation is equal to the square root of variance, and therefore the standard deviation of employee bonuses is equal to the square root of 12,500 or $111.8

Q.340 During a disease outbreak, the probability of surviving after an infection is 60%. Determine the probability that at least 8 out of a group of 9 infected persons will survive.

A. 0.7

B. 0.07

C. 0.007

D. 0.06

The correct answer is **B.**

We note the following:

I. That the infection of any single individual is independent of all other infections for other individuals.
II. The trials are identical (Probability of survival is 60% every time)

Therefore, the number of survivors takes on a binomial distribution with n = 9 and p = 0.6

If X stands for the number of survivors, then:

$$P(X \geq 8) = P(X = 8 \text{ or } 9) = \frac{9!}{8!1!}(0.6)^8 0.4^1 + \frac{9!}{9!0!}(0.6)^9 0.4^0$$
$$= 0.071$$

Note: Under the binomial distribution,

$$P(X = x) = {}^nC_x p^x q^{n-x}$$

Q.351 The normal distribution and the lognormal distribution are related in such a way that:

A. If a random variable X follows a lognormal distribution, ln X is normally distributed.

B. If a random variable X follows a normal distribution, ln X is said to have a lognormal distribution.

C. The mean and variance of a lognormal distribution are twice that of the normal distribution, provided the value of *n* is the same.

D. The mean and variance of the normal distribution are twice that of the lognormal distribution, provided the value of *n* is the same.

The correct answer is **A.**

A random variable X is said to follow a lognormal distribution if its natural logarithm, ln X, is normally distributed. This is a fundamental property of the lognormal distribution and is the reason why it is named as such. The lognormal distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed. Thus, if the random variable X is lognormally distributed, then the distribution of ln X is a normal distribution. This property is particularly useful in various fields, including finance and economics, where lognormal distributions are often used to model variables that are always positive, such as asset prices and exchange rates. The lognormal distribution is preferred in these cases because it can capture the positive skewness and leptokurtosis (fat tails) often observed in financial data, which the normal distribution cannot.

Choice B is incorrect. This statement reverses the actual relationship between the normal and lognormal distributions. If a random variable X follows a normal distribution, it does not imply that ln X will have a lognormal distribution. Instead, if X is lognormally distributed, then ln X will be normally distributed.

Choice C is incorrect. The mean and variance of a lognormal distribution are not necessarily twice that of the normal distribution even if the value of n is the same. The mean and variance of these two distributions depend on their respective parameters (mean and standard deviation for normal; shape and scale for lognormal) rather than being directly proportional to each other.

Choice D is incorrect. Similar to choice C, this statement incorrectly suggests that there's a direct proportionality between the mean and variance of these two distributions which isn't true in general cases as they depend on their specific parameters rather than being directly proportional to each other.

Things to Remember

- The normal distribution is a symmetric bell-shaped distribution that is characterized by its mean and standard deviation.
- The lognormal distribution is a skewed distribution where the logarithm of the variable follows a normal distribution.
- Lognormal distributions are commonly used in finance to model asset prices and other variables that cannot be negative.
- The lognormal distribution is often preferred over the normal distribution for modeling certain financial variables due to its ability to capture positive skewness and fat tails.
- The parameters of the lognormal distribution are the shape parameter (related to the

mean of the logarithm of the variable) and the scale parameter (related to the standard deviation of the logarithm of the variable).

Q.352 A motor vehicle production company based in California is assembling its first batch of fully electric cars. After inspecting about 100 newly assembled units, engineers establish that there are a total of 40 defects. While some units have no defects, others have one, two, or more defects. Assume that the distribution of mechanical defects follows a Poisson distribution. Drawing on the first 100 units produced, how many cars, out of every 10,000 units assembled, would we expect to have at least one defect?

A. About 330

B. About 0.330

C. About 3,300

D. About 1250

The correct answer is **C.**

Let’s use X to denote the number of defects in a car.

$X \sim Poi(40/100)$ i.e. $\lambda = 0.4$

$$P(X \text{ is at least } 1) = 1 - P(X = 0) = 1 - \exp(-0.4) = 0.330$$

This is the probability of at least 1 defect in a car. Therefore, for every 10,000 cars, we would expect 0.330 * 10,000 = 3,300 units to have one or more defects.

Further explanation

From the question, the most important information is that we have 40 defect vehicles in every 100 vehicles manufactured by the company. This implies that, probability of defect is equal to 40/100.

Now, to solve the foe the probability that there is at least one defect vehicle, putting in mind that this follows a Poisson distribution, with $\lambda = \frac{40}{100} = 0.4$,

By having at least one defect, means that we expect either 1 defect or more. So that, we don't expect that there will be zero defects, therefore,

$$P(\text{at least one defect}) = 1 - P(\text{No defect}) = 1 - P(x = 0)$$

Now, recall that the pmf of a Poisson distribution is,

$$P(x;\lambda) = P(x;\lambda) = \frac{(e^{-\lambda})(\lambda^x)}{x!}$$

So that,

$$P(\text{at least one defect}) = 1 - P(\text{No defect}) = 1 - P(x = 0) = 1 - \frac{(e^{-0.4})(0.4^0)}{0!} = 1 - \frac{(e^{-0.4})(1)}{1} = 0.330$$

Now, to get the number of defects out of 10,000 manufactured vehicles, we will multiply that number by the defect probability, i.e., $0.330 * 10,000 = 3,330$

Q.354 Insurance claims in a certain class of business are modeled using a normal distribution with mean $3,000 and a standard deviation of $400. Calculate the probability that the next claim received will exceed $3,500. Please click here to view the standard normal table

A. 0.8944

B. 0.25

C. 0.75

D. 0.1056

The correct answer is **D.**

$$X \sim N(3000, 400^2)$$
$$P(X > 3500) = P\left[Z > \frac{(3500-3000)}{400}\right]$$
$$= P(Z > 1.25)$$
$$= 1 - P(Z < 1.25) = 1 - 0.8944 = 0.1056$$

Note that the probability, $P(Z < 1.25)$, can be read directly from the standard normal table.

Q.3264 As a portfolio analyst, you're directed to label a fund consisting of 9 stocks out of which 4 stocks should be small-cap stocks, 3 stocks should be blue-chips and 2 stocks should be from emerging markets. Determine how many ways these 9 stocks can be labeled.

A. 1260

B. 362880

C. 60480

D. 112840

The correct answer is **A.**

There is a total of $9! = 362,880$ ways in which these 9 stocks can be sequenced.
However, the number of ways these 9 stocks can be labeled according to the required three categories = $\frac{9!}{(4!*3!*2!)} = 1,260$.

Things to Remember

- Factorial (!) notation is used to calculate the number of ways to arrange a set of items.
- When dealing with combinations, the order of selection does not matter.
- In this question, the concept of combinations is used to determine the number of ways the 9 stocks can be labeled according to the specified categories.
- Combinations involve selecting items from a larger set without considering the order in which they are selected.
- It is important to understand the difference between permutations and combinations in probability and combinatorics.

Q.3265 A teacher wants to select groups of 3 students out of 15 for group work. How many different groups of 3 are possible?

A. 25

B. 45

C. 225

D. 455

The correct answer is **D.**

Counting problems involve determining the exact number of ways two or more operations or events can be performed together. This particular counting problem can be solved using a combination, which is basically a selection of some given items where the order does not matter. The formula is:

$$ {}_{n}C_{r} = \frac{n!}{((n-r)!r!} $$

Where

n=15; and

r=3

$$ {}_{n}C_{r} = \frac{15!}{((15-3)! * 3!} = 455 $$

Q.3270 A trader purchases one single stock every day during five working days. His risk manager believes that the probability of selecting an underpriced stock at any given time is 52%. Assuming a binomial distribution, what is the probability of selecting exactly two underpriced stocks during the week out of the universe of underpriced and overpriced stocks?

A. 0.395

B. 0.208

C. 0.327

D. 0.299

The correct answer is **D.**

Since it's a binomial distribution, we will solve the question with the help of the Bernoulli trial method.

The probability of having exactly 2 underpriced stocks in 5 trials (5 days), given that the probability of selecting an underpriced stock at any time is 52%, can be expressed as:

$$\left(\frac{n!}{x!(n-x)!}\right) * p^x * (1-p)^{n-x}$$
$$= \frac{5!}{(2! * 3!)} * 0.52^2 * (1-0.52)^3$$
$$= \left(\frac{120}{12}\right) * 0.2704 * (0.110592) = 0.2990$$

Q.3271 As an investment analyst, your job is to determine how many companies will announce IPOs out of 50 virtual reality startup companies operating in Palo Alto. The annual IPO rate in high-tech industries in all other states of the U.S. is 7.85%. Using a binomial model, what is the standard deviation of the number of virtual reality company IPOs in Palo Alto?

A. 3.616

B. 1.902

C. 1.38

D. 2.125

The correct answer is **B.**

Variance according to the Binomial model $= n * p * (1 - p) = 50 * 0.0785 * (1 - 0.0785) = 3.616$

$$\text{Standard deviation} = \sqrt{3.616} = 1.9018$$

Q.3272 As a research analyst, you're analyzing the probability that the prices of copper will be set below $44/kg after the upcoming government elections. Suppose that the prices of copper are uniformly distributed with a floor at $38/kg and a ceiling at $54/kg imposed by the government, then what is the probability that the prices of copper will be set below $44/kg?

A. 0.815

B. 0.625

C. 0.375

D. 0.429

The correct answer is **C.**

Since the government has set a floor of $38/kg (*a* or the lower boundary) and the ceiling of $54/kg (*b* or the upper boundary.

Thus, $n = 54 - 38 = 16$

The possible outcomes (prices) of copper that fall below $44 is $\$44 - \$38 = \$6$.

Therefore, the probability that the prices of copper will be set under $44 is:

$$\frac{(X - a)}{(b - a)} = \frac{(44 - 38)}{(54 - 38)} = 0.375$$

Q.3274 In Toronto, Canada, there is a 90% chance of having a sunny day. What is the probability that there will be exactly 3 sunny days in the next 7 days?

A. 0.9

B. 0.00255

C. 0.0625

D. 0.00125

The correct answer is **B.**

To calculate the probability of having exactly 3 sunny days in the next 7 days, we can use the Binomial Distribution formula:

$$P(3) = \frac{7!}{((7-3)! * 3!)} * 0.9^3 * (1-0.9)^{7-3} = 0.00255$$

Q.3275 A portfolio has an expected return of 9% with a standard deviation of 7%. If the returns are normally distributed, then what is the probability that the return will be greater than 16%? Click here to view the normal distribution table.

A. 0.1052

B. 0.2241

C. 0.1228

D. 0.1587

The correct answer is **D.**

A 16% return is 1 standard deviation above the mean of 9%, since the standard deviation is 7% (9% + 7% = 16%). The probability of getting a result more than 1 standard deviation above the mean is 1 - Prob($Z \leq 1$) = 1 - 0.8413 = 0.1587 or 15.87%.
Note: 0.8413 is obtained from the Z-table.

Things to Remember

- Standard deviation is a measure of the dispersion or spread of a set of values. In finance, it is used to measure the volatility of an investment.
- Expected return is the mean or average return that an investment is expected to generate over a period of time.
- Normal distribution is a continuous probability distribution that is symmetric around the mean, with the data being distributed in a specific pattern.
- Z-score is a statistical measurement that describes a value's relationship to the mean of a group of values. It is used in hypothesis testing and to determine probabilities.
- The Z-table is a table that provides the percentage of values (probabilities) that fall below a given Z-score in a standard normal distribution.

Q.3276 The population living in Calgary, Canada has a mean income of CAD 55,000 with a standard deviation of CAD 10,000. If the distribution is assumed to be normal, what is the percentage of the population that makes between CAD 45,000 and CAD 50,000?
Click here to view the normal distribution table.

A. 0.1498

B. 0.1511

C. 0.1624

D. 0.2014

The correct answer is **A.**

$\frac{(45{,}000-55{,}000)}{10{,}000} = -1$, which corresponds to a Z-score of 0.1587.

$\frac{(50{,}000-55{,}000)}{10{,}000} = -0.5$, which corresponds to a Z-score of 0.3085.

The difference is $0.3085 - 0.1587 = 0.1498$ or 14.98%.

Things to Remember

- The normal distribution is a continuous probability distribution that is symmetrical around the mean, with the majority of the values falling close to the mean and fewer values farther away from the mean.
- The standard deviation is a measure of the dispersion or spread of a set of values. In a normal distribution, about 68% of values fall within one standard deviation of the mean, 95% within two standard deviations, and 99.7% within three standard deviations.
- Z-scores are a way to standardize values in a normal distribution, allowing for comparison of values from different normal distributions. A Z-score indicates how many standard deviations a value is from the mean.
- When calculating probabilities in a normal distribution, Z-scores are used to find the corresponding probabilities from a standard normal distribution table.

Q.3277 A portfolio's expected return is 17% and its standard deviation is 4%. If the returns are normally distributed, then what is the probability that the returns will be greater than 29%? Use the following standard normal table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							

A. 0.0013

B. 0.01

C. 0.13

D. 0.0525

The correct answer is **A.**

A 29% return is 3 standard deviations above the mean of 17% (17% + 3*4% = 29%). The probability of getting 3 standard deviations above the mean is 1 - Prob(Z≤3) = 1 - 0.9987 = 0.0013 or 0.13%.
Note: 0.9987 is obtained from the Z-table.

Q.3278 A portfolio manager's bonus depends on the return generated by the fund. The different bonus bands are listed below:

Band	Bonus %
Return > 5%	2%
Return > 8%	4%
Return > 12%	10%
Return > 20%	14%
Return > 25%	20%

The mean return and the standard deviation of the fund managed by the portfolio manager stood at 8% and 2%, respectively. Assuming that mutual fund returns are normally distributed, what is the probability that the portfolio manager earns a bonus of between 4% to 10% this year? Click here to view the normal distribution table.

A. 0.6737

B. 0.5

C. 0.4226

D. 0.4772

The correct answer is **D.**

The bonus of 4% corresponds to the fund return of greater than 8%, and the bonus of 10% corresponds to the fund return of greater than 12%. Therefore, the task is to calculate the probability of the return being between 8% and 12%.

To calculate this probability, we need to convert these returns into z-scores using the z-score formula:

$$z = \frac{X - \mu}{\sigma}$$

where X is the value of interest, μ is the mean, and σ is the standard deviation.

For a return of 8%, the z-score is:

$$z_1 = \frac{0.08 - 0.08}{0.02} = 0$$

For a return of 12%, the z-score is:

$$z_2 = \frac{0.12 - 0.08}{0.02} = 2$$

We then use these z-scores to find the corresponding probabilities from the standard normal distribution table.

For a z-score of 0, the probability is 0.5 (the mean of the standard normal distribution splits the area under the curve into two equal halves).

For a z-score of 2, the probability is approximately 0.9772 (this is the cumulative probability up to 2 standard deviations above the mean).

The probability of the return being between 8% and 12% is then the difference between these two probabilities:

$$P = P(Z < 2) - P(Z < 0) = 0.9772 - 0.5 = 0.4772$$

So, there's about a 47.72% chance that the portfolio manager earns a bonus of between 4% and 10% this year, given the distribution of the fund's returns.

Q.3628 If the rate at which accidents occur in a U.S. city is, on average, three every day, calculate the probability that more than 5 accidents occur on a single day.

A. 0.15

B. 0.06

C. 0.05

D. 0.084

The correct answer is **D.**

The number of accidents in a city can be modeled as a Poisson distribution with mean λ
The probability of n accidents can be given by:

$$P(N = n) = \frac{e^{-\lambda}\lambda^n}{n!}$$

We can use the recurrence relationship given:

$$\begin{aligned}
P(N > 5) &= 1 - P(N \leq 5) \\
&= 1 - [P(N = 0) + P(N = 1) + P(N = 2) + P(N = 3) + P(N = 4) + P(N = 5)] \\
&= 1 - \frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} + \frac{e^{-3}3^2}{2!} + \frac{e^{-3}3^3}{3!} + \frac{e^{-3}3^4}{4!} + \frac{e^{-3}3^5}{5!} \\
&= 1 - (0.0498 + 0.1494 + 0.2240 + 0.2240 + 0.1680 + 0.1008) \\
&= 0.084
\end{aligned}$$

Q.3629 Given a binomial random variable with P(X=0) = .20 and P(X=1) = .35 and E(X) = 1.5 calculate Var(X).

A. 0.4

B. 0.6

C. 1

D. 1.3

The correct answer is **D.**

We are given that:

- $P(X = 0) = 0.20$

- $P(X = 1) = 0.35$

- $E(X) = 1.5$

Equation for the probability mass function (PMF) of a binomial random variable:

$p(x) = \binom{n}{x} p^x (1-p)^{n-x}$

Using the PMF to find $p(0)$:

$$p(0) = \binom{n}{0} p^0 (1-p)^n = 1 \cdot 1 \cdot (1-p)^n = (1-p)^n = 0.20 \ldots\ldots\ldots \text{equation 1}$$

Using the PMF to find $p(1)$:

$$p(1) = \binom{n}{1} p^1 (1-p)^{n-1} = np(1-p)^{n-1} = 0.35\ldots\ldots\ldots\text{equation 2}$$

Given that $E(X) = np = 1.5$, we use this to solve for equation 2\):

$$(1.5)(1-p)^{n-1} = 0.35$$
$$\Rightarrow (1-p)^{n-1} = \frac{0.35}{1.5} = 0.2333\ldots\ldots\ldots\text{equation 3}$$

But we can simplify $(1-p)^{n-1} = \frac{(1-p)^n}{(1-p)}$ So that equation 3 becomes:

$$\frac{(1-p)^n}{(1-p)} = 0.2333$$

And from equation 1, $(1-p)^n = 0.20$

$$\Rightarrow \frac{(1-p)^n}{(1-p)} = \frac{0.20}{(1-p)} = 0.233$$
$$\Rightarrow \frac{0.20}{(1-p)} = 0.2333$$
$$\Rightarrow (1-p) = \frac{0.20}{0.2333} = 0.86$$

Finally, using the formula for variance of a binomial distribution $Var(X) = np(1-p)$, we substitute the known values knowing that $np = 1.5$ and $(1-p) = 0.86$:

$Var(X) = (1.5)(0.86) = 1.3$

Therefore, the variance of X is 1.3.

Q.3630 Given a binomial random variable with E(X) = 2 and Var(X) = 1.5, calculate P(X=5).

A. 0.015

B. 0.023

C. 0.039

D. 0.047

The correct answer is **B.**

We know that the mean and variance of a binomial distribution are given by:

$$E(X) = np = 2 \ldots\ldots\ldots\ldots \text{Equation 1}$$
$$Var(X) = np(1-p) = 1.5 \ldots\ldots\ldots\ldots \text{Equation 2}$$

Now, since from Equation 1, np = 2, then we can express Equation 2 as:

$$Var(X) = 2 * (1-p) = 1.5$$

Now, solving the above equation, we get that $p = .25$ and thus $0.25n = 2, \Rightarrow n = 2/0.25 = 8$

Now, using the binomial distribution formula, we have:

$$P(X = 5) = \binom{8}{5}(.25^5)(.75^3) = .023$$

Q.3631 Given a Poisson random variable with E(X) = .25, calculate P(X=1)

A. 0.1

B. 0.15

C. 0.19

D. 0.25

The correct answer is **C.**

$$E(X) = \lambda = .25$$
$$p(1) = \frac{e^{-\lambda}\lambda^{1}}{1!} = e^{-.25} * (.25) = .19$$

Things to Remember

- The Poisson distribution is used to model the number of events occurring in a fixed interval of time or space.
- The mean of a Poisson distribution is equal to its rate parameter, denoted by λ.
- The probability mass function of a Poisson random variable X is given by P(X=k) = (e^(-λ) * λ^k) / k!, where k is the number of events.
- Poisson distribution is often used in risk management to model rare events such as insurance claims, credit defaults, or operational failures.

Q.3632 Given a Poisson random variable with P(X=1) = 0.09 and P(X=2) = 0.0045, calculate Var(X).

A. 0.009

B. 0.3162

C. 0.05

D. 0.1

The correct answer is **D.**

Since we have a Poisson random variable with P(X=1) = 0.09 and P(X=2) = 0.0045, then we can start by writing the Poisson probability mass function:

$$P(X = 1) = 0.09 = \frac{e^{-\lambda}\lambda^1}{1!} \implies 0.09 = e^{-\lambda}\lambda \quad ...(i)$$

$$P(X = 2) = 0.0045 = \frac{e^{-\lambda}\lambda^2}{2!} \implies 0.009 = e^{-\lambda}\lambda^2 \quad ...(ii)$$

Dividing equation (ii) by equation (i):

$$\frac{0.009}{0.09} = \frac{e^{-\lambda}\lambda^2}{e^{-\lambda}\lambda} \implies \frac{1}{10} = \lambda \implies \lambda = 0.1$$

Therefore, the variance of the Poisson random variable X is:

$\text{Var}(X) = \lambda = 0.1$

Things to Remember

- A Poisson random variable is used to model the number of events occurring in a fixed interval of time or space.
- The Poisson distribution is characterized by a single parameter, lambda (λ), which represents the average rate of occurrence of the event.
- The variance of a Poisson random variable is equal to its parameter λ.
- The mean of a Poisson random variable is also equal to its parameter λ.
- Poisson distribution is often used in risk management to model rare events such as insurance claims, credit defaults, or operational failures.

Q.3737 The supermarket sells discounted items for an excellent price; this price is modeled uniformly on the interval [0,500].
Calculate the difference between the median and the 10th percentile.

A. 250

B. 200

C. 240

D. 50

The correct answer is **B.**

The probability distribution function of the price is given by

$$f(x) = \frac{1}{500-0} = \frac{1}{500}$$

We can find the median by solving for k in:

$$P(X \text{ less than } k) = (k-0) * f(x) = 0.5$$
$$\Rightarrow \frac{1}{500}k = 0.5$$
$$\Rightarrow k = 0.5 \times 500 = 250$$

Similarly, we can find the 10th percentile by solving for k in:

$$P(X \text{ less than } k) = (k-0) * f(x) = 0.10$$
$$\Rightarrow \frac{1}{500}k = 0.10$$
$$\Rightarrow k = 0.10 \times 500 = 50$$

The difference between these amounts is:

$$250 - 50 = 200$$

Reading 15: Multivariate Random Variables

Q.332 Two random variables X and Y are such that V[X] = 4V[Y] and Cov[X,Y] = V[Y]
Let E = X + Y and F = X – Y

Find Cov[E, F].

A. V[Y] – V[X]

B. Cov[X,Y]

C. V[Y]

D. 3V[Y]

The correct answer is **D.**

$$\begin{aligned} \text{Cov}[E,F] &= \text{Cov}[X+Y, X-Y] \\ &= \text{Cov}[X,X] - \text{Cov}[X,Y] + \text{Cov}[Y,X] - \text{Cov}[Y,Y] \\ &= V[X] - V[Y] \\ &= 4V[Y] - V[Y] \\ &= 3V[Y] \end{aligned}$$

Logic applied:

I. Given a random variable X, the covariance between X and itself is simply its variance
II. Cov[X,Y] = Cov[Y,X]

Q.338 Two stocks, X and Y, have a correlation of 0.50. Stock Y's return has a standard deviation of 0.26. Given that the covariance between X and Y is 0.005, determine the variance of returns for stock X.

A. 0.13

B. 0.00148

C. 0.0385

D. 0.0148

The correct answer is **B.**

Correlation between X and Y,

$$\begin{aligned}\text{Corr}(X,Y) &= \frac{\text{Cov}(X,Y)}{(\sigma_X * \sigma_Y)}\\ 0.50 &= \frac{0.005}{(\sigma_X * 0.26)}\\ 0.13\sigma_X &= 0.005\\ \sigma_X &= 0.0385\\ \text{Variance}(X) &= \sigma^2 = 0.0385^2 = 0.00148\end{aligned}$$

Q.349 Which of the following *best describes* the central limit theorem?

A. When the sample size is large, the sum of independent and identically distributed (i.i.d.) random variables are normally distributed.

B. The sum of *n* independent and identically distributed random variables approaches the normal distribution as *n* becomes large.

C. For simple random samples of size *n* from a population with mean μ and finite variance σ^2, the sampling distribution approaches the normal distribution with mean μ and variance $\frac{\sigma^2}{n}$, as the sample size becomes large.

D. For simple random samples of size *n* from a population with mean μ and finite variance σ^2, the sampling distribution of the sample mean approaches the normal distribution with mean μ and variance $\frac{\sigma^2}{n}$, as the sample size becomes large.

The correct answer is **D.**

The Central Limit Theorem (CLT) is a fundamental theorem in statistics that describes the shape of the distribution of sample means. It states that if you have a population with mean μ and standard deviation σ and take sufficiently large random samples from the population with replacement, then the distribution of the sample means will approach a normal distribution with a mean (μ) and standard deviation ($\sigma/\sqrt{n}$). This will hold true regardless of the shape of the population distribution. The key condition is that the sample size should be sufficiently large, typically $n > 30$. This theorem forms the basis of many statistical procedures and concepts, including confidence intervals and hypothesis testing.

Choice A is incorrect. While it is true that the sum of independent and identically distributed (i.i.d.) random variables tends to be normally distributed when the sample size is large, this statement does not fully encapsulate the essence of the Central Limit Theorem (CLT). The CLT also specifies that the mean of these sums will approach the population mean and their variance will approach $\frac{\sigma^2}{n}$, which are not mentioned in this choice.

Choice B is incorrect. This option only talks about how the sum of i.i.d random variables approaches a normal distribution as n becomes large. However, it fails to mention anything about how their mean and variance behave, which are crucial aspects of CLT.

Choice C is incorrect. This choice incorrectly states that it's the sampling distribution itself that approaches a normal distribution with mean μ and variance $\frac{\sigma^2}{n}$ as n becomes large. In reality, according to CLT, it's specifically the sampling distribution of sample means that exhibits this behavior.

Things to Remember

- The Central Limit Theorem (CLT) is a fundamental concept in statistics that states that the distribution of sample means approaches a normal distribution as the sample size becomes large.
- CLT applies to the sampling distribution of the sample mean, not the individual observations.
- CLT is essential for understanding the behavior of sample means and is used in various statistical analyses and hypothesis testing.
- Sample size plays a crucial role in the application of CLT, with larger sample sizes leading to a closer approximation to a normal distribution.
- CLT is not limited to normally distributed populations; it applies to a wide range of population distributions.

Q.3263 Assuming that the covariance of returns of Stock X and Stock Y is $\text{Cov}(R_X, R_Y) = 0.093$, the variance of $R_X = 0.69$, and the variance of $R_Y = 0.36$, the correlation of returns of Stock X and Stock Y is *closest to*:

A. 0.155.

B. 0.1865.

C. 0.1119.

D. 0.2133

The correct answer is **B.**

$$\text{Corr}(R_X, R_Y) = \frac{\text{Cov}(R_X, R_Y)}{\sigma(R_X)\sigma(R_Y)}$$

Since

$$\begin{aligned} \text{Variance} &= \sigma^2 \\ \sigma(R_X) &= \sqrt{0.69} = 0.8306, \text{ and} \\ \sigma(R_Y) &= \sqrt{0.36} = 0.6 \\ \text{Corr} R_X, R_Y &= \frac{\text{Cov} R_X, R_Y}{\sigma(R_X)\sigma(R_Y)} = \frac{0.093}{0.8306 * 0.6} = 0.1865 \end{aligned}$$

Q.3266 The covariance matrix of two stocks is given in the following exhibit.

Exhibit: Covariance Matrix

Stock	X	Y
X	650	120
Y	120	450

What is the correlation of returns for stocks X and Y?

A. 0.45

B. 0.22.

C. 0.37.

D. 0.33

The correct answer is **B.**

We can start by calculating the standard deviation from the given variance.

$$\begin{aligned}\sigma(X) &= (650)^{0.5} = 25.50\\ \sigma(Y) &= (450)^{0.5} = 21.21\\ \text{Covariance}(X,Y) &= 120\\ \text{Correlation}(X, Y) &= \frac{\text{Cov}(R_X, R_Y)}{\sigma(R_X)\sigma(R_Y)} = \frac{(120)}{(25.50 * 21.21)} = 0.22\end{aligned}$$

Q.3267 A portfolio consists of two funds A and B. The weights of the two funds in the portfolio and the covariance matrix of the two funds are given in the following two exhibits.

Exhibit 1: Weight of the Funds in the Portfolio

Fund	A	B
Weight	60%	40%

Exhibit 2: Covariance Matrix

Fund	A	B
A	700	200
B	200	500

What is the portfolio variance?

A. 428

B. 500

C. 324

D. 328

The correct answer is **A.**

Based on the covariance matrix:

$$\begin{aligned}\text{Variance}_{\text{Portfolio}} &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2 \\ &= 0.60^2 \times 700 + 0.40^2 \times 500 + 2 \times 0.60 \times 0.40 \times 200 \\ &= 428\end{aligned}$$

Fund	A	B
A	700	200
B	200	500

Q.3627 Living in a certain city is really expensive. The mean spent by a citizen monthly is $3,000 with a variance of $500. Nonetheless, the city mayor decided to put a tax in order to make people regulate their monthly expenses adding 20% to all daily living articles. Find the new variance of this city.

A. 520

B. 580

C. 600

D. 720

The correct answer is **D.**

For this exercise, we need to find the variance of new living costs. Let X be the old living costs and Y the new living costs, then we have:

$$\begin{aligned} Y &= 1.2X \\ Var(Y) &= Var(1.2X) \\ &= 1.44Var(X) \\ &= 1.44(500) = 720 \end{aligned}$$

Note: $Var(aX) = a^2Var(X)$

Things to Remember

- Variance is a measure of how spread out the values in a data set are. It is the average of the squared differences from the mean.
- When a constant is added to each data point in a set, the variance remains the same.
- When a constant is multiplied to each data point in a set, the variance is multiplied by the square of that constant.
- Understanding how to calculate variance is crucial in risk management and financial analysis.

Q.3743 The yearly profits of the two firms A and B can be summarized in the following probability matrix.

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

What is the marginal distribution of company A?

Table A: Marginal Distribution of Company A

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 = x_1)$	0.0697	0.4274	0.3436	0.1598

Table B: Marginal Distribution of Company A

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 = x_1)$	0.0697	0.5274	0.3436	0.0593

Table C: Marginal Distribution of Company A

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 = x_1)$	0.0593	0.5274	0.3436	0.0697

Table D: Marginal Distribution of Company A

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 = x_1)$	0.0697	0.5274	0.1235	0.2794

A. Table A

B. Table B

C. Table C

D. Table D

The correct answer is **A.**

Recall that the marginal distribution gives the distribution of a single component in a joint distribution. In the case of bivariate distribution, the marginal PMF of X_1 is computed by summing up the probabilities for X_1 across all the values at which X_2 is realized.

For example, the first probability $P(X_1 = -1M)$ is calculated as:

$$0.0197 + 0.0390 + 0.011 + 0 = 0.0697$$

So, there is 6.97% that company A will go at a loss of 1 million.

Note that we are adding along the columns. For the second probability $P(X_1 = 0M)$ is calculated as:

$$= 0.0395 + 0.230 + 0.127 + 0.0309 = 0.4274$$

The third and the fourth probabilities are calculated in a similar manner.

Q.3744 The yearly profits of the two firms A and B can be summarized in the following probability matrix.

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

What is the marginal distribution of company B?

A.

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.1325	0.4244	0.3599	0.0832

B.

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.0235	0.4856	0.3254	0.1655

C.

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.0712	0.4228	0.3482	0.1583

D.

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.0633	0.4423	0.3658	0.1286

A. Table A

B. Table B

C. Table C

D. Table D

The correct answer is **C.**

To compute the marginal distribution for company B, we need to sum across the rows. For instance, $P(X_2 = -50)$ is given by:

$$0.0197 + 0.0395 + 0.010 + 0.002 = 0.0712$$

So, there is a 7.12% chance that company B will go at a loss of 50 million.

Similarly, $P(X_2 = 0)$:

$$0.0390 + 0.230 + 0.124 + 0.0298 = 0.4228$$

That is, there is 42.28% chance that company B will not make any profit.

Q.3745 The yearly profits of the two firms A and B can be summarized in the following probability matrix.

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

What is the conditional distribution of company A given that company B made a profit of 100 Million?

A.

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 \vert X_2 = 100)$	0.0697	0.4274	0.3436	0.1598

B.

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 \vert X_2 = 100)$	0.0697	0.5274	0.6436	0.1598

C.

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 \mid X_2 = 100)$	0.0697	0.5274	0.3436	0.2598

D.

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 \mid X_2 = 100)$	0	0.1952	0.4144	0.3904

A. Table A

B. Table B

C. Table C

D. Table D

The correct answer is **D.**

The conditional distributions describe the probability of an outcome of a random variable conditioned on the other random variable taking a particular value.

Recall that given any two events A and B, then:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

This result can be applied in bivariate distributions. That is, the conditional distribution of X_1 given X_2 is defined as:

$$f_{(X_1 \mid X_2)}(x_1 \mid X_2 = x_2) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{X_2}(x_2)}$$

So, in this case, we need:

$$f_{(X_1 \mid X_2)}(x_1 \mid X_2 = 100) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{X_2}(X_2 = 100)}$$

The marginal distribution of company B($f_{X_2}(x_2)$) given by:

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.0712	0.4228	0.3482	0.1583

We are simply summing up the probabilities given in the table from left to right:
Thus:
P(Profit = -50) = 0.0197 + 0.0395 + 0.010 + 0.002 = 0.0712

P(Profit = 0) = 0.0390 + 0.230 + 0.124 + 0.0298 = 0.4228

P(Profit = 10) = 0.011 + 0.127 + 0.144 + 0.0662 = 0.3482

P(Profit = 100) = 0 + 0.0309 + 0.0656 + 0.0618 = 0.1583

The joint distribution is the following table (given at the introduction of this question)

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

To calculate the conditional distribution, we divide the individual components of the last column by the the marginal distribution(0.1583). So, the conditional distribution is:

Company A(X_1) Profits	-1 Million	0 million	2 Million	4 Million
$P(X_1 \mid X_2 = 100)$	$\frac{0}{0.1583} = 0$	$\frac{0.0309}{0.1583} = 0.1952$	$\frac{0.0656}{0.1583} = 0.4144$	$\frac{0.0618}{0.1583} = 0.3904$

Q.3747 The yearly profits of the two firms A and B can be summarized in the following probability matrix.

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

What is the covariance of company A and B given that $E(X_1X_2) = 43.23$?

A. 24.56

B. 23.43

C. 21.45

D. 22.45

The correct answer is **B.**

We know the covariance is given by:

$$\text{Cov}(X_1, X_2) = E(X_1X_2) - E(X_1)E(X_2)$$

Now,

Using the marginal distribution:

Company A(X_1) Profits	-1 Million	0 Million	2 Million	4 Million
$P(X_1 = x_1)$	0.0697	0.4274	0.3436	0.1598

$$\begin{aligned} E(X_1) &= \sum_{\forall x_1} x_1 P(X_1 = x_1) \\ &= -1 \times 0.0697 + 0 \times 0.4274 + 2 \times 0.3436 + 4 \times 0.1598 \\ &= 1.2567 \end{aligned}$$

Similarly, using the marginal distribution:

Company B(X_2) Profits	-50 Million	0 Million	10 Million	100 Million
$P(X_2 = x_2)$	0.0712	0.4228	0.3482	0.1583

$$\begin{aligned} E(X_2)E(X_1) &= \sum_{\forall x_2} x_2 P(X_2 = x_2) \\ &= -50 \times 0.0712 + 0 \times 0.4228 + 10 \times 0.3482 + 100 \times 0.1583 \\ &= 15.752 \end{aligned}$$

So that the covariance of companies A and B is given by:

$$\text{Cov}(A, B) = E(X_1X_2) - E(X_1)E(X_2) = 43.23 - 1.2567 \times 15.752 = 23.43$$

Note that the following steps were used to calculate the marginal distributions of A and B.

Marginal distribution of Company A (X_1):
Sum the probabilities across each column to find the probabilities for each profit level of

Company A.

$$P(X_1 = -1 \text{ Million}) = 0.0197 + 0.0390 + 0.011 + 0 = 0.0697$$
$$P(X_1 = 0 \text{ Million}) = 0.0395 + 0.230 + 0.124 + 0.0309 = 0.4274$$
$$P(X_1 = 2 \text{ Million}) = 0.010 + 0.124 + 0.144 + 0.0656 = 0.3436$$
$$P(X_1 = 4 \text{ Million}) = 0.002 + 0.0298 + 0.0662 + 0.0618 = 0.1598$$

Marginal distribution of Company B (X_2):
Sum the probabilities across each row to find the probabilities for each profit level of Company B.

$$P(X_2 = -50 \text{ Million}) = 0.0197 + 0.0395 + 0.010 + 0.002 = 0.0712$$
$$P(X_2 = 0 \text{ Million}) = 0.0390 + 0.230 + 0.124 + 0.0298 = 0.4228$$
$$P(X_2 = 10 \text{ Million}) = 0.011 + 0.127 + 0.144 + 0.0662 = 0.3482$$
$$P(X_2 = 100 \text{ Million}) = 0 + 0.0309 + 0.0656 + 0.0618 = 0.1583$$

Q.3748 The yearly profits of the two firms A and B can be summarized in the following probability matrix.

		Company A (X_1) Profits			
		-1 Million	0 Million	2 Million	4 Million
Company B (X_2) Profits	-50 Million	0.0197	0.0395	0.010	0.002
	0 Million	0.0390	0.230	0.124	0.0298
	10 Million	0.011	0.127	0.144	0.0662
	100 Million	0	0.0309	0.0656	0.0618

What is the correlation coefficient between the two companies A and B if $\text{Cov}(A, B) = 23.43$?

A. 0.4553

B. 0.3827

C. 0.4562

D. 0.5651

The correct answer is **B.**

Recall that the correlation coefficient is given by:

$$\text{Corr}(X_1, X_1) = \rho_{X_1X_2} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{s_1^2}\sqrt{s_2^2}} = \frac{\sigma_{12}}{\sigma_1\sigma_2}$$

We have $\text{Cov}(A, B) = \text{Cov}(X_1, X_2) = 23.43$. We need to calculate the variance of each component, and we are good to go.

Recall that:

$$Var(X) = \sigma_X^2 = E(X^2) - [E(X)]^2$$

So that,

$$Var(X_1) = \sigma_{X_1}^2 = E(X_1^2) - [E(X_1)]^2$$

Using the marginal distribution of company, A:

$$E(X_1^2) = (-1)^2 \times 0.0697 + 0^2 \times 0.4274 + 2^2 \times 0.3436 + 4^2 \times 0.1598 = 4.0009$$

We had calculated $E(X_1) = 1.2567$ then:

$$Var(X_1) = 4.0009 - [1.2567]^2 = 2.4216$$

Similarly, using the marginal distribution for company B

$$Var(X_2) = \sigma_{X_2}^2 = E(X_2^2) - [E(X_2)]^2$$

$$\begin{aligned} E(X_2^2) &= (-50)^2 \times 0.0712 + 0 \times 0.4228 + 10^2 \times 0.3482 + 100^2 \times 0.1583 \\ &= 1795.82 \end{aligned}$$

We had calculated $E(X_2) = 15.752$

So, that:

$$Var(X_2) = 1795.82 - [15.752]^2 = 1547.6945$$

So, if we substitute these in our correlation coefficient, we get:

$$\text{Corr}(X_1, X_1) = \rho_{X_1X_2} = \frac{23.43}{\sqrt{2.4216}\sqrt{1547.6945}} = 0.3827$$

Q.3749 An investor invests 30% of his assets in security A and 70% in security B. The variance of returns for security in security A is 1234.56, and that of B is 243.56. The covariance between securities A and B is 25.56. What is the standard deviation of the combined returns from these securities?

A. 18.89

B. 14.78

C. 15.53

D. 13.45

The correct answer is **C.**

The standard deviation is just the square root of the variance. The variance is given by:

$$\begin{aligned}\sigma^2_{A+B} &= w^2_A\sigma^2_A + w^2_B\sigma^2_B + 2w_A w_B \mathrm{Cov}(A, B)\\ &= 0.3^2 \times 1234.56 + 0.7^2 \times 243.56 + 2 \times 0.3 \times 0.7 \times 25.56\\ &= 241.19\end{aligned}$$

So the standard deviation of the portfolio (combined returns from these securities) is given by:

$$\sigma_{A+B} = \sqrt{241.19} = 15.53$$

Q.3750 The resulting probability matrix displays the amount of returns of two income-generating sections of bank: Loans and Stock Market

Loans	Returns(X_1)	−20%	0%	20%
Return	Probability	30%	55%	15%
Stock Market	Returns(X_2)	−5%	0%	9%
Returns	Probability	40%	31%	29%

Assuming that the two income-generating avenues are independent of each other, what is the joint probability distribution (matrix)?

A.

	Loan Return (X_1)			
		−20%	0%	20%
Stock	−5%	12%	22%	6%
Market	0%	9.3%	17.05%	4.65%
Returns(X_2)	9%	8.7%	15.95%	4.35%

B.

	Loan Return (X_1)			
		−20%	0%	20%
Stock	−5%	12%	12%	7%
Market	0%	10.3%	17.05%	4.65%
Returns(X_2)	9%	8.7%	15.95%	4.35%

C.

	Loan Return (X_1)			
		−20%	0%	20%
Stock	−5%	12%	12%	6%
Market	0%	9.3%	27.05%	5.65%
Returns(X_2)	9%	8.7%	25.95%	4.35%

D.

	Loan Return (X_1)			
		−20%	0%	20%
Stock	−5%	12%	22%	6%
Market	0%	9.3%	14.05%	4.65%
Returns(X_2)	9%	7.7%	55.95%	4.35%

A. Table A

B. Table B

C. Table C

D. Table D

The correct answer is **A.**

Recall that if two marginal distributions are independent then:

$$f_{(X_1,X_2)}(x_1,x_2) = f_{(X_1)}(x_1)f_{(X_2)}(x_2)$$

We are given the marginal distributions so that the joint distributions are given by multiplying their corresponding PMFs. For example, the joint probability that loan return is -20%, and the stock return is -5% is 30%×40%=12.

The other joint distributions are given in the table below:

Loans Return	Returns(X_1)	−20%	0%	20%
	Probability	30%	55%	15%
Stock Market Returns	Returns(X_2)	−5%	0%	9%
	Probability	40%	31%	29%

		Loan Return(X_1)		
		−20%	0%	20%
Stock	5%	40% × 30% = 12%	40% × 55% = 22%	40% × 15% = 6%
Market	0%	31% × 30% = 9.3%	31% × 55% = 17.05%	31% × 15% = 4.65%
Returns(X_2)	9%	29% × 30% = 8.7%	29% × 55% = 15.95%	29% × 15% = 4.35%

The resulting probability matrix displays the amount of returns of two income-generating sections of bank: Loans and Stock Market

	Loan	Return	(X_1)	
		−20%	0%	20%
Stock	−5%	12%	22%	6%
Market	0%	9.3%	17.05%	4.65%
Returns(X_2)	9%	8.7%	15.95%	4.35%

Q.3751 The resulting probability matrix displays the amount of returns of two independent income-generating sections of bank: Loans and Stock Market

Loans Return	Returns(X_1)	−20%	0%	20%
	Probability	30%	55%	15%
Stock Market Returns	Returns(X_2)	−5%	0%	9%
	Probability	40%	31%	29%

What is the conditional distribution of loan returns given that the return from the stock market is 9%?

A.

Loans Return(X_1)	−20%	0%	20%
$P(X_1 \mid X_2 = 9\%)$	40%	31%	29%

B.

Loans Return(X_1)	−20%	0%	20%
$P(X_1 \mid X_2 = 9\%)$	9.3%	17.05%	4.65%

C.

Loans Return(X_1)	-20%	0%	20%
$P(X_1 \mid X_2 = 9\%)$	30%	55%	15%

D.

Loans Return(X_1)	-20%	0%	20%
$P(X_1 \mid X_2 = 9\%)$	8.7%	15.95%	4.35%

A. Table A

B. Table B

C. Table C

D. Table D

The correct answer is **C.**

The conditional distribution of X_1 given X_2 is defined as:

$$f_{(X_1 \mid X_2)}(x_1 \mid X_2 = x_2) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{(X_2)}(x_2)}$$

So, in this case, we need:

$$f_{(X_1 \mid X_2)}(x_1 \mid X_2 = 10\%) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{(X_2)}(X_2 = 9\%)}$$

Note that we are given the marginal distribution of stock market return $f_{(X_2)}(x_2)$ given by:

Stock Market	Returns(X_2)	-5%	0%	9%
Returns	Probability	40%	31%	29%

We calculated the joint distribution as:

		Loan Return (X_1)		
		-20%	0%	20%
Stock	-5%	12%	22%	6%
Market	0%	9.3%	17.05%	4.65%
Returns(X_2)	9%	8.7%	15.95%	4.35%

To calculate the conditional distribution, we divide the last column by the corresponding marginal distribution(29%). So, the conditional distribution is:

Loans Return(X_1)	-20%	0%	20%
$P(X_1 \mid X_2 = 9\%)$	$\frac{8.7\%}{29\%} = 30\%$	$\frac{15.95\%}{29\%} = 55\%$	$\frac{4.35\%}{29\%} = 15\%$

Q.3752 Three random variables X, Y, and Z have the equal variance of $\sigma^2 = 2$. X is independent of both Y and Z, and that Y and Z are correlated with a correlation coefficient of 0.8. What is the covariance between X and K given that K=Y+Z?

A. 0

B. 2

C. 3

D. 4

The correct answer is **A.**

Recall from the properties of the covariance that:

$$\text{Cov}(A, B + C) = \text{Cov}(A, B) + \text{Cov}(A, C)$$

So, we need:

$$\text{Cov}(X, K) = \text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z) = 0 + 0 = 0$$

This is true because the X in uncorrelated with Y and Z!

Q.3753 Three random variables X, Y, and Z have equal variance of $\sigma^2 = 2$. X is independent of both Y and Z, and that Y and Z are correlated with a correlation coefficient of 0.8. What is the covariance between Z and V given that $V = 3X - 2Y$.

A. 4.1

B. -4.3

C. 3.2

D. -3.2

The correct answer is **D.**

Using the same property that:

$$Cov(A, B + C) = Cov(A, B) + Cov(A, C)$$

Then,

$$Cov(Z, V) = Cov(Z, 3X - 2Y) = 3Cov(Z, X) - 2Cov(Z, Y)$$

But $Cov(Z, X) = 0$. So,

$$Cov(Z, 3X - 2Y) = 0 - 2Cov(Z, Y) = -2Cov(Z, Y)$$

Also, recall that the correlation coefficient between two random variables A and B

$$\rho_{AB} = \frac{Cov(A, B)}{\sigma_A \sigma_B} \Rightarrow Cov(A, B) = \rho_{AB} \times \sigma_A \sigma_B$$

We are given $\rho_{YZ} = 0.8$ with equal variances. So, using the above reasoning, we have:

$$Cov(Z, Y) = \rho_{ZY} \times \sigma_Z \sigma_Y = \rho_{ZY} \times \sigma^2 = 0.8 \times 2 = 1.6$$

Therefore,

$$Cov(Z, 3X - 2Y) = -2Cov(Z, Y) = -2 \times 1.6 = -3.2$$

Note that if X and Y are independent variables, then their covariance is 0.

Q.3754 The amount of profit (X) for a sales company is continuously distributed uniformly with the parameters 0 and 1,500. However, a financial analyst believes that the actual profit (Y) is a minimum of X. What is the conditional distribution of X given X<1,300?

A. Continuous uniform with parameters 0 and 1,300.

B. Continuous uniform with parameters 0 and 1,500.

C. Continuous uniform with parameters 0 and 1,000.

D. Continuous uniform with parameters 0 and 2,800.

The correct answer is **A.**

We are given that: $X \sim U(0, 1500)$ and that $Y = \min(X, 1300)$

We need:

$$P(X < x | X < 1300)$$

Recall the conditional distribution is given by:

$$f_{(X_1 | X_2)}(x_1 | X_2 = x_2) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{(X_2)}(x_2)}$$

So that,

$$P(X < x | X < 1300) = \frac{P(X < x, X < 1300)}{P(X < 1300)}$$
$$= \frac{P(X < x)}{\frac{1300}{1500}} = \frac{\left(\frac{x}{1500}\right)}{\left(\frac{1300}{1500}\right)} = \frac{x}{1300}$$

Therefore, the marginal distribution is a continuous uniform with the parameters 0 and 1300. This should be intuitive since we are computing conditional distribution, which is the scaled version of the original distribution.

This can be seen where $P(X < x, X < 1300) = P(X < x)$ since 1300 is less than 1500.

Q.3758 Let X represent the age of an insured automobile involved in an accident. Let Y denote the length of time the insurance contract has been in place at the time of the accident. X and Y have joint probability density function

$$f(x,y) = \begin{cases} \frac{1}{64}(10 - xy^2), & 2 < x < 10, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

What is the expected length of time the contract has been in place for an insured automobile involved in an accident?

A. 0.4563

B. 0.55

C. 0.4375

D. 0.201

The correct answer is **C.**

We need to calculate the marginal distribution of Y.

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(xy)\,dx = \int_{2}^{10} \frac{1}{64}(10 - xy^2)dx = \frac{1}{64}[10x - \frac{x^2y^2}{2}]_{2}^{10} = \frac{1}{64}[80 - 48y^2]$$

So,

$$f(y) = \begin{cases} \frac{1}{64}[80 - 48y^2], & 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

We need to calculate the expectation. Recall that for a continuous random variable, the expectation is given by:

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$$

So, in this case,

$$E(Y) = \int_{0}^{1} y.(\frac{1}{64}[80 - 48y^2])\,dy = \frac{1}{64}[40y^2 - 12y^4]_0^1 = \frac{1}{64}[40 - 12] = \frac{28}{64} = \frac{7}{16} = 0.4375$$

Q.3760 Given the following joint probability density function of two random variables:

$$f(x_1, x_2) = \begin{cases} 4x_1x_2, & 0 < x_1 < 1, 0 < x_2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $E(X_1|X_2 = 0.5)$

A. 0.01

B. 0.54

C. 0.33

D. 0.67

The correct answer is **D.**

We need to calculate the conditional probability $f(x_1|X_2 = 0.5)$. Recall that:

The conditional distribution is given by:

$$f_{(X_1|X_2)}(X_1|X_2 = x_2) = \frac{f_{(X_1,X_2)}(x_1, x_2)}{f_{(X_2)}(x_2)}$$

So,

$$f(x_1|X_2 = 0.5) = \frac{f_{(X_1,X_2)}(x_1, x_2 = 0.5)}{f_{(X_2)}(x_2 = 0.5)}$$

The denominator $f_{(X_2)}(x_2 = 0.5)$ requires that we calculate the marginal distribution of X_2 which is given by:

$$\begin{aligned} f_{(X_2)}(x_2) &= \int_{-\infty}^{\infty} f_{(X_1,X_2)}(x_1, x_2)dx_1 \\ &= \int_0^1 4x_1x_2dx_1 = [2x_1^2x_2]_0^1 = 2x_2 - 0 = 2x_2 \end{aligned}$$

So, $f_{(X_2)}(x_2 = 0.5) = 2 \times 0.5 = 1$

Thus,

$$f(x_1|X_2 = 0.5) = \frac{f_{(X_1,X_2)}(x_1, x_2 = 0.5)}{f_{(X_2)}(x_2 = 0.5)} = \frac{(4x_1 \times 0.5)}{1} = 2x_1$$

Given that

$$f(x_1 \mid X_2 = 0.5) = 2x_1, \quad 0 < x_1 < 1$$

We know that $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$. Then:

$$E(X_1 \mid X_2 = 0.5) = \int_{-\infty}^{\infty} x_1 f_{X_1|X_2=0.5}(X_1 \mid X_2 = 0.5) dx_1 = \int_0^1 x_1 \times 2x_1 dx_1 = \left[\frac{2}{3}x_1^3\right]_0^1 = \frac{2}{3} - 0 = \frac{2}{3} \approx 0.67$$

Q.3761 A financial risk manager believes that the prevailing interests rate (X_1) and the return in a stock market (X_2) can be modeled using the following joint probability function:

$$f(x_1, x_2) = \begin{cases} \frac{1}{8}x_1x_2, & 0 \le x_1 \le 1, 0 \le x_2 \le 4\sqrt{2} \\ 0, & \text{elsewhere} \end{cases}$$

What is the covariance between the interest rate and the return in the stock market?

A. 0.0972

B. 0.0444

C. 0.2222

D. 0

The correct answer is **D.**

We know that,

$$Cov(X_1, X_2) = E(X_1X_2) - E(X_1) * E(X_2)$$

Now,

$$
\begin{aligned}
E(X_1) &= \int_0^{4\sqrt{2}} \int_0^1 x_1 \frac{1}{8} x_1 x_2 dx_1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \int_0^1 x_1^2 x_2 dx_1 dx_2 \\
&= \frac{1}{8} \int \left[\frac{x_1^3}{3} x_2\right]_0^1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \frac{x_2}{3} dx_2 = \frac{1}{8}\left[\frac{x_2^2}{6}\right]_0^{4\sqrt{2}} = \frac{1}{8}\frac{16}{3} = \frac{2}{3}
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
E(X_2) &= \int_0^{4\sqrt{2}} \int_0^1 x_2 \frac{1}{8} x_1 x_2 dx_1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \int_0^1 x_1 x_2^2 dx_1 dx_2 \\
&= \frac{1}{8} \int \left[\frac{x_1^2}{2} x_2^2\right]_0^1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \frac{x_2^2}{2} dx_2 = \frac{1}{8}\left[\frac{x_2^3}{6}\right]_0^{4\sqrt{2}} = \frac{1}{8}\frac{64\sqrt{2}}{3} = \frac{8\sqrt{2}}{3}
\end{aligned}
$$

Similarly,

$$
\begin{aligned}
E(X_1 X_2) &= \int_0^{4\sqrt{2}} \int_0^1 x_1 x_2 \frac{1}{8} x_1 x_2 dx_1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \int_0^1 x_1^2 x_2^2 dx_1 dx_2 \\
&= \frac{1}{8} \int \left[\frac{x_1^3}{3} x_2^2\right]_0^1 dx_2 \\
&= \frac{1}{8} \int_0^{4\sqrt{2}} \frac{x_2^2}{3} dx_2 = \frac{1}{8}\left[\frac{x_2^3}{9}\right]_0^{4\sqrt{2}} = \frac{1}{8}\frac{128\sqrt{2}}{9} = \frac{16\sqrt{2}}{9}
\end{aligned}
$$

$$
Cov(X_1, X_2) = E(X_1 X_2) - E(X_1) * E(X_2) = \frac{16\sqrt{2}}{9} - \frac{2}{3} \times \frac{8\sqrt{2}}{3} = 0
$$

Q.325 Compute the sample standard deviation given the following sample data: $\sum x = 31,353$ $n = 100$ $\sum x^2 = 10,687,041$

A. 86

B. 93

C. 71

D. 75

The correct answer is **B.**

The formula for calculation of sample variance is:

$$s^2 = \frac{1}{(n-1)}[\sum_{i=1}^{n} x^2 - n\bar{x}^2]$$

where:

$\sum x$ = sum of x

n = sample size

$\sum x^2$ =sum of x squared.

From the data,sample mean is given by:

$$\bar{x} = \frac{31353}{100} = 313.53$$
$$\Rightarrow s^2 = \frac{1}{(99)}[10,687,041 - 100 \times 313.53^2] = 8655.9082$$

The sample standard deviation is given by:

$$= \sqrt{s^2} = \sqrt{8655.9082} = 93.0371 \approx 93$$

Q.326 On Tuesday, an insurance company receives a total of 10 claims for automobile policies. After the first-round assessment, it's found that the mean claim amount of the 10 claims is $426 while the standard deviation is 112. On Tuesday, the chief claims analyst authorizes the removal of one of the claims for $545 from the list on grounds that it's fraught with fraud. Compute the standard deviation for the remaining set of 9 claims.

A. 110.2

B. 12145.2

C. 421.8

D. 420

The correct answer is **A.**

$\sum x$ = the total of the original set of 10 claims = $426 \times 10 = 4260$

After removing the claim worth $545, the new value of $\sum x = 4260 - 545 = 3,715$

Thus, the new mean = $\frac{3715}{9} = \$412.8$

Since $s^2 = \frac{1}{(n-1)}[\sum x^2 - n\bar{x}^2]$, using data for all the ten claims,

$$112^2 = \frac{1}{9}[\sum x^2 - 10 \times 426^2]$$

Making $\sum x^2$ the subject of the formula,

$$\sum x^2 = 112^2 \times 9 + 10 \times 426^2 = 1,927,656$$

Removing the fraudulent claim gives,

$$\sum x^2 = 1,927,656 - 545^2 = 1,630,631$$

Now, using the data for the remaining 8 claims,

$$S^2 = \frac{1}{(9-1)}[1,630,631 - 9 \times 412.8^2] = 12,145.2$$

Therefore,

$$s = \sqrt{12,145.2} = 110.2$$

Q.328 A renowned economist has calculated that the Canadian economy will be in one of 3 possible states in the coming year: Boom, Normal, or Slow. The following table gives the returns of stocks A and B under each economic state.

State	Probability State	Return for Stock A	Return for Stock B
Boom	40%	12%	18%
Normal	35%	10%	15%
Slow	25%	8%	12%

Which of the following is closest to the covariance of the returns for stocks A and B?

A. 0.103

B. 0.0001734

C. 0.1545

D. 0.0003765

The correct answer is **D.**

$$\text{Cov}(A,B) = \sum P(s) * [R_A - E(R_A)] * [R_B - E(R_B)]$$

First, you have to determine the expected return for every stock:

$$E(R_A) = 0.4 * 0.12 + 0.35 * 0.1 + 0.25 * 0.08 = 0.103$$
$$E(R_B) = 0.4 * 0.18 + 0.35 * 0.15 + 0.25 * 0.12 = 0.1545$$

State	PS	RA	RB	P(S) * [RA − E(RA)] * [RB − E(RB)]
Boom	0.40	0.12	0.18	$0.4 * [0.12 – 0.103] * [0.18 – 0.1545] = 0.0001734$
Normal	0.35	0.1	0.15	$0.35 * [0.1 – 0.103] * [0.15 – 0.1545] = 0.000004725$
Slow	0.25	0.08	0.12	$0.25 * [0.08 – 0.103] * [0.12 – 0.1545] = 0.0001984$
				$\text{Cov}(A,B) = 0.0001734 + 0.000004725 + 0.0001984 = 0.0003765$

Q.331 The mean height of female FRM exam candidates over the years is 1.671m while that of male candidates is 1.757. Given that the mean height of ALL of the exam candidates is 1.712m, calculate the percentage of the candidates who are female:

A. 0.523

B. 0.46

C. 0.087

D. 0.477

The correct answer is **A.**

Let F be the proportion of females.

Applying the idea of a weighted mean,

$$\begin{aligned} 1.671F + 1.757(1 - F) &= 1.712 \\ \Rightarrow 1.671F + 1.757 - 1.757F &= 1.712 \\ \Rightarrow 1.757 - 1.712 &= 1.757F - 1.671F \\ \Rightarrow 0.045 &= 0.086F \\ \Rightarrow F &= 0.523 \text{ or } 52.3\% \end{aligned}$$

Q.370 At a certain investment firm, each of the firm's 5 managers is tasked with overseeing a project. During a given one-year period, the managers reported the following individual returns from their projects:
[24%, 26%, 30%, 18%, 20%]

Calculate the population variance of these returns.

A. 0.001824

B. 0.1824

C. 0.228

D. 0.00228

The correct answer is **A.**

Note that the data given is comprised of the entire population and NOT a sample. As such, we should use the formula for calculating the population variance.

We know that $\sigma^2 = \frac{[\sum(X_i - \mu)^2]}{N}$ where N is the size of the population

And $\mu = \frac{(\sum(X_i))}{N} = \frac{(0.24+0.26+0.30+0.18+0.20)}{5} = 0.236$

$$\sigma^2 = \frac{[(0.24-0.236)^2 + (0.26-0.236)^2 + (0.30-0.236)^2 + (0.18-0.236)^2 + (0.20-0.236)^2]}{5}$$
$$= \frac{(0.000016 + 0.000576 + 0.004096 + 0.003136 + 0.001296)}{5}$$
$$= 0.001824$$

Note: Had we been given sample data, the formula for the mean would remain unchanged but when calculating the variance, we would divide the sum of squared deviations by (n – 1) to remove bias.

Q.3258 The returns generated by a sample of five stocks from the Karachi Stock Exchange are given in the exhibit below.

Stock	Return
A	12%
B	13%
C	5%
D	4%
E	20%

What is the standard deviation of this sample?

A. 6.53%.

B. 5.84%.

C. 7.53%.

D. 8.34%

The correct answer is **A.**

$$\text{Mean} = \frac{(0.12 + 0.13 + 0.05 + 0.04 + 0.2)}{5} = 0.108$$

Stock	Return	$X - \text{Mean}$	$(X - \text{Mean})^2$
A	12%	1.2%	0.000144
B	13%	2.2%	0.000484
C	5%	−5.8%	0.003364
D	4%	−6.9%	0.004624
E	20%	9.2%	0.008464
Total			0.017080

$$\text{Sample deviation} = \left(\frac{1}{(n-1)}(X - \text{Mean})^2\right)^{\frac{1}{2}}$$

$$= \left(\frac{0.017080}{4}\right)^{\frac{1}{2}} = 6.53\%$$

Note: The standard deviation calculated with a divisor of $n - 1$ is a standard deviation calculated from the sample as an estimate of the standard deviation of the population from which the sample was drawn.

Q.3261 Which of the following statements is the *most* accurate? Lognormal Distributions are:

A. Skewed to the left and rarely used to model asset prices.

B. Skewed to the left and often used to model asset prices.

C. Skewed to the right and often used to model asset prices.

D. Skewed to the right and rarely used to model asset prices.

The correct answer is **C.**

Lognormal distributions are indeed skewed to the right and are frequently used to model asset prices. The reason for this is that lognormal distributions are bound by zero on the left side, which means they cannot take negative values. This characteristic makes them particularly suitable for modeling asset prices, as these prices also cannot take negative values. Furthermore, the right skewness of lognormal distributions allows them to capture the potential for large positive returns, which is a feature often observed in asset prices. Therefore, lognormal distributions are a popular choice in finance for modeling asset prices.

Choice A is incorrect. The Lognormal Distribution is not skewed to the left, but rather to the right. Furthermore, it is frequently used in financial modeling for asset prices due to its properties of non-negativity and skewness.

Choice B is incorrect. While it's true that the Lognormal Distribution is often used to model asset prices, this distribution isn't skewed to the left as stated in this option. It's actually skewed to the right.

Choice D is incorrect. This statement incorrectly suggests that Lognormal Distribution isn't commonly used for modeling asset prices and that it's skewed towards the right side which contradicts with its frequent usage in financial modeling due to its beneficial properties such as non-negativity and skewness towards right.

Things to Remember

- Lognormal distributions are commonly used in finance to model asset prices due to their properties of non-negativity and right skewness.
- Lognormal distributions are bounded by zero on the left side, making them suitable for modeling assets that cannot have negative prices.
- Skewness in a distribution refers to the asymmetry of the distribution, with right skewness indicating a longer right tail.
- Lognormal distributions are often used in option pricing models, risk management, and portfolio optimization in finance.

Q.3762 The following are the data on the financial analysis of a sales company's income over the last 200 months:

$$n = 200, \sum_{i=1}^{n} (x_i - \hat{\mu})^2 = 774,759.90 \text{ and } \sum_{i=1}^{n} (x_i - \hat{\mu})^3 = -13,476.784$$

What is the value of skewness?

A. --0.0002774

B. -0.00051738

C. -0.00031736

D. -0.00021733

The correct answer is **A.**

The skewness is given by:

$$\frac{\hat{\mu}^3}{\hat{\sigma}^3} = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^3}{\left[\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \hat{\mu})^2\right]^{\frac{3}{2}}} = \frac{\frac{1}{200}(-13,476.784)}{\left[\frac{1}{200-1} \times 774,759.90\right]^{\frac{3}{2}}} = -0.000274$$

The value of the skewness is slightly negative and not far from 0, implying the data is symmetrical.

Q.3763 A sample of 100 monthly profits gave out the following data:

$$\sum_{i=1}^{100} x_i = 3,453 \text{ and } \sum_{i=1}^{100} x_i^2 = 800,536$$

What is the sample mean and standard deviation of the monthly profits?

A. Sample Mean=33.53, Standard deviation=85.99

B. Sample Mean=34.53, Standard deviation=82.96

C. Sample Mean=43.53, Standard deviation=89.99

D. Sample Mean=33.63, Standard deviation=65.99

The correct answer is **B.**

Recall that the sample mean is given by:

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\Rightarrow \bar{X} = \frac{1}{100} \times 3,453 = 34.53$$

The variance is given by:

$$s^2 = \frac{1}{n-1}\{\sum_{i=1}^{n} X_i^2 - n\hat{\mu}^2\} = \frac{1}{99}(800,536 - 100 \times 34.53^2) = 6881.857677$$

So that the standard deviation is given to be:

$$s = \sqrt{6881.857677} = 82.96$$

Q.3766 The following data represents a sample of daily profit of a sales company for six weeks in a particular year.

Week	Amount of the Profit($)
1	3,800
2	2,800
3	2,700
4	9,900
5	2,600
6	4,300

What is the median of the profit?

A. 3,000

B. 5,000

C. 3,300

D. 3,700

The correct answer is **C.**

The first step would be to sort the data as follows:

Week	Amount of the Profit($)
5	2,600
3	2,700
2	2,800
1	3,800
6	4,300
4	9,900

Since the number of observations is even, the median is estimated by averaging the two consecutive central points. That is:

$$\text{Med}(x) = \frac{1}{2}[x_{\frac{6}{2}} + x_{\frac{6}{2}+1}] = \frac{1}{2}[x_3 + x_4] = \frac{1}{2}[2800 + 3800] = 3,300$$

Q.3767 The following data represents a sample of daily profit of a sales company for six weeks in a particular year.

Week	Amount of the Profit($)
1	3,800
2	2,800
3	2,700
4	9,900
5	2,600
6	4,300

What is the 25% quantile of the profits?

A. 2750

B. 2,800

C. 2,725

D. 2,625

The correct answer is **C.**

Method: Anchor the first observation at 0% and the last at 100%, equally spacing the in-between values with each division being 100/(n - 1) = 20%.

1. **Sort the Data:** The profits in ascending order are $2,600, $2,700, $2,800, $3,800, $4,300, and $9,900.
2. **Assign Percentiles:** Each observation is assigned a percentile:
    - 0% - $2,600
    - 20% - $2,700
    - 40% - $2,800
    - 60% - $3,800
    - 80% - $4,300
    - 100% - $9,900
3. **Find the 25% Quantile:** The 25% quantile falls between the 20% and 40% points, which are $2,700 and $2,800 respectively.
4. **Interpolate:** Calculate the value that is 25% of the way between $2,700 (20%) and $2,800 (40%):
    - Difference between these two values is $2,800 - $2,700 = $100.

- 25% of $100 is $25.
- Therefore, the 25% quantile is $2,700 + $25 = $2,725.

Q.3768 The following data represents a sample of daily profit of a sales company for six weeks in a particular year.

Week	Amount of the Profit($)
1	3,800
2	2,800
3	2,700
4	9,900
5	2,600
6	4,300

What is the 75% percentile profit?

A. 4,175

B. 3,925

C. 4,055

D. 2,720

The correct answer is **A.**

The first step would be to sort the data as follows:

Week	Amount of the Profit($)
5	2,600
3	2,700
2	2,800
1	3,800
6	4,300
4	9,900

Now, to find the 75% percentile, we need to find the 75% level, which is 75% of the way between the 4th and the 5th observations. Therefore,

$$q_{75} = 0.25 \times 3,800 + 0.75 \times 4,300 = 4,175$$

Q.3769 The following data represents a sample of daily profit of a sales company for six weeks in a particular year.

Week	Amount of the Profit($)
1	3,800
2	2,800
3	2,700
4	9,900
5	2,600
6	4,300

What is the interquartile range?

A. 1,600

B. 1,550

C. 1,475

D. 1,450

The correct answer is **D.**

Recall that:

$$IQR = q_3 - q_1$$

The first step would be to sort the data as follows:

Week	Amount of the Profit($)
5	2,600
3	2,700
2	2,800
1	3,800
6	4,300
4	9,900

We estimate the α-quantile using the data point in location $alpha \times n$.

Now, to find the 25% quantile, we need to find the 25% level, which is 25% of the way between ranked observations 2 and 3. Therefore,

$$q_{25} = 0.75 \times 2,700 + 0.25 \times 2,800 = 2,725$$

Similarly, to find q_3, we need to find the 75% level, which is 75% of the way between the 4th and the 5th observations. Therefore,

$$q_{75} = 0.25 \times 3,800 + 0.75 \times 4,300 = 4,175$$

Therefore,

$$IQR = 4,175 - 2,725 = 1,450$$

Please note that the method used here is linear interpolation.

Q.3770 What are the conventional values of skewness and kurtosis of a normal random variable?

A. Skewness=0, kurtosis=3

B. Skewness=1, kurtosis=3

C. Skewness=0, kurtosis=2

D. Skewness=0, kurtosis=4

The correct answer is **A.**

In a normal distribution, the skewness is 0. This is because skewness measures the asymmetry of a distribution, and a normal distribution is perfectly symmetrical. Therefore, there is no skewness. The kurtosis of a normal distribution is 3. Kurtosis measures the 'tailedness' of a distribution. A kurtosis of 3 indicates that the distribution has neither fat nor thin tails, which is characteristic of a normal distribution. Therefore, the conventional values of skewness and kurtosis for a normal random variable are 0 and 3, respectively.

Choice B is incorrect. Skewness measures the asymmetry of a probability distribution about its mean. A skewness of 1 implies a significant degree of skewness, which contradicts the symmetric property of a normal distribution.

Choice C is incorrect. Kurtosis measures the "tailedness" or outliers of a probability distribution. A kurtosis value less than 3 indicates that the distribution has lighter tails and fewer outliers than the normal distribution, which is not characteristic of a normal random variable.

Choice D is incorrect. While it's true that higher kurtosis can indicate more extreme values or "fat tails," for a standard normal random variable, excess kurtosis (kurtosis above that of the normal distribution) should be zero, making total kurtosis equal to 3, not 4.

Things to Remember

- Skewness measures the asymmetry of a distribution around its mean. A skewness of 0 indicates a perfectly symmetrical distribution.
- Kurtosis measures the "tailedness" of a distribution. A kurtosis of 3 indicates a normal distribution with neither fat nor thin tails.
- For a normal random variable, the conventional values of skewness and kurtosis are 0 and 3, respectively.
- Skewness and kurtosis are important measures in understanding the shape and characteristics of a probability distribution.

Q.3771 A sample amount of profit for a certain company for the first 15 weeks of the year is given below:

Weeks	Amount of the Profit($)
1	0
2	7,000
3	13,000
4	13,000
5	20,000
6	23,000
7	25,000
8	27,000
9	34,000
10	41,000
11	60,000
12	66,000
13	76,000
14	77,000
15	96,000

What is the difference between the biased and an unbiased estimator of the sample mean?

A. 0

B. 1.0

C. 3.2

D. 4.6

The correct answer is **A.**

This question does not require any calculation whatsoever since the sample always mean unbiased. So, the difference is just by itself, which is zero.

Q.3772 A sample amount of profit for a certain company for the first 15 weeks of the year is given below:

Weeks	Amount of the Profit($)
1	0
2	70
3	130
4	130
5	200
6	230
7	250
8	270
9	340
10	410
11	600
12	660
13	760
14	770
15	960

What is the difference between the biased and unbiased estimators of the variance?

A. 5,867.50

B. 5,503.90

C. 5,767.49

D. 5,867.87

The correct answer is **C.**

Study the prefilled table below

Weeks	Amount of the Profit($) = X_i	$(X_i - \hat{\mu})^2$
1	0	148481.778
2	70	99435.111
3	130	65195.111
4	130	65195.111
5	200	34348.444
6	230	24128.444
7	250	18315.111
8	270	13301.778
9	340	2055.111
10	410	608.444
11	600	46081.778
12	660	75441.778
13	760	140375.111
14	770	147968.444
15	960	330241.778
	385.3333	1211173.333

We know that biased estimator of the sample variance is given by:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

Therefore, we need to calculate the sample mean of the data:

$$\bar{X} = \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i = \$385.33$$

So that,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 = \frac{1}{15} \times 1211173.333 = 80744.89$$

Recall that the unbiased estimator of the sample variance is given by:

$$s^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{15}{14} \times 80744.89 = 86512.38$$

So, the difference is given by:

$$86512.38 - 80744.89 = 5,767.49$$

Q.3773 The following are the two series of data X and Y.

X	Y
700	318
130	304
140	317
200	305
230	309
250	307
270	316
340	309
400	315
620	327
620	450
760	324
650	500
750	699

Assuming the Central limit theorem, what is the correct representation of the sample means distributions of the random variables above?

A.

$$\begin{bmatrix} 432.86 \\ 264.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} 4058.71 & 1106.37 \\ 3106.37 & 914.81 \end{bmatrix}\right)$$

B.

$$\begin{bmatrix} 432.86 \\ 464.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} 4058.71 & 1106.37 \\ 2106.37 & 914.81 \end{bmatrix}\right)$$

C.

$$\begin{bmatrix} 432.86 \\ 364.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} 4058.71 & 1106.37 \\ 1106.37 & 914.81 \end{bmatrix}\right)$$

D.

$$\begin{bmatrix} 432.86 \\ 324.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} 4458.71 & 1146.37 \\ 1406.37 & 914.81 \end{bmatrix}\right)$$

The correct answer is **C.**

Recall that according to CLT for bivariate data, the vector of means is usually distributed. That is:

$$\begin{bmatrix} \hat{\mu}_x \\ \hat{\mu}_y \end{bmatrix} \rightarrow N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \frac{\sigma_X^2}{n} & \frac{\sigma_{XY}}{n} \\ \frac{\sigma_{XY}}{n} & \frac{\sigma_Y^2}{n} \end{bmatrix}\right)$$

This implies that we need to calculate the mean and variance of each series and their covariance. Study the following table:

X	Y	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
700	318	71365.3061	2142.36735	−12364.9
130	304	91722.449	3634.36735	18257.96
140	317	85765.3061	2235.93878	13847.96
200	305	54222.449	3514.79592	13805.1
230	309	41151.0204	3056.5102	11215.1
250	307	33436.7347	3281.65306	10475.1
270	316	26522.449	2331.5102	7863.673
340	309	8622.44898	3056.5102	5133.673
400	315	1079.59184	2429.08163	1619.388
620	327	35022.449	1390.22449	−6977.76
620	450	35022.449	7346.93878	16040.82
760	324	107022.449	1622.93878	−13179.2
650	500	47151.0204	18418.3673	29469.39
750	699	100579.592	112033.653	106152.2

Starting with the mean, we have:

$$\hat{\mu}_X = \bar{X}_X = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{6060}{14} = 432.8571$$

$$\hat{\mu}_Y = \bar{X}_Y = \frac{1}{n}\sum_{i=1}^{n} Y_i = \frac{5100}{14} = 364.2857$$

We go to the variances (we shall use the unbiased estimator)

$$\hat{\sigma}_X^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \bar{X})^2 = \frac{738685.714}{13} = 56821.978$$

$$\hat{\sigma}_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \frac{166494.857}{13} = 12807.2967$$

The last part involves computing the sample covariance given by:

$$\text{Cov}(X, Y) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) = \frac{201358.60}{13} = 15489.12$$

We can now apply the CLT:

$$\begin{bmatrix} 432.86 \\ 364.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \frac{56821.978}{14} & \frac{15489.12}{14} \\ \frac{15489.12}{14} & \frac{12807.2967}{14} \end{bmatrix}\right)$$

$$\begin{bmatrix} 432.86 \\ 364.29 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} 4058.71 & 1106.37 \\ 1106.37 & 914.81 \end{bmatrix}\right)$$

Q.3777 An analyst gathers monthly data about the returns of a stock for the past five years. If the mean monthly return is 6% and the standard deviation of the series of returns is 1.8%, then what is the standard deviation of the mean over the period?

A. 6.24%

B. 0.23%

C. 13.94%

D. 4.02%

The correct answer is **B.**

To calculate the standard deviation of the mean (also known as the standard error), you can use the following formula:

Standard Error (SE) = (Standard Deviation) / sqrt(n)

Where *n* is the number of observations.

Since the analyst has collected monthly data for five years, there are a total of 5 years * 12 months/year = 60 observations.

Given the standard deviation of the series of returns is 1.8%, we can now calculate the standard error:

$SE = \frac{1.8\%}{\sqrt{60}} \approx \frac{1.8\%}{7.75} \approx 0.232\%$ (approximately)

So, the standard deviation of the mean over the period is approximately 0.232%.

Things to Remember

- The standard deviation of a series of returns measures the dispersion of those returns around the mean.
- The standard error of the mean (SE) is a measure of how spread out the sample means are from the population mean.
- The formula for calculating the standard error of the mean is SE = (Standard Deviation) / sqrt(n), where n is the number of observations.
- When calculating the standard error, it is important to use the correct units and ensure consistency in the calculations.
- Understanding the concept of standard deviation and standard error is crucial in risk management and statistical analysis.

Q.3778 Assume we have equally invested in two different companies; ABC and XYZ. We anticipate that there is a 15% chance that next year's stock returns for ABC Corp will be 6%, a 60% probability that they will be 8% and a 25% probability that they will be 10%. In addition, we already know the expected value of returns is 8.2%, and the standard deviation is 1.249%. We also anticipate that the same probabilities and states are associated with a 4% return for XYZ Corp, a 5% return, and a 5.5% return. The expected value of returns is then 4.975%, and the standard deviation is 0.46%. Calculate the portfolio standard deviation:

A. 5.61e-05

B. 0

C. 0.00849

D. 0.00897

The correct answer is **C.**

The portfolio variance is given by:

$$\text{Portfolio variance} = W_A^2 \times \sigma^2(R_A) + W_B^2 \times \sigma^2(R_B) + 2 \times (W_A) \times (W_B) \times \text{Cov}(R_A, R_B)$$

First, we must calculate the covariance between the two stocks. Now using

$$\begin{aligned}
\sigma_{R_i R_j} &= \sum_{i=1}^{n} P(R_i)\,[R_i - E(R_i)]\,[R_j - E(R_j)] \\
\text{cov}(ABC, XYZ) &= 0.15(0.06 – 0.082)(0.04 – 0.04975) \\
&+ 0.6(0.08 – 0.082)(0.05 – 0.04975) \\
&+ 0.25(0.10 – 0.082)(0.055 – 0.04975) \\
&= 0.0000555
\end{aligned}$$

Since we already have the weight and the standard deviation of each asset, we can proceed and calculate the portfolio variance:

$$\begin{aligned}
&= 0.5^2 \times 0.01249^2 + 0.5^2 \times 0.0046^2 + 2 \times 0.5 \times 0.5 \times 0.0000555 \\
&= 0.000072040
\end{aligned}$$

Therefore, the standard deviation is $(0.000072040)^{\frac{1}{2}} = 0.00849$

Q.3779 A sample of 36 working days was analyzed; for the amount of income of a company. If the income has a standard deviation of 7, what is the approximate probability that the mean of this sample is greater than 44.50, assuming that the mean of the yearly income is $\mu=42$? Please click here to see the standard normal table.

A. 0.045

B. 0.016

C. 0.065

D. 0.042

The correct answer is **B.**

According to the Central Limit Theorem,

$$\frac{\hat{\mu}-\mu}{\frac{\sigma}{\sqrt{n}}} \rightarrow N(0,1)$$

Note that $\hat{\mu} = \bar{X}$ = Sample Mean

We need:

$$P(\bar{X} > 44.5) = P\left[\frac{\hat{\mu}-\mu}{\frac{\sigma}{\sqrt{n}}} > \frac{44.5-42}{\frac{7}{\sqrt{36}}}\right]$$
$$\Rightarrow P(\bar{X} > 44.5) \approx P(Z > 2.143) = 1-\varphi(2.143) = 1-0.9839 = 0.0161$$

Q.3797 A portfolio is composed of 60% equities and 40% bonds. The variance of equities is 320, the variance of bonds is 110, and the covariance is 90. What is the portfolio's variance?

A. 154.4

B. 176

C. 192

D. 279.2

The correct answer is **B.**

$$\begin{aligned}\text{Portfolio variance} &= w_A^2 * s^2(R_A) + w_B^2 * s^2(R_B) + 2 * (w_A) * (w_B) * \text{Cov}(R_A, R_B) \\ &= (0.6)^2 * (320) + (0.4)^2 * (110) + 2 * (0.6) * (0.4) * (90) = 115.2 + 17.6 + 43.2 \\ &= 176\end{aligned}$$

Q.3798 Alan West, a portfolio manager, created the following portfolio:

Security	Security Weight (%)	Expected Standard deviation(%)
A	20	4
B	80	10

If the correlation of returns between the two securities is 0.60, then what is the expected standard deviation of the portfolio?

A. 9.50%

B. 8.10%

C. 9.15%.

D. 8.50%.

The correct answer is **D.**

$$\begin{aligned}\text{Portfolio standard deviation} &= [(0.2)^2(4\%)^2 + (0.8)^2(10\%)^2 + 2(0.2)(0.8)(0.6)(4\%)(10\%)]^{0.5} \\ &= 8.50\%\end{aligned}$$

Here, we're simply using the portfolio variance formula and raising it to the 0.5 power to get the portfolio standard deviation.

Q.3799 Raul Perez, a portfolio manager, created the following portfolio:

Security	Security Weight (%)	Expected Standard deviation(%)
A	40	7
B	60	12

If the covariance of returns between the two securities is -0.004, then what is the expected standard deviation of the portfolio?

A. 6.36%

B. 6.56%

C. 8.14%

D. 6.10%

The correct answer is **A.**

$$\begin{aligned}\sigma^2{}_P &= w^2{}_1\sigma^2{}_1 + w^2{}_2\sigma^2{}_2 + 2w_1w_2Cov(R_1, R_2)\\ &= [(0.4)^2(7\%)^2 + (0.6)^2(12\%)^2 + 2(0.4)(0.6)(-0.004)]\\ &= 0.004048\\ \sigma_P &= 0.004048^{0.5} = 0.063624\end{aligned}$$

Q.3800 Tina Fer, a portfolio manager, created the following portfolio:

Security	Security Weight (%)	Expected Standard deviation(%)
A	10	6
B	90	15

If the standard deviation of the portfolio is 14.1%, then what is the covariance between the two securities?

A. 0.009

B. 0.09

C. 0.008

D. -0.011

The correct answer is **A.**

We know that:

$$Var(A,B) = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2W_AW_BCov(A,B)$$

Where
w_A^2 = weight of security A.
w_B^2 = weight of security B.
σ_A^2 = variance of security A
σ_B^2 = variance of security B
$Cov(A,B)$ = covariance between securities A and B.
Plugging in the values given in the table, we have:

$$(0.141)^2 = 0.10^2 \times 0.06^2 + 0.90^2 \times 0.15^2 + 2 \times 0.10 \times 0.90 \times Cov(A,B)$$
$$\Rightarrow Cov(A,B) = \frac{(0.141)^2 - (0.10^2 \times 0.06^2 + 0.90^2 \times 0.15^2)}{2 \times 0.10 \times 0.90} = 0.009$$

Q.3801 Carla Mayes, a portfolio manager created the following portfolio:

Security	Expected Return (%)	Expected Standard Deviation(%)
A	5	8
B	10	14

If the correlation of returns between the two securities is -0.20, then what is the standard deviation of a portfolio invested 75% in Security A and 25% in Security B?

A. 0.51%

B. 0.81%

C. 5.12%

D. 6.31%

The correct answer is **D.**

We know that:

$$\begin{aligned}\sigma_P^2 &= w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\rho_{AB}\sigma_A\sigma_B \\ &= (0.75)^2(0.08)^2 + (0.25)^2(0.14)^2 + 2(0.75)(0.25)(-0.20)(0.08)(0.14) \\ &= 0.003985 = 0.3985\%\end{aligned}$$

The standard deviation of the portfolio = 6.3120%

Q.3805 Assuming that the covariance of returns of Stock X and Stock Y is $Cov(R_X, R_Y) = 0.093$, the variance of $R_X = 0.69$, and the variance of $R_Y = 0.36$, what is the correlation of returns of Stock X and Stock Y?

A. 0.155

B. 0.1865

C. 0.1713

D. 0.1119

The correct answer is **B.**

$Corr\ (R_X, R_Y) = Cov\ R_X, R_Y\ /\ \sigma(R_X)\ \sigma(R_Y)$

Since Variance = σ^2,
$\sigma(R_X) = \sqrt{0.69} = 0.8306$, and
$\sigma(R_Y) = \sqrt{0.36} = 0.6$

$Corr\ R_X, R_Y = Cov\ R_X, R_Y\ /\ \sigma(R_X)\ \sigma(R_Y) = 0.093/0.8306 * 0.6 = 0.1865$

Things to Remember

- Covariance measures the relationship between the returns of two assets. A positive covariance indicates that the returns of the two assets move together, while a negative covariance indicates they move in opposite directions.
- Variance is a measure of the dispersion of returns for a single asset. It shows how much the returns deviate from the average return.
- Correlation is a standardized measure of the relationship between two assets. It ranges from -1 to 1, where 1 indicates a perfect positive correlation, -1 indicates a perfect negative correlation, and 0 indicates no correlation.
- Correlation helps in diversifying a portfolio as assets with low or negative correlation can reduce overall portfolio risk.

Q.3807 A junior fund manager at Dapper Assets Management is constructing a portfolio consisting of two large-cap stocks that trade on the London stock exchange. In a meeting with the investment committee, the manager was asked to present the covariance of both stocks. Using the data given in the following table, calculate the covariance if the population mean is unknown.

Year	Stock A Return	Stock B Return
1	17%	45%
2	21%	20%
3	-8%	-2%
4	-1%	2%
5	4%	-19%
6	19%	2%
7	-7%	13%

A. 0.0113

B. 0.101

C. 0.1156

D. 0.00907

The correct answer is **A.**

Below are the calculations:

$$\bar{X}_A = \frac{0.17 + 0.21 - 0.08 - 0.01 + 0.04 + 0.19 - 0.07}{7} = 0.0642$$

$$\bar{X}_B = \frac{0.45 + 0.20 - 0.02 + 0.02 - 0.19 + 0.02 - 0.13}{7} = 0.0871$$

Stock A Return Mean Return of A	Stock B Return minus Mean Return of B	Covariance = (Stock A Return minus Mean Return of A) × (Stock B Return minus Mean Return of B)
0.1057	0.3629	0.0384
0.1457	0.1129	0.0164
-0.1443	-0.1071	0.0155
-0.0743	-0.0671	0.0050
-0.0243	-0.2771	0.0067
0.1257	-0.0671	-0.0084
-0.1343	0.0429	-0.0058

$0.0384 + 0.0164 + 0.0155 + 0.0050 + 0.0067 - 0.0084 - 0.0058$

$$\begin{aligned} Cov &= \frac{0.0384 + 0.0164 + 0.0155 + 0.0050 + 0.0067 - 0.0084 - 0.0058}{(n-1)} \\ &= \frac{0.0384 + 0.0164 + 0.0155 + 0.0050 + 0.0067 - 0.0084 - 0.0058}{6} \\ &= 0.0113 \end{aligned}$$

The reason we use "n-1" and not "n" is essentially that the population mean E(X) is not known and is replaced by the sample mean x-bar.
Note: You can also do the problem with the help of the financial calculator by using the STAT function and using the sample standard deviation.

Q.3809 Hakim Ahmed has recently joined Lampard Investment Inc. He was given the data related to the assets of a portfolio provided in the following table. If the weight of Asset X is 35% and the weight of Asset Z is 65%, then what is the variance of the portfolio?

Variance Asset X	0.1225
Variance Asset Z	0.4225
Covariance	0.19

A. 0.28

B. 0.1156

C. 0.2245

D. 0.2587

The correct answer is **A.**

$$\begin{aligned} \text{Portfolio Variance} &= w_X^2\sigma_X^2 + w_Z^2\sigma_Z^2 + 2 * w_X w_Z * Cov(XZ) \\ &= 0.35^2 * 0.1225 + 0.65^2 * 0.4225 + 2 * 0.35 * 0.65 * 0.19 \\ &= 0.2800 \end{aligned}$$

Q.3810 Hakim Ahmed has recently joined Lampard Investment Inc. He has been given data related to the assets of a client's portfolio provided in the following table:

Variance Asset X	0.1225
Variance Asset Z	0.3721
Covariance	0.19

If the weight of Asset X is 35% and the weight of Asset Z is 65%, then what is the correlation coefficient between Assets X and Z?

A. 0.8899

B. 0.0469

C. 0.4412

D. 0.4168

The correct answer is **A.**

tandard deviation of X = $0.1225^{1/2}$ = 0.35

Standard deviation of Z = $0.3721^{1/2}$ = 0.61

Correlation coefficient = Covariance(X,Z)/(Standard deviation of X * Standard deviation of Z) = 0.19/(0.35 * 0.61) = 0.8899

Note: The correlation coefficient does not take into account the weights.

Q.3812 A portfolio consists of two funds A and B. The weights of the two funds in the portfolio and the covariance matrix of the two funds are given in the following two exhibits.

Exhibit 1: Weight of the Funds in the Portfolio

Fund	A	B
Weight	60%	40%

Exhibit 2: Covariance Matrix

Fund	A	B
A	700	200
B	200	500

What is the portfolio variance?

A. 428

B. 500

C. 512

D. 324.8

The correct answer is **A.**

$$\begin{aligned}\text{Portfolio Variance} &= (w_A^2 \times \sigma_A^2) + (w_B^2 \times \sigma_B^2) + (2 \times w_A \times w_B \times \text{Cov}_{AB}) \\ &= (0.6^2 \times 700) + (0.4^2 \times 500) + (2 \times 0.6 \times 0.4 \times 200) \\ &= (0.36 \times 700) + (0.16 \times 500) + (0.48 \times 200) \\ &= 252 + 80 + 96 \\ &= 428\end{aligned}$$

The portfolio variance is 428.

Reading 17: Hypothesis Testing

Q.371 The mean hourly wage for coal workers in the U.S. is \$15.5 with a population standard deviation of \$3.2. Calculate the standard error of the sample mean if the sample size is 30.

A. 3.2

B. 0.3413

C. 0.1067

D. 0.5842

The correct answer is **D.**

Since the standard deviation for the population is known,
Standard error of the mean = $\frac{\sigma}{\sqrt{n}} = \frac{3.2}{\sqrt{30}} = 0.5842$
Interpretation: If we were to take a number of samples of size 30 from U.S. coal workers population and proceed to prepare a sampling distribution of the sample means, the distribution would have a mean of \$15.5 and a standard error of \$0.58.

Note: In most cases, σ is unknown in which case we work with the sample standard deviation, s.

Things to Remember

- Standard Error of the Mean (SEM) is a measure of how much the sample mean is expected to vary from the population mean.
- SEM is calculated as the population standard deviation divided by the square root of the sample size.
- As the sample size increases, the standard error of the mean decreases, indicating a more precise estimate of the population mean.
- Standard error is a crucial concept in hypothesis testing and confidence interval construction.

Q.373 An auto insurance company intends to establish the mean claim amount demanded by policyholders who own SUVs. After extensive analysis of its records, the company believes the standard deviation of such claims is about $200.
The company wishes to construct a 95% confidence interval for the mean claim amount such that the interval is of width "±$50". Determine the value of *n*, the sample size that would be required to achieve this.

A. 62

B. 100

C. 30

D. 124

The correct answer is **A.**

A 95% CI for the mean μ, assuming normal distribution = $\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$

Since σ = 200, we need to find the value of *n* such that:

$$\begin{aligned} 1.96 \times \frac{\sigma}{\sqrt{n}} &= 50 \\ 1.96 \times 200 &= 50\sqrt{n} \\ 392 &= 50\sqrt{n} \\ \sqrt{n} &= 7.84 \\ n &= 62 \end{aligned}$$

Q.374 After 72 FRM Part 1 students took a mock exam, the mean score was 75. Assuming that the population standard deviation is 10, construct a 99% confidence interval for the mean score on the mock exam.

A. (75, 85)

B. (65, 75)

C. (71.96, 78.04)

D. (75, 78.04)

The correct answer is **C.**

A 99% CI for the mean, $\mu = \bar{x} \pm Z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$ where $\alpha = 0.01$

From the normal dist. table, $Z_{0.005} = 2.58$

$$\mu = 75 \pm 2.58 \times \frac{10}{\sqrt{72}} = 75 \pm 3.04$$

Giving us a CI = $71.96 \leq \mu \leq 78.04$

Q.375 Which of the following best defines the term “hypothesis” as used in statistics?

A. An assumption about a problem.

B. The current state of knowledge or belief about the value of a population parameter.

C. A test that divides the sample space into a region of acceptance and the critical region.

D. A statement about the value of a population parameter developed to test a theory or belief.

The correct answer is **D.**

A hypothesis in statistics is a statement about the value of a population parameter that is developed to test a theory or belief. It describes an assumption about the true value of the parameter, such as the mean or standard deviation. For example, a hypothesis may be formulated as "the mean monthly return on stock options is positive" or "the mean monthly return on stock options is greater than 10%". The purpose of a hypothesis is to test the validity and accuracy of a belief about the parameter, and to determine whether the observed data supports or contradicts the hypothesis.

Option A is incorrect. While an assumption may be made about a problem, it does not necessarily constitute a hypothesis. A hypothesis is a specific statement about the value of a population parameter that is developed to test a theory or belief.

Option B is incorrect. The current state of knowledge or belief about the value of a population parameter is not a hypothesis. A hypothesis is a specific statement about the value of the parameter that is developed to test a theory or belief.

Option C is incorrect. A test that divides the sample space into a region of acceptance and the critical region is a hypothesis test, which is a statistical method used to determine whether the observed data supports or contradicts a hypothesis. However, this option does not define the term "hypothesis" itself.

Things to Remember

- Null Hypothesis vs. Alternative Hypothesis: In hypothesis testing, the null hypothesis (H0) is a statement that there is no significant difference or relationship between variables, while the alternative hypothesis (Ha) is a statement that there is a significant difference or relationship.

- Type I Error and Type II Error: Type I error occurs when the null hypothesis is rejected when it is actually true, while Type II error occurs when the null hypothesis is not rejected when it is actually false.

- Significance Level: The significance level (alpha) is the probability of committing a Type I error, typically set at 0.05 or 0.01 in hypothesis testing.

- P-value: The p-value is the probability of obtaining results as extreme as the observed data, assuming that the null hypothesis is true. A lower p-value indicates stronger evidence against the null hypothesis.

- One-Tailed vs. Two-Tailed Tests: In hypothesis testing, a one-tailed test is used when the alternative hypothesis specifies the direction of the effect (e.g., greater than, less than), while a two-tailed test is used when the alternative hypothesis does not specify a direction.

Q.378 A random sample of 50 FRM exam candidates was found to have an average IQ of 125. The standard deviation among candidates is known (approximately 20). Assuming that IQs follow a normal distribution, carry out a statistical test (5% significance level) to determine whether the average IQ of FRM candidates is greater than 120. Compute the test statistic and give a conclusion.

A. Test statistic: 1.768; Reject H_0

B. Test statistic: 2.828; Reject H_0

C. Test statistic: 1.768; Fail to reject H_0

D. Test statistic: 1.0606; Fail to reject H_0

The correct answer is **A.**

The first step: Formulate H_0 and H_1

$H_0 : \mu = 120$

$H_1 : \mu > 120$

Note that this is a one-sided test because H_1 explores a change in one direction only

Under H_0, $\frac{(\bar{x}-120)}{\left(\frac{\sigma}{\sqrt{n}}\right)} \sim N(0, 1)$

Next, compute the test statistic:

$$= \frac{(125 - 120)}{\left(\frac{20}{\sqrt{50}}\right)} = 1.768$$

Next, we can confirm that P(Z > 1.6449) = 0.05, which means our critical value is the upper 5% point of the normal distribution i.e. 1.6449. Since 1.768 is greater than 1.6449, it lies in the rejection region. As such, we have sufficient evidence to reject H_0 and conclude that the average IQ of FRM candidates is indeed greater than 120.

Alternatively, we could go the "p-value way"

$$P(Z > 1.768) = 1 - P(Z < 1.768) = 1 - 0.96147 = 0.03853 \text{ or } 3.853\%$$

This probability is less than 5% meaning that we have sufficient evidence against H_0. This approach leads to a similar conclusion.

Q.379 A stock has an initial market price of $80. Exactly one year from now, its price will be given by:
P = 80 * exp(i) where *i* is the rate of return.

i is normally distributed with mean 0.2 and standard deviation 0.3. Construct a 95% confidence interval for the price of the stock after one year.

A. ($54.27, $80)

B. ($80, $175.91)

C. ($54.27, $175.91)

D. ($54.27, $140)

The correct answer is **C.**

A 95% CI for the price $= \mu \pm Z_{\frac{\alpha}{2}} \times \sigma$ where $\alpha = 0.05$

$$\begin{aligned} &= \mu \pm 1.96 \times 0.3 \\ &= \mu \pm 0.588 \end{aligned}$$

Therefore,

$$\begin{aligned} P &= 80 * \exp(0.2 \pm 0.588) \\ &= (\$54.27, \$175.91) \end{aligned}$$

Note: "exp" stands for "exponential"

Note:

When considering the confidence interval for the price of an asset, we only consider the assets expected price, its standard deviation, and critical values from the normal distribution, i.e.

CI = Expected price +/- critical value * standard deviation.

Q.380 A manager conducts a hypothesis test at the 1% significance level. What does this mean?

A. P(reject H_0 | H_0 is false) = 0.01

B. P(reject H_0 | H_0 is true) = 0.01

C. P(not reject H_0 | H_0 is false) = 0.01

D. P(not reject H_0 | H_0 is true) = 0.01

The correct answer is **B.**

The significance level represents the probability of committing a type I error or rejecting a correct model. This is equivalent to P(reject H_0 | H_0 is true).

Option A is incorrect. P(reject H_0 | H_0 is false) represents the power of the test, which is the probability of correctly rejecting a false null hypothesis.

Option C is incorrect. P(not reject H_0 | H_0 is false) represents the complement of the significance level and is equivalent to the confidence level of the test.

Option D is incorrect. P(not reject H_0 | H_0 is true) represents the power of the test, which is the probability of correctly not rejecting a true null hypothesis.

Things to Remember

- Hypothesis testing is a statistical method used to make inferences about a population parameter based on sample data.
- The null hypothesis (H_0) represents the default assumption that there is no significant difference or relationship.
- The alternative hypothesis (H_1 or H_a) represents the claim that is being tested.
- The significance level, denoted by α, is the probability of rejecting the null hypothesis when it is actually true.
- A common significance level used in hypothesis testing is 0.05 or 5%, but other levels such as 1% or 10% can also be chosen based on the context of the problem.
- Type I error occurs when the null hypothesis is rejected when it is actually true, and the probability of committing a Type I error is equal to the significance level.
- Type II error occurs when the null hypothesis is not rejected when it is actually false, and the probability of committing a Type II error is denoted by β.

Q.381 Suppose you conducted a hypothesis test. What would happen if you decrease the level of significance of the test?

A. The likelihood of committing a type II error decreases

B. The likelihood of a type I error increases

C. The likelihood of rejecting the null hypothesis when it's in fact true decreases

D. The likelihood of frequently committing a type I error increases, even when it's in fact true

The correct answer is **C.**

When conducting a hypothesis test, the level of significance (denoted by α) represents the probability of rejecting the null hypothesis when it is actually true. Therefore, decreasing the level of significance would decrease the likelihood of rejecting the null hypothesis when it is actually true, reducing the probability of committing a type I error.

Option A is incorrect. The likelihood of committing a type II error (failing to reject a false null hypothesis) is not directly affected by the level of significance. However, decreasing the level of significance may increase the probability of committing a type II error, as it reduces the power of the test to detect a true alternative hypothesis.

Option B is incorrect. Decreasing the level of significance would decrease the probability of committing a type I error (rejecting a true null hypothesis), not increase it.

Option D is incorrect. The likelihood of frequently committing a type I error is not directly related to the level of significance. However, decreasing the level of significance may increase the probability of committing a type I error, as it reduces the threshold for rejecting the null hypothesis.

Things to Remember

- Level of significance (α) in hypothesis testing represents the probability of rejecting the null hypothesis when it is actually true.
- Type I error occurs when the null hypothesis is rejected when it is actually true.
- Type II error occurs when the null hypothesis is not rejected when it is actually false.
- Power of a test is the probability of correctly rejecting a false null hypothesis (1 -Type II error).
- Decreasing the level of significance reduces the probability of committing a Type I error but may increase the probability of committing a Type II error.

Q.383 At a 95% confidence interval, the value at risk (VaR) of a portfolio is approximately $10 million. During 100 days, the VaR was exceeded on 9 different occasions. Based on this information:

A. This model is overestimating risk.

B. This model is underestimating risk.

C. This model is appropriate for estimating the risk.

D. The model is accurate.

The correct answer is **B.**

The 95% confidence interval (CI) indicates that we expect losses exceeding $10 million to occur just 5% of the time. Therefore, during a 100-day period, we would expect a maximum of 5 exceedances. The fact that 9 exceedances were recorded suggests that the model is underestimating risk, and is therefore not appropriate for estimating the risk accurately.

Things to Remember

- Value at Risk (VaR) is a measure used to estimate the potential loss on an investment or portfolio over a specific time period under normal market conditions.
- Confidence interval (CI) is a range of values that is likely to contain the true value of an unknown population parameter.
- When VaR is exceeded more frequently than expected based on the confidence level, it indicates that the model is underestimating risk.
- It is important to regularly review and update risk models to ensure they accurately reflect the current market conditions and risks.

Q.584 Construct a 95% confidence interval for the future value of a pension fund where the number of simulations is 100, the mean ending value is $400,000, and the standard deviation is $23,300.

A. ($395,433.2, $404,566.8)

B. ($400,000, $404,613)

C. ($395,456, $404,456)

D. ($395, $404)

The correct answer is **A.**

The confidence interval is constructed using the normal distribution, not the student's t-distribution because n is large (in line with the central limit theorem)

The interval is given by:

$$\bar{X} - 1.96 * \left(\frac{s}{\sqrt{N}}\right), \bar{X} + 1.96 * \left(\frac{s}{\sqrt{N}}\right)$$

Thus,

$$\begin{aligned} CI &= \$400,000 - 1.96\left(\frac{\$23,300}{\sqrt{100}}\right), \$400,000 + 1.96\left(\frac{\$23,300}{\sqrt{100}}\right) \\ &= (\$395,433.2, \$404,566.8) \end{aligned}$$

Q.3281 A sample of 121 applicants received the Canadian travel visa in 45 days on average. Suppose the population is normally distributed, and the standard deviation of the sample is 19, then what is the 95% confidence interval for the population mean?

A. [44.7 days; 45.3 days]

B. [41.6 days; 48.4 days]

C. [42.2 days; 47.8 days]

D. [40.1 days; 49.8 days]

The correct answer is **B.**

Before calculating the confidence interval, we will calculate the Standard error of the sample mean:
Standard error of the sample mean = Standard deviation of the sample mean/Sample size = 19/ √121 = 1.73
The z-static at the 95% confidence interval is 1.96.

The confidence interval is calculated as Mean +/- Reliability factor * Standard error
Lower limit of the confidence interval = 45 - (1.96 * 1.73) = 41.60
Upper limit of the confidence interval = 45 + (1.96 * 1.73) = 48.40

Things to Remember

- Population mean and sample mean
- Standard deviation and standard error
- Normal distribution and z-score
- Confidence intervals and reliability factors

Q.3282 A sample of 100 students is currently renting rooms in the mean distance of 18 miles from a small U.S. College. Assuming that the population is normally distributed and the standard deviation of the sample is 14 miles, what is the 99% confidence interval for the population mean?

A. [15.26 miles; 20.74 miles]

B. [16.6 miles; 19.4 miles]

C. [14.4 miles; 21.6 miles]

D. [12.8 miles; 23.6 miles]

The correct answer is **C.**

$$\text{Standard error of the sample} = \frac{\text{Standard deviation of sample mean}}{\sqrt{\text{Sample size}}} = \frac{14}{\sqrt{100}} = 1.4$$

Z-static (Reliability factor) at the 99% confidence interval = 2.58

Lower limit of the confidence interval = $18 - (2.58 * 1.4) = 14.39$ miles

Upper limit of the confidence interval = $18 + (2.58 * 1.4) = 21.61$ miles

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Q.3283 The mean return of a sample of 28 BB+ corporate bonds is 7.5%, and the sample's standard deviation is 14%. Assuming that the population is normally distributed and the population variance is unknown, what is the 95% confidence interval for the population mean? *Here is a t distribution table: t distribution table*

A. [2.77%; 12.23%]

B. [2.07%; 12.93%]

C. [2.93%; 11.43%]

D. [4.12%; 13.3%]

The correct answer is ***B.***

Since the population variance is unknown and the population is normally distributed, and the sample size is less than 30, we will use a t-statistic. The t-statistic for a 95% confidence interval and 27 degrees of freedom (df=n-1) is 2.052.

$$\text{The standard error of the sample} = \frac{\text{Standard Deviation of sample mean}}{\sqrt{\text{Sample size}}}$$

$$= \frac{14}{\sqrt{28}} = 2.646$$

The confidence interval is $7.5 - (2.052 * 2.646) = 2.07$ *and* $7.5 + (2.052 * 2.646) = 12.93$

Using a reliability factor based on the t-distribution is essential for a small sample size. Using a t reliability factor is appropriate when the population variance is unknown, and when the sample size is less or equal to 30.

Q.3287 Which of the following best represents a 99 percent confidence interval if the mean score from 40 students in an exam is 85 and the population's standard deviation is 18?

A. [77.658; 92.342]

B. [73.658; 92.342]

C. [77.658; 90.342]

D. [80.212; 93.526]

*The correct answer is **A.***

The mean is 85, the standard deviation is 18, and the sample size is 40.

The 99% z-value is 2.58.

$$\text{Confidence interval} = \bar{x} \pm z * \left(\frac{\sigma}{\sqrt{n}}\right)$$

$$= 85 \pm 2.58\left(\frac{18}{\sqrt{40}}\right) = [77.658; 92.342]$$

Q.3288 While performing a hypothesis test, Albert Khan is told that his analysis suffers from a Type I error. We can therefore conclude that:

A. Khan rejected the null hypothesis when it was actually false.

B. Khan failed to reject the null hypothesis when it was actually false.

C. Khan failed to reject the null hypothesis when it was actually true.

D. Khan rejected the null hypothesis when it was actually true.

The correct answer is **D.**

A Type I error occurs when the null hypothesis is rejected even though it is true. In other words, the test concludes that there is a significant difference or effect when there is actually none. This error is also known as a false positive. In this case, Khan rejected the null hypothesis when it was actually true, which means that he made a Type I error.

Option A is incorrect. This statement describes a correct decision, where the null hypothesis is rejected when it is actually false. This decision is known as a true positive.

Option B is incorrect. This statement describes a failure to reject the null hypothesis when it is actually false. This decision is known as a Type II error.

Option C is incorrect. This statement describes a correct decision, where the null hypothesis is not rejected when it is actually true. This decision is known as a true negative.

Things to Remember

- Type I error is also known as a false positive, where the null hypothesis is incorrectly rejected.
- Type II error is also known as a false negative, where the null hypothesis is incorrectly not rejected.
- The significance level (alpha) of a hypothesis test determines the probability of making a Type I error.
- Type I error is denoted by the symbol α (alpha).
- Controlling Type I error is important in hypothesis testing to ensure the validity of the conclusions drawn from the analysis.

Q.3289 A two-tailed hypothesis test at the 95% confidence level has a p-value of 2.14%. This means that:

A. At a 5% significance level, we cannot reject the null hypothesis.

B. At a 2% significance level, we can reject the null hypothesis.

C. At a 5% significance level, we can reject the null hypothesis.

D. We have insufficient information to provide any kind of conclusion.

The correct answer is **C.**

The p-value is the probability of obtaining results at least as extreme as the observed results of a hypothesis test, assuming that the null hypothesis is correct. In other words, the p-value is the lowest level at which we can reject the null hypothesis. At a 95% confidence level, the significance level is 5% (1-95%). Therefore, any p-value less than 5% will lead to rejection of the null hypothesis. In this case, the p-value of 2.14% is less than 5%, which means that we can reject the null hypothesis at a 95% confidence level. Therefore, option C is the correct answer.

Things to Remember

- A p-value is the probability of obtaining results as extreme as the observed results, assuming the null hypothesis is true.
- The significance level (alpha) is the threshold used to determine if the null hypothesis should be rejected.
- In hypothesis testing, if the p-value is less than the significance level, we reject the null hypothesis.
- A two-tailed hypothesis test considers both directions of the alternative hypothesis.
- Confidence level is 1 minus the significance level, and it represents the probability that the confidence interval includes the true population parameter.

Q.3291 A survey is conducted to determine if the average starting salary of investment bankers is equal to or greater than $57,000 per year. Given a sample of 115 newly employed investment bankers with a mean starting salary of $65,000 and a standard deviation of $4,500, and assuming a normal distribution, what is the test statistic?

A. 204.44

B. 19.06

C. 1.78.

D. 746

The correct answer is **B.**

$$\begin{aligned}
\text{Standard error of the sample mean} &= \frac{\text{Standard deviation}}{\sqrt{\text{Sample size}}} \\
&= \frac{\$4,500}{\sqrt{115}} \\
&= 419.6272 \\
\text{Test statistic} &= \frac{(\text{Sample mean} - \text{Hypothesized value})}{\text{Standard error of the sample mean}} \\
&= \frac{(65,000 - 57,000)}{(419.6272)} = 19.06
\end{aligned}$$

Q.3292 Hilda believes that the average return on equity in the consumer durables industry is greater than 80%. What are the null (H_0) and the alternative (H_1) hypotheses for this study?

A. $H_0 : M = 0.8$ versus $H_1 : M \neq 0.8$

B. $H_0 : M \geq 0.8$ versus $H_1 : M < 0.8$

C. $H_0 : M \leq 0.8$ versus $H_1 : M > 0.8$

D. $H_0 : M < 0.8$ versus $H_1 : M \geq 0.8$

The correct answer is **C.**

The null hypothesis (H_0) is that the average return on equity in the consumer durables industry is less than or equal to 80%. The alternative hypothesis (H_1) is that the average return on equity in the consumer durables industry is greater than 80%. This is a one-sided test because the belief is that the return is greater than 80%, and we expect to reject the null hypothesis if the sample data provides sufficient evidence to support the alternative hypothesis.

Things to Remember

- Null Hypothesis (H_0) is the default assumption that there is no significant difference or relationship between variables.
- Alternative Hypothesis (H_1) is the hypothesis that contradicts the null hypothesis, suggesting that there is a significant difference or relationship.
- In hypothesis testing, the null hypothesis is typically denoted by H_0 and the alternative hypothesis by H_1.
- A one-sided test is used when the alternative hypothesis specifies that the parameter is either greater than or less than the value stated in the null hypothesis.
- A two-sided test is used when the alternative hypothesis specifies that the parameter is simply not equal to the value stated in the null hypothesis.

Q.3293 The average return on the Dow Jones Industrial Average for 121 quarterly observations is 1.5%. If the standard deviation of the returns can be assumed to be 8%, what is the 99% confidence interval for the quarterly returns of the Dow Jones?

A. [-0.4%; 3.4%]

B. [0.1%; 2.9%]

C. [-6.5%; 9.5%]

D. [-0.1%; 2.9%]

The correct answer is **A.**

$$\begin{aligned}\text{Standard error of the sample mean} &= \frac{\text{Standard deviation}}{\sqrt{\text{Sample size}}}\\ &= \frac{8\%}{\sqrt{121}}\\ &= 0.00727\end{aligned}$$

The critical value for the z-statistic for a 99% confidence interval is 2.575.

$$\text{Confidence Interval} = 1.5\% \pm 2.575[0.00727] = [-0.37\%; 3.37\%]$$

Q.3295 If a researcher wants to test that the mean return of 50 small-cap stocks from the Singapore Exchange is greater than 14%, which of the following would be the alternative hypothesis for the test?

A. $H_1 : \mu \neq 14\%$

B. $H_1 : \mu > 14\%$

C. $H_1 : \mu < 14\%$

D. $H_0 : \mu < 14\%$

The correct answer is **B.**

Since the researcher wants to test that if the mean of 50 small-cap stocks is greater than 14%, the null hypothesis is $H_0 : \mu \leq 14\%$ and the alternative hypothesis is $H_1 : \mu > 14\%$. We always want to reject the null hypothesis and accept the alternative. Since the researcher wants to prove that the mean returns are greater than 14%, the alternative hypothesis is $H_1 : \mu > 14\%$.

Things to Remember

- Null Hypothesis (H_0) is the default assumption that there is no significant difference or relationship between variables.
- Alternative Hypothesis (H_1) is the hypothesis that contradicts the null hypothesis, suggesting that there is a significant difference or relationship.
- In hypothesis testing, the researcher aims to gather evidence to reject the null hypothesis in favor of the alternative hypothesis.
- One-tailed tests, like the one in this question where the researcher is interested in whether the mean return is greater than 14%, have the alternative hypothesis focused on one direction (greater than or less than).
- Two-tailed tests have alternative hypotheses that consider the possibility of a difference in either direction (greater than or less than).

Q.3300 An investor is planning to invest in mutual funds. He intends to maximize his chances of earning a return in excess of 20%. The list of mutual funds available to the investor is listed in exhibit 1.

Exhibit 1: Potential List of Mutual Funds

Fund	Mean Return	Std. Dev. of Return
X	25%	2.%
Y	24.8%	3%
Z	26%	4%

Assuming that mutual fund returns are normally distributed and using a z-table, what is the correct probability of earning a return in excess of 20%? Please click here to see the standard normal table.

A. 1.60% for Fund Y.

B. 94.52% for Fund Z.

C. 99.38% for Fund X.

D. 11.6% for Fund Y.

The correct answer is **C.**

To find the probability of earning a return in excess of 20% for each mutual fund, we need to calculate the z-score for each fund and use the z-table to find the corresponding probability.

The z-score is calculated as:
Z = (Desired Return - Mean Return) / Std. Dev. of Return
Let's calculate the z-score for each mutual fund:
Fund X:
Z = (20% - 25%) / 2% = -5% / 2% = -2.5

Fund Y:
Z = (20% - 24.8%) / 3% = -4.8% / 3% = -1.6

Fund Z:
Z = (20% - 26%) / 4% = -6% / 4% = -1.5

Now, we look up these z-scores in the z-table to find the probability of earning less than 20%. However, we want the probability of earning more than 20%, so we'll subtract each probability from 1.

Fund X:
P(X > 20%) = 1 - P(Z < -2.5) ≈ 1 - 0.0062 ≈ 0.9938

Fund Y:
P(Y > 20%) = 1 - P(Z < -1.6) ≈ 1 - 0.0548 ≈ 0.9452

Fund Z:

P(Z > 20%) = 1 - P(Z < -1.5) ≈ 1 - 0.0668 ≈ 0.9332

Based on these probabilities, Fund X has the highest probability of earning a return in excess of 20% with a probability of approximately 99.38%.

Q.3330 For a sample of the past 28 monthly stock returns for Bidco Inc., the mean return is 5% and the sample standard deviation is 15%. Assume that the population variance is unknown.
The related t-table values are given below, where (t_{ij}) denotes the $(100-j)^{th}$ percentile of t-distribution value with i degrees of freedom):

$t_{27,0.025}$	2.05
$t_{27,0.05}$	1.70
$t_{26,0.025}$	2.06
$t_{26,0.05}$	1.71

What is the 95% confidence interval for the mean monthly return?

A. [0.00181, 0.0989]

B. [-0.0084, 0.1084]

C. [-0.00811, 0.10811]

D. [0.02135, 0.07835]

The correct answer is **C.**

$$\text{CI} = \text{Mean return} \pm (\text{Reliability factor} \times \text{Standard error})$$

Here, we use the t-reliability factor since the population variance is unknown. Since there are 28 observations, the degrees of freedom are $28-1=27$. The t-test is a two-tailed test, and confidence intervals are two-sided. So the correct critical t-value is $t_{27,0.025} = 2.05$

$$\text{CI} = 0.05 \pm 2.05 \times \frac{0.15}{\sqrt{28}} = 0.05 \pm 0.05811 = [-0.00811, 0.10811]$$

Q.3331 Using returns observed over the past 18 monthly, an analyst has estimated the mean monthly return of stock A to be 2.85% with a standard deviation of 1.6%.

One-tailed t-distribution table

Degrees of Freedom		a	
	0.1	0.05	0.025
14	1.35	1.78	2.15
15	1.34	1.75	2.13
16	1.34	1.75	2.12
17	1.33	1.74	2.11
18	1.33	1.73	2.10

Using the t-table above, the 95% confidence interval for the mean return is between:

A. [0.02031, 0.03688]

B. [0.02051, 0.03650]

C. [0.02194, 0.03506]

D. [0.02054, 0.03646]

The correct answer is **D.**

$$\text{CI} = \text{Mean return} \pm (\text{Reliability factor} \times \text{Standard error})$$

Here, we use the t-reliability factor since the population variance is unknown. Since there are 18 observations, the degrees of freedom are $18 - 1 = 17$.
This t-test is two-tailed and confidence intervals are two-sided, implying that the 0.025 column must be used. So the correct critical t-value is $t_{17,0.025} = 2.11$

$$\text{CI} = 0.0285 \pm 2.11 \times \frac{0.016}{\sqrt{18}} = 0.0285 \pm 0.00796 = [0.02054, 0.03646]$$

Reading 18: Linear Regression

Q.388 A hospital uses ultrasound technology to measure the weight of unborn babies as follows:

Gestation period in weeks	30	32	34	36	38	40
Estimated weight of foetus	1.6	1.7	2.5	2.8	3.2	3.5

Further information: $S_{XX} = 70, S_{YY} = 3.015, S_{XY} = 14.3$ Calculate the least square estimator of the slope and the Y-intercept (in that order).

A. Least square estimator: 0.2043; Y-intercept: -4.6

B. Least square estimator: 0.20; Y-intercept: -4

C. Least square estimator: 2.55; Y-intercept: 35

D. Least square estimator: 0.2043; Y-intercept: 35

The correct answer is **A.**

Under OLS estimation,
The regression equation is of the form: $y_i = \alpha + \beta x_i$
Where Y is the dependent variable (foetal weight), X is the independent variable (gestation period), α = the y-intercept, and beta = the slope.

$$\beta = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2} = \frac{S_{XY}}{S_{XX}} = \frac{14.3}{70} = 0.2043$$

$$\alpha = \bar{y} - \beta\bar{x}$$

$$\bar{y} = \frac{(1.6 + 1.7 + 2.5 + 2.8 + 3.2 + 3.5)}{6} = 2.55$$

$$\bar{x} = \frac{(30 + 32 + 34 + 36 + 38 + 40)}{6} = 35$$

Therefore,

$$\alpha = 2.55 - 0.2043 \times 35 = -4.60$$

Q.389 Given a regression equation is of the form: $y_i = \alpha + \beta x_i$ where Y is the dependent variable (foetal weight), X is the independent variable (gestation period, in weeks), α = the y-intercept, and β = the slope.
If $\alpha = -4.60$ and $\beta = 0.2043$, then estimate the weight of the foetus at exactly 31 weeks.

A. 1.533kg

B. 1.733kg

C. 1.722kg

D. 1.8kg

The correct answer is **B.**

The regression equation is of the form: $y_i = \alpha + \beta x_i$

Therefore, weight = -4.6 + 0.2043*31 = 1.733kgs

Things to Remember

- Regression analysis is a statistical technique used to understand the relationship between a dependent variable and one or more independent variables.
- The y-intercept (α) in a regression equation represents the value of the dependent variable when all independent variables are equal to zero.
- The slope (β) in a regression equation represents the change in the dependent variable for a one-unit change in the independent variable.
- Regression analysis helps in predicting the value of the dependent variable based on the values of the independent variables.
- It is important to understand the assumptions of regression analysis, such as linearity, independence, homoscedasticity, and normality of residuals.

Q.391 A limited liability company uses the ordinary least squares method to estimate a linear relationship between total monthly revenue and the total promotional expenditure. The linear function is found to have a positive slope that's significantly different from zero. Assuming that other variables, like product price and supply region, remain constant during the period covered by the data set, this implies that:

A. The company should significantly increase promotional expenditure.

B. The company should significantly reduce promotional expenditure.

C. Promotional expenditures have no effect on demand.

D. Promotional expenditures have a significant influence on demand.

The correct answer is **D.**

The positive slope found using the ordinary least squares method to estimate a linear relationship between total monthly revenue and the total promotional expenditure implies that promotional expenditures have a significant influence on demand. This means that an increase in promotional expenditure is likely to result in an increase in demand for the product. However, it is important to note that other variables, such as product price and supply region, should also be taken into consideration when making business decisions. Therefore, the decision to increase promotional expenditure should be reviewed in the context of the overall business strategy and goals.

Option A is incorrect. While a positive slope suggests that promotional events have an influence on demand, it does not necessarily mean that the company should significantly increase promotional expenditure. The decision to increase promotional expenditure should be based on a careful analysis of the potential benefits and costs, as well as the overall business strategy and goals.

Option B is incorrect. A positive slope suggests that promotional events have an influence on demand, so reducing promotional expenditure may not be the best course of action. Again, the decision to reduce promotional expenditure should be based on a careful analysis of the potential benefits and costs, as well as the overall business strategy and goals.

Option C is incorrect. A positive slope indicates that promotional expenditures have an effect on demand, so it is unlikely that promotional expenditures have no effect on demand. However, the magnitude of the effect may vary depending on other factors, such as product price and supply region.

Things to Remember

- Ordinary Least Squares (OLS) method is a common technique used in regression analysis to estimate the relationship between variables by minimizing the sum of the squares of the differences between the observed and predicted values.

- A positive slope in a linear regression model indicates a positive relationship between the independent and dependent variables, suggesting that an increase in one variable is associated with an increase in the other variable.

- Other factors, such as product price and supply region, can also impact demand and should be considered alongside promotional expenditure when making business decisions.

- It is important to conduct a cost-benefit analysis before significantly increasing or reducing promotional expenditure to ensure that the decision aligns with the overall business strategy and goals.

Q.392 Ordinary least squares is used to estimate the relationship between foetal weight and the number of weeks of gestation in a group of women. The exercise gives the following results:

- Total sum of squares, $ST_{TT} = 3.125$
- Regression sum of squares, $SS_{REG} = 2.925$
- Residual sum of squares, $SS_{RES} = 0.2$

This implies that:

A. The variability explained by the model is 0.2

B. The variability unexplained by the model is 3.125

C. The variability explained by the model is 2.925

D. The variability explained by the model is 3.125

The correct answer is **C.**

The variability explained by the model is 2.925. In the context of the ordinary least squares method, the regression sum of squares (SSREG) represents the variability that is explained by the model. This is the portion of the total variability that can be attributed to the relationship between the dependent variable (in this case, foetal weight) and the independent variable (in this case, the number of weeks of gestation). In this study, the SSREG is 2.925, which means that the model explains 2.925 units of the total variability in foetal weight. This is a significant portion of the total variability, indicating that the number of weeks of gestation has a strong influence on foetal weight.

Choice A is incorrect. The variability explained by the model is not 0.2, but rather 2.925 as indicated by the Regression sum of squares (SSREG). The value of 0.2 represents the Residual sum of squares (SSRES), which is the variability that remains unexplained by the model.

Choice B is incorrect. The total variability in the data set, represented by Total sum of squares

(STTT), is 3.125 and not the variability unexplained by the model. The unexplained variability or error term in this case would be given by Residual sum of squares (SSRES) which equals to 0.2.

Choice D is incorrect. While it's true that Total Sum of Squares (STTT) equals to 3.125, this does not represent 'the variability explained by model'. Instead, it represents total variation in dependent variable without considering any independent variable effect or simply put, it's a measure of total variance in observed data from its mean value.

Things to Remember

- Total Sum of Squares (STTT) is the total variation in the dependent variable without considering any independent variable effects.
- Regression Sum of Squares (SSREG) represents the variability explained by the model, indicating the portion of total variability attributed to the relationship between the dependent and independent variables.
- Residual Sum of Squares (SSRES) is the unexplained variability by the model, showing the portion of total variability that remains unaccounted for.
- R-squared (R^2) is a measure of how well the independent variables explain the variability of the dependent variable in a regression model.
- Coefficient of determination (R^2) is the square of the correlation coefficient between the observed and predicted values in a regression model.

Q.394 The annual returns of two stocks, X and Y, are jointly normally distributed. Each stock has a marginal distribution with mean = 2%, standard deviation = 10%. The correlation coefficient between X and Y is 0.9. If the annual return on stock X is 4%, what is the expected annual return on stock Y?

A. 0.038

B. 0.029

C. 0.0038

D. 0.4

The correct answer is **A.**

We can use the information given above to construct a regression model of Y on X.

$$R_Y = 2\% + 0.9\left(\frac{10}{10}\right) * (R_X - 2\%) + \epsilon$$

Replacing R_X with 4% gives:

$$R_Y = 2\% + 0.9\left(\frac{10}{10}\right) * (4\% - 2\%) = 3.8\%$$

Note: The intercept is simply the marginal distribution of each stock.

Remember that by definition, the marginal distribution gives the probabilities of various values of the variables in the subset without reference to the values of the other variables.

Alternative Approach

This is a classic example of exam-style questions that test your understanding of concepts not the mere application of formulas.

We have two stocks whose performance appears to be linearly related. With a positive correlation of 0.9, the returns of the two stocks move in the same direction; when the return of stock 1, the return of stock 2 also increases, and the opposite is true. This begs the question, can we try to model this as a simple linear regression? Yes

The general format of simple regression is:

$$Y = a + bX$$

Where Y and X are our variables, a is the intercept, and b is the slope

What we are trying to model here is that the return on Y has two components:

(I) a = intercept, a component that independent of X's performance, i.,e, the return earned by Y even if X earns a zero return. That's effectively the marginal distribution

(II) bX = a component that depends on the performance of X multiplied by a factor b

If we know the values of a and b, we can easily get the expected annual return on stock Y.

Recall that in linear regression,

$$
\begin{aligned}
a &= Y_{\text{mean}} - b(X_{\text{mean}}) \\
Y_{\text{mean}} &= X_{\text{mean}} = 2\% \\
b &= \text{slope} = \text{beta (Y with respect to X)} = \frac{\text{cov(X,Y)}}{\text{var(X)}} \\
&= \text{correlation(X,Y)} * \text{standard dev(X)} * \frac{\text{standard dev(Y)}}{\text{variance(X)}} \\
&= \text{correlation(X,Y)} * \frac{\text{standard dev(Y)}}{\text{standard dev(X)}} \\
&= 0.9 * \frac{10\%}{10\%} = 0.9
\end{aligned}
$$

Thus, $a = 2\% - 0.9(2\%) = 0.2\%$

With these, we can comfortably predict the expected return on Y

$$
Y = a + bX = 0.2\% + 0.9(4\%) = 3.8\%
$$

Q.395 If the value of the independent variable = 0, what would be the expected value of the dependent variable?

A. The slope coefficient

B. The residual value

C. The intercept coefficient

D. 0

The correct answer is **C.**

When the value of the independent variable is 0, the expected value of the dependent variable can be calculated using the regression equation:

$$E(Y|X) = b_0 + b_1 * X$$

Therefore, $E(Y|0) = b_0$, which is the y-intercept or intercept coefficient. The intercept coefficient represents the expected value of the dependent variable when the independent variable is equal to 0.

A is incorrect. The slope coefficient represents the change in the expected value of the dependent variable for a one-unit increase in the independent variable. It does not provide information about the expected value of the dependent variable when the independent variable is equal to 0.

B is incorrect. The residual value is the difference between the actual value of the dependent variable and the predicted value of the dependent variable based on the regression equation. It does not provide information about the expected value of the dependent variable when the independent variable is equal to 0.

D is incorrect. Assuming that the intercept coefficient is not equal to 0, the expected value of the dependent variable when the independent variable is equal to 0 cannot be 0.

Q.396 A graduate school constructs a linear regression model to estimate the effect of increased study hours on the average performance of students in a test. The slope coefficient is found to be equal to 2. What does this mean?

A. The average score when the number of study hours is zero is 2.

B. The predicted score when the number of study hours is zero is 2%.

C. For every one unit change in the number of study hours, the model predicts that the average score will change by 2 units.

D. For every one unit change in the number of study hours, the model predicts that the average score will change by 2%.

The correct answer is **C.**

The slope coefficient represents the expected change in the dependent variable for a 1-unit change in the independent variable. If the coefficient is 2 and the independent variable changes by 1 unit, the dependent variable will change by 2 units. Candidates usually have difficulties differentiating between the slope coefficient and the intercept coefficient. The latter is simply the value of the dependent variable (Y) when the independent variable (X) is equal to zero.

Things to Remember

- Linear regression is a statistical method that models the relationship between a dependent variable and one or more independent variables.
- The slope coefficient in a linear regression model represents the change in the dependent variable for a one-unit change in the independent variable.
- The intercept coefficient in a linear regression model represents the value of the dependent variable when the independent variable is zero.
- R-squared value in regression analysis indicates the proportion of the variance in the dependent variable that is predictable from the independent variable(s).
- Assumptions of linear regression include linearity, independence of errors, homoscedasticity, and normality of errors.

Q.397 The estimated slope coefficient (β_1) for a certain stock is 0.8823 with a standard error equal to 0.0931. Assuming that the sample had 10 observations, carry out a statistical test to determine if the slope coefficient is statistically different than zero. Quote the test statistic and the decision rule using a 5% level of significance. Please click here to use the t-distribution table

A. 9.477, reject H_0

B. 9.477, do not reject H_0

C. 2.307, reject H_0

D. 2.307, do not reject H_0

The correct answer is **A.**

The first step entails formulating the hypothesis:

$$H_0 : \beta_1 = 0 \text{ vs } H_a : \beta_1 \neq 0$$

The test statistic for the slope coefficient takes the form: $t_{\alpha,n-2} = \frac{(\beta_1 - \beta_0)}{se(\beta_1)}$

In this case, $t_{0.025,8} = \frac{(0.8823-0)}{0.0931} = 9.477(\alpha = \frac{5\%}{2}$ since this is a 2-tailed test)

The critical 2-tailed values ($t_{0.025,8}$) are ± 2.306

Our test statistic lies outside the non-rejection region (-2.306, 2.306). As such, we have sufficient evidence to reject the null hypothesis and conclude that the slope coefficient is statistically different than zero.

Q.398 An analyst obtained the following linear regression relationship between 2 variables, X and Y:

$$Y = \alpha + \beta_1 X$$

where $\alpha = 0.45$ and $\beta = 0.8823$He proceeded to construct a 2-sided 95% confidence interval for the slope coefficient (β_1) and obtained the following interval:

$$\beta = 0.8823 \pm 0.2147$$

Suppose the analyst decided to test the hypothesis $H_0 : \beta_1 = 1$ vs $H_a : \beta_1 \neq 1$ at 5% significance, what would be the inference?

A. Reject H_0

B. Do not reject H_0

C. The slope coefficient is statistically different than "1"

D. Cannot tell from the information provided

The correct answer is **B.**

The 95% 2-sided confidence interval contains value "1". Therefore, if the analyst were to conduct a 2-sided test at the 5% level, he would end up not rejecting the null hypothesis.
Further explanation
In general, if the Y% confidence interval contains the hypothesized parameter, then a hypothesis test at the 1-Y% level of significance will always fail to reject the null hypothesis, affirming the assumption that the parameter can take on that particular value. If the interval does not contain the hypothesized parameter, then the hypothesis test at the 1-Y% level of significance will always reject the null hypothesis, discrediting the assumption that the parameter can take on that particular value.

Q.399 Two variables have a linear relationship of the form:

$$Y = -4.6 + 0.2043X$$

The slope coefficient has a standard error equal to 0.01828. Carry out a test of $H_0 : \beta_1 = 0$ vs $H_a : \beta_1 > 0$ at 0.5% significance, quoting the test statistic and the conclusion if the number of observations, n, is 6.

A. $11.2, \beta > 0$

B. $11.2, \beta = 0$

C. $0.01828, \beta \neq 0$

D. $11.2, \beta \neq 0$

The correct answer is **A.**

The test statistic for the slope coefficient takes the form: $t_{\alpha,n-2} = \frac{(\beta_1 - \beta_0)}{se(\beta_1)}$
In this case, $t_{0.005,4} = \frac{(0.2043-0)}{0.01828} = 11.1761$

Since the alternative hypothesis has a ">" sign, this is a one-tailed test. We should, therefore, compare our test statistic to the upper 0.5% point of the t-distribution with 4 degrees of freedom. $T_{0.005,4} = 4.604$

Since 11.1761 is greater than 4.604, we have **sufficient evidence** to reject H_0 and conclude that $\beta_1 > 0$

Q.400 During a statistical test to determine if the mean return on an asset is different from zero, an FRM Part 1 candidate obtains a p-value of 1.4%. With a significance level of 1%, she would:

A. Reject the null hypothesis.

B. Fail to reject the null hypothesis.

C. Conclude that the mean return is different from zero.

D. Conclude that the mean return is negative (loss).

The correct answer is **B.**

The candidate would fail to reject the null hypothesis. In statistical hypothesis testing, the p-value is the smallest level of significance at which we can reject the null hypothesis. The p-value is a measure of the probability that an observed difference could have occurred just by random chance. In this case, the p-value of 1.4% means that there is a 1.4% chance that the observed difference (between the mean return and zero) occurred by chance. The significance level is a threshold set by the researcher (in this case, 1%) to decide when to reject the null hypothesis. If the p-value is less than or equal to the significance level, we reject the null hypothesis. If the p-value is greater than the significance level, we fail to reject the null hypothesis. Since the p-value (1.4%) is greater than the significance level (1%), the candidate would fail to reject the null hypothesis. This means that there is not enough evidence to conclude that the mean return on the asset is different from zero.

Choice A is incorrect. The p-value of 1.4% is greater than the significance level of 1%. Therefore, we do not have enough evidence to reject the null hypothesis.

Choice C is incorrect. The p-value only tells us about the probability that we would observe a result as extreme as, or more extreme than, our observed result if the null hypothesis were true. It does not provide any information about whether the mean return is different from zero.

Choice D is incorrect. The p-value and statistical test do not provide any information about whether the mean return is negative or positive; they only help us determine if it deviates significantly from zero.

Things to Remember

- Null Hypothesis (H0) and Alternative Hypothesis (Ha)
- Significance Level (alpha)
- p-value
- Type I error and Type II error
- Interpreting statistical significance

Q.402 An organization estimates that the effect of increasing the number of qualified Financial Risk Managers hired by 1 will improve the stock's annual return by 2.8% with a standard error of 0.52%. Construct a 90% 2-sided confidence interval for the size of the slope coefficient, assuming the stock's returns are normally distributed.

A. (1.9%, 2.8%)

B. (1.4%, 3.1%)

C. (1.9%, 3.5%)

D. (1.9%, 3.7%)

The correct answer is **D.**

The CI will take the form:

$$\begin{aligned} &\beta_1 \pm Z_{\frac{\alpha}{2}} \times se(\beta_1) \\ =&\beta_1 \pm Z_{0.05} = 2.8\% \pm 1.645 \times 0.52\% = 2.8\% \pm 0.8554\% = (1.9446\%, 3.6554\%) \end{aligned}$$

Q.404 A linear regression model gave the following results:

$$S_{yy} = 10.6; S_{xx} = 12.0; S_{xy} = 8.0; n = 18$$

Test (at 1% significance) whether β is significantly different from zero, given that its standard error = 0.16 and give the value of the test statistic and the conclusion.

A. 0.667, β is not significantly different from zero

B. 0.4169, β is not significantly different from zero

C. 0.667, β is significantly different from zero

D. 4.169, β is significantly different from zero

The correct answer is **D.**

We wish to test:

$$H_0 : \beta = 0 \text{ vs } H_1 \neq 0$$

Under H_0, $\frac{(\beta - \beta_0)}{se(\beta)}$ has a t_{n-2} distribution.

$$\beta = \frac{S_{xy}}{S_{xx}} = \frac{8}{12} = 0.667$$

The observed test statistic is:

$$\frac{(0.667 - 0)}{0.16} = 4.169$$

This is a two-tailed test and therefore we should compare the test statistic to the upper 0.5% point of the t-distribution.

$$T_{0.005,16} = 2.921$$

4.169 > 2.921 so we have sufficient evidence to reject H_0 at the 1% level of significance. What's the conclusion? β is significantly different from zero.

Note that, we can tell whether to use one-tail or two-tail tests by considering what the question wants you to get.

A two-tailed test is used if you want to determine if there is any difference between the groups. For example, let's say you want to determine if Group X performed higher or lower than Group Y, then you should use a two-tailed test. In our case, we have been asked to determine whether β is significantly different from zero. In this case, we will use a two-tailed test.

One-tailed test, on the other hand, is used when you want to determine if there is any difference between two groups, but now in a given particular direction. Let's say you are only interested in determining if Group X performed lower than group Y, in such a case, you will need to use a one-tailed test.

Q.406 A 95% confidence interval for β_1 is determined to be (20, 25). This means that:

A. We can be 95% confident that the mean value of Y lies between 20 and 25 units.

B. We can be 95% confident that the value of X will increase by between 20 and 25 units for every one unit increase in Y.

C. We can be 95% confident that the value of Y will increase by between 20 and 25 units for every one unit increase in X.

D. At the 5% level of significance, we would not find evidence of a linear relationship between X and Y.

The correct answer is **C.**

The parameter β_1 in a linear regression model represents the change in the dependent variable (Y) for every one unit increase in the independent variable (X). Therefore, a 95% confidence interval for β_1 being (20, 25) means that we can be 95% confident that for every one unit increase in X, the value of Y will increase by between 20 and 25 units. This interpretation is based on the fundamental understanding of the slope in a linear regression model, which quantifies the rate of change in the dependent variable as the independent variable changes. The confidence interval provides a range of plausible values for this rate of change, and the level of confidence (95% in this case) indicates the degree of certainty that the true value of β_1 lies within this range.

Choice A is incorrect. The confidence interval provided does not pertain to the mean value of Y, but rather to the slope of the regression line, β_1. This means that we are 95% confident that for every one unit increase in X, Y will increase by between 20 and 25 units.

Choice B is incorrect. This statement incorrectly interprets the directionality of the relationship between X and Y. In a linear regression model, β_1 represents how much Y changes with a one unit increase in X, not vice versa.

Choice D is incorrect. The confidence interval does not provide information about whether there is a linear relationship between variables at any level of significance. It only provides an estimate range for β_1, which indicates how much we expect Y to change as X increases by one unit.

Things to Remember

- Confidence Interval: A range of values that is likely to contain the true value of a parameter with a certain level of confidence.
- Linear Regression: A statistical method used to model the relationship between a dependent variable and one or more independent variables.
- Slope (β_1): In the context of linear regression, the slope represents the change in the dependent variable for a one unit increase in the independent variable.
- Level of Significance: The probability of rejecting the null hypothesis when it is actually true, typically denoted by α.
- Null Hypothesis: A statement that there is no significant difference or relationship between variables.

Q.409 Use the regression equation " $\overline{\text{WPO}}$ = -3.2% + 0.49($\overline{\text{S\&P 500}}$) " to calculate a 95% confidence interval on the predicted value of WPO. You have been given that n = 30, the standard error of the forecast is 3.76%, and the forecasted value of S&P 500 excess return is 10%.

A. (1.7%, 9.37%)

B. (-5.97%, 1.7%)

C. (4.9%, 9.37%)

D. (-5.97%, 9.37%)

The correct answer is **D.**

The predicted value for WPO is:

$$\overline{WPO} = -3.2\% + 0.49 * 10\% = 1.7\%$$

$$T_{0.025,28} = 2.04 \text{(Degrees of freedom = n– 2)}$$

Therefore,

$$\begin{aligned} CI_{95\%} &= \overline{WPO} \pm T_{0.025,30} * \text{Standard error of forecast} = 1.7\% \pm 2.04 * 3.76\% \\ &= 1.7\% \pm 7.67 \text{ or } (-5.97\%, 9.37\%) \end{aligned}$$

Q.410 Sometimes the explanatory power of regression analysis can be overstated. Under which of the following scenarios would that most likely happen?

A. If the residual term is normally distributed.

B. If the explanatory variables are not correlated with one another.

C. The omission of a crucial explanatory variable, which has significant influence on the explanatory variables included as well as the dependent variable..

D. If there are only two explanatory variables.

The correct answer is **C.**

The omission of a crucial explanatory variable, which has significant influence on the explanatory variables included as well as the dependent variable, can lead to an overstatement of the explanatory power of regression analysis. This is because the omitted variable can have a significant effect on both the dependent variable and the included explanatory variables. This violates one of the assumptions of the Ordinary Least Squares (OLS) model, which states that the error term should be conditionally independent of the explanatory variables. When this assumption is violated, the error term can become correlated with the explanatory variables, leading to biased and inconsistent estimates. This can incorrectly increase the explanatory power of the regression, leading to misleading conclusions about the relationships between the variables.

Choice A is incorrect. The normal distribution of the residual term does not lead to an overstatement of the explanatory power of regression analysis. In fact, it is one of the assumptions in a standard linear regression model.

Choice B is incorrect. The lack of correlation among explanatory variables does not result in an overstatement of the explanatory power. On contrary, multicollinearity (high correlation among predictors) can inflate the variance of the coefficient estimates and make them unstable and difficult to interpret.

Choice D is incorrect. The number of explanatory variables itself doesn't lead to an overstatement or understatement of regression's explanatory power. It's more about whether these variables are relevant and correctly specified in relation to dependent variable.

Things to Remember

- Assumptions of Ordinary Least Squares (OLS) model
- Effects of omitted variables on regression analysis
- Multicollinearity and its impact on regression analysis
- Interpretation of coefficient estimates in regression

Q.3333 An analyst is trying to establish the relationship between the return on stock $X(R_X)$ and the return on stock $S(R_S)$. Stock X is listed on the Bombay Stock Exchange (BSE). The analyst has assumed a linear relationship as follows.

$$R_X = a + b \times R_S + \epsilon_t$$

Furthermore, the analyst has gathered the following historical data.

Expected return on stock X	15%
Expected return on S	10%
Standard deviation of return on stock X	20%
Standard deviation of return on stock S	15%
Correlation between returns on stock X and S	0.3

Which of the following is the correct model that can be deduced using the ordinary least squares technique?

A. $E(R_X) = 0.40 + 0.40 \times E(R_S)$

B. $E(R_X) = 0.11 + 0.40 \times E(R_S)$

C. $E(R_X) = 0.40 + 0.11 \times E(R_S)$

D. None of the above

The correct answer is **B.**

As we know that the expected value of the error is zero, the equation from the question is reduced to:

$$E(R_X) = a + b \times E(R_S)$$

$$b = \frac{Cov(S,X)}{Var(S)} = \frac{[Corr(S,X) * SD(S) * SD(X)]}{Var(S)} = Corr(S,X) * \frac{SD(X)}{SD(S)}$$

$$b = 0.3 * \frac{0.20}{0.15} = 0.40$$

$$a = \bar{x} - b\bar{s} = 0.15 - 0.4 * 0.10 = 0.11$$

Note: You may be used to the use of variables coded X and Y, where we regress Y on X (use the values of variable X to predict those of Y, with X as the independent variable). In such a case, we usually formulate the regression as,

$$Y = a + bX$$

The same logic applies here only that X has been used as the dependent variable ($X = a + bS$). The question presents a good way to test your understanding not just the application of formulas.

Q.3336 The return on a stock (R) exhibits the following relationship with the market return (MR).

$$R = -1.15 \times MR + 2\%; R^2 = 81\%$$

Assuming all coefficients are significant, which of the following interpretations is correct?

A. A 1% increase in MR results into a 1.15% increase in R.

B. A 1% increase in MR results into 2% increase in R.

C. The correlation between the return on the stock and the return on the market is 0.81.

D. The correlation between the return on the stock and the return on the market is -0.90.

The correct answer is **D.**

The correlation between the return on the stock and the return on the market is -0.90. The correlation coefficient, denoted as r, is a measure of the strength and direction of the linear relationship between two variables. In this case, the variables are the return on the stock (R) and the return on the market (MR). The correlation coefficient is calculated as $r = [\text{Sign of estimated slope coefficient}]\sqrt{R^2}$. Given that the estimated slope coefficient is -1.15 (a negative value), the sign of r will also be negative. The square root of R^2 (which is 0.81) is approximately 0.90. Therefore, the correlation coefficient r is -0.90, indicating a strong negative linear relationship between R and MR. This means that as MR increases, R tends to decrease, and vice versa.

Choice A is incorrect. The relationship between the return on the stock and the market return is not positive as suggested by this option. The coefficient of MR in the given equation is -1.15, which means that a 1% increase in MR results in a 1.15% decrease in R, not an increase.

Choice B is incorrect. This choice incorrectly interprets the constant term (2%) as the change in R for a 1% change in MR. In reality, this constant term represents the expected value of R when MR equals zero.

Choice C is incorrect. The coefficient of determination (R^2) measures how well our model explains variation in our dependent variable (in this case, R), but it does not directly give us correlation between variables. Correlation would be square root of R^2, but since we have negative relationship it should be -0.9 and not +0.81.

Q.3337 The return on a stock (R) exhibits the following relationship with the market return (MR).

$$R = -1.15 \times MR + 2\%; R^2 = 81\%$$

Compute the ratio of the standard deviation of stock return to standard deviation of the market return.

A. 1.15

B. 1.28

C. 0.81

D. 0.9

The correct answer is **B.**

The coefficient of determination (R^2) measures the fraction of the total variation in the dependent variable that is explained by the independent variable. The return on R explains approximately 81% of the variation from the return on MR.

The correlation coefficient is given by:

$$r = [\text{Sign of estimated slope coefficient}]\sqrt{R^2} = -\sqrt{0.81} = -0.9$$

From OLS, the slope coefficient (b) is given by:

$$b = \frac{Cov(MR, R)}{Var(MR)} = \frac{[Corr(MR, R) * SD(MR) * SD(R)]}{Var(MR)} = Corr(MR, R) * \frac{SD(R)}{SD(MR)}$$

$$-1.15 = -0.9 * \frac{SD(R)}{SD(MR)}$$

$$\frac{SD(R)}{SD(RM)} = \frac{1.15}{0.9} = 1.28$$

Note: Given a regression $Y = a + bX$, the sign of r depends on the sign of the estimated slope coefficient b:

- If b is negative, then r takes a negative sign.
- If b is positive, then r takes a positive sign.

Q.3338 An analyst has attempted to get some insight into the relationship between the return on stock A($R_{A,t}$) and the return on the Nasdaq Composite index ($R_{NC,t}$). The analyst gathers historical data and comes up with the following estimates:

Expected mean return for A	10%
Annual mean return for Nasdaq Composite	6%
Annual volatility for Nasdaq Composite	15%
Covariance between the returns of A and Nasdaq Composite	5%

The analyst goes ahead and formulates the following regression model using the data:

$$R_{A,t} = \alpha + \beta R_{NC,t} + \epsilon_t$$

Using the ordinary least squares technique, which of the following models will the analyst obtain?

A. $R_{A,t} = -0.03333 + 2.2222 R_{NC,t} + \epsilon_t$

B. $R_{A,t} = -0.05 + 2.2222 R_{NC,t} + \epsilon_t$

C. $R_{A,t} = 2.2222 + 0.06 R_{NC,t} + \epsilon_t$

D. $R_{A,t} = 1.80 - 0.05 R_{NC,t} + \epsilon_t$

The correct answer is **A.**

The regression coefficients for a model specified by $Y = \hat{a} + \hat{b}X + \varepsilon$, where ε represents the error term, are obtained using the formula:

$$\hat{b} = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} = \frac{0.05}{0.15^2} = 2.2222$$

$$\hat{\alpha} = E(Y) - \hat{b}E(X) = 0.1 - 2.2222(0.06) = -0.03333$$

Q.3340 The return on a stock R exhibits the following relationship with the market return (MR).

$$R = \hat{a} + \hat{b} \times MR$$

Where $\hat{b}$ is the slope coefficient and $\hat{a}$ is the intercept. After gathering 36 observations, an analyst computed the estimated slope coefficient as 0.6 with a standard error of 0.2. Determine whether the estimated slope coefficient is different from 0 at a 95% confidence level with reference to the critical t-value. Click here to see critical values of the t-distribution.

A. The slope coefficient is not significant.

B. The slope coefficient is statistically significant with a t-statistic of 2.03.

C. The slope coefficient is statistically significant with a t-statistic of 3.

D. The slope coefficient is statistically significant with a t-statistic of 1.015.

The correct answer is **C.**

Null Hypothesis: $b = 0$

Alternate Hypothesis: $b \neq 0$

This is a two-tailed test.

$$\text{Test Statistic} = \frac{\text{Sample statistic} - \text{Hypothesized value}}{\text{Standard error of the sample statistic}}$$

$$t = \frac{(0.6 - 0)}{0.2 = 3}$$

Critical test-statistic for two-tailed test is ± 2.03

We need to consider the t-statistic in a 2-tailed test.

$\alpha = 5$

$\frac{\alpha}{2} = 0.025;$

Degrees of freedom $= n - 2 = 36 - 2 = 34;$

$t_{0.025,34} = \pm 2.03$ from statistics table.

As the test statistic is greater than its critical value, we can reject the null hypothesis and conclude that the slope coefficient is statistically different from zero. In other words, we can say that slope coefficient is statistically significant.

Q.3341 An analyst has regressed the annual return on a stock (R_{stock}) against the annual return on the NIFTY 50(R_{index}) for 30 years. The NIFTY is the National Stock Exchange (NSE) index in India.The results are as shown below. Regression equation:

$$R_{index,\,t} = \hat{a} + \hat{b} \times R_{stock,\,t} + \varepsilon_t$$

Coefficient	Coefficient Estimate	Standard Error
a	0.002	0.001
b	1.223	0.063

Interpret whether the regression coefficients are statistically different from zero at a 95% confidence level?\ Click here to see critical values of the t-distribution.

A. Intercept term (a): Yes; Slope coefficient (b): Yes

B. Intercept term (a): No; Slope coefficient (b): No

C. Intercept term (a): No; Slope coefficient (b): Yes

D. Intercept term (a): Yes; Slope coefficient (b): No

The correct answer is **C.**

The relevant hypotheses are:

$H_0 : a = 0$ vs. $H_1 : a \neq 0$, and

$H_0 : b = 0$ vs. $H_1 : b \neq 0$

Degrees of freedom $= n - 2 = 30 - 2 = 28$

Critical value for t-test = 2.048 under 2-tailed 5% significance level.

$$\text{Test Statistic} = \frac{\text{Sample Statistic} - \text{Hypothesized Value}}{\text{Standard Error of Sample Statistic}}$$

The decision rule is to reject H_0 if test statistic > upper critical value or if test statistic < lower critical value

$$\text{Estimated t-statistic for the intercept} = \frac{0.002 - 0}{0.001} = 2(< 2.048)$$
$$\text{Estimated t-statistic for the slope} = \frac{1.223 - 0}{0.063} = 19.413(> 2.048)$$

The intercept is not statistically significant while the slope is statistically significant at the 5%

level.

Q.3342 An analyst has regressed the annual return on a stock (R_{stock}) against the annual return on the NIFTY 50 (R_{index}) for 30 years. The NIFTY is the index of the National Stock Exchange (NSE), India. Results are shown below. Regression equation:

$$R_{index,\ t} = \hat{a} + \hat{b} \times R_{stock,\ t} + \varepsilon_t$$

Coefficient	Coefficient Estimate	Standard Error
a	0.002	0.001
b	1.223	0.063

What is the 90% confidence interval for the slope coefficient? Click here to see critical values of the t-distribution.

A. [1.1165; 1.3295]

B. [1.223; 1.3295]

C. [1.1158; 1.3301]

D. [0.063; 1.223]

The correct answer is **C.**

$$\text{CI} = \text{point estimate} \pm (\text{reliability factor x standard error})$$

When constructing confidence intervals for the slope coefficient for a sample less than or equal to 30, we make use of the t-distribution to come up with the reliability factor.

$$\text{CI} = \hat{b} \pm t_{\frac{\alpha}{2}, n-2} s_{\hat{b}}$$

$\alpha = 1 - 0.90 = 0.10$ which implies that, $\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$;

Degrees of freedom $= n - 2 = 30 - 2 = 28$;

$t_{0.1,28} = \pm 1.701$ from statistical tables.

The interval of the slope coefficient is $1.223 \pm 1.701 * 0.063 = 1.1158$ to 1.3301.

Q.3343 An analyst has regressed the annual return on a stock (R_{stock}) against the annual return on the NIFTY 50(R_{index}) for 36 years. The NIFTY is the National Stock Exchange (NSE) index in India. Results are shown below. Regression equation:

$$R_{index,\,t} = \hat{a} + \hat{b} \times R_{stock,\,t} + \varepsilon_t$$

Coefficient	Coefficient Estimate	Standard Error
a	0.002	0.001
b	1.223	0.063

An analyst wants to test the hypothesis given below at 5% significance level: $H_0 : b \leq 1$ $H_a : b > 1$
Which of the following statement is correct about slope coefficient?

A. Estimated t-statistic: 1.223; Hypothesis: Fail to reject H_0

B. Estimated t-statistic: 3.54; Hypothesis: Reject H_0

C. Estimated t-statistic: 3.54; Hypothesis: Fail to reject H_0

D. Estimated t-statistic: 1.223; Hypothesis: Reject H_0

The correct answer is **B.**

Degrees of freedom = $n - 2 = 36 - 2 = 34$
Estimated t-statistic for the slope coefficient = $\frac{1.223-1}{0.063} = 3.54$
As an upper one-tailed test, the decision rule is to reject H_0 if test statistic > upper critical value from the t-distribution. Critical value for a one-tailed t-test at a 5% significance level = 1.69

The estimated t-statistic (3.54) is greater than the upper 95% point of the t-distribution (1.69), so the analyst would reject the null hypothesis.

Q.3344 An analyst wishes to establish the relationship between corporate revenue (Y_t) and the average years of experience per employee (X_t) and comes up with the following model.

$$Y_t = 0.45 + 0.78X_t$$

The analyst also observes that the standard error of the coefficient of the average years of experience per employee is 0.65. In order to test the null hypothesis that the average years of experience per employee have no effect on corporate revenue, what is the correct statistic to calculate?

A. F-test

B. t-test

C. Chi-square test

D. Durbin Watson test

The correct answer is **B.**

In both single and multiple regression, we use the t-test to determine whether a specific independent variable has a significant effect on the dependent variable at a given level of confidence. If we take the coefficient of the average years of experience per employee to be β_1, then the test would be formulated as follows:
$H_0 : \beta_1 = 0$
$H_0 : \beta_1 \neq 0$

In this case, the test statistic would be computed as follows:

$$\begin{aligned} t &= \frac{\text{estimated value of } \beta_1 - \text{value of } \beta_1 \text{ under } H_0}{\text{standard error of estimated value of } \beta_1} \\ &= \frac{0.78 - 0}{0.65} = 1.2 \end{aligned}$$

Q.3966 A financial analyst develops a Capital Pricing Model that regresses the expected monthly return of a company on the prevailing interest rates. The coefficients are $\beta_0 = 0.064$ and $\beta = 0.65$ where β_0 is the intercept. What is the value of the monthly expected return for the company if the interest rate at a particular month is 5%?

A. 0.0965

B. 0.0856

C. 0.0778

D. 0.0567

The correct answer is **A.**

Using the stated regression parameters, the regression equation is:

$$\text{Expected Monthly Return} = \beta_0 + \beta(\text{Interest rates})$$

So, when the interest rate is 5%, then the corresponding expected return is:

$$\text{Expected Monthly Return} = 0.064 + 0.05 \times 0.65 = 0.0965 = 9.65\%$$

Q.3968 A regression analysis of monthly returns of a sales company on the market return over ten years gives an intercept of $\hat{\beta}_0 = 0.65$, the slope $\hat{\beta} = 1.65$. Other quantities include: $s^2 = 20.45$, $\hat{\sigma}_X^2 = 18.65$ and $\hat{\mu}_X = 0.61$. What is the standard error estimate of $\hat{\beta}_0$?

A. 0.5463

B. 0.56435

C. 0.4552

D. 0.4169

The correct answer is **D.**

We know that:

$$\text{SEE}_{\beta_0} = \sqrt{\frac{s^2(\hat{\mu}_X^2 + \hat{\sigma}_X^2)}{n\hat{\sigma}_X^2}}$$

Therefore,

$$\text{SEE}_{\beta_0} = \sqrt{\frac{20.45(0.61^2 + 18.65)}{120 \times 18.65}} = 0.4169$$

Q.3969 A regression analysis of monthly returns of a sales company on the market return over ten years gives an intercept of $\hat{\beta}_0 = 0.65$, the slope $\hat{\beta} = 1.65$. Other quantities include: $s^2 = 20.45$, $\hat{\sigma}_X^2 = 18.65$ and $\hat{\mu}_X = 0.61$. The analyst wishes to test whether the slope coefficient is different from 0. What is the test statistic of $\hat{\beta}$?

A. 17.2594

B. 10.1891

C. 24.3234

D. 20.3232

The correct answer is **A.**

We start by starting the hypothesis:

$$H_0 : \beta = 0 \text{ vs } H_0 \neq 0$$

We know that:

$$T = \frac{\hat{\beta} - \beta_{H_0}}{SEE_{\hat{\beta}}}$$

Where:

$$SEE_{\beta} = \sqrt{\frac{s^2}{n\hat{\sigma}_X^2}} = \sqrt{\frac{20.45}{120 \times 18.65}} = 0.0956$$

So,

$$T = \frac{\hat{\beta} - \beta_{H_0}}{SEE_{\hat{\beta}}} = \frac{1.65 - 0}{0.0956} = 17.2594$$

Q.3970 A regression analysis of the monthly returns of a sales company on the market return over ten years gives an intercept of $\hat{\beta}_0 = 0.65$, the slope $\hat{\beta} = 1.65$. Other quantities include: $s^2 = 20.45$, $\hat{\sigma}_X^2 = 18.65$ and $\hat{\mu}_X = 0.61$. The analyst wishes to test whether the slope coefficient is different from 0. What is the 99% confidence interval for $\hat{\beta}$?

A. [1.6034,1.8906]

B. [1.3034,1.8966]

C. [1.4748,1.8252]

D. [1.3998,1.9002]

The correct answer is **D.**

The confidence interval is given by:

$$[\hat{\beta} - C_t \times SEE_{\hat{\beta}}, \hat{\beta} + C_t \times SEE_{\hat{\beta}}]$$

10 years have 120 months.

The critical value (C_t) at 1% (0.5% on each tail) level with 118 degrees of freedom (i.e., n-2) is 2.617. (From the t-distribution table)

Now, $SEE_{\hat{\beta}}$ is given by,

$$SEE_{\beta} = \sqrt{\frac{s^2}{n\hat{\sigma}_X^2}} = \sqrt{\frac{20.45}{120 \times 18.65}} = 0.0956$$

So,

$$\begin{aligned}[\hat{\beta} - C_t \times SEE_{\hat{\beta}}, \hat{\beta} + C_t \times SEE_{\hat{\beta}}] &= [1.65 - 2.617 \times 0.0956, 1.65 + 2.617 \times 0.0956] \\ &= [1.3998, 1.9002]\end{aligned}$$

Q.3971 You want to develop a model that regresses the number of options on the number of underlying stocks using a large data set. You estimate the following regression equation.

$$\text{Number of Options} = \hat{\alpha} + \hat{\beta}\,(\text{amount of underlying stock})$$

The statistical software you used returns a 90% confidence interval of [0.30,1.60] for the slope coefficient. If the 10% critical value for the t-test is 1.70, what is the likely value of the p-value corresponding to your slope coefficient if you wanted to test whether the slope is different from 0?

A. 0.0132

B. 0.0164

C. 0.0192

D. 0.0186

The correct answer is **A.**

Clearly, this a two-tailed test due to confidence interval. So we need:

$$p-\text{value} = 2[1 - \Phi(|T|)]$$

Recall that the confidence interval is given by:

$$[\hat{\beta} - C_t \times SEE_{\hat{\beta}}, \hat{\beta} + C_t \times SEE_{\hat{\beta}}]$$

$\hat{\beta}$ is equivalent to the midpoint of the confidence interval. That is:

$$\frac{1}{2}(\hat{\beta} - C_t \times SEE_{\hat{\beta}}, \hat{\beta} + C_t \times SEE_{\hat{\beta}}) = \hat{\beta}$$

So,

$$\hat{\beta} = \frac{1}{2}(0.30 + 1.60) = 0.95$$

The critical value C_t is for two-tailed 10% is 1.70. So, using the lower bound of the confidence interval then:

$$0.95 - 1.70 \times SEE_{\hat{\beta}} = 0.30 \Rightarrow SEE_{\hat{\beta}} = \frac{0.95 - 0.30}{1.70} = 0.3824$$

We know that the test statistic is given by:

$$T = \frac{\hat{\beta} - \beta_{H_0}}{SEE_{\hat{\beta}}} = \frac{0.95 - 0}{0.3824} = 2.4843$$

We need:

$$2[1-\Phi(|T|)] = 2[1-\Phi(|2.4843|)] = 2[1-0.9934] = 0.0132$$

Q.3972 Which of the following is the correct sequence of steps in hypothesis testing?

A. State the hypothesis, select the level of significance, compute the test statistic, formulate the decision rule, and make a decision.

B. State the hypothesis, select the level of significance, formulate the decision rule, compute the test statistic, and make a decision.

C. State the hypothesis, formulate the decision rule, select the level of significance, compute the test statistic, and make a decision.

D. Formulate the decision rule, state the hypothesis, select the level of significance, compute the test statistic, and make a decision.

The correct answer is **B.**

Hypothesis testing is a systematic method used in statistics to test the validity of a claim or hypothesis about a parameter. The process of hypothesis testing involves several steps. The first step is to state the null hypothesis and the alternative hypothesis. The null hypothesis is a statement about the population that either is believed to be true or is used to put forth an argument unless it can be shown to be incorrect beyond a reasonable doubt. The alternative hypothesis is a statement that will be accepted in place of the null hypothesis if the null hypothesis is rejected. The second step is to select a level of significance. The level of significance is the maximum probability of rejecting the null hypothesis when it is true. The third step is to formulate a decision rule. The decision rule is a statement that tells under what circumstances to reject the null hypothesis. The decision rule is based on the level of significance and the statistical test being used. The fourth step is to compute the test statistic. The test statistic is a quantity derived from the sample and used in the test of hypothesis. The final step is to make a decision. If the value of the test statistic is unlikely, based on the null hypothesis, reject the null hypothesis.

Choice A is incorrect. The sequence of steps in hypothesis testing is not correct. After stating the hypothesis and selecting the level of significance, it's important to formulate the decision rule before computing the test statistic. This is because the decision rule helps determine whether to reject or fail to reject the null hypothesis based on the computed test statistic.

Choice C is incorrect. In this option, formulating a decision rule before selecting a level of significance is not appropriate in hypothesis testing. The level of significance, which determines how much risk you are willing to take that you will reject a true null hypothesis, should be selected prior to formulating a decision rule.

Choice D is incorrect. Formulating a decision rule should not be done before stating the hypothesis and selecting a level of significance in hypothesis testing process. Stating your hypotheses first allows for clear understanding and direction for your statistical analysis.

Things to Remember

- Null Hypothesis (H0) and Alternative Hypothesis (Ha)
- Level of Significance
- Decision Rule
- Test Statistic
- Type I and Type II errors

Understanding these concepts is crucial for conducting hypothesis testing accurately and interpreting the results correctly.

Q.3973 The covariance between the 10-year money supply growth rates and the inflation rate is 0.007668, and the variance of the money supply growth rates is 0.02320. An investment analyst wants to explain the inflation rates using the money supply growth rates and predict the inflation rate when the money supply rate is 25%. The 10-year means for the money supply growth rate and inflation rate are 9% and 3%, respectively. The predicted inflation rate is *closest* to:

A. 7.234%

B. 8.289%

C. 6.345%

D. 8.756%

The correct answer is **B.**

Intuitively, the dependent variable (Y) is the inflation rate, and the independent variable (X) is the money supply growth rate.

We know that:

$$\hat{\beta} = \frac{\hat{\sigma}_{XY}}{\hat{\sigma}_X^2} = \frac{0.007668}{0.02320} = 0.33051$$

And

$$\hat{\beta}_0 = \bar{Y} - \beta\bar{X} = 0.03 - 0.33051 \times 0.09 = 0.0002541$$

So, the estimated regression equation is:

$$\hat{Y} = 0.0002541 + 0.33051X$$

Now, the analyst wants to predict the inflation rate ($\hat{Y}$) when the money supply growth rate (X) is 0.25 (25%). That is:

$$\hat{Y} = 0.0002541 + 0.33051 \times 0.25 = 0.0828816 \approx 8.289\%$$

Reading 19: Regression with Multiple Explanatory Variables

Q.436 Which of the following best explains why multiple linear regression analysis may be preferred to single linear regression?

A. It is simpler to model using modern software and computer programming.

B. It reduces the omitted variable bias.

C. It's easier to model and establishes the relationship between the dependent variable and important independent variables.

D. It requires multiple data inputs

The correct answer is **B.**

Multiple linear regression analysis is often preferred over single linear regression because it reduces the omitted variable bias. Omitted variable bias occurs when a regression model leaves out one or more independent variables that have a significant impact on the dependent variable. This omission can lead to an overstatement of the explanatory power of the regression analysis, as it fails to account for all the factors that influence the dependent variable. By including multiple independent variables, multiple linear regression analysis can capture a more comprehensive picture of the relationships at play, thereby reducing the omitted variable bias.

Choice A is incorrect. While modern software and computer programming have made it easier to perform multiple linear regression analysis, this does not inherently make it a preferred method over single linear regression. The choice of method depends on the nature of the data and the research question at hand, not on the ease of use of software or programming.

Choice C is incorrect. Multiple linear regression may not necessarily be easier to model than single linear regression. In fact, multiple linear regression can be more complex due to the inclusion of multiple independent variables and potential interactions between them. Furthermore, establishing relationships between dependent and independent variables is a feature common to both single and multiple regressions.

Choice D is incorrect. While it is true that multiple linear regression requires more data inputs (i.e., multiple independent variables), this requirement is a characteristic of the method rather than a rationale for its preference.

Things to Remember

- Omitted variable bias can lead to biased and inefficient estimates in regression analysis.
- Multiple linear regression allows for the inclusion of multiple independent variables to better capture the relationships in the data.

- Covariance matrix plays a crucial role in estimating the coefficients in multiple linear regression.
- R-squared value in multiple linear regression helps in understanding the proportion of variance explained by the model.
- Multicollinearity can be a challenge in multiple linear regression when independent variables are highly correlated.

Q.440 Assume we have the following multiple regression model:

$$Y = b_0 + 0.25X_1 + 0.14X_2 + \epsilon$$

It would be correct to say that:

A. If the independent variable X_1 increases by 1 unit, we would expect Y to increase by 0.25 units.

B. If the independent variable X_1 increases by 1 unit, we would expect Y to increase by 0.25/0.14 units.

C. If independent variable X_1 increases by 1 unit, we would expect Y to increase by 0.25 units, holding X_2 constant.

D. If independent variable X_2 increases by 1 unit, we would expect Y to increase by 0.25 + 0.14 units.

The correct answer is **C.**

If independent variable X_1 increases by 1 unit, we would expect Y to increase by 0.25 units, holding X_2 constant. This is the correct interpretation of the coefficient of X_1 in the given multiple regression model. In multiple regression models, each slope coefficient represents the estimated change in the dependent variable for a one-unit change in that independent variable, holding the other independent variables constant. This is also known as the partial effect of the independent variable on the dependent variable. Therefore, if X_1 increases by 1 unit, we would expect Y to increase by 0.25 units, assuming that X_2 remains constant.

Choice A is incorrect. While it is true that an increase in X_1 by 1 unit would lead to an increase in Y by 0.25 units, this statement does not take into account the effect of the other independent variable, X_2. In a multiple regression model, the change in the dependent variable due to a unit change in one independent variable is calculated while holding all other variables constant.

Choice B is incorrect. The ratio of coefficients (0.25/0.14) does not represent any meaningful relationship between Y and X_1. Each coefficient represents the expected change in Y for a unit change in its corresponding independent variable, holding all other variables constant.

Choice D is incorrect. The sum of coefficients (0.25 + 0.14) does not represent any meaningful relationship between Y and either of the independent variables (X_1,X_2). Each coefficient represents the expected change in Y for a unit change in its corresponding independent variable, holding all other variables constant.

Q.441 Which of the following best describes how OLS estimators are derived in multiple regression models?

A. By minimizing the absolute difference of the residuals.

B. By minimizing the sum of squared deviations.

C. By minimizing the distance between the predicted values and the fitted values.

D. By equating the sum of squared errors to zero.

The correct answer is **B.**

The Ordinary Least Squares (OLS) method in multiple regression models is derived by minimizing the sum of squared deviations. This method is used to estimate the parameters of a linear regression model. The objective of OLS is to find the values of the model's parameters that minimize the sum of the squared residuals, where a residual is the difference between the observed and predicted values of the dependent variable. The residuals are squared to ensure that positive and negative values do not cancel each other out, which allows the model to capture the magnitude of the deviations regardless of their direction. This method is widely used in econometrics and statistics due to its desirable properties, such as unbiasedness and efficiency, under certain assumptions.

Choice A is incorrect. While minimizing the absolute difference of residuals is a method used in some statistical models, it is not the method used to derive OLS estimators in multiple regression models. The OLS method specifically minimizes the sum of squared deviations.

Choice C is incorrect. Minimizing the distance between predicted values and fitted values does not accurately describe how OLS estimators are derived in multiple regression models. This statement may be misleading as it could be interpreted as minimizing the difference between each individual predicted value and its corresponding fitted value, which isn't what happens in OLS estimation.

Choice D is incorrect. Equating the sum of squared errors to zero does not accurately describe how OLS estimators are derived in multiple regression models either. In fact, if this were true, it would imply that there's no error or deviation from our model's predictions and actual observations which isn't possible with real-world data.

Things to Remember

- OLS (Ordinary Least Squares) method is a common technique used in regression analysis to estimate the parameters of a linear regression model.
- The OLS method minimizes the sum of squared residuals, where a residual is the difference between the observed and predicted values of the dependent variable.
- OLS estimators are derived by finding the values of the model's parameters that minimize the sum of squared deviations.
- OLS estimators have desirable properties such as unbiasedness and efficiency under certain assumptions.

- OLS is widely used in econometrics and statistics for estimating linear regression models.

Q.443 In a problem involving three independent variables and one dependent variable, assume that the computed coefficient of determination is 0.59. This result means that:

A. 59% of the total variation is explained by the dependent variable.

B. The correlation coefficient is 0.59 as well.

C. At least two of the three independent variables are highly correlated.

D. 59% of the total variation in the dependent variable is explained by the independent variables.

The correct answer is **D.**

The coefficient of determination, often denoted as R-squared, is a statistical measure that shows the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model. In the context of multiple regression, where there are more than one independent variables, the coefficient of determination refers to the proportion of the variance in the dependent variable that is predictable from the independent variables. Therefore, a coefficient of determination of 0.59 means that 59% of the total variation in the dependent variable can be explained by the independent variables. This does not necessarily mean that the independent variables cause the dependent variable to change, but rather, they provide a measure of how well observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model.

Choice A is incorrect. The coefficient of determination, also known as R-squared, does not measure the percentage of total variation explained by the dependent variable. Instead, it measures the proportion of variance in the dependent variable that can be predicted from the independent variables.

Choice B is incorrect. The correlation coefficient and the coefficient of determination are not necessarily equal. The correlation coefficient measures how closely two variables move together, while R-squared measures how much variation in one variable can be explained by another.

Choice C is incorrect. A high R-squared value does not imply that at least two independent variables are highly correlated with each other. It only indicates that a significant portion of the variance in our dependent variable can be predicted from our independent variables.

Things to Remember

- R-squared (coefficient of determination) ranges from 0 to 1, where 0 indicates that the model does not explain any of the variability of the response data around its mean, and

1 indicates that the model explains all the variability of the response data around its mean.

- R-squared is a key output of regression analysis and is used to determine the goodness-of-fit of a model.

- Adjusted R-squared is a modified version of R-squared that adjusts for the number of predictors in the model, providing a more accurate measure of goodness-of-fit for models with multiple predictors.

- R-squared can be influenced by outliers, multicollinearity, and overfitting, so it is important to consider these factors when interpreting the results.

Q.444 Under multiple linear regression models, there’s always the risk of overestimating the impact of additional variables on the explanatory power of the resulting model, which is why most researchers recommend using the adjusted R^2, $\overline{R^2}$, instead of R^2 itself. This adjusted R^2:

A. Is always positive.

B. Will never be greater than the regression R-Squared.

C. Cannot increase when an additional independent variable is incorporated into the model.

D. Is always negative.

The correct answer is **B.**

The adjusted R-squared, $\overline{R^2}$, will never be greater than the regression R-Squared, R^2. This is because the adjusted R-squared is a modification of the R-squared that adjusts for the number of predictors in the model. The adjusted R-squared increases only if the new variable improves the model more than would be expected by chance. It decreases when the variable improves the model by less than expected by chance. Therefore, the adjusted R-squared is always less than or equal to the R-squared.

Choice A is incorrect. The adjusted R-squared, $\overline{R^2}$, is not always positive. It can be negative if the model's explanatory power is poor and the number of predictors (independent variables) in the model is large relative to the number of observations.

Choice C is incorrect. The adjusted R-squared, $\overline{R^2}$, can increase when an additional independent variable that significantly improves the model's fit is incorporated into it. However, unlike R^2, $\overline{R^2}$ penalizes for adding variables that do not improve the model significantly.

Choice D is incorrect. The adjusted R-squared, $\overline{R}^2$, does not have to be always negative. It can range from 1 to negative infinity depending on how well or poorly the regression line fits the data and how many predictors are in your model relative to your sample size.

Things to Remember

- R-squared (R^2) is a measure of how well the independent variables explain the variability of the dependent variable in a regression model.
- Adjusted R-squared ($\overline{R}^2$) penalizes for adding unnecessary variables to the model that do not significantly improve the model's fit.
- Adjusted R-squared is always less than or equal to the R-squared value in multiple linear regression models.
- Adding more variables to a model can lead to overfitting, where the model performs well on the training data but poorly on new data.
- Model selection techniques like AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) can help in choosing the best model among competing models.

Q.445 When an important variable is omitted from a regression model, the assumption that $E(\epsilon_i|X_i) = 0$ is violated. This implies that:

A. The OLS estimator is biased.

B. The product of the residuals and any of the independent variables is no longer zero.

C. The sum of the residuals is no longer equal to zero.

D. The coefficient of determination is zero.

The correct answer is **A.**

The Ordinary Least Squares (OLS) estimator is biased. In the context of regression analysis, bias refers to the difference between the expected value of an estimator and the true value of the parameter it is estimating. The OLS estimator is designed to minimize the sum of the squared residuals, and it assumes that the expected value of the error term, given the independent variables, is zero. This assumption is crucial for the unbiasedness of the OLS estimator. If a significant variable is omitted from the model, this assumption is violated, which implies that the OLS estimator is biased. This is because the omitted variable might be correlated with the included variables, causing the error term to also be correlated with the included variables. This correlation between the error term and the independent variables leads to a biased estimate of the regression coefficients, meaning that the OLS estimator will systematically overestimate or underestimate the true parameter values.

Choice B is incorrect. The product of the residuals and any of the independent variables being zero is not necessarily a result of omitting a key variable from the model. This statement refers to orthogonality condition in OLS regression, which states that each explanatory variable should be uncorrelated with the error term. However, this does not directly relate to bias caused by omitted variables.

Choice C is incorrect. The sum of residuals being equal to zero is an inherent property of OLS estimators and does not depend on whether a significant variable has been omitted or not. Even if a key variable is omitted, it doesn't imply that sum of residuals will no longer be zero.

Choice D is incorrect. The coefficient of determination (R-squared) being zero implies that none of the independent variables in the model explain any variation in dependent variable which isn't necessarily due to omission of key variables from model. It could also be due to poor selection or irrelevance of included independent variables.

Things to Remember

- Omitted Variable Bias: Omitted variable bias occurs when a relevant variable is left out of a regression model, leading to biased and inconsistent parameter estimates.
- Endogeneity: Omission of important variables can lead to endogeneity issues, where the independent variables are correlated with the error term.
- Multicollinearity: Omitting important variables can exacerbate multicollinearity issues, where independent variables are highly correlated with each other.
- Adjusted R-squared: Adjusted R-squared is a modified version of R-squared that adjusts for the number of predictors in the model, providing a more accurate measure of the goodness of fit.
- Model Specification: Ensuring the correct specification of the regression model is crucial to avoid bias and obtain reliable estimates.

Q.447 An analyst uses the following regression model to explain stock returns:
Dependent variable:
ASR = Annual stock returns (%)

Independent variables:
MCP = Market capitalization (divided by $1million to simplify modeling)
SEF = Stock exchange firm, where SEF = 1 if the stock is that of a firm listed on the New York Stock Exchange and SEF = 0 if not listed
FMR = Forbes magazine ranking (FMR = 4 is the highest ranking)

The following table presents the regression results:

	Coefficient	Standard Error
Intercept	0.6330	1.11
MCP	0.0840	0.0130
SEF	0.5101	0.1235
FMR	0.7000	0.3241

Based on the results in the table above, which of the following is the correct regression equation?

A. 0.0840(MCP) + 0.5101(SEF) + 0.7(FMR)

B. 0.6330 + 0.0840(MCP) + 0.5101(SEF) + 0.7(FMR)

C. 1.11 + 0.0840(MCP) + 0.5101(SEF) + 0.7(FMR)

D. 1.11 + 0.0130(MCP) + 0.1235(SEF) + 0.3241(FMR)

The correct answer is **B.**

All the regression parameters are contained in the coefficients column. Standard errors would only be used to conduct hypothesis tests or construct confidence intervals about the independent variables.

Q.448 An analyst uses the following regression model to explain stock returns:
Dependent variable:
ASR = Annual stock returns (%)

Independent variables:
MCP = Market capitalization (divided by $1million to simplify modeling)
SEF = Stock exchange firm, where SEF = 1 if the stock is that of a firm listed on the New York Stock Exchange and SEF = 0 if not listed

FMR = Forbes magazine ranking (FMR = 4 is the highest ranking)

If the regression equation is 0.6330 + 0.0840(MCP) + 0.5101(SEF) + 0.7(FMR), then what is the expected amount of stock return that would be attributed to it being a listed stock?

A. 1.11 + 0.5101

B. 0.1235

C. 0.5101

D. 1.11 + 0.1235

The correct answer is **C.**

The regression equation is 0.6330 + 0.0840(MCP) + 0.5101(SEF) + 0.7(FMR)
The coefficient on SEF is the amount of stock return that would be attributed to it being a listed stock.

Things to Remember

- Regression analysis is a statistical technique used to understand the relationship between a dependent variable and one or more independent variables.
- Coefficient estimates in a regression model represent the change in the dependent variable for a one-unit change in the independent variable, holding all other variables constant.
- R-squared (R2) is a measure of how well the independent variables explain the variability of the dependent variable in a regression model.
- T-statistics and p-values are used to test the significance of the coefficients in a regression model.
- Interpreting regression coefficients involves understanding the direction and magnitude of the relationship between variables.

Q.451 An analyst believes that future 15-year real earnings of the S&P 500 are a function of the trailing dividend payout ratio of the stocks in the index (DB) and the yield curve slope (YC). She collects data and obtains the following multiple regression results:

	Coefficient	Standard Error
Intercept	−10.8%	1.567%
DB	0.27	0.029
YC	0.12	0.210

Test the statistical significance of the independent variable DB at the 5% level of significance, quoting the value of the test statistic and the conclusion. (Number of observations = 43)

A. Test statistic = 2.021; DB regression coefficient is statistically different from zero

B. Test statistic = 9.310; DB regression coefficient is statistically different from zero

C. Test statistic = 0.018; DB regression coefficient is not statistically different from zero

D. Test statistic = 9.310; DB regression coefficient has little effect on the returns of S&P 500

The correct answer is **B.**

We are testing the following hypothesis:

$$H_0 : \beta_{DB} = 0 \text{vs} H_1 : \beta_{DB} \neq 0$$

The test statistic is $\frac{0.27}{0.029} = 9.310$

$t_{\frac{a}{2},n-k-1} = t_{0.025,40} = 2.021$

The test statistic (9.310) is greater than the upper critical value (2.021) of the t-distribution with 40 degrees of freedom. Therefore, we reject the null hypothesis and conclude that the DB regression coefficient is statistically different from zero at the 5% level of significance.

Q.454 A multiple regression model has 3 independent variables such that:

$$Y_i = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

An analyst carries out a joint hypothesis test to determine the statistical significance of the independent variable coefficients, incorporating all the 3 variables. The null hypothesis is such that each variable coefficient is equated to zero. The results reveal that the F-statistic is greater than the one-tailed critical F-value. This implies that:

A. At least one of the coefficients is statistically significantly different from zero.

B. Each of the independent variable coefficients is statistically significantly different from zero.

C. None of the coefficients is statistically different from zero.

D. Only one of the independent variable coefficients is statistically different from zero.

The correct answer is **A.**

The F-statistic is a measure used in statistical analysis to assess the significance of the overall regression model. In the context of a joint hypothesis test, the F-statistic is used to determine whether at least one of the independent variables in the model has a significant effect on the dependent variable. The null hypothesis in this case is that all the coefficients of the independent variables are equal to zero, implying that none of the variables have a significant effect. If the calculated F-statistic is greater than the critical F-value, the null hypothesis is rejected. This means that at least one of the coefficients is statistically significantly different from zero, indicating that at least one of the independent variables has a significant effect on the dependent variable.

Choice B is incorrect. The F-statistic being greater than the critical F-value does not imply that each of the independent variable coefficients is statistically significantly different from zero. It only indicates that at least one of the coefficients is statistically significant, but it does not specify which or how many.

Choice C is incorrect. If none of the coefficients were statistically different from zero, then we would expect our F-statistic to be less than or equal to the critical F-value, not greater. Therefore, this statement contradicts our observation.

Choice D is incorrect. While it's possible that only one coefficient could be statistically significant (which would still lead to a larger F-statistic), this choice incorrectly assumes that only one coefficient must be significant based on our observation. In reality, any number (one or more) of coefficients could be significantly different from zero.

Q.455 Peter Bridge, FRM, runs a regression of monthly stock returns on four independent variables over 65 months. The total sum of squares is 540, and the sum of squared residuals is

250. Carry out a statistical test at the 5% significance level with the null hypothesis that all four of the independent variables are equal to zero. Quote the F-statistic and the conclusion.

A. F-statistic = 17.40; At least one of the 4 independent variables is significantly different from zero.

B. F-statistic = 2.525; At least one of the 4 independent variables is significantly different from zero.

C. F-statistic = 72.5; All the 4 independent variables are significantly different from zero.

D. F-statistic = 17.40; None of the independent variables is significantly different from zero.

The correct answer is **A.**

The set of hypotheses is such that:

$H_0 : B_1 = B_2 = B_3 = B_4 = 0$ vs H_1 At least one $B_j \neq 0$

The F-statistic is given by:

$$F = \frac{\left[\frac{ESS}{k}\right]}{\left[\frac{SSR}{(n-k-1)}\right]}$$

Where:

TSS =total sum of squares;

ESS = explained sum of squares;

RSS = residual sum of squares;

k = number of independent variabls; and

n=number of observations.

$$ESS = TSS – SSR = 540 – 250 = 290$$

Therefore,

$$F = \frac{\left[\frac{290}{4}\right]}{\left[\frac{250}{(60)}\right]} = 17.4$$

The one-tailed critical F-value at 5% level with 4 and 60 degrees of freedom in the numerator and denominator respectively, is approximately 2.525. Since 17.40 > 2.525, we can reject the null hypothesis and conclude that at least one of the four independent variables is significantly different from zero.

Q.459 Elizabeth Graham, FRM, runs a regression of monthly stock returns on five independent variables over 66 months. The explained sum of squares is 270, and the sum of squared residuals is 250. Graham then performs a statistical test at the 10% significance level with the null hypothesis that all five of the independent variables are equal to zero. Quote the F-statistic and the conclusion.

A. F-statistic = 12.96; At least one of the 5 independent variables is significantly different from zero.

B. F-statistic = 1.946; At least one of the 5 independent variables is significantly different from zero.

C. F-statistic = 72.5; All the 5 independent variables are significantly different from zero.

D. F-statistic = 17.40; None of the independent variables is significantly different from zero.

The correct answer is **A.**

The set of hypotheses is such that:

$H_0 : B_1 = B_2 = B_3 = B_4 = 0$ vs H_1 At least one $B_j \neq 0$

$$\text{The F-statistic} = \frac{\left[\frac{ESS}{k}\right]}{\left[\frac{SSR}{(n-k-1)}\right]}$$

$$F = \frac{\left[\frac{270}{5}\right]}{\left[\frac{250}{60}\right]} = 12.96$$

The one-tailed critical *F*-value at 10% level with 5 and 60 degrees of freedom in the numerator and denominator respectively, is approximately 1.946. Since 12.96 > 1.946, we can reject the null hypothesis and conclude that at least one of the five independent variables is significantly different from zero.

Q.460 An analyst runs a regression of monthly value-stock returns on 8 independent variables . Given the following information: Explained Sum of Squares=1435 Residual sum of Squares=1335 Number of obsrvations=28 R^2 and the F-statistic, respectively, are closest to:

A. 53%, 4

B. 51%, 3.8

C. 52%, 2.6

D. 50%, 4

The correct answer is **C.**

$$R^2 = \frac{ESS}{TSS} = \frac{1435}{2770} = 52\%$$

$$F = \frac{\left[\frac{ESS}{df}\right]}{\left[\frac{SSR}{n-(df+1)}\right]} = \frac{\left[\frac{1435}{8}\right]}{\left[\frac{1335}{28-8-1}\right]} = 2.5529$$

Where,

Total Sum of Squares(TSS)=Explained Sum of Squares+Residual Sum Squares

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Q.462 Under multiple linear regression, if an estimator is said to be consistent, what does this imply?

A. On average, the estimated values of the coefficients will be equal to the true values.

B. The coefficient estimates will be as close to their true values as possible, regardless of the sample size.

C. The estimates will converge upon the true values as the sample size, n, increases.

D. The OLS estimator will also be unbiased and efficient.

The correct answer is **C.**

The term 'consistent' in the context of estimators in multiple linear regression analysis refers to a specific property of these estimators. A consistent estimator is one where the estimates produced by the estimator will converge to the true population values as the sample size increases. This means that as we collect more and more data, the estimates provided by a consistent estimator will get closer and closer to the actual values in the population. This property is crucial in statistical analysis, as it ensures that our estimates become more accurate and reliable as we gather more data. Therefore, choice C accurately captures the definition of a consistent estimator.

Choice A is incorrect. While it's true that an unbiased estimator will have expected values equal to the true values, this does not necessarily mean the estimator is consistent. Consistency refers to the property of an estimator where its estimates converge to the true value as sample size increases, which is not guaranteed by unbiasedness alone.

Choice B is incorrect. This statement describes efficiency rather than consistency. An efficient estimator provides estimates that are as close as possible to their true values regardless of sample size, but this doesn't imply consistency. A consistent estimator may not always provide estimates close to their true values for small sample sizes, but it will converge towards these values as sample size increases.

Choice D is incorrect. The properties of being unbiased and efficient do not automatically imply consistency in an OLS (Ordinary Least Squares) estimator or any other type of estimator for that matter. These are separate statistical properties with different implications about an estimator's behavior and performance.

Things to Remember

- Consistency is a desirable property of estimators as it ensures that our estimates become more accurate and reliable as we gather more data.
- Consistency implies that as the sample size increases, the estimates provided by the estimator will converge to the true population values.
- Unbiasedness, efficiency, and consistency are three important properties of estimators, each with its own implications and criteria.
- Efficiency refers to how close the estimates are to the true values, regardless of sample size, while unbiasedness refers to the expected value of the estimator being equal to the true value.
- Consistency is particularly important in regression analysis, where we aim to estimate the relationships between variables accurately.

Q.463 Assume that after constructing a multiple linear regression model, you find that $R^2 = 0.97$. How confident would you be in using the line of best fit for the purposes of prediction?

A. Not confident.

B. Very confident.

C. The relationship would be too weak at predicting using a linear function.

D. The relationship would be random and hence impossible to predict.

The correct answer is **B.**

A high R^2 value of 0.97 indicates that 97% of the variability in the dependent variable can be explained by the independent variables in the model. This means that the model fits the data very well and the line of best fit accurately represents the data. Therefore, we can be very confident in using this model for prediction. The closer the R^2 value is to 1, the better the model fits the data. An R^2 value of 0.97 is considered very high, indicating a strong relationship between the independent and dependent variables. This means that the model is likely to make accurate predictions.

Choice A is incorrect. The high R^2 value of 0.97 indicates a strong relationship between the dependent and independent variables in the regression model, which should increase our confidence in using this model for predictive purposes, not decrease it.

Choice C is incorrect. An R^2 value of 0.97 suggests a very strong linear relationship between the dependent and independent variables, contrary to what this option suggests about a weak predictive power using a linear function.

Choice D is incorrect. A high R^2 value like 0.97 implies that there's little randomness in the data and that most of it can be explained by our regression model, making predictions far from impossible.

Things to Remember

- Coefficient of Determination (R^2) measures the proportion of the variance in the dependent variable that is predictable from the independent variables.
- An R^2 value closer to 1 indicates a better fit of the model to the data.
- Adjusted R^2 should be considered when comparing models with different numbers of independent variables.
- Residual analysis is important to check the assumptions of the linear regression model.
- Outliers and influential points can greatly affect the regression results.

Q.482 A market analyst has established that future 10-year growth of earnings in the S&P 500 can be explained by a combination of two factors: the slope of the yield curve (YCS) and the preceding dividend payout ratio (PR) of stocks that have been featured in the index. The analyst carries out a regression and obtains the following results:

	Coefficient	Standard error
Intercept	−10.6%	1.525%
YCS	0.20	0.024
PR	0.12	0.230

Test the statistical significance of YCS at the 10% level of significance, quoting the t-statistic and the conclusion if n = 46. Please click here to see the t-table

A. 16.60; The YCS coefficient is statistically significantly different from zero.

B. 16.60; The PR coefficient is statistically significantly different from zero.

C. 8.333; The YCS coefficient is statistically significantly different from zero.

D. 1.68; The YCS coefficient is not statistically significantly different from zero.

The correct answer is **C.**

You should test the hypothesis:

$$H_0 : PR = 0 \text{ vs } H_1 : PR \neq 0$$

The t-statistic = $\frac{0.20}{0.024} = 8.333$ $

From statistical tables, we find that the 10% 2-tailed critical t-value with a total of (46 – 2 – 1) degrees of freedom = 1.68 (approximately). We should reject H_0 only if the t-statistic is greater than the tabulated t-value.

$$8.333 > 1.68$$

Therefore, we can reject H_0 and conclude that the YCS regression coefficient is statistically significantly different from zero.

Q.3299 A portfolio manager believes that returns on pharmaceutical stocks are more volatile than the returns generated on e-commerce stocks. To check this hypothesis, the portfolio manager collects the data summarized in exhibit 1.

Exhibit 1: Volatility in Pharmaceutical vs. e-Commerce Stocks

	Pharma Stock	e-Commerce Stocks
Standard Deviation	1.50%	2.10%
Sample Size	20	25

What is the value of the test statistic?

A. 1.51

B. 1.96

C. 1.7

D. 2.14

The correct answer is **B.**

As the test requires testing the equality of variances of two populations, the appropriate test is the F-test.

$$\text{Test statistic} = \frac{(\text{std. dev. ecommerce})^2}{(\text{std. dev. pharma})^2} = \frac{(2.10\%)^2}{(1.50\%)^2} = 1.96$$

Note: A convention, or usual practice, is to use the larger of the two standard deviations on top (in the numerator). When we follow this convention, the value of the test statistic is always greater than or equal to 1; tables of critical values of F then need to include only values greater than or equal to 1. Under this convention, the rejection point for any formulation of hypotheses is a single value on the right-hand side of the relevant F-distribution. However, even without following this convention, we would still arrive at the same conclusion (on whether or not to reject the null).

Q.3334 Tom Well, FRM, works for a trading company. Using historical data, he has computed the following variables considering one independent and one dependent variable.

- Explained Sum of Squares (ESS) = 60
- Sum of Squared Residuals (SSR) = 15

If we are dealing with a sample size of 62 observations, what is the coefficient of determination and the standard error of the estimate, respectively.

A. Coefficient of Determination = 0.50, Standard Error of the Estimate = 0.50

B. Coefficient of Determination = 0.80, Standard Error of the Estimate = 0.80

C. Coefficient of Determination = 0.80, Standard Error of the Estimate = 0.50

D. Coefficient of Determination = 0.25, Standard Error of the Estimate = 0.25

The correct answer is **C.**

$$\text{Total variation (TSS)} = \text{Explained variation (ESS)} + \text{Unexplained variation (SSR)}$$
$$\text{TSS} = 60 + 15 = 75$$

$$\text{Coefficient of determination: R-square} = \frac{\text{ESS}}{\text{TSS}} = \frac{60}{75} = 0.80$$

Standard Error of the Estimate (SEE) is given by:

$$\text{SEE} = \sqrt{\frac{\text{Unexplained variation}}{n-2}} = \sqrt{\frac{15}{62-2}} = 0.5$$

Additional tips:

$$\text{Total variation} = \sum\left(Y - \overline{Y}\right)^2$$
$$\text{Unexplained variation} = \sum\left(Y - \hat{Y}\right)^2 = \sum\left(\hat{\epsilon}\right)^2$$
$$\text{SEE} = \sqrt{\frac{\sum(\text{Actual Y} - \text{Predicted Y})^2}{n-2}} = \sqrt{\frac{\sum\left(Y - \hat{Y}\right)^2}{n-2}} = \sqrt{\frac{\sum\left(\hat{\epsilon}\right)^2}{n-2}}$$

Q.3345 During the course of building a model using multiple linear regression, an analyst tried to judge the model based on its coefficient of determination (R^2) and adjusted R^2. Which of the following interpretation is correct?

A. The adjusted R^2 is always greater than the R^2

B. Both the adjusted R^2 and the R^2 always have positive values

C. The adjusted R^2 is always less than or equal to the R^2

D. The adjusted R^2 always increases with an increase in the number of independent

The correct answer is **C.**

The adjusted R^2 is always less than or equal to the R^2. The coefficient of determination, R^2, is a measure of how well the regression predictions approximate the real data points. An R^2 of 100% indicates that all changes in the dependent variable are completely explained by changes in the independent variable(s). However, R^2 has a significant limitation: it always increases as more independent variables are added to the model, even if those variables are only weakly associated with the response. This can lead to overfitting, where the model describes random error or noise instead of the underlying relationship. To counteract this problem, the adjusted R^2 is used. The adjusted R^2 accounts for the number of predictors in the model. Unlike R^2, the adjusted R^2 increases only if the new variable improves the model more than would be expected by chance, and it can decrease if a variable improves the model less than expected by chance. Therefore, the adjusted R^2 is always less than or equal to the R^2.

Choice A is incorrect. The adjusted R^2 is not always greater than the R^2. In fact, it's usually less than or equal to the R^2. This is because the adjusted R^2 takes into account the number of predictors in the model and adjusts for it, which can lead to a lower value compared to R^2.

Choice B is incorrect. While it's true that both metrics are non-negative and can take values between 0 and 1, they do not always have positive values. Both metrics can be zero if there's no linear relationship between dependent and independent variables.

Choice D is incorrect. The adjusted R^2 does not always increase with an increase in the number of independent variables. Unlike R-squared, which only increases or stays constant as you add more X variables to your model, Adjusted R-squared increases only when new predictors enhance the model above what would be obtained by probability and decreases when they do not.

Things to Remember

- R-squared (R^2) is a measure of how well the regression predictions approximate the real data points.
- Adjusted R^2 accounts for the number of predictors in the model to prevent overfitting.
- Adjusted R^2 increases only if the new variable improves the model more than expected by chance.

- Adjusted R^2 can decrease if a variable improves the model less than expected by chance.
- Both R^2 and adjusted R^2 can range between 0 and 1, with 1 indicating a perfect fit.
- Adjusted R^2 is a more reliable metric when comparing models with different numbers of predictors.

Q.3346 An analyst performed a regression of monthly returns on a stock with 4 independent variables over a 50 month period. The analyst calculated the total sum of squares (TSS) and the sum of square residuals or error (SSR) as 500 and 100, respectively.
What is the adjusted R^2?

A. 0.8

B. 0.78

C. 0.2

D. 0.75

The correct answer is **B.**

$$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = 1 - \frac{\text{Unexplained variation}}{\text{Total variation}} = 1 - \frac{100}{500} = 0.8$$

$$\overline{R}^2 = 1 - \left(\frac{n-1}{n-k-1}\right)(1 - R^2) = 1 - \left(\frac{50-1}{50-4-1}\right)(1 - 0.8) = 1 - \frac{49}{45} \times 0.2 = 0.7822 \approx 0.78$$

Q.3350 In a hypothetical world, GDP is regressed against interest rate and inflation, and regression results are shown below.

$$\text{GDP} = \text{a} + \text{b (Interest rate)} + \text{c (Inflation)} + \text{Error term}$$

	Coefficient	p-Value
a	9	0.042
b	2	0.035
c	1.5	0.012

ANOVA	df	SS
Regression	2	240
Residual	37	1070
Total	39	1300
Total	0.428	
R2	0.183	
Observation	40	

Which of the test is relevant to determine whether the regression model as a whole is significant?

A. $\text{F}-\text{test}; \text{H}_0$: All slope coefficients = 0; H_a: At least one slope coefficient ≠ 0

B. $\text{F}-\text{test}; \text{H}_0$: All slope coefficients ≥ 0; H_a: At least one slope coefficient < 0

C. $\text{t}-\text{test}; \text{H}_0$: All slope coefficients = 0; H_a: At least one slope coefficient ≠ 0

D. $\text{t}-\text{test}; \text{H}_0$: All slope coefficients ≥ 0; H_a: At least one slope coefficient < 0

The correct answer is **A.**

An F-test is used to test whether any of the independent variables explain the variation in the dependent variable (test of overall model significance). It is used to determine whether the regression model as a whole is significant.

H_0: All slope coefficients = 0

H_a: At least one slope coefficient ≠ 0

Q.3352 In a hypothetical world, GDP is regressed against interest rate, and inflation and regression results are shown below.

$$\text{GDP} = \text{a} + \text{b (Interest Rate)} + \text{c (Inflation)} + \text{Error Term}$$

	Coefficient	p-Value
a	9	0.042
b	2	0.035
c	1.5	0.012

ANOVA	df	SS
Regression	2	240
Residual	37	1070
Total	39	1310
Total	0.428	
R2	0.183	
Observation	40	

Assume that on a certain significance level, the critical value of the F-statistic is 4. Which of the following is correct?

A. The value of F-statistic is less than its critical value.

B. We fail to reject the null hypothesis: H_0: All slope coefficients = 0.

C. The model as a whole is statistically significant.

D. The model as whole is not statistically significant.

The correct answer is **C.**

The F-statistic is given by:

$$F = \frac{\left[\frac{ESS}{k}\right]}{\left[\frac{SSR}{(n-k-1)}\right]}$$

Where:

TSS =total sum of squares;

ESS = explained sum of squares;

RSS = residual sum of squares;

k = number of independent variables; and

n=number of observations.

Therefore,

$$F = \frac{\frac{240}{2}}{\frac{1070}{(40-2-1)}} = \frac{120}{\frac{1070}{37}} = 4.1495$$

The F-statistic is 4.1495 and its critical value is 4.

If F-statistic > F critical, we reject the null hypothesis.

H_0: All slope coefficients = 0

H_a: At least one slope coefficient ≠ 0

Model as a whole is significant.

Q.3353 Suppose that the following regression is estimated using 25 quarterly observations:

$$y_t = \beta_1 + \beta_2 x_{2t} + \beta_3 x_{3t} + u_t$$

What is the appropriate critical value for a 2-sided 5% size of test of $H_0 : \beta_2 = 1$?

A. 1.72

B. 2.06

C. 2.07

D. 1.64

The correct answer is **C.**

When we have a total of k "regressors" (including a constant) and n observations, the t-test statistic will follow a t-distribution with $n - k - 1$ degrees of freedom.
In this case, $n = 25, k = 2$

Also, this is a two-tailed test.

Thus, we would be interested in $t_{0.025,22}$. We would be looking in the t-tables in the degrees of freedom = 22 rows and the 2.5% column (so that 2.5% is in each tail for a 5% 2-tailed test). The critical value would be 2.07.

Note: the fact that observations are quarterly is irrelevant in our calculations.

Also, note that when testing a hypothesis about a single coefficient in a model with multiple regressors is just the same as testing in a model with a single explanatory variable. Test of the null hypothesis, $H_0 : \beta_i = \beta_\delta$ can be achieved using a t-test. However, the t-test is **NOT** suitable to test hypotheses that involve more than one parameter. In such scenarios, then we have to use the F-test.

Q.3354 Consider the following 2 regression models built to estimate a common phenomenon:

Model	R^2	Adjusted R^2
Model 1	0.75	0.73
Model 2	0.81	0.72

If a variable x_4 is introduced in model 2 and the coefficient estimate β_4 is non-zero, which one of the following is most likely correct?

A. The researcher must have made a mistake because the adjusted R^2 for model 2 must be greater than the adjusted R^2 for model 1.

B. Variable x_4 is statistically significant.

C. The coefficient estimate β_4 is non-zero but not significant.

D. There is no sufficient information to determine.

The correct answer is **C.**

The coefficient estimate β_4 is non-zero but not significant. The fact that the R^2 value for Model 2 is higher than that for Model 1, but the Adjusted R^2 value for Model 2 is lower, suggests that the additional variable x_4 in Model 2 is not statistically significant. In other words, it does not have a substantial impact on the dependent variable. This is because the Adjusted R^2 value takes into account the number of predictors in the model, unlike the R^2 value. Therefore, even though the coefficient estimate β_4 for the variable x_4 is non-zero, it does not significantly contribute to the prediction of the dependent variable, hence it is not significant.

Choice A is incorrect. The adjusted R^2 for Model 2 being less than that of Model 1 does not necessarily indicate a mistake by the researcher. The adjusted R^2 takes into account the number of predictors in the model and penalizes for unnecessary complexity. Therefore, it's possible for a model with more predictors (like Model 2) to have a lower adjusted R^2 if the additional predictors do not significantly improve the model's predictive power.

Choice B is incorrect. Just because variable x_4's coefficient estimate β_4 is non-zero, it doesn't automatically mean that variable x_4 is statistically significant. Statistical significance depends on various factors such as sample size, variability in data etc., and cannot be determined solely based on whether or not a coefficient estimate is zero.

Q.3355 Consider the following 2 regression models:

$$\text{Model 1}: y_t = \beta_1 + \beta_2 x_{2_t} + u_t$$
$$\text{Model 2}: y_t = \beta_1 + \beta_2 x_{2_t} + \beta_3 x_{3_t} + u_t$$

A researcher determines that the two models have identical R-squared values. This most likely implies that:

A. Model 2 must have a lower value of adjusted R-squared.

B. Model 2 must have a higher value of adjusted R-squared.

C. Model 1 and 2 will also have identical values of adjusted R-squared.

D. Variable x_3 is statistically significant.

The correct answer is **A.**

If two regression models have identical R-squared values, it implies that the additional variable in the second model, in this case x_3, does not contribute to the explanatory power of the model for the dependent variable y. This is because the R-squared value represents the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model. Therefore, if the R-squared values are identical for both models, it means that the additional variable x_3 in the second model does not improve the model's ability to explain the variance in y. As a result, the coefficient estimate for x_3, denoted as β_3, must be exactly zero, indicating that x_3 is not statistically significant. Furthermore, the adjusted R-squared value, which takes into account the number of predictors in the model, will be lower for the second model. This is because the adjusted R-squared value penalizes the model for including predictors that do not improve the model's ability to explain the variance in the dependent variable. In this case, since x_3 does not improve the model's explanatory power, the adjusted R-squared value for the second model will be lower than that of the first model.

Choice B is incorrect. The adjusted R-squared takes into account the number of predictors in the model. It increases only if the new predictor improves the model more than would be expected by chance, and it decreases when a predictor improves the model by less than expected by chance. In this case, since both models yield same R-squared values, adding another variable x_3 in Model 2 does not improve the prediction power of Model 1. Therefore, adjusted R-squared for Model 2 will not be higher.

Choice C is incorrect. As explained above, since adding another variable x_3 in Model 2 does not improve its prediction power compared to Model 1 (as indicated by same R-squared values), their adjusted R-squares will not be identical either because adjusted R-square takes into account number of predictors in a model.

Choice D is incorrect. The fact that both models have same R-square value implies that addition of variable x_3 did not contribute significantly to explaining variation in dependent variable y. Hence we cannot conclude that variable x_3 is statistically significant.

Q.3356 For a sample of 40 years, the relationship between GDP growth (Y_i), inflation (X_1) and interest rates (X_2) is modeled as follows:

$$y_t = \beta_1 + \beta_2 x_{1_t} + \beta_3 x_{2_t} + u_t$$

An economist wishes to test the joint hypothesis that $\beta_1 = 0$, $\beta_2 = 0$, and $\beta_3 = 0$ at the 90% confidence level.
The p-value for the t-statistic for β_1 is 0.11, and the p-value for the t-statistic for β_2 is 0.12. The p-value for the F-statistic for the regression is 0.09. Which of the following statements is correct?

A. We cannot reject the null hypothesis because none of the β_s is different from zero at 90% confidence.

B. We can reject the null hypothesis because each of the β_s is different from zero at 90% confidence.

C. We cannot reject the null hypothesis because the F-statistic is not significant at 90% confidence.

D. We can reject the null hypothesis because the F-statistic is significant at 90% confidence.

The correct answer is **D.**

To test, the joint hypothesis, we need the F-test. The t-tests are not sufficient - individual tests do not account for the effects of interactions among the independent variables Thus, the information on the p-value of each β is not relevant in this case.
P-value is the smallest level of significance at which we can reject the null hypothesis. In other words, we always reject the null hypothesis whenever the p-value is less than the level of significance of the test. In this case, the P-value of the F-test is 0.09 which is less than 10% (level of significance). As such, we can reject the null hypothesis and conclude that at least one β is different from zero. Note that a p-value greater than the significance level is not statistically significant and indicates strong evidence for retaining the null hypothesis.

Reading 20: Regression Diagnostics

Q.401 A financial risk analyst has been investigating a linear regression model that explains the return of a stock based on a set of macroeconomic variables. Upon analyzing the residuals, the analyst suspects the presence of heteroskedasticity in the model. Which of the following consequences of heteroskedasticity should the analyst be most concerned about?

A. Bias in the regression coefficients.

B. Inaccurate goodness-of-fit measures.

C. Biased standard errors of the coefficients.

D. Violation of the normality assumption.

The correct answer is **C.**

Heteroskedasticity in a regression model refers to a situation where the variability of the error terms is not constant across all levels of the independent variables. This inconsistency in the error variance can lead to biased standard errors of the coefficients. The standard errors are crucial in determining the significance of the regression coefficients. If these are biased, it can lead to incorrect inferences about the significance of the regression coefficients. This can further result in misleading results from t-tests and F-tests that are based on these standard errors. Therefore, the most significant concern with heteroskedasticity is the potential bias it introduces to the standard errors of the coefficients, which can compromise the reliability of the statistical inferences drawn from the model.

Choice A is incorrect. Heteroskedasticity does not cause bias in the regression coefficients. The presence of heteroskedasticity does not affect the unbiasedness property of the OLS estimators, meaning that even if heteroskedasticity is present, on average, we can expect our estimates to be correct.

Choice B is incorrect. While heteroskedasticity may impact certain aspects of a regression model's performance, it does not directly lead to inaccurate goodness-of-fit measures. Goodness-of-fit measures like R-squared are unaffected by heteroskedasticity.

Choice D is incorrect. Heteroskedasticity refers to a situation where the variability of a variable is unequal across different ranges of values of another variable with which it has been paired for analysis. It doesn't necessarily imply violation of normality assumption in regression analysis.

Things to Remember

- Homoskedasticity: This is the opposite of heteroskedasticity, where the variance of the error terms is constant across all levels of the independent variables.

- White's Test: A statistical test used to detect heteroskedasticity in a regression model.

- Robust Standard Errors: An alternative to correct for heteroskedasticity in regression analysis by providing standard errors that are robust to violations of the homoskedasticity assumption.

- Weighted Least Squares (WLS): A method used to address heteroskedasticity by assigning different weights to observations based on the variance of the error terms.

Q.437 One of the following assumptions is applied in the multiple least squares regression model. Which one?

A. The independent variables included in the model are homoskedastic.

B. The residual terms are heteroskedastic.

C. The dependent variable is unique and stationary.

D. There is no perfect multi-collinearity.

The correct answer is **D.**

The assumption of no perfect multicollinearity is indeed a fundamental assumption in multiple least squares regression. Multicollinearity refers to a situation where two or more independent variables in a regression model are highly linearly related. In the case of perfect multicollinearity, the independent variables are perfectly linearly related, meaning that one can be expressed exactly as a linear combination of the others. This poses a problem as it undermines the statistical significance of an independent variable. While multicollinearity can inflate the variance of the regression coefficients, making them unstable and difficult to interpret, perfect multicollinearity makes it impossible to estimate the regression coefficients using ordinary least squares. Therefore, the assumption of no perfect multicollinearity is crucial in multiple regression analysis to ensure that the model can provide meaningful and interpretable results.

Choice A is incorrect. The assumption of homoskedasticity applies to the error terms, not the independent variables in a multiple least squares regression model. Homoskedasticity means that the variance of the error terms is constant across all levels of the independent variables.

Choice B is incorrect. In fact, it's quite opposite to one of the key assumptions in multiple least squares regression model which assumes that residuals are homoskedastic, not heteroskedastic. Heteroskedasticity refers to a situation where there are sub-populations that have different variabilities from others.

Choice C is incorrect. The assumption about dependent variable being unique and stationary does not apply in multiple least squares regression model. The dependent variable need not be stationary; it can change over time or with changes in other variables.

Things to Remember

- Homoskedasticity: The assumption that the variance of the error terms is constant across all levels of the independent variables.
- Heteroskedasticity: The situation where the variance of the error terms is not constant across all levels of the independent variables.
- Perfect multicollinearity: When independent variables in a regression model are perfectly linearly related, making it impossible to estimate the regression coefficients.
- Stationarity: A time series is said to be stationary if its statistical properties such as mean, variance, and autocorrelation structure do not change over time.

Q.442 Jessica Pearson, FRM, builds a model to study the annual salaries of individuals in a certain developed country. The model incorporates just 2 independent variables – age and experience. She is surprised for ending up with a negative value for the coefficient of experience, which seems to be somewhat counterintuitive. Furthermore, the coefficients have low t-statistics but otherwise the model has a high R^2. Which of the following is the most likely cause of such results?

A. Heteroskedasticity

B. Multicollinearity

C. Homoskedasticity

D. Serial correlation

The correct answer is **B.**

Multicollinearity is a statistical phenomenon where two or more independent variables in a multiple regression model are highly correlated with each other. This high correlation distorts the standard error of the regression and the standard errors for the coefficients, which can cause issues when conducting tests for significance. In the context of Jessica Pearson's model, the independent variables are age and experience. These two variables are likely to be highly correlated because as a person ages, they also gain more experience. This high correlation between age and experience is causing multicollinearity in the model, which is leading to the unexpected results. Despite the high R^2 value, multicollinearity can still exist if the standard errors for the coefficients are high. Therefore, the most likely cause of the negative coefficient for experience and the low t-statistics, despite a high R^2, is multicollinearity.

Choice A is incorrect. Heteroskedasticity refers to the circumstance in which the variability of a variable is unequal across the range of values of another variable to which it is compared. In this case, heteroskedasticity would imply that the variance of salaries changes with age or experience, but it does not explain why the coefficient for experience is negative or why t-statistics are low.

Choice C is incorrect. Homoskedasticity refers to a situation where the error term (that is, the "noise" or random disturbance in the relationship between independent variables and dependent variable) has constant variance across all levels of independent variables. This condition does not explain why there would be a negative coefficient for experience or low t-statistics.

Choice D is incorrect. Serial correlation (also known as autocorrelation) occurs when residuals from an estimated regression equation are correlated with each other. This typically applies to time series data and would not be relevant in this context where Jessica Pearson's model analyzes cross-sectional data on individuals' annual salaries.

Things to Remember

- Multicollinearity can lead to inflated standard errors and unreliable coefficient estimates in regression models.
- High multicollinearity can make it difficult to determine the individual impact of each independent variable on the dependent variable.
- Variance inflation factor (VIF) is a common measure used to detect multicollinearity in regression analysis.
- Addressing multicollinearity can involve dropping one of the correlated variables, combining them into a single variable, or using regularization techniques like ridge regression.
- High R^2 values do not necessarily indicate the absence of multicollinearity, as it is possible to have a good overall fit while still having issues with individual coefficient estimates.

Q.3347 What will be the properties of the OLS estimator in the presence of near multicollinearity?

A. It will be consistent, unbiased, and efficient.

B. It will not be consistent.

C. It will be consistent, unbiased, but not efficient.

D. It will be consistent but not unbiased.

The correct answer is **C.**

Near multicollinearity in the context of an Ordinary Least Squares (OLS) regression model refers to a situation where one or more predictor variables are highly correlated with each other. This condition doesn't perfectly violate the assumption of no multicollinearity, but it comes close, causing practical problems in estimation. Here's how it affects the properties of the OLS estimators:

Consistency: Multicollinearity does not affect the consistency of the OLS estimators. As long as the sample size grows, the estimates converge to the true parameters, assuming that the model is correctly specified and the usual least squares assumptions are met except for perfect multicollinearity.

Unbiasedness: The OLS estimators remain unbiased under multicollinearity. Multicollinearity does not cause bias in the coefficients; the expected values of the estimators are still the true parameter values, provided the model is correctly specified.

Efficiency: This is where multicollinearity impacts the OLS estimators the most. Efficiency, in the sense of the Gauss-Markov theorem, is compromised. The standard errors of the estimators increase, which means the estimators have higher variance. This makes the estimates less precise, and hence, not the best linear unbiased estimators (BLUE) under the presence of multicollinearity.

Things to Remember

- Perfect multicollinearity occurs when one or more independent variables in a regression model can be exactly predicted from the others, leading to singular or non-invertible matrices.
- Adjusted R-squared is a useful metric to assess the goodness-of-fit of a regression model, especially in the presence of multicollinearity.
- Variance Inflation Factor (VIF) is a measure to detect multicollinearity by quantifying how much the variance of an estimated regression coefficient is increased due to multicollinearity.
- Ridge regression and Lasso regression are techniques used to address multicollinearity issues by introducing a penalty term in the regression estimation process.

Q.3780 When is a set of data termed as homoscedastic? When:

A. the variance of the error terms is the same across all the observations.

B. the observations are i.i.d. random variables.

C. the variance of the errors varies with the independent variables.

D. the variance of the error terms is zero.

The correct answer is **A.**

Homoscedasticity refers to the condition where the variance of the error terms (or residuals) in a regression model is constant across all levels of the independent variables. This implies that the spread of the residuals remains consistent regardless of the value of the independent variables. Homoscedasticity is a key assumption in linear regression analysis because it ensures that the model's predictions are equally reliable across all values of the independent variables.

B is incorrect. The term "i.i.d." refers to each observation being drawn from the same probability distribution and being independent of each other, which is crucial for many statistical methods but does not specifically address the variance of the error terms in a regression model.

C is incorrect. This statement describes heteroscedasticity, not homoscedasticity. Heteroscedasticity occurs when the variance of the error terms changes at different levels of the independent variables, meaning that the spread of the residuals is not constant.

D is incorrect. If the variance of the error terms were zero, it would imply that there are no errors at all in the regression model. This would mean that the model perfectly predicts the dependent variable for all observations, which is highly unlikely in real-world data

Things to Remember

- Homoscedasticity is a key assumption in linear regression analysis to ensure the reliability of the model's predictions.
- Heteroscedasticity, on the other hand, refers to the condition where the variance of the error terms changes at different levels of the independent variables.
- The assumption of homoscedasticity is important for the validity of statistical inferences and hypothesis testing in regression analysis.
- Tests such as the Breusch-Pagan test and White test can be used to detect heteroscedasticity in regression models.

Q.3782 Assume that you want to test for heteroskedasticity in a model with one explanatory variable, using a sample size of 100. What is the value of R^2 at which the null hypothesis will be rejected at a 5% level of significance? Please click here to access statistical tables

A. 0.0399

B. 0.0112

C. 0.0563

D. 0.0599

The correct answer is **D.**

We use the White test statistic calculated as nR^2 where R^2 is calculated in the second regression. The test statistic has a $\chi^2_{\frac{k(k+3)}{2}}$ (chi-distribution), where k is the number of explanatory variables in the first-step model.

In this case, we have one explanatory variable so that $\chi^2_{\frac{k(k+3)}{2}} = \chi^2_2$ whose critical value at 5% size is 5.99.

Therefore, we would not reject the null hypothesis

$$\Rightarrow nR^2 = 5.99$$

$$R^2 = \frac{5.99}{100} = 0.0599$$

A detailed breakdown of the chi-square test for heteroskedasticity

R^2 refers to the coefficient of determination. The question is built upon the chi-square test for heteroskedasticity, which follows the following steps:

I. Regress Y on Xs (independents) and generate squared residuals
II. Regress squared residuals on Xs (or a subset of Xs) (This is what is called the second regression).
III. Calculate the R^2 for the model in step 2 and use it to calculate the chi-squared test statistic: $\chi^2 = nR^2$
IV. The chi-squared statistic is compared to its critical value with $[k \times \frac{(k+3)}{2}]$ degrees of freedom, where k = number of independent variables.
V. If the calculated χ^2 > critical χ^2, we reject the null hypothesis of no conditional heteroskedasticity

Q.3783 A regression model is estimated as:

$$Y_i = 3 + 1.5X_{1i} - 2X_{2i} + \epsilon_i$$

What is the value of $\hat{\beta}_1$ if the model is reduced to $Y_i = \alpha + \hat{\beta}_1 X_1 + \epsilon_i$ given that $\rho_{X_1X_2} = 0.7$, $\sigma^2_{X_1} = 25$ and $\sigma^2_{X_2} = 36$?

A. -0.45

B. 0.67

C. -0.18

D. 0.23

The correct answer is **C.**

Recall the formula of the omitted variable.Given large samples sizes, the OLS estimator $\hat{\beta}_1$ converges to:

$$\beta_1 + \beta_2\delta$$

Where:

$$\delta = \frac{Cov(X_1, X_2)}{Var(X_1)}$$

So, we need to calculate the value of δ. Recall also that:

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X\sigma_Y} \Rightarrow Cov(X, Y) = \rho_{XY}\sigma_X\sigma_Y$$

Where

ρ_{XY} = the correlation between X and Y

σ_X =standard deviation of X

σ_Y =standard deviation of Y

$Cov(X, Y)$=covariance between X and Y

Using this analogy,

$$\Rightarrow \delta = \frac{Cov(X_1, X_2)}{Var(X_1)} = \frac{\rho_{X_1X_2}\sigma_{X_1}\sigma_{X_2}}{Var(X_1)} = \frac{0.7 \times \sqrt{25} \times \sqrt{36}}{25} = 0.84$$

So,

$$\hat{\beta}_1 = \beta_1 + \beta_2\delta = 1.5 + (-2 \times 0.84) = -0.18$$

Note that β_1 and β_2 are obtained from the original model

Q.3785 Given a model with two independent variables: $Y_i = \alpha + \beta_1 X_1 i + \beta_2 X_2 i + \epsilon_i$, what is the value of the correlation coefficient between X_1 and X_2 so that the value of the variance inflation factor is 15?

A. 0.9595

B. 0.9654

C. 0.9661

D. 0.4563

The correct answer is **C.**

Recall that the VIF is given by:

$$\frac{1}{1 - R_j^2}$$

Where R_j^2 measures how other variables explain a variable j. But in this case, we have two explanatory variable such that:

$$R_j^2 = \rho_{X_1X_2}^2$$

Thus:

$$\frac{1}{1 - R_j^2} = \frac{1}{1 - \rho_{X_1X_2}^2} = 15$$
$$\therefore \rho_{X_1X_2} = \sqrt{1 - \frac{1}{15}} = 0.9661$$

Q.3786 Consider the following data sets (We are using a small sample size for illustration purposes. In an exam situation, it might involve large sample sizes)

Y	X_1	X_2
−2	−0.41	−0.01
−0.11	0.40	−1.2
−1.68	−0.86	−0.91
−0.36	1.69	0.37
−0.08	0.46	−0.64
−0.74	1.40	−1.09

What are the estimated values of the parameters ($\hat{\alpha}$ and $\hat{\beta}_1$) in the model:

$$Y = \alpha + \beta_1 X_1$$

A. $\hat{\alpha}$=-1.080 ,$\hat{\beta}_1$=0.5633

B. $\hat{\alpha}$=-1.280 ,$\hat{\beta}_1$=0.3433

C. $\hat{\alpha}$=-1.5797 ,$\hat{\beta}_1$=0.6633

D. $\hat{\alpha}$=-1.780 ,$\hat{\beta}_1$=0.9933

The correct answer is **A.**

We need to use the standard formula:

$$\hat{\beta}_1 = \frac{Cov(Y, X_1)}{Var(X_1)}$$

We need to calculate:

$$Cov(Y, X_1) = \frac{1}{(n-1)} \sum_{i=1}^{n} (Y - \bar{Y})(X_1 - \bar{X}_1)$$

Consider the following table:

Y	X_1	X_2	$Y-\bar{Y}$	$X_1-\bar{X}_1$	$(Y-\bar{Y})(X_1-\bar{X}_1)$
−2	−0.41	−0.01	−1.17167	−0.85667	1.003728
−0.11	0.4	−1.2	0.718333	−0.04667	−0.03352
−1.68	−0.86	−0.91	−0.85167	−1.30667	1.112844
−0.36	1.69	0.37	0.468333	1.243333	0.582294
−0.08	0.46	−0.64	0.748333	0.013333	0.009978
−0.74	1.4	−1.09	0.088333	0.953333	0.084211
Mean= −0.82833	Mean= 0.446667	Mean= −0.58		Total	2.759533

$$Cov(Y, X_1) = \frac{1}{6-1}(2.7595) = 0.5519$$

And if you calculate correctly,

$$Var(X_1) = \frac{1}{n-1}\sum_{i=1}^{n}(X_1 - \bar{X}_1)^2 = 0.9797$$

So that:

$$\hat{\beta}_1 = \frac{0.5519}{0.9797} = 0.5633$$

And from the regression equation:

$$\hat{\alpha} = \bar{Y} - \hat{\beta}_1\bar{X}_1 = -0.82833 - 0.5633 \times 0.446667 = -1.079937 \approx -1.080$$

Q.3787 Consider the following data sets (We are using a small sample size for illustration purposes. In an exam situation, it might involve large sample sizes)

Y	X_1	X_2
−2	−0.41	−0.01
−0.11	0.40	−1.2
−1.68	−0.86	−0.91
−0.36	1.69	0.37
−0.08	0.46	−0.64
−0.74	1.40	−1.09

What is the estimated regression equation

$$\hat{Y} = \hat{\alpha} + \hat{\beta}_1 X_1$$

A. $\hat{Y}$=0.8967+0.9633X_1

B. $\hat{Y}$=-1.0799 +0.5633X_1

C. $\hat{Y}$=-1.8967+0.7633X_1

D. $\hat{Y}$=-1.5745+0.6633X_1

The correct answer is **B.**

We need to calculate the estimated parameters first

$$\hat{\beta}_1 = \frac{Cov(Y, X_1)}{Var(X_1)}$$

We need to calculate:

$$Cov(Y, X_1) = \frac{1}{(n-1)} \sum_{i=1}^{n} (X_1 - \bar{X}_1)(Y - \bar{Y})$$

Consider the following table:

Y	X_1	X_2	$Y - \bar{Y}$	$X_1 - \bar{X}_1$	$(Y - \bar{Y})(X_1 - \bar{X}_1)$
-2	-0.41	-0.01	-1.17167	-0.85667	1.003728
-0.11	0.4	-1.2	0.718333	-0.04667	-0.03352
-1.68	-0.86	-0.91	-0.85167	-1.30667	1.112844
-0.36	1.69	0.37	0.468333	1.243333	0.582294
-0.08	0.46	-0.64	0.748333	0.013333	0.009978
-0.74	1.4	-1.09	0.088333	0.953333	0.084211
Mean= -0.82833	Mean= 0.446667	Mean= -0.58		Total	2.759533

$$Cov(Y, X_1) = \frac{1}{6-1}(2.7595) = 0.5519$$

And if you calculate correctly,

$$Var(X_1) = \frac{1}{n-1} \sum_{i=1}^{n} (X_1 - \bar{X}_1)^2 = 0.9797$$

So that:

$$\hat{\beta}_1 = \frac{0.5519}{0.9797} = 0.5633$$

And from the regression equation:

$$\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X}_1 = 0.82833 - 0.5633 \times 0.446667 = -1.0799$$

So that the estimated regression equation is stated as:

$$\hat{Y} = -1.0799 + 0.5633X_1$$

Q.3788 Assume that you have estimated two regression equations: $\hat{Y} = 0.5767 + 0.5633X_1$ and $\hat{Y} = 0.6767 - 0.7633X_2$ and that covariance between explanatory variables X_1 and X_2 is 0.603 ($Cov(X_1, X_2) = 0.603$) and $Var(X_1) = 0.874$ and $Var(X_2) = 0.75$ What is the estimated expression the intercept ($\hat{\alpha}$) for $Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \epsilon_i$?

A. $\hat{\alpha} = \hat{Y}_i - 4.875X_1i + 1.627X_2i$

B. $\hat{\alpha} = \hat{Y}_i - 3.585X_1i + 1.907X_2i$

C. $\hat{\alpha} = \hat{Y}_i - 2.585X_1i + 3.827X_2i$

D. $\hat{\alpha} = \hat{Y}_i - 2.4476X_1i + 2.7312X_2i$

The correct answer is **D.**

This question that needs application of the omitted variables formula. Recall that if the regression model is stated as:

$$Y_i = \alpha + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

If we omit X_2 from the estimated model, then the model is given by:

$$Y_i = \alpha + \beta_1 X_{1i} + \epsilon_i$$

Now, in large samples sizes, the OLS estimator $\hat{\beta}_1$ converges to:

$$\beta_1 + \beta_2 \delta_1$$

Where:

$$\delta_1 = \frac{Cov(X_1, X_2)}{Var(X_1)}$$

Maintaining this line of thought, the first regression equation suggests that:

$$\beta_1 + \beta_2\delta_1 = 0.5633$$

And the second equation suggests that:

$$\beta_2 + \beta_1\delta_2 = -0.7633$$

Moreover,

$$\delta_1 = \frac{Cov(X_1, X_2)}{Var(X_1)} = \frac{0.603}{0.874} = 0.6899$$

And

$$\delta_2 = \frac{Cov(X_1, X_2)}{Var(X_2)} = \frac{0.603}{0.750} = 0.804$$

So that:

$$\beta_1 + 0.6899\beta_2 = 0.5633 \ldots.(1)$$
$$\beta_2 + 0.804\beta_1 = -0.7633 \ldots..(2)$$

If we solve the two equations simultaneously, we get:

$$\beta_1 = 2.4476 \text{ and } \beta_2 = -2.7312$$

So that the combined regression model is:

$$Y_i = \alpha + 2.4476X_{1i} - 2.7312X_{2i}$$

So that the expression of the intercept is:

$$\hat{\alpha} = \hat{Y}_i - 2.4476X_{1i} + 2.7312X_{2i}$$

Q.3789 A financial analyst wishes to come up with Ordinary Least Squares Estimation (OLS) regression model to analyze the financial performance of a company. However, the analyst is aware that some of the explanatory variables might be excluded (and hence omitted variable bias). What is the cause of omitted variables bias?

A. Omitted variable bias happens when the omitted variable is independent of the included independent variables but is not a determinant of the dependent variable.

B. Omitted variable bias occurs when the omitted variable is correlated with all of the included independent variables and is a determinant of the dependent variable.

C. Omitted variable bias occurs when the omitted variable is independent of the included independent variables and is a determinant of the dependent variable.

D. Omitted variable bias occurs when the omitted variable is correlated with at least one of the included independent variables and is a determinant of the dependent variable.

The correct answer is **D.**

Omitted variable bias occurs when a variable that should have been included in a regression model is left out. This omitted variable is both a significant determinant of the dependent variable and is correlated with at least one of the independent variables that are included in the model. When this happens, the estimated coefficients of the included variables become biased, leading to inaccurate predictions. This is because the effect of the omitted variable is falsely attributed to the included variables with which it is correlated. Therefore, it is crucial to include all relevant variables in a regression model to avoid omitted variable bias and ensure the accuracy of the model's predictions.

Choice A is incorrect. Omitted variable bias does not occur when the omitted variable is independent of the included independent variables and is not a determinant of the dependent variable. In this case, omitting such a variable would not cause any bias as it does not affect the dependent variable nor is it correlated with other independent variables.

Choice B is incorrect. While it's true that an omitted variable can be correlated with all of the included independent variables, this alone doesn't lead to omitted variable bias. The key factor for this bias to occur is that the omitted variable must also be a determinant of the dependent variable.

Choice C is incorrect. Even though an omitted variable can be a determinant of the dependent variable, if it's completely uncorrelated with all other included independent variables, its omission will not result in any bias in estimating parameters for those included variables.

Things to Remember

- Including irrelevant variables in a regression model can lead to multicollinearity, which can affect the accuracy and interpretation of the model.
- R-squared value in regression analysis helps to understand the proportion of the

variance in the dependent variable that is predictable from the independent variables.

- Adjusted R-squared is a modified version of R-squared that adjusts for the number of predictors in the model, providing a more accurate measure of the model's goodness of fit.
- Interaction terms in regression models capture the combined effect of two or more independent variables on the dependent variable.
- Heteroscedasticity in regression analysis refers to the situation where the variance of the residuals is not constant across all levels of the independent variables.

Q.3790 Consider the following data sets:

Observation	Y	X
1	3.67	1.85
2	1.88	0.65
3	1.35	0.63
4	0.34	1.24
5	0.89	2.45

The regression analysis was done on the entire data set, and the regression equation was estimated as:

$$\hat{Y} = 1.4110 + 0.1512X_1$$

Additionally, first four observations were used, leading to the following estimated regression equation:

$$\hat{Y} = 0.3169 + 1.3667X_1$$

What is Cook's distance for the 5th observation?

A. 3.3923

B. 1.6268

C. 0.6458

D. 1.3667

The correct answer is **B.**

Cook's distance is given by:

$$D_j = \frac{\sum_{i=1}^{n} \left(\hat{Y}_i^{(-j)} - \hat{Y}_i\right)^2}{ks^2}$$

Where:

$\hat{Y}_i^{(-j)}$=fitted value of $\hat{Y}_i$ when the observed value j is excluded, and the model is approximated using $n-1$ observations.

k=number of coefficients in the regression model

s^2=estimated error variance from the model using all observations

So, in our case we have:

$$\hat{Y}_i = 1.4110 + 0.1512X_1$$

And

$$\hat{Y}_i^{(-j)} = 0.3169 + 1.3667X_1$$

Now study the following table:

Oservation	Y	X	$\hat{Y}_i$	$\hat{Y}_i^{(-j)}$	$\left(\hat{Y}_i^{(-j)} - \hat{Y}_i\right)^2$	$(\hat{Y}_i - y)^2$
1	3.67	1.85	1.69072	2.845295	1.333043	3.917549
2	1.88	0.65	1.50928	1.205255	0.092431	0.137433
3	1.35	0.63	1.506256	1.177921	0.107804	0.024416
4	0.34	1.24	1.598488	2.011608	0.170668	1.583792
5	0.89	2.45	1.78144	3.665315	3.548985	0.794665
				Sum	5.252932	6.457855

Now,

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\hat{Y}_i - y\right)^2 = \frac{1}{4} \times 6.457856 = 1.614464$$

And:

$$\sum_{i=1}^{5}\left(\hat{Y}_i^{(-j)} - \hat{Y}_i\right)^2 = 5.252932$$

And thus:

$$D_j = \frac{5.252932}{2 \times 1.614464} = 1.626834$$

The value of the Cook's distance is greater than 1 implying that the 5^{th} observation is an outlier.

Q.3791 What is the extraneous variable in regression diagnostics?

A. It is a variable which is eliminated to increase the effectiveness of a model.

B. It is one that is unnecessarily included in the model, whose actual coefficient and consistently approximated value is 0 in large sample sizes. If we add these variables is costly.

C. It is a variable that is included in a model in case the sample size is not appropriate.

D. It is a variable that is necessary to control for confounding effects in the model.

The correct answer is **B.**

An extraneous variable, in the context of regression diagnostics, is a variable that is unnecessarily included in the model. Its actual coefficient and consistently approximated value is 0 in large sample sizes. This means that the variable does not contribute significantly to the model's predictive power or accuracy. Including such variables in the model can be costly in terms of computational resources and time. It can also lead to overfitting, where the model becomes too complex and performs poorly on new, unseen data. Therefore, it is often beneficial to identify and eliminate extraneous variables from a regression model to improve its effectiveness and efficiency.

Choice A is incorrect. An extraneous variable is not necessarily eliminated to increase the effectiveness of a model. While it's true that unnecessary variables can reduce the accuracy and efficiency of a model, an extraneous variable refers specifically to one that has no real effect on the dependent variable in large sample sizes, not just any variable that might be removed for optimization purposes.

Choice C is incorrect. The inclusion of an extraneous variable in a model does not depend on the appropriateness of the sample size. Rather, it refers to a variable which doesn't have any significant impact on the dependent variable when considering large samples.

Choice D is incorrect. A variable that is necessary to control for confounding effects is not an extraneous variable; rather, it is an important variable that must be included to ensure the accuracy and validity of the model.

Things to Remember

- Overfitting: Overfitting occurs when a model is too complex and fits the training data too closely, leading to poor performance on new, unseen data.

- Confounding variables: These are variables that are related to both the independent and dependent variables in a study, leading to spurious relationships if not properly controlled for.

- Model optimization: Removing extraneous variables is part of the process of optimizing a regression model for better predictive power and efficiency.

- Sample size considerations: Sample size plays a crucial role in the generalizability and reliability of regression models, but the presence of extraneous variables is not directly related to sample size adequacy.

Q.3793 Multicollinearity occurs when several variables can significantly explain one or more independent variables. Which one of the following is most likely to be true about multicollinearity?

A. Multicollinearity does pose technical problems in parameter estimation, and data modeling.

B. When there is multicollinearity in a model, the coefficients tend to be jointly statistically significant.

C. Multicollinearity can be detected using the Variance Inflation Factor, where a variable with high VIF is considered for inclusion in the model.

D. Multicollinearity ensures that the model is unbiased and provides precise estimates of the coefficients.

The correct answer is **B.**

When multicollinearity is present in a model, the coefficients of the variables tend to be jointly statistically significant. This is often evidenced by the F-statistic of the regression. The F-statistic is a measure of how well the regression model fits the data. A high F-statistic indicates that the model explains a large amount of the variation in the dependent variable. When multicollinearity is present, the variables are highly correlated, which means they collectively contribute to explaining the dependent variable. This results in the coefficients of these variables being jointly significant.

Choice A is incorrect. While multicollinearity can pose technical problems in parameter estimation, it does not necessarily affect data modeling. The presence of multicollinearity can make the estimates of the parameters unstable and unreliable, but it does not inherently distort the model itself.

Choice C is incorrect. The Variance Inflation Factor (VIF) is indeed a tool used to detect multicollinearity; however, a variable with high VIF indicates that it has strong correlation with other variables and thus may need to be removed from the model rather than included.

Choice D is incorrect. Multicollinearity does not ensure that the model is unbiased or provides precise estimates of the coefficients. In fact, multicollinearity can lead to large standard errors for the estimated coefficients, making them unstable and imprecise. While the model itself may remain unbiased, the precision of the coefficient estimates is compromised, which can lead to difficulties in interpreting the results.

Things to Remember

- Multicollinearity can lead to inflated standard errors of the coefficients, making them less precise and potentially unreliable.
- High multicollinearity can make it difficult to determine the individual impact of each independent variable on the dependent variable.
- Principal Component Analysis (PCA) is a technique that can be used to address multicollinearity by transforming the original variables into a new set of uncorrelated variables.
- Ridge regression and Lasso regression are regularization techniques that can also help mitigate the effects of multicollinearity in regression models.
- It is important to assess the presence of multicollinearity before interpreting the results of a regression model to ensure the reliability of the coefficients and statistical significance.

Reading 21: Stationary Time Series

Q.505 A time series is said to be stationary if:

A. Its statistical properties including the mean and variance do not change over time.

B. Its mean, variance, and covariances with lagged and leading values change over time.

C. Its mean remains constant but variance and covariances with lagged and leading values change over time.

D. Its mean and variance are variables but covariances with lagged and leading values do not change over time.

The correct answer is **A.**

A time series is said to be stationary if its statistical properties including the mean and variance do not change over time. This is because the fundamental assumption behind a stationary time series is that the properties that are learned from the data in one time period are consistent and can be applied to any other period. This includes the mean and variance of the series. If these properties remain constant over time, then the series can be said to be stationary. This is important because many statistical modeling techniques require the time series to be stationary to make effective and reliable predictions.

Choice B is incorrect. A time series is considered stationary when its statistical properties do not change over time. This includes the mean, variance, and covariances with lagged and leading values. If these properties change over time, as suggested in this option, the time series would not be stationary.

Choice C is incorrect. While it's true that a constant mean is one of the conditions for stationarity, it's not sufficient on its own. The variance and covariances with lagged and leading values also need to remain constant for a time series to be considered stationary.

Choice D is incorrect. This choice suggests that only the covariances with lagged and leading values need to remain constant for a time series to be stationary. However, this isn't accurate - both the mean and variance also need to remain constant over time.

Things to Remember

- Stationary time series are important in time series analysis as they allow for more reliable and accurate forecasting models.
- Non-stationary time series can exhibit trends, seasonality, and other patterns that make forecasting more challenging.
- Differencing is a common technique used to transform a non-stationary time series into

a stationary one by removing trends or seasonality.

- Unit root tests, such as the Augmented Dickey-Fuller (ADF) test, are often used to determine if a time series is stationary or non-stationary.
- Stationarity is a key assumption in many time series models, such as ARIMA (AutoRegressive Integrated Moving Average) models.

Q.506 Financial asset return time series have one common characteristic. Which one?

A. They are not weakly stationary.

B. They are highly correlated.

C. They do not exhibit any trend.

D. Their distributions have very thin tails.

The correct answer is **C.**

Financial asset return time series do not exhibit any trend. This means that there is no consistent pattern of increase or decrease in the returns over time. The returns are random and unpredictable, which is a key characteristic of financial markets. This lack of trend is due to the efficient market hypothesis, which states that current prices fully reflect all available information. Therefore, future price movements are independent of past price movements, resulting in a lack of trend in asset returns. This characteristic is observed across different types of financial assets, making it a common characteristic of financial asset return time series.

Choice A is incorrect. Financial asset return time series are often weakly stationary, meaning that their mean and variance remain constant over time and the value of covariance between two-time periods depends only on the distance or gap between the two-time periods, not on the actual time at which the covariance is computed.

Choice B is incorrect. The correlation of financial asset return time series can vary greatly depending on a multitude of factors such as market conditions, type of assets, geographical location etc. Therefore, it cannot be generalized that they are highly correlated.

Choice D is incorrect. Financial asset return time series often exhibit distributions with fat tails (kurtosis), indicating a higher probability of extreme outcomes compared to a normal distribution. This means they do not have thin tails as suggested in this option.

Things to Remember

- Weakly stationary time series have constant mean and variance over time.
- Efficient Market Hypothesis (EMH) suggests that asset prices reflect all available information, leading to random and unpredictable returns.
- Correlation of financial asset return time series can vary based on multiple factors.
- Financial asset return time series often exhibit fat-tailed distributions, indicating higher probabilities of extreme outcomes.

Q.508 Distinguish between independent white noise and normal (Gaussian white noise).

A. An independent white noise is a time series that exhibits both serial independence and a lack of serial correlation while a normal white process is a time series that’s serially independent, serially uncorrelated, and is normally distributed.

B. A normal white noise is a time series that exhibits both serial independence and a lack of serial correlation while an independent white noise is a time series that’s serially independent, serially uncorrelated, and is normally distributed.

C. An independent white noise is a time series with equal mean and variance while a normal white noise is a time series where the mean is not equal to the variance.

D. An independent white noise is discrete while a normal white noise is continuous.

The correct answer is **A.**

Independent white noise and normal (Gaussian) white noise are both types of time series data. An independent white noise is a time series that exhibits both serial independence and a lack of serial correlation. This means that the values in the series are not influenced by the values that came before them, and there is no predictable pattern or trend in the data. On the other hand, a normal white noise process is a time series that’s serially independent, serially uncorrelated, and is normally distributed. This means that in addition to the properties of independent white noise, the values in a normal white noise series also follow a normal distribution. This distribution is characterized by its bell-shaped curve and is defined by its mean and standard deviation.

Choice B is incorrect. This choice incorrectly swaps the definitions of independent white noise and normal (Gaussian) white noise. A normal white noise is a time series that’s serially independent, serially uncorrelated, and is normally distributed while an independent white noise exhibits both serial independence and a lack of serial correlation.

Choice C is incorrect. This choice inaccurately defines the characteristics of both types of white noises. While it's true that a time series with equal mean and variance can be considered as an independent white noise, this does not define it completely. Similarly, for normal (Gaussian) white noise, the mean not being equal to variance doesn't necessarily make it so.

Choice D is incorrect. The distinction between discrete and continuous data does not differentiate between these two types of noises in time series analysis. Both independent and normal (Gaussian) white noises can be either discrete or continuous depending on the context.

Things to Remember

- White noise: A white noise process is a sequence of uncorrelated random variables with a constant mean and variance. It is an important concept in time series analysis.
- Serial independence: In time series analysis, serial independence refers to the lack of a relationship between the values at different time points in the series.

- Serial correlation: Serial correlation, also known as autocorrelation, is the correlation between the values of a time series at different time points.

- Normal distribution: A normal distribution is a probability distribution that is symmetric and bell-shaped, characterized by its mean and standard deviation.

- Mean and variance: The mean of a distribution represents its average value, while the variance measures the spread of the values around the mean.

Q.510 Which of the following statements is most likely correct regarding lag operators? Lag operators:

A. only use lagged future values.

B. are of limited use in modeling a time series.

C. consider only infinite-order polynomials

D. quantify how a time series evolves by lagging a data series.

The correct answer is **D.**

Lag operators indeed quantify how a time series evolves by lagging a data series. In time series analysis, a lag operator, also known as a backshift operator, is a mathematical tool that shifts the index of a time series backwards. The lag operator operates on an element of a time series to produce the previous element. For instance, if we denote the lag operator as L, then $Ly_t = y_{t - 1}$, where y_t is the current value of the time series and $y_{t - 1}$ is the previous value. This operation is crucial in time series analysis as it allows us to understand the evolution of the series over time, which is fundamental for forecasting future values. Therefore, lag operators are an essential tool in time series modeling and forecasting.

Choice A is incorrect. Lag operators do not use lagged future values. Instead, they use past values in a time series to model and forecast future data points.

Choice B is incorrect. Lag operators are not of limited use in modeling a time series. On the contrary, they are fundamental tools used extensively in time series analysis for forecasting and understanding the evolution of a data series over time.

Choice C is incorrect. Lag operators do not consider only infinite-order polynomials. They can be applied to any order of polynomial or non-polynomial functions as well, depending on the specific requirements of the model being used.

Things to Remember

- Lag operators are denoted by the symbol L and are used to shift the index of a time series backwards.
- They are essential in time series analysis for modeling, forecasting, and understanding the evolution of a data series over time.
- Lag operators are not limited to a specific order of polynomials and can be applied to various types of functions.
- Understanding lag operators is crucial for building autoregressive and moving average models in time series analysis.
- Lag operators are commonly used in econometrics, finance, and other fields to analyze and predict future values based on historical data.

Q.511 Unlike structural models, pure time series models do not incorporate any explanatory variable. Which of the following is a disadvantage of pure time series models when compared to the structural models?

A. They are not theoretically motivated.

B. They cannot produce forecasts easily.

C. They cannot be used when the data has a very high frequency.

D. It’s difficult to select the most appropriate explanatory variables to include in a pure time-series model.

The correct answer is **A.**

Pure time series models are not theoretically motivated. This means that they lack a solid theoretical foundation that explains why the current value of a variable should be related to its past values and to the values of a random error process. For instance, in the context of stock returns, it might be difficult to understand why the current value of a stock return should be related to its past values and to the values of a random error process without any theoretical backing. It would be more theoretically sound to explain return fluctuations using some macroeconomic variables that influence profitability, such as the state of the economy as a whole. This lack of theoretical motivation is a significant disadvantage of pure time series models when compared to structural models, which incorporate explanatory variables and are therefore more theoretically grounded.

Choice B is incorrect. Pure time series models can indeed produce forecasts. In fact, they are often used for this purpose as they rely on historical data to predict future trends.

Choice C is incorrect. The frequency of the data does not limit the use of pure time series models. These models can handle both high and low frequency data.

Choice D is incorrect. This statement is misleading because in a pure time-series model, there are no explanatory variables to select from as it relies solely on past values of the variable being forecasted.

Things to Remember

- Structural models incorporate explanatory variables to explain the relationship between variables, while pure time series models do not.
- Pure time series models rely on historical data to forecast future values.
- Structural models are more theoretically grounded compared to pure time series models.
- Time series models can handle both high and low frequency data.
- Selection of explanatory variables is not a concern in pure time series models as they do not include any.

Q.512 The following sample autocorrelation estimates are obtained using 200 data points:

Lag	1	2	3
Coefficient	0.3	−0.15	−0.10

Compute the value of the Box-Pierce Q-statistic.

A. 12.6

B. 28.0

C. 18.2

D. 24.5

The correct answer is **D.**

The Box-Pierce Q-statistic is given by:

$$Q_{BP} = n \sum \rho_k^2$$

Where n is the sample size and ρ represents the autocorrelations.

Therefore, $Q_{BP} = 200(0.3^2 + (-0.15)^2 + (-0.10)^2) = 24.5$

Q.513 The following sample autocorrelation estimates are obtained using 250 data points:

Lag	1	2	3
Coefficient	0.3	−0.15	−0.10

Compute the value of the Ljung Box Q statistic.

A. 18

B. 31

C. 40

D. 35

The correct answer is **B.**

The Ljung Box statistic, also known as the modified Box Pierce statistic, is a function of the accumulated autocorrelations, ρ_i, up to time lag m. It's calculated as:

$$Q(m) = n(n+2)\sum_{i=1}^{\tau}\left(\frac{{\rho_i}^2}{n-i}\right)$$

In this case, time lag = 3

Thus,

$$Q(3) = 250 \times 252\left(\frac{0.3^2}{249}\right) + 250 \times 252\left(\frac{(-0.15)^2}{248}\right) + 250 \times 252\left(\frac{(-0.1)^2}{247}\right) = 31.04$$

Q.518 The following sample autocorrelation estimates are obtained using 300 data points:

Lag	1	2	3
Coefficient	0.25	−0.1	−0.05

Compute the value of the Ljung Box Q statistic.

A. 20

B. 22.74

C. 24

D. 18.45

The correct answer is **B.**

$$Q(m) = n\sum_{j=1}^{h}\left(\frac{n+2}{n-j}\right)\rho_j^2 = n(n+2)\sum_{j=1}^{h}\left(\frac{\rho_j^2}{n-j}\right)$$

In this case, time lag = 3
Thus,

$$Q(3) = 300(302)\left[\frac{0.25^2}{299} + \frac{(-0.1)^2}{298} + \frac{(-0.05)^2}{297}\right] = 22.74$$

Q.519 The following sample autocorrelation estimates are obtained using 300 data points:

Lag	1	2	3
Coefficient	0.25	−0.1	−0.05

Calculate the value of the Box Pierce Q-statistic.

A. 22.5

B. 22.74

C. 21.5

D. 18

The correct answer is **A.**

$$Q_{BP} = n\sum \rho_k^2$$

Where n is the sample size and ρ represents the autocorrelations.

Therefore,

$$Q_{BP} = 300(0.25^2 + (-0.1)^2 + (-0.05)^2) = 22.5$$

Note: Provided the sample size is large, the Box Pierce and the Ljung Box tests typically arrive at the same result.

Q.524 Which of the following statements regarding the Ljung-Box Q statistic is true?

A. It can only be used for detecting autocorrelation in autoregressive models.

B. It is not affected by the number of observations in the time series data.

C. It's widely used to check the independence of residuals in time series models.

D. It can only be applied to stationary time series data.

The correct answer is **C.**

The Ljung-Box Q statistic is indeed widely used to check the independence of residuals in time series models. This is a crucial step in time series analysis as it helps to ensure that the model's residuals, or the differences between the observed and predicted values, are random and not correlated. If the residuals are not independent, it indicates that there is some pattern in the time series data that the model has not captured. This could lead to inaccurate predictions and forecasts. Therefore, the Ljung-Box Q statistic plays a vital role in validating the adequacy of time series models.

Choice A is incorrect. The Ljung-Box Q statistic is not limited to detecting autocorrelation in autoregressive models only. It can be used to detect autocorrelation in any time series model, not just autoregressive ones.

Choice B is incorrect. The Ljung-Box Q statistic does get affected by the number of observations in the time series data. As the number of observations increases, the power of this test also increases.

Choice D is incorrect. The Ljung-Box Q statistic can be applied to both stationary and non-stationary time series data for checking independence of residuals and presence of autocorrelations among them.

Things to Remember

- Autocorrelation: The Ljung-Box Q statistic is used to detect autocorrelation in time series data, which is the correlation between a variable's current value and its past values.
- Residuals: In time series modeling, residuals are the differences between the observed values and the values predicted by the model. Checking the independence of residuals is crucial for model validation.
- Time Series Models: These are statistical models used to analyze and forecast time-dependent data. The Ljung-Box Q statistic helps in assessing the adequacy of these models.
- Stationarity: A time series is said to be stationary if its statistical properties such as mean, variance, and autocorrelation structure do not change over time. The Ljung-Box Q statistic can be applied to both stationary and non-stationary time series data.

Q.528 An autoregressive process of order q is considered stationary if:

A. The roots of the characteristic equation lie on the unit circle.

B. The roots of the characteristic equation lie outside the unit circle.

C. The roots of the characteristic equation lie inside the unit circle.

D. The characteristic equation is of order 1.

The correct answer is **B.**

An autoregressive process of order q (AR(q)) is considered stationary if the roots of its characteristic equation lie outside the unit circle. This means that all the roots have an absolute value greater than 1. The characteristic equation of an AR(q) model is derived from the autoregressive equation itself. For instance, in a simple AR(1) model, the equation is given by

$$Y_t = \alpha + \rho_1 Y_{t-1} + \epsilon_t$$

where Y_t is the variable of interest at time t, α is the intercept, ρ_1 is the AR parameter, and ϵ_t is the white-noise shock. The associated first-order polynomial is

$$1 - \rho_1 L$$

, and the characteristic equation is

$$1 - \rho_1 z = 0$$

. The solution to this equation is $z = 1/\rho_1$. If $|\rho_1| < 1$, then the AR(1) model is considered stationary. This is because the effects of shocks to the system will diminish over time, leading to a stable process with consistent properties such as a constant mean and constant variance. This property of stationarity is crucial for many statistical analyses and forecasting methods.

Choice A is incorrect. If the roots of the characteristic equation lie on the unit circle, it implies that the AR(q) model is non-stationary. This is because for an AR(q) model to be stationary, its roots must lie outside the unit circle.

Choice C is incorrect. The statement that the roots of the characteristic equation lie inside the unit circle does not hold true for a stationary AR(q) model. In fact, this condition would imply a non-stationary process as well.

Choice D is incorrect. The order of characteristic equation does not determine stationarity of an AR(q) model. It's about where its roots are located in relation to unit circle which determines whether it's stationary or not.

Things to Remember

- Stationarity is a key concept in time series analysis, indicating that the statistical properties of a process do not change over time.
- Autoregressive (AR) models are a type of time series model where the current value of a variable is regressed on past values of itself.

- The characteristic equation of an AR(q) model is derived from the autoregressive equation and its roots determine the stationarity of the process.
- For an AR(q) model to be stationary, all the roots of its characteristic equation must lie outside the unit circle, indicating a stable process.
- Non-stationary processes can exhibit trends, cycles, or other patterns that make statistical analysis and forecasting challenging.

Q.529 Which of the following statements best explains the main setback of the moving average representation of a first-order moving average process, MA(1)?

A. It does not incorporate observable shocks, so the solution is to use an autoregressive representation.

B. It does not show evidence of autocorrelation cutoff.

C. It only incorporates observable shocks, so the solution is to use an autoregressive representation.

D. The process is highly complicated and can only be carried out using a computer program.

The correct answer is **A.**

The moving average representation of a first-order moving average process, MA(1), does not incorporate observable shocks. This is a significant limitation because it attempts to estimate a variable in terms of random, unobservable white shocks. This makes the process less useful for estimation. The solution to this problem is to use an autoregressive representation. In an autoregressive representation, an observable item can be used, making it more useful for estimation. Therefore, the statement in choice A accurately describes the main setback of the moving average representation of an MA(1) process.

Choice B is incorrect. The moving average representation of an MA(1) process does show evidence of autocorrelation cutoff. This is because the autocorrelation function for an MA(1) process cuts off after lag 1, which means that there are no correlations for lags greater than 1.

Choice C is incorrect. The statement contradicts the nature of a moving average (MA) model. An MA model does incorporate observable shocks, but it's not a limitation; rather, it's one of its characteristics that differentiates it from autoregressive models.

Choice D is incorrect. While time series analysis can be complex and often requires computational tools, this isn't a specific limitation to the moving average representation of an MA(1) process. Both simple and complex statistical analyses can be performed manually or with computer software depending on their complexity and data size.

Things to Remember

- Moving Average (MA) process is a type of time series model that uses the past forecast errors in a regression-like model to predict future values.
- Autoregressive (AR) process is another type of time series model that uses past values of the variable in a regression-like model to predict future values.
- Autocorrelation function (ACF) is a function that represents the correlation between a time series and a lagged version of itself.
- Autocorrelation cutoff refers to the phenomenon where the autocorrelation function of a time series cuts off after a certain lag, indicating no further correlations beyond that point.
- White noise is a type of random signal with a constant power spectral density and a flat frequency spectrum.

Q.530 The key difference between a moving average representation and an autoregressive process is that:

A. An autoregressive process is never covariance stationary.

B. An autoregressive process shows evidence of autocorrelation cutoff.

C. Unlike the autoregressive process, a moving average representation shows evidence of gradual decay.

D. A moving average representation shows evidence of autocorrelation cutoff.

The correct answer is **D.**

A moving average representation shows evidence of autocorrelation cutoff. This means that the correlation between a variable and its lagged values abruptly drops to zero after a certain point. This is a key characteristic of moving average models. In contrast, an autoregressive process shows a gradual decay in autocorrelations, meaning the correlation between a variable and its lagged values decreases slowly over time. This difference is crucial in determining the appropriate model to use when analyzing time series data. Understanding this distinction can help in selecting the right model for forecasting and interpreting the data accurately.

Choice A is incorrect. An autoregressive process can be covariance stationary if the absolute value of its root is greater than one. Therefore, it's not accurate to say that an autoregressive process is never covariance stationary.

Choice B is incorrect. While it's true that an autoregressive process may show evidence of autocorrelation cutoff, this isn't a distinguishing feature between AR and MA models as both can exhibit this characteristic under certain conditions.

Choice C is incorrect. This statement incorrectly characterizes the behavior of a moving average representation. In fact, it's the autoregressive process that typically shows evidence of gradual decay in autocorrelations, not the moving average representation.

Things to Remember

- Covariance stationary: A time series is covariance stationary if its mean and variance are constant over time and the covariance between two time periods depends only on the lag between them.
- Autoregressive (AR) process: A time series model where the current value of a variable is regressed on its lagged values.
- Moving Average (MA) process: A time series model where the current value of a variable is a linear combination of past white noise error terms.
- Autocorrelation: The correlation of a variable with itself at different time lags.
- Autocorrelation cutoff: A phenomenon where the autocorrelation between a variable and its lagged values abruptly drops to zero after a certain lag.
- Gradual decay: Refers to the slow decrease in autocorrelations between a variable and its lagged values over time.

Q.533 What is the purpose of a q^{th}-order moving average process?

A. To add a fifth error term to an MA(1) process.

B. To add a third error term to an MA(1) process.

C. To add as many additional lagged variables as needed so as to produce a robust set of estimates for the time series.

D. To invert the moving average representation and make it more useful.

The correct answer is **C.**

The primary purpose of a q^{th}-order moving average process is to add as many additional lagged variables as needed to produce a robust set of estimates for the time series. In time series analysis, the moving average (MA) model is a common approach for modeling univariate time series. The MA model involves the assumption that the time series is a linear function of the residual errors from previous time steps, which are referred to as lagged variables. The order of the MA model, denoted as q^{th}, refers to the number of lagged variables included in the model. By increasing the order of the MA model, we can include more lagged variables, which can help to capture more complex patterns in the time series data and thus produce a more robust set of estimates.

Choice A is incorrect. The order of the moving average process, denoted as q^{th}, does not specifically aim to add a fifth error term to an MA(1) process. The order of the moving average process determines the number of lagged error terms in the model, but it is not limited to adding a fifth error term.

Choice B is incorrect. Similar to choice A, this option incorrectly suggests that the primary purpose of a q^{th}-order moving average process is to add a third error term to an MA(1) process. While it's true that increasing the order can add more lagged error terms, it's not confined only to adding a third one.

Choice D is incorrect. The primary purpose of a q^{th}-order moving average process isn't about inverting the moving average representation and making it more useful. In fact, inversion doesn't necessarily make an MA model more useful; rather, its usefulness depends on how well it fits and predicts time series data.

Things to Remember

- Moving Average (MA) model is a common approach for modeling univariate time series data.
- The order of the MA model, denoted as q^{th}, refers to the number of lagged variables included in the model.
- Increasing the order of the MA model allows for capturing more complex patterns in the time series data.
- The primary purpose of a q^{th}-order moving average process is to add as many additional lagged variables as needed to produce a robust set of estimates for the time series.
- Inverting the moving average representation is not the main purpose of increasing the order of the MA model.

Q.3367 The following sample autocorrelation estimates have been obtained using 200 data points:

Lag	1	2	3	4
Coefficient	0.15	−0.14	−0.1	−0.08

What is the value of the Box-Pierce Q-statistic?

A. 1.05

B. 17.1

C. 11.7

D. 12.5

The correct answer is **C.**

$$Q_{BP} = T\sum_{\tau=1}^{m}\hat{\rho}^2(\tau)$$

- T = Sample size
- $\hat{\rho}^2(\tau)$ = Sample auto correlation function for τ lags
- m = number of lags under observation

$$Q_{BP} = 200(0.15^2 + -0.14^2 + -0.1^2 + -0.08^2) = 11.7$$

Q.3374 Assume the shock in a time series is approximated by Gaussian white noise. Yesterday's realization, $y_{(t-1)}$ was 0.015 and the lagged shock was -0.160. Today's shock is 0.170.
If the weight parameter theta, θ, is equal to 0.70, determine today's realization under a first-order moving average, MA(1), process.

A. 0.254

B. 0.075

C. 0.062

D. 0.058

The correct answer is **D.**

An MA(1) process is defined as:

$$y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$$

Where:

- y_t is the value of the time series at time t.
- μ is the mean of the process (often assumed to be 0 for simplicity).
- ε_t is the white noise error term (shock) at time t.
- ε_{t-1} is the white noise error term at time $t-1$.
- θ is the moving average parameter.

Applying the Given Values

We are given:

- ε_t (today's shock) = 0.170
- ε_{t-1} (yesterday's shock) = -0.160
- θ = 0.70
- We will assume $\mu = 0$ (as is common unless otherwise specified).

Therefore, today's realization (y_t) is calculated as follows:

$$y_t = 0 + 0.170 + (0.70 \times -0.160)$$

$$y_t = 0.170 - 0.112$$

$$y_t = 0.058$$

Today's realization under the MA(1) process is 0.058.

Q.3375 Consider the following AR(1) model with the disturbances having zero mean and unit variance

$$y_t = 0.2 + 0.3y_{t-1} + u_t$$

The (unconditional) variance of y will be given by:

A. 1.15

B. 1.0989

C. 0.2198

D. 0.2145

The correct answer is **B.**

The (unconditional) variance of an AR(1) process is given by the variance of the disturbances divided by (1 minus the square of the autoregressive coefficient), which in this case is $1/(1-0.3^2) = 1.0989$

Q.3376 An analyst is analyzing the sales and fitting time series model and found that sales are periodically spiked in autocorrelations as they gradually decay. Such behavior is most likely indicative of:

A. Seasonality in sales data

B. A regime change structural sales data series

C. A structural shift in sales data series

D. A differencing lag

The correct answer is **A.**

Seasonality in sales data refers to the presence of variations that occur at specific regular intervals less than a year, such as weekly, monthly, or quarterly. Seasonality may be caused by various factors, such as weather, holidays, and the like. In the context of time series analysis, if there is significant seasonality in the data, the autocorrelation plot should show spikes at lags equal to the period. For instance, for monthly data, a seasonality effect would yield significant peaks at lag 12, 24, 36, 48, and so forth. This is because the sales data is repeating a specific pattern every specific period (e.g., every 12 months for yearly seasonality). Therefore, the autocorrelation will be high at these lags, indicating a strong relationship between the sales data points 12, 24, 36, etc. months apart.

Choice B is incorrect. A regime change in a structural sales data series would not result in periodic spikes that gradually diminish over time. Instead, it would likely cause a sudden and significant shift in the data, which may or may not be temporary.

Choice C is incorrect. A structural shift in the sales data series implies a one-time change in the underlying process generating the data, such as a policy change or market disruption. This would not lead to periodic spikes but rather to an abrupt and lasting alteration of the level or trend of the series.

Choice D is incorrect. Differencing lag refers to an operation used to make a time series stationary by subtracting each observation from its previous one at a certain lag order. This process does not generate periodic spikes that gradually diminish over time; instead, it aims at removing trends and seasonality from non-stationary time series.

Things to Remember

- Seasonality in time series data refers to patterns that repeat at regular intervals, such as daily, weekly, monthly, or yearly.
- Autocorrelation plots are used to identify patterns in time series data, with spikes at specific lags indicating relationships between observations at those intervals.
- Regime change in a time series refers to a shift in the underlying structure or behavior of the data, often leading to sudden and significant changes.
- Structural shifts in time series data imply a one-time change in the data-generating process, such as policy changes or market disruptions.
- Differencing lag is a technique used to make time series data stationary by removing trends and seasonality through lagged differencing.

Q.3960 The AR(2) model is defined as: $Y_t = 0.4 + 1.5Y_{t-1} - 0.7Y_{t-2} + \epsilon_t$ where ϵ_t is a white noise. What is the long-term mean of the time series?

A. 3.0

B. 1.0

C. 2.1

D. 2.0

The correct answer is **D.**

We know that for an AR(p) model $Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-1} + \ldots + \beta_p Y_{t-p} + \epsilon_t$, the mean is given by

$$E(Y_t) = \frac{\alpha}{1 - \beta_1 - \beta_2 - \ldots - \beta_p}$$

So, in this case:

$$E(Y_t) = \frac{\alpha}{1 - (\beta_1 + \beta_2)} = \frac{0.4}{1 - (1.5 - 0.7)} = 2$$

Q.3962 The lag operator is applied to the AR times series as follows:

$$(1-0.2L)(1-0.6L^4)Y_t = \epsilon_t$$

What is the resulting time series?

A. $0.2Y_{t-1} + 0.6Y_{t-4} - 0.12Y_{t-5} + \epsilon_t$

B. $Y_{t-2} + 0.6Y_{t-4} - 0.12Y_{t-5} + \epsilon_t$

C. $Y_{t-2} + 0.6Y_{t-4} - 0.12Y_{t-4} + \epsilon_t$

D. $0.1Y_{t-1} + 0.6Y_{t-4} - 0.2Y_{t-5} + \epsilon_t$

The correct answer is **A.**

This question requires the application of the properties of the Lag operator. Recall that:

$$LY_t = Y_{t-1}$$

And that the lag operator is multiplicative property. So,

$$\begin{aligned}(1-0.2L)(1-0.6L^4)Y_t &= (1-0.6L^4-0.2L+0.12L^5)Y_t \\ &= Y_t - 0.6Y_tL^4 - 0.2Y_tL + 0.12Y_tL^5 = \epsilon_t \\ &= Y_t - 0.6Y_{t-4} - 0.2Y_{t-1} + 0.12Y_{t-5} = \epsilon_t\end{aligned}$$

Rearranging we get:

$$Y_t = 0.2Y_{t-1} + 0.6Y_{t-4} - 0.12Y_{t-5} + \epsilon_t$$

Q.3963 The MA(2) model is defined as $Y_t = 0.1 + 0.8\epsilon_{t-1} + 0.16\epsilon_{t-2} + \epsilon_t$. Rewrite the model using a lag polynomial.

A. $\epsilon_t(0.4L^2 + 0.16L^2 + 3)$

B. $\epsilon_t(0.5L + 0.15L^2 + 3)$

C. $0.1 + \epsilon_t(0.8L + 0.16L^2 + 1))$

D. $\epsilon_t(0.2L + 0.14L^2 + 2)$

The correct answer is **C.**

Recall that

$$LY_t = Y_{t-1}$$

So that

$$\begin{aligned} LY_t &= L(0.1) + 0.8L(\epsilon_{t-2}) + 0.16L(\epsilon_{t-3}) + L(\epsilon_{t-1}) \\ &= L(0.1) + 0.8(L\epsilon_t) + 0.16L^2(\epsilon_t) + L^0(\epsilon_t) \\ &= 0.1 + \epsilon_t(0.8L + 0.16L^2 + 1) \end{aligned}$$

Note: Note that 0.1 is a constant and hence is not affected by the lag operator.

Q.3964 The sample autocorrelations for a time series are estimated to be $\hat{\rho}_1 = 0.20, \hat{\rho}_2 = -0.03$ and $\hat{\rho}_3 = 0.05$ from a sample size of 90. What is the value Ljung-Box Q statistic?

A. 3.45

B. 3.87

C. 20.54

D. 4.04

The correct answer is **D.**

The Ljung Box statistic, also known as the modified Box Pierce statistic, is a function of the accumulated autocorrelations, ρ_i, up to time lag m. It's calculated as:

$$Q(m) = n(n+2)\sum_{i=1}^{\tau}\left(\frac{\rho_i^{\,2}}{n-i}\right)$$

In this case, time lag = 3
Thus,

$$Q(3) = 90 \times 92\left(\frac{0.2^2}{89}\right) + 90 \times 92\left(\frac{-0.03^2}{88}\right) + 90 \times 92\left(\frac{0.05^2}{87}\right) = 4.04$$

Q.3965 An investment analyst wishes to forecast the future returns based on the prevailing interest rate then. The analyst chooses AR times series to model the monthly interest rates movement over 20 years. The equivalent AR(1) model has an intercept of 0.24 and an AR parameter of 0.65. What is the mean-reverting value of the times series used by the analyst?

A. 0.69

B. 0.56

C. 0.65

D. 0.54

The correct answer is **A.**

Recall that a stationary time series tend to revert to its historical mean. So the mean-reverting level of the AR series is its mean:

$$\lim_{\tau \to \infty} ?E_T(Y_{T+\tau}) = E(Y_t)$$

$$\Rightarrow \text{Mean-reverting level} = \frac{0.24}{1 - 0.65} = 0.6857$$

Reading 22: Nonstationary Time Series

Q.470 Forecasting involves using sample data to predict future movements. Which of the following is correct regarding forecasting?

A. Forecasts are only possible in the presence of time-series data.

B. Forecasts will always improve whenever the number of parameters is increased.

C. As the number of variables incorporated in a regression equation increases, the risk of over-fitting the in-sample data reduces.

D. In-sample forecasting ability is a very poor test of model appropriateness and adequacy.

The correct answer is **D.**

The statement that 'In-sample forecasting ability is a very poor test of model appropriateness and adequacy' is accurate. In-sample forecasting refers to the practice of using the same data set (the 'in-sample' data) to both develop and test a forecasting model. While this approach may seem intuitive, it can lead to misleading results. This is because the model is essentially 'cheating' by using knowledge of the data it is supposed to predict. As a result, the model's performance on the in-sample data is likely to be overly optimistic, providing a poor indication of how well the model will perform on new, unseen data (the 'out-of-sample' data). This is a fundamental principle of forecasting and model validation, and it underscores the importance of using separate data sets for model development and testing.

Choice A is incorrect. Forecasts are not only possible in the presence of time-series data. While time-series data is often used for forecasting, other types of data such as cross-sectional and panel data can also be used for this purpose.

Choice B is incorrect. It's not true that forecasts will always improve whenever the number of parameters is increased. In fact, adding too many parameters can lead to overfitting, which means that the model may fit the sample data very well but perform poorly on new, unseen data.

Choice C is incorrect. The statement contradicts a fundamental principle in regression analysis known as Occam's razor: simpler models are preferable to more complex ones if they explain the same amount of variance in the dependent variable. As we increase the number of variables incorporated in a regression equation, we risk over-fitting our model to our sample (in-sample) data and reducing its predictive power on out-of-sample or future observations.

Things to Remember

- Forecasting can be done using various types of data, including time-series, cross-sectional, and panel data.
- Overfitting occurs when a model is too complex and fits the sample data very well but

performs poorly on new, unseen data.

- Occam's razor principle states that simpler models are preferred over complex ones if they explain the same amount of variance in the dependent variable.
- Model validation is crucial in forecasting to ensure that the model performs well on new, unseen data.

Q.472 A Financial Risk Manager exam candidate suggests that a model based on financial theory is likely to lead to a high degree of out-of-sample forecast accuracy. Which of the following best explains why the candidate is correct?

A. A solid financial background significantly increases the chances of the model working in the out-of-sample period as well as for the sample data used to estimate the model's parameters.

B. A financial background increases the chances of use of authentic input data.

C. Financial theory incorporates industry-wide variables.

D. Financial theory would be easy to understand and research on.

The correct answer is **A.**

A model that is based on a solid financial background is more likely to produce accurate out-of-sample forecasts. This is because a strong financial background provides a comprehensive understanding of the financial markets, including the various factors that influence market trends and the relationships between these factors. This understanding can be used to develop a model that accurately represents the complexities of the financial markets, thereby increasing the likelihood that the model will perform well not only on the sample data used to estimate the model's parameters, but also on out-of-sample data. Furthermore, a solid financial background can help in identifying and addressing potential issues and limitations of the model, thereby further enhancing its predictive accuracy.

Choice B is incorrect. While a financial background may increase the chances of using authentic input data, it does not necessarily guarantee that a model based on financial theory will yield accurate out-of-sample forecasts. The accuracy of these forecasts depends more on the robustness and applicability of the model itself rather than just the authenticity of input data.

Choice C is incorrect. Although financial theory does incorporate industry-wide variables, this alone does not ensure high accuracy in out-of-sample forecasts. The inclusion of such variables can enhance the comprehensiveness and relevance of a model, but it doesn't directly translate into forecast accuracy.

Choice D is incorrect. The ease with which financial theory can be understood and researched has no direct bearing on its ability to produce accurate out-of-sample forecasts. A model's predictive power comes from its theoretical soundness, empirical validity, and appropriate parameter estimation - not simply from being easy to understand or research.

Things to Remember

- Out-of-sample forecast accuracy is crucial in risk management as it helps in assessing the reliability and effectiveness of models in real-world scenarios.
- Financial theory provides the foundation for understanding the relationships and dynamics in financial markets, which is essential for developing accurate models.
- Parameter estimation in financial models is a critical aspect that affects the model's performance in both in-sample and out-of-sample scenarios.
- Model validation techniques, such as backtesting and stress testing, are used to evaluate the accuracy and robustness of financial models.
- Overfitting is a common issue in model development where a model performs well on historical data but fails to generalize to new data, highlighting the importance of using sound financial theory.

Q.477 Joel Matip, FRM, is running a regression model to forecast in-sample data. He’s worried about data mining and over-fitting the data. The criterion that provides the highest penalty factor based on degrees of freedom is the:

A. Schwarz information criterion.

B. Akaike information criterion.

C. Unbiased mean squared error.

D. Mean squared error.

The correct answer is **A.**

The Schwarz Information Criterion (SIC), also known as the Bayesian Information Criterion (BIC), is a criterion for model selection among a finite set of models. It is based on the likelihood function, and it also includes a penalty term for complexity. The SIC has a higher penalty factor for complexity (i.e., the number of parameters in the model) compared to other criteria. This makes it more conservative and less likely to overfit the data. The penalty factor for the SIC is

$T^{\frac{k}{T}}$, where T is the sample size and k is the number of parameters in the model. This penalty factor increases with the number of parameters, thus discouraging the use of overly complex models.

Choice B is incorrect. The Akaike information criterion (AIC) does impose a penalty factor for the number of parameters in the model, but it is less severe than the Schwarz information criterion (SIC). The SIC imposes a larger penalty for model complexity, making it more conservative and better at avoiding overfitting.

Choice C is incorrect. The Unbiased mean squared error (UMSE) does not impose any penalty based on degrees of freedom or model complexity. It simply measures the average squared difference between predicted and actual values, without considering whether this accuracy was achieved through overfitting.

Choice D is incorrect. Like UMSE, Mean Squared Error (MSE) also does not impose any penalty based on degrees of freedom or model complexity. It only measures the average squared difference between predicted and actual values, without taking into account whether this accuracy was achieved through overfitting.

Things to Remember

- Model Selection Criteria: Criteria like AIC, BIC, and others help in selecting the best model by balancing goodness of fit and model complexity.
- Overfitting: Overfitting occurs when a model learns the noise in the data rather than the underlying pattern, leading to poor generalization to new data.
- Penalty for Complexity: Including a penalty for model complexity helps prevent overfitting by discouraging overly complex models.
- Degrees of Freedom: Degrees of freedom in a model represent the number of values in the final calculation of a statistic that are free to vary.
- Sample Size: The size of the sample used in the analysis affects the penalty factor for complexity in criteria like BIC.

Q.485 Define trend as used in business and economics.

A. A curve that represents the change in a series of data points collected over a given period of time.

B. A straight line extrapolated from past and present values to forecast future values of a given variable.

C. A pattern of gradual change in output as a result of data points moving in a given direction over time, and which can be represented by a line, curve, or graph.

D. A pattern of gradual change in output as a result of data points moving in a positive direction over time, and which can be represented by a line, curve, or graph.

The correct answer is **C.**

A trend in business and economics is a pattern of gradual change in a variable's output over time. This change is a result of data points moving in a specific direction, which could be either positive or negative. The trend can be represented in various forms, such as a line, curve, or graph. This definition encapsulates the essence of a trend in these fields. It highlights the gradual change, the direction of the data points, and the various forms in which a trend can be represented. This understanding of trends is crucial in making informed decisions and predictions in business and economics.

Choice A is incorrect. While a trend can be represented by a curve, it does not necessarily have to be. A trend can also be represented by a straight line or even a graph. Furthermore, the definition of 'trend' is not limited to changes in data points over time; it also includes the direction and pattern of these changes.

Choice B is incorrect. This option describes one method of forecasting future values based on past and present data, but this does not encompass the full meaning of 'trend'. Trends do not always follow a straight line and they are not solely used for forecasting purposes.

Choice D is incorrect. The direction of data points (whether positive or negative) does not define a trend. A trend refers to any consistent pattern in data over time, regardless of whether that pattern involves an increase or decrease in values.

Things to Remember

- Time Series Analysis: Trends are often analyzed in the context of time series data, where observations are recorded at regular intervals over time.
- Seasonality: Trends can be influenced by seasonal patterns, which are recurring fluctuations that occur at specific time intervals.
- Cyclical Trends: In addition to gradual changes, trends can also exhibit cyclical patterns that repeat over longer time frames.
- Extrapolation: Forecasting future values based on trends involves extrapolating past and present data points to predict future outcomes.
- Regression Analysis: Statistical techniques like regression analysis can be used to quantify and analyze trends in data.

Q.486 An FRM exam candidate studies labor participation rate for males 18 years and over, and obtains the following linear model:

$$P_t = \beta_0 + \beta_1 t$$

Where p_t stands for participation rate and t = time in years. The fact that β_1 (the regression slope) is positive means:

A. Participation rate decreases with increase in time.

B. Participation rate increases with time.

C. Time is the only explanatory variable.

D. As males grow older, less and less of them engage in labor.

The correct answer is **B.**

The regression slope, denoted by β_1, is a measure of the rate of change in the dependent variable (in this case, the participation rate) for each unit change in the independent variable (in this case, time). A positive regression slope indicates that the dependent variable increases as the independent variable increases. Therefore, if β_1 is positive, it means that the participation rate increases with time. This is consistent with the concept of a positive linear relationship, where an increase in one variable is associated with an increase in the other variable. This relationship is fundamental to regression analysis, which is a statistical method used to understand and quantify the relationships between variables.

Choice A is incorrect. A positive regression slope, β_1, indicates that the participation rate increases with time, not decreases. The sign of the slope coefficient determines the direction of the relationship between the independent variable (time) and dependent variable (participation rate). If β_1 was negative, then we could say that participation rate decreases with increase in time.

Choice C is incorrect. Although time is used as an explanatory variable in this model, it does not imply that it's the only explanatory variable. Other variables could also be included in a more complex model to explain changes in labor participation rates over time.

Choice D is incorrect. The given linear model does not provide any information about individual age effects on labor participation rates for males aged 18 years and above. It only shows how overall labor participation rates change over time for this group as a whole.

Q.487 In the study of financial markets, understanding the nature of relationships between different financial metrics and variables is crucial for effective analysis and decision-making. Among the following relationships, which one is most likely to be characterized by a linear trend according to foundational financial theories?

A. The relationship between the default probability of a bond and its credit rating.

B. The relationship between the price of a call option and the underlying asset's price.

C. The relationship between the required rate of return and the beta of a stock.

D. The relationship between the expected return of a portfolio and the standard deviation of its returns.

The correct answer is **C.**

This is the most likely to be linear among the given options. According to the Capital Asset Pricing Model (CAPM), there is a linear relationship between a stock's beta and its expected return. The formula is:

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$$

Where:
$E(R_i)$ is the expected return on the asset
R_f is the risk-free rate
β_i is the beta of the asset
$E(R_m)$ is the expected return of the market

This equation describes a straight line, with beta as the independent variable and expected return as the dependent variable.

A is incorrect. The relationship between default probability and credit rating is typically not linear. Credit ratings are ordinal categories, and the increase in default probability is often not uniform between rating steps.

B is incorrect. Due to the nature of option pricing models like the Black-Scholes, the delta (rate of change of the option price with respect to changes in the underlying asset price) changes as the underlying's price changes, resulting in a curved relationship.

D is incorrect. The relationship between expected return and standard deviation of returns is typically represented by a curved efficient frontier in modern portfolio theory, not a straight line.

Things to Remember

- The Capital Asset Pricing Model (CAPM) is a foundational financial theory that describes the relationship between risk and expected return for an asset.
- CAPM assumes a linear relationship between a stock's beta and its expected return, making it essential for calculating the cost of equity and evaluating investment performance.

- Beta (β) measures the sensitivity of a stock's returns relative to market returns, indicating its systematic risk.
- The Security Market Line (SML) is the graphical representation of CAPM, illustrating the expected return of an asset as a function of its beta.

Q.490 A seasonal time series component:

A. Reflects variability due to natural disasters.

B. Reflects variability during a single year.

C. Reflects a regular, multi-year pattern of being above and below the trend line.

D. Reflects gradual variability over a long time period.

The correct answer is **B.**

A seasonal time series component reflects variability during a single year. This means that the data exhibits a pattern of change that repeats annually. For example, retail sales may increase during the holiday season each year, or energy consumption may rise during the summer months due to increased use of air conditioning. This predictable and recurring pattern within each year is what defines a seasonal time series component. It's important to note that the pattern must repeat with a fixed and known frequency within each year for it to be considered a seasonal component. This component is crucial in time series analysis as it helps in forecasting and understanding the underlying patterns in the data.

Choice A is incorrect. The variability due to natural disasters is not a seasonal component of a time series. Natural disasters are unpredictable and do not follow a regular pattern, hence they cannot be considered as seasonal components which are expected to occur at specific intervals.

Choice C is incorrect. A regular, multi-year pattern of being above and below the trend line does not describe the characteristics of a seasonal time series component. Seasonal components reflect variability within a single year, whereas this choice describes cyclical patterns that can span multiple years.

Choice D is incorrect. Gradual variability over a long time period refers to the trend component of a time series, not the seasonal component. The trend represents long-term progression of the series while seasonality refers to short-term, predictable and regular patterns within specific periods (usually within an annual cycle).

Things to Remember

- Seasonal time series components are recurring patterns that repeat within a single year.

- These components help in understanding and forecasting the short-term variations in data.
- Seasonal components are different from trend components, which represent long-term progression.
- Identifying and accounting for seasonal components is important in time series analysis for accurate forecasting.

Q.492 An FRM exam candidate wishes to determine the type of variation in a time series and singles out non-seasonal variation while at the same time discarding seasonal variation. The candidate's approach:

A. Is appropriate since seasonality has minimal impact on most phenomena, including economic ones.

B. Is appropriate because seasonal variation is very easy to establish without the need for intensive statistical analysis.

C. Is inappropriate because seasonality may account for a large part of the variation.

D. Is inappropriate because non-seasonal variation is irrelevant while studying economic relationships.

The correct answer is **C.**

The candidate's approach is inappropriate because seasonality may account for a large part of the variation. In time series analysis, both seasonal and non-seasonal variations play a significant role. Seasonal variation refers to fluctuations in the data that occur in regular, predictable intervals, such as monthly, quarterly, or annually. These variations can be due to seasonal factors like weather, holidays, and school schedules. On the other hand, non-seasonal variation refers to fluctuations that do not follow a particular pattern. These can be due to factors like economic conditions, political events, and technological changes. When analyzing a time series, it is crucial to consider both types of variations as they provide a comprehensive understanding of the data. Ignoring seasonal variation can lead to inaccurate conclusions as it might account for a significant portion of the variation in the data. Therefore, the candidate's approach of disregarding seasonal variation is inappropriate.

Choice A is incorrect. Seasonality can have a significant impact on many phenomena, including economic ones. Therefore, disregarding it may lead to inaccurate analysis and conclusions.

Choice B is incorrect. While it might be easier to establish seasonal variation compared to non-seasonal variation, this does not justify ignoring it in the analysis. The ease of establishing a factor should not determine its relevance or importance in an analysis.

Choice D is incorrect. Non-seasonal variation can be very relevant when studying economic relationships as they may reflect underlying trends or cycles that are independent of seasonality. Ignoring non-seasonal variations could potentially miss out on important insights about the data being analyzed.

Things to Remember

- Seasonal variation refers to fluctuations in the data that occur in regular, predictable intervals, such as monthly, quarterly, or annually.
- Non-seasonal variation refers to fluctuations that do not follow a particular pattern and can be due to factors like economic conditions, political events, and technological changes.
- Both seasonal and non-seasonal variations play a significant role in time series analysis and should be considered for a comprehensive understanding of the data.
- Ignoring seasonal variation can lead to inaccurate conclusions as it might account for a significant portion of the variation in the data.
- Seasonality can have a significant impact on many phenomena, including economic ones, and should not be disregarded in analysis.

Q.493 An analyst wishes to use seasonal dummy variables (0s and 1s) to model seasonality. Suppose there are four seasons in a year. Which of the arrangements below would represent the third season (third quarter)?

A. 1, 0, 0, 0,

B. 0, 0, 1, 0,

C. 0, 0, 0, 1

D. 0, 1, 0, 0,

The correct answer is **B.**

Dummy variables are binary (0 or 1) variables used in regression models to represent categorical data, such as seasonality, by assigning values that indicate the presence or absence of a specific category. In the context of seasonality, dummy variables can help capture the effects of different seasons on the dependent variable. For example, let's say we are analyzing monthly sales data, and we suspect that sales are affected by seasonality. We can create dummy variables to represent each quarter of the year: Q1 (January, February, March): Dummy variable D1, Q2 (April, May, June): Dummy variable D2, Q3 (July, August, September): Dummy variable D3, Q4 (October, November, December): Implicit dummy variable. To represent these quarters using dummy variables, we assign the value of 1 to the dummy variable corresponding to the relevant quarter and 0 to the others. For Q3: D1 = 0, D2 = 0, D3 = 1, D4 = 0. The arrangement for the third season (third quarter) would therefore be: 0, 0, 1, 0. The fourth season (fourth quarter) is implicitly represented when all dummy variables (D1, D2, and D3) are set to 0. We do this to avoid the dummy variable trap, which occurs when dummy variables are multicollinear.

Choice A is incorrect. This representation would be used for the first season or quarter, not the third one. The '1' in the first position indicates that it's the first season, while all other seasons are represented by '0'.

Choice C is incorrect. This representation would be used for the fourth season or quarter, not the third one. The '1' in the last position indicates that it's the fourth season, while all other seasons are represented by '0'.

Choice D is incorrect. This representation would be used for second season or quarter and not for third one as indicated by 1 at second place.

Things to Remember

- Dummy variables are used in regression models to represent categorical data by assigning binary values (0 or 1) to indicate the presence or absence of a specific category.
- Seasonal dummy variables are used to capture the effects of different seasons on the dependent variable in time series analysis.
- To represent seasons using dummy variables, assign the value of 1 to the relevant season and 0 to the others.
- The dummy variable trap occurs when dummy variables are multicollinear, which can lead to issues in the regression analysis.
- Implicit dummy variables are used to represent the omitted category when creating dummy variables for categorical data.

Q.494 Which one of the following statements best explains the idea of holiday variation?

A. The dates of certain holidays may change from one country to another.

B. The number of holidays in a year differs from one country to another.

C. The number of holidays varies from year to year.

D. The dates of certain holidays change from year to year.

The correct answer is **D.**

The term 'holiday variation' refers to the phenomenon where the dates of certain holidays change from year to year. While these holidays occur approximately in the same period each year, their exact dates can vary. This is particularly true for holidays that are based on lunar calendars or other non-Gregorian calendar systems. For example, Easter, a Christian holiday, does not have a fixed date in the Gregorian calendar (which is widely used today) because it is based on the lunar calendar. Instead, it is observed on the first Sunday after the first full moon following the vernal equinox. As a result, its date can vary between March 22 and April 25. This variation in holiday dates can have significant implications in various fields, such as business and economics, where accurate forecasting and planning are crucial.

Choice A is incorrect. While it's true that the dates of certain holidays may change from one country to another, this does not define the concept of holiday variation in the context of calendar variations. Holiday variation refers specifically to changes in dates within a single calendar or country, not differences between countries.

Choice B is incorrect. The number of holidays in a year differing from one country to another does not describe holiday variation either. This choice refers more to cultural or national differences rather than variations within a single calendar year.

Choice C is incorrect. Although the number of holidays can vary from year to year due to factors such as leap years or specific cultural events, this isn't what is meant by 'holiday variation'. The term specifically refers to the changing dates of certain holidays within a given year.

Things to Remember

Additional important concepts related to holiday variation:

- Some holidays are based on lunar calendars or other non-Gregorian calendar systems, leading to variations in their dates from year to year.
- Holiday variation can impact various fields such as business and economics, requiring accurate forecasting and planning.
- Easter, a Christian holiday, is an example of a holiday with variable dates due to its

dependence on the lunar calendar.

- Understanding holiday variation is important for scheduling, resource planning, and cross-cultural communication.

Q.497 Quarter 1 sales for an automobile manufacturer were $180 million. If the quarter 1 seasonal index was 1.8, what is the estimate of annual sales for this firm?

A. $400 million

B. $180 million

C. $720 million

D. $100 million

The correct answer is **A.**

An index of 1.8 means that quarter 1 sales are 80% above the norm (average).

Thus, each quarter has average sales worth $\frac{\$180{,}000{,}000}{1.8} = \$100,000,000$.

Therefore, the annual estimate = 4 * Quarter average = 4 * $100,000,000 = $400,000,000.

Things to Remember

- Seasonal index is used to adjust historical data to account for regular, predictable fluctuations in data due to seasonal factors.
- Calculating annual estimates based on seasonal indices involves adjusting quarterly or monthly data to estimate annual figures.
- Seasonal indices help in identifying patterns and trends in data, allowing for better forecasting and planning.
- Understanding seasonal variations is crucial for industries where sales or demand fluctuate throughout the year, such as retail, tourism, and agriculture.

Q.499 Which one of the following is correct? A time series data set with quarterly seasonality can be handled by:

A. Using three dummy variables.

B. Using two dummy variables – one for each season.

C. Deseasonalizing the data and then applying non-seasonal methods.

D. Using only one dummy variable.

The correct answer is **A.**

When dealing with time series data that exhibits seasonality, one common approach is to use dummy variables. A dummy variable is a numerical variable used in regression analysis to represent subgroups of the sample in your study. In this case, the subgroups are the different seasons in the data. The number of dummy variables used should be one fewer than the number of seasons. This is because one season can be represented as the absence of all other seasons. Therefore, if there are four seasons in the data, three dummy variables should be used. This method allows for the effects of the different seasons to be accounted for in the analysis, providing a more accurate representation of the data.

Choice B is incorrect. Using two dummy variables for each season would not be appropriate for a time series data set that shows quarterly seasonality. This is because there are four quarters in a year, and therefore, four seasons to account for in the data. Using only two dummy variables would not capture the full extent of the seasonal variation present in the data.

Choice C is incorrect. Deseasonalizing the data and then applying non-seasonal methods may not be an appropriate method for handling this type of time series data either. While deseasonalizing can help to remove seasonal patterns from the data, it does not necessarily mean that non-seasonal methods will be effective at analyzing this type of time series data. The underlying structure of the time series may still exhibit other forms of dependence or patterns that are better handled by other methods.

Things to Remember

- **Seasonality in time series data refers to patterns that repeat at regular intervals, such as daily, weekly, monthly, or quarterly.**
- **Dummy variables are used in regression analysis to represent categorical data, such as seasons, in a numerical way.**
- **When dealing with quarterly seasonality, it is important to use dummy variables to capture the effects of each season on the data.**
- **Deseasonalizing data involves removing the seasonal component from the time series to focus on the underlying trend and irregular components.**

- **Non-seasonal methods are statistical techniques that do not take into account seasonal patterns in the data.**

Q.502 An analyst applies the following time series model to daily data:R_t = returnsD_1 – dummy variable for Monday and zero otherwiseD_2 – dummy variable for Tuesday and zero otherwiseD_3 – dummy variable for Wednesday and zero otherwiseD_4 – dummy variable for Thursday and zero otherwiseWhat is the interpretation of the parameter estimate for the intercept?

A. It’s the average return for the 5 days from Monday to Friday.

B. It’s the Friday deviation from the mean return for the week.

C. It’s the average return on Monday.

D. It’s the average return on Friday.

The correct answer is **D.**

The intercept in this time series model represents the average return on Friday. This is because, for any observation made on a Friday, all the dummy variables (D_1, D_2, D_3, and D_4) will be zero. Therefore, the estimated value of the intercept, denoted as β_0, will provide an estimate of the average return on Friday. This interpretation is based on the structure of the model, where the intercept represents the baseline return (in this case, for Friday) and the coefficients of the dummy variables represent the deviation of the returns on the other days from this baseline.

Choice A is incorrect. The intercept in this model does not represent the average return for the 5 days from Monday to Friday. Instead, it represents the average return on a specific day, which in this case is Friday.

Choice B is incorrect. The intercept does not represent the deviation of Friday's returns from the mean return for the week. Rather, it signifies the average return on Fridays.

Choice C is incorrect. The intercept does not denote Monday's average returns but rather indicates Friday's average returns.

Things to Remember

- Dummy variables are used in regression analysis to represent categorical data or to capture the effect of different categories on the dependent variable.
- The intercept in a regression model represents the value of the dependent variable when all independent variables are zero.
- In time series analysis, dummy variables are often used to account for the effect of

specific time periods or events on the dependent variable.

- Interpreting coefficients in regression models involves understanding the impact of each independent variable on the dependent variable, holding other variables constant.
- Friday's average return being captured by the intercept in this model highlights the importance of correctly interpreting coefficients and dummy variables in time series analysis.

Q.523 An analyst intends to use linear regression to model the relationship between two-time series. After some testing, she finds out that one of the time series has a unit root. She should:

A. Not use linear regression if the two time series are not co-integrated.

B. Not use linear regression.

C. Perform another test on a higher level of significance before proceeding to use linear regression.

D. Only use linear regression if the time series are co-integrated.

The correct answer is **B.**

The analyst should not use linear regression. A unit root in a time series indicates the presence of a stochastic or random trend. This means that the series is non-stationary, and its mean and variance change over time. Linear regression assumes that the data is stationary, meaning that the mean, variance, and autocorrelation structure do not change over time. If these assumptions are violated, as they would be in the presence of a unit root, the results of the linear regression would be unreliable. Therefore, the analyst should not use linear regression to model the relationship between the two time series. Instead, she should first difference the series to remove the trend and make the series stationary before proceeding with any form of modeling.

Choice A is incorrect. While it's true that if two time series are co-integrated, linear regression can be used, this choice does not address the issue of the unit root found in one of the time series. The presence of a unit root indicates that the time series is non-stationary and using linear regression on such data can lead to spurious results.

Choice C is incorrect. Performing another test at a higher level of significance does not resolve the problem posed by a unit root in one of the time series. Even if another test were performed and passed, it would not change the fact that one of our variables has a unit root and thus violates an assumption necessary for valid linear regression analysis.

Choice D is incorrect. Similar to Choice A, this option suggests using linear regression only if both time series are co-integrated. However, it fails to address how to handle situations where there's a unit root present in one or both variables which makes them non-stationary and unsuitable for standard linear regression analysis.

Things to Remember

- Stationarity in time series analysis is a key concept. A stationary time series is one whose statistical properties such as mean, variance, and autocorrelation are constant over time.
- Unit root tests, such as the Augmented Dickey-Fuller (ADF) test, are used to determine if a time series is stationary or non-stationary.
- Cointegration is a statistical property of a collection of time series variables that have a common stochastic trend, which means they move together over time.
- Differencing is a common technique used to make a time series stationary by computing the differences between consecutive observations.
- Spurious regression results can occur when using linear regression on non-stationary data, leading to unreliable and misleading conclusions.

Q.3363 A mortgage analyst produced a model to predict housing starts (given in thousands) within Florida in the US. The time series model contains both a trend and a seasonal component and is given by the following:

$$y_t = 0.2 \times \text{Time}_t + 10.5 + 3.0 \times D_{2t} + 5.4 \times D_{3t} + 0.7 \times D_{4t}$$

The trend component is reflected in variable TIME_t, where (t) = month.
Seasons are defined as follows :

Season	Months	Dummy
Winter	December,January and February	−
Spring	March, April and May	D_{2t}
Summer	June, July and August	D_{3t}
Fall	September, October and November	D_{4t}

The model starts in May 2019, i.e., $y_{(T+1)}$ refers to June 2019. What does the model predict for September 2020?

A. 23

B. 14

C. 13

D. 16

The correct answer is **B.**

The model is given as:

$$y_t = 0.2 \times \text{Time}_t + 10.5 + 3.0 \times D_{2t} + 5.4 \times D_{3t} + 0.7 \times D_{4t}$$

Since we have three dummies and an intercept, quarterly seasonality is reflected by the intercept (10.5) plus the three seasonal dummy variables (D_2, D_3 and D_4)
If $y_{(T+1)}$ = June 2019, then Sept 2020 = $y_{(T+16)}$

Finally, note that Sept. falls under D_{4t}

$$y_{(T+16)} = 0.20 \times 16 + 10.5 + 0.7 \times 1 = 14.4$$

Thus, the model predicts 14 housing starts in Sept. 2020.

Q.3364 A ski resort has come up with a model to predict the number of guests (given in hundreds) checking in throughout the year. The time series model contains both a trend and a seasonal component and is given by the following:

$$y_t = 0.2 \times \text{Time}_t + 10.5 + 3.0 \times D_{2t} + 2.1 \times D_{3t} + 2.8 \times D_{4t}$$

The trend component is reflected in variable $TIME_{(t)}$, where (t) = month.
Seasons are defined as follows :

Season	Months	Dummy
Winter	December,January and February	−
Spring	March, April and May	D_{2t}
Summer	June, July and August	D_{3t}
Fall	September, October and November	D_{4t}

The model starts in April 2019, i.e., $y_{(T+1)}$ refers to May 2019. How many more guests are expected in April 2020 than in July of the same year?

A. 10

B. 30

C. 9

D. 15

The correct answer is **B.**

The model is given as:

$$y_t = 0.2 \times \text{Time}_t + 10.5 + 3.0 \times D_{2t} + 2.1 \times D_{3t} + 2.8 \times D_{4t}$$

Since we have three dummies and an intercept, quarterly seasonality is reflected by the intercept (10.5) plus the three seasonal dummy variables (D_2, D_3 and D_4)
If $y_{(T+1)}$ = May 2019, then April 2020 = $y_{(T+12)}$ and July 2020 = $y_{(T+15)}$

Finally, note that April falls under $D_{(2t)}$, while July falls under $D_{(3t)}$

$$y_{(T+12)} = 0.20 \times 12 + 10.5 + 3.0 \times 1 = 15.9$$
$$y_{(T+15)} = 0.20 \times 15 + 10.5 + 2.1 \times 1 = 15.6$$
$$y_{(T+12)} - y_{(T+15)} = 15.9 – 15.6 = 0.3$$

Thus, the model predicts 1,590 guests in April and 1,560 guests in July, representing a difference of 30.

Q.3366 A pure seasonal dummy model is constructed as below:

$$y_t = \sum_{i=1}^{s} \gamma_i D_{i,t} + \varepsilon_t$$

If all seasonal factors (γ_i) in the model are equal, it can be concluded that:

A. A seasonally adjusted time series is to be constructed

B. There is a lack of seasonality

C. There is a need for additional dummy variables

D. Both A and C

The correct answer is **B.**

If all seasonal factors (γ_i) in the model are equal, it implies that there is a lack of seasonality. In the context of a time series model, seasonality refers to the presence of variations that occur at specific regular intervals less than a year, such as weekly, monthly, or quarterly. Seasonality can be caused by various factors such as weather, holidays, and the like. However, if all the seasonal factors are equal, it means that these factors do not change or vary with the seasons. Therefore, there is no effect of seasonality in the time series data. For instance, if we are analyzing sales data, this could mean that the sales are equal in all four quarters of the year, indicating no seasonality.

Choice A is incorrect. The construction of a seasonally adjusted time series is not necessarily inferred from the observation that all seasonal factors are equal. Seasonal adjustment involves removing the seasonal component of a time series, but this does not directly relate to the equality of all seasonal factors in a model.

Choice C is incorrect. The need for additional dummy variables cannot be inferred from the observation that all seasonal factors are equal. Dummy variables are used to incorporate categorical data into a regression model, and their necessity would depend on whether there are other categorical variables that have not been included in the model, rather than on whether existing dummy variable coefficients are equal.

Choice D is incorrect. As explained above, neither constructing a seasonally adjusted time series nor adding more dummy variables can be inferred from all seasonal factors being equal.

Q.3953 An investment analyst wants to fit the weekly sales (in millions) of his company by using the sales data from Jan 2018 to Feb 2019. The regression equation is defined as:

$$\ln Y_t = 5.105 + 0.044t, \quad t = 1, 2, \ldots, 100$$

What is the trend value estimate of the sales in the 90^{th} week?

A. 8,647.28 Million

B. 7,947.26 Million

C. 8,537.38 Million

D. 8,237.48 Million

The correct answer is **A.**

From the regression equation, $\hat{\beta}_0 = 5.105$ and $\hat{\beta} = 0.044$. We know that, under log-linear trend models, the predicted trend value is given by:

$$\begin{aligned} \ln Y_t &= \hat{\beta}_0 + \hat{\beta}_1 t \\ \Rightarrow Y_t &= e^{\hat{\beta}_0 + \hat{\beta}_1 t} \\ \therefore Y_{90} &= e^{5.105 + 0.044 \times 90} \\ &= 8,647.28 \text{ Million} \end{aligned}$$

Q.3954 A financial analyst wishes to conduct an ADF test on the log of 20-year real GDP from 1999 to 2019. The result of the tests is shown below:

Deterministic	γ	δ_0	δ_1	Lags	5%CV	1%CV
None	−0.004 (−1.555)			8	−1.940	−2.570
Constant	−0.008 (−1.422)	0.010 (1.025)		4	−2.860	−3.445
Trend	−0.084 (−4.376)	0.188 (−4.110)		3	−3.420	−3.984

The output of the ADF reports the results at the different number deterministic terms (first column), and the last three columns indicate the number of lags according to AIC and the 5% and 1% critical values that are appropriate to the underlying sample size and the deterministic terms. The quantities in the parenthesis (below the parameters) are the test-statistics. What deterministic term is most preferred for this model?

A. Constant

B. Trend

C. None (No deterministic components)

D. Both the constant and the trend

The correct answer is **B.**

The Augmented Dickey-Fuller (ADF) test is a type of unit root test that is used to determine the presence of unit roots in a time series sample. The presence of a unit root indicates that the time series is non-stationary. The ADF test uses an autoregressive model and optimizes an information criterion across multiple different lag lengths.

In the given question, the ADF test results are presented for three different deterministic terms: None, Constant, and Trend. The deterministic term is a component of the model that is not random and does not change over time. The choice of deterministic term can significantly impact the results of the ADF test.

The recommended method for choosing the appropriate deterministic term is to include the terms that are significant at the 10% level. In this case, the Trend deterministic term is chosen because the absolute value of its test statistic is greater than 1.645 (i.e., |-4.376| > |1.645|). This indicates that the Trend term is statistically significant.

Furthermore, since the absolute value of the Trend test statistic is greater than both the 5% critical value (CV) and the 1% CV, there is no evidence of unit roots. This means that the null hypothesis of the presence of a unit root is rejected, indicating that the time series is stationary. Therefore, the Trend deterministic term is the most preferred for this model.

Choice A is incorrect. Although the constant deterministic term has a smaller absolute value of the test statistic (-1.422) compared to the trend deterministic term (-4.376), it is still greater than both 5% and 1% critical values, indicating that we cannot reject the null hypothesis of a unit root at these significance levels. Therefore, this model does not provide sufficient evidence to support that there is no unit root in the series.

Choice C is incorrect. The test statistic for no deterministic components (-1.555) is greater than both 5% and 1% critical values, suggesting that we cannot reject the null hypothesis of a unit root at these significance levels. Hence, this model does not provide enough evidence to conclude that there's no unit root in the series.

Choice D is incorrect. The question doesn't provide any information or results for a model with both constant and trend as deterministic terms; therefore, we can't make any conclusions about its suitability based on given data.

Q.3955 The seasonal dummy model is generated on the quarterly growth rates of mortgages. The model is given by:

$$Y_t = \beta_0 + \sum_{j=1}^{s-1} \gamma_j D_{jt} + \epsilon_t$$

The estimated parameters are $\hat{\gamma}_1 = 6.25, \hat{\gamma}_2 = 50.52, \hat{\gamma}_3 = 10.25$ and $\hat{\beta}_0 = -10.42$ using the data up to the end of 2019. What is the difference between the forecasted value of the growth rate of the mortgages in the second and third quarters of 2020?

A. 24.56

B. 32.45

C. 40.27

D. 30.32

The correct answer is **C.**

For the second quarter, define the set of dummy variables:

$$D_{jt} = \begin{cases} 1, & \text{for } Q_2 \\ 0, & \text{for } Q_1, Q_3 \text{ and } Q_4 \end{cases}$$

So,

$$E(\hat{Y}_{Q2}) = \beta_0 + \sum_{j=1}^{3} \gamma_j D_{jt} = -10.42 + 0 \times 6.25 + 1 \times 50.52 + 0 \times 10.25 = 40.1$$

Analogously, the dummy variables for the third quarter is defined as:

$$D_{jt} = \begin{cases} 1, & \text{for } Q_3 \\ 0, & \text{for } Q_1, Q_2 \text{ and } Q_4 \end{cases}$$

$$E(\hat{Y}_{Q3}) = \beta_0 + \sum_{j=1}^{3} \gamma_j D_{jt} = -10.42 + 0 \times 6.25 + 0 \times 50.52 + 1 \times 10.25 = -0.17$$

So that:

$$Q_2 - Q_3 = 40.10 - -0.17 = 40.27$$

Q.3957 A log trend model is approximated on the interest rate (in %) movement in a certain market based on data from 2000 until 2020. The estimated model is given as:

$$\ln Y_t = -0.1567 + 0.00134t + \hat{\epsilon}_t$$

The standard deviation of the residual is 0.0342. Assuming that the residuals are normally distributed, what is the 95% confidence interval for interest rate movement for year 2023?

A. [12.030, 13.754]

B. [12.018, 13.739]

C. [-0.0584, 0.0756]

D. [11.994, 13.713]

The correct answer is **A.**

In this case,

$$E_T[Y_T] = \exp(E_T[\ln Y_T] + \frac{\sigma^2}{2})$$

And the error bounds on the In are $\pm 1.96 * 0.0342$.

Thus the bounds-multiplier is given by $\exp(\pm 1.96 * 0.0342) = 0.935, 1.069$

E[ln Y_T] can be calculated as follows:

$$E_T[\ln Y_T] = -0.1567 + 0.00134 * 2023 = 2.554$$

We also need

$$\frac{\sigma^2}{2} = \frac{0.0342^2}{2} = 0.0006$$

Thus,

$$E[Y_{2023}] = \exp(2.554 + 0.0006) = 12.866$$

Thus, the 95% confidence interval for interest rate movement 3 years from now, is given by:

$$95\%CI_{2023} = [0.935 * 12.866, 1.069 * 12.866] = [12.030, 13.754]$$

Q.3959 Which of the following statements best describes the time series with a unit-roots?

A. It is a random walk time series with a drift described using AR(1) model whose lag coefficient is 1

B. It is a random walk time series with a drift described using AR(1) model whose lag coefficient is 0

C. Time series with unit roots are covariance stationary.

D. It is a stationary time series that reverts to its mean over time.

The correct answer is **A.**

A time series with a unit-root is indeed a random walk time series with a drift, described using an AR(1) model where the lag coefficient is 1. In time series analysis, a unit-root test checks whether a time series variable is non-stationary and possesses a unit-root. The presence of a unit-root indicates that the time series does not have a tendency to return to a long-run mean and has a high level of persistence. The AR(1) model, or first-order autoregressive model, is a common approach for modeling time series data. In this model, the value at a given time is a function of the value at the previous time. If the lag coefficient is 1, it indicates that the series is a random walk, meaning that the current value is equal to the previous value plus a random error. The 'drift' term in a random walk model represents a constant trend over time. Therefore, a unit-root time series can be accurately described as a random walk with a drift, modeled using an AR(1) model with a lag coefficient of 1.

Choice B is incorrect. A random walk time series with a drift described using an AR(1) model whose lag coefficient is 0 does not characterize a unit-root time series. In fact, the lag coefficient of 0 would imply that the value at any given point in time does not depend on its previous value, which contradicts the concept of a unit-root where there is persistence in values over time.

Choice C is incorrect. Time series with unit roots are not covariance stationary. Covariance stationarity requires that mean and variance are constant over time and covariances for different intervals depend only on the length of the interval, but not on where they are in time. However, a unit root process violates these conditions as it has a changing mean and variance over time.

Choice D is incorrect. A unit-root time series is not stationary and does not revert to its mean over time. Instead, it follows a random walk, which means it tends to wander away from its initial value and does not exhibit mean reversion. This is contrary to the properties of stationary time series, which do revert to their mean over time.

Things to Remember

- Unit-root tests are commonly used in time series analysis to determine if a time series is stationary or non-stationary.
- A random walk is a type of non-stationary time series where the differences between consecutive observations are not constant.

- The presence of a unit-root in a time series indicates that the series has a stochastic trend and does not exhibit mean reversion.
- Covariance stationarity requires that the mean, variance, and autocovariance structure of a time series remain constant over time.
- Stationary time series are characterized by a constant mean and variance, and exhibit mean reversion behavior.

Reading 23: Measuring Return, Volatility, and Correlation

Q.553 Distinguish between historical and implied volatility.

A. Historical volatility measures the standard deviation of past price movements while implied volatility gives an estimate of future volatility in the price of the asset.

B. Historical volatility measures the standard deviation of past price movements while implied volatility is the immeasurable volatility in the future price of an asset.

C. Historical volatility is the volatility of an asset that has been recorded as at present, while implied volatility is the future volatility.

D. Historical volatility measures the total standard deviation of past price movements while implied volatility gives the historical volatility beyond a given benchmark.

The correct answer is **A.**

Historical volatility and implied volatility are two different measures used to understand the price movements of an asset. Historical volatility, as the name suggests, is based on historical data. It measures the standard deviation of past price movements, which means it quantifies the variation or dispersion of an asset's prices in the past. On the other hand, implied volatility is forward-looking. It gives an estimate of future volatility in the price of the asset. This is derived from the price of an option and reflects the market's expectation of future volatility. It is important to note that implied volatility does not predict the direction of future price movements, only the magnitude of the movements.

Choice B is incorrect. While it is true that historical volatility measures the standard deviation of past price movements, implied volatility is not immeasurable. Implied volatility can be estimated using option pricing models such as the Black-Scholes model.

Choice C is incorrect. Historical volatility does not refer to the present recorded volatility of an asset but rather refers to the standard deviation of past price movements. Implied volatility, on the other hand, does indeed provide an estimate for future price fluctuations based on current market conditions and investor sentiment.

Choice D is incorrect. The statement incorrectly suggests that implied volatility provides a measure of historical volatility beyond a given benchmark which isn't accurate. Implied Volatility gives an estimate of future expected fluctuations in the price of an asset based on current market conditions and investor sentiment.

Things to Remember

- Historical volatility is also known as statistical volatility and is calculated using historical price data.
- Implied volatility is often used in options pricing models, such as the Black-Scholes

model, to determine the fair value of an option.

- Implied volatility can be influenced by factors such as market sentiment, upcoming events, and changes in interest rates.
- Historical volatility is backward-looking and provides insights into how volatile an asset has been in the past.
- Implied volatility is forward-looking and helps traders and investors gauge market expectations for future price movements.

Q.554 Suppose that we know from experience that $\alpha = 4$ for a certain financial variable and we observe that $P(X > 20) = 0.01$. Apply the Power Law and find the probability that $X > 10$.

A. 22

B. 16,000

C. 20

D. 0.16

The correct answer is **D.**

The power law tails are such that, the probability of observing a random variable, X, greater than a given value x is defined as:

$$P(X > x) = kx^{-\alpha}$$

Where k and α are constants.

Therefore,$0.01 = k * 20^{-4}$

k=1600

Thus, $P(X > 10) = 1600 * 10^{-4} = 0.16$

Q.3381 Assumed that asset prices are normally distributed. The expected value of an asset price is $80 with daily volatility of 2%. Compute the 95% confidence interval of the asset price at the end of 4 days.

A. 80 ± 2.000

B. 80 ± 3.200

C. 80 ± 6.272

D. 80 ± 3.136

The correct answer is **C.**

4-day Volatility = $2 * \sqrt{4} = 4\%$
Compute one standard deviation move in dollar = $80 * 0.04 = 3.2$

95% confidence interval of asset price = $80 \pm 1.96 * 3.2 = 80 \pm 6.272$

Things to Remember

- Normal distribution is a key concept in finance and risk management, often used to model asset prices and returns.
- Expected value represents the mean or average value of a random variable, such as an asset price.
- Volatility measures the degree of variation of asset prices and is a key input in risk management models.
- Confidence intervals provide a range of values within which we are confident a parameter, such as an asset price, will fall.
- The z-score of 1.96 corresponds to the 95% confidence level in a standard normal distribution.

Q.3382 A zero correlation between two variables indicates that:

A. They are independent of each other

B. They do not have any linear relationship between them

C. They are linearly dependent on each other

D. They have a negative linear relationship

The correct answer is **B.**

A zero correlation between two variables indicates that there is no linear relationship between them. In other words, changes in one variable do not correspond to changes in the other variable in a consistent, predictable manner. This does not mean that the variables are completely unrelated. There could be a non-linear relationship between the variables, such as a quadratic or exponential relationship. However, the correlation coefficient only measures linear relationships, so it would be zero in these cases.

Choice A is incorrect. Zero correlation does not imply independence between two variables. Two variables can be independent and yet have a non-zero correlation, or they can be dependent and have a zero correlation. Independence is a stronger condition than zero correlation.

Choice C is incorrect. Zero correlation does not mean that the variables are linearly dependent on each other. Linear dependence would imply a perfect positive or negative correlation, i.e., the absolute value of their correlation would be 1.

Choice D is incorrect. A negative linear relationship between two variables would result in a negative value for the correlation coefficient, not zero.

Things to Remember

- Correlation measures the strength and direction of a linear relationship between two variables.
- A correlation coefficient of 0 indicates no linear relationship between the variables.
- Correlation does not imply causation; just because two variables are correlated does not mean that one causes the other.
- Correlation can range from -1 to 1, where -1 indicates a perfect negative linear relationship, 0 indicates no linear relationship, and 1 indicates a perfect positive linear relationship.
- Non-linear relationships between variables may exist even when the correlation coefficient is 0.

Q.4036 After arranging the annual data of a portfolio comprising of two assets X and Y from the period 2012 to 2016, it is found that the number of concordant data pairs is 2 and the number of discordant pairs is 8. On the basis of this information, which of the following is closest to the Kendall τ?

A. -0.6

B. -0.2

C. -0.08

D. 0.6

The correct answer is **A.**

Since the data is analyzed from 2012 to 2016, we have five observation pairs and therefore $5 * \frac{(5-1)}{2} = 10$ combinations of pairs to evaluate. By using the Kendall equation:

$$\tau = \frac{(n_c - n_d)}{(n(n-1)/2)}$$

$$\tau = \frac{(2 - 8)}{10} = -0.6$$

Q.4038 A portfolio manager is trying to determine the correlation between the return of two assets. Given the following data about the yearly returns of the stocks, he decides to calculate the Kendall's τ correlation coefficient for the returns of these assets.

Year	Return of Asset X	Return of asset Y
1	3%	8%
2	1%	5%
3	−4%	6%
4	5%	2%
5	2%	9%

What is Kendall's τ correlation coefficient for the returns of the two assets?

A. -0.1

B. -0.2

C. 0

D. 0.1

The correct answer is **B.**

First: order the return set pairs of X and Y with respect to the set X. This is done in columns 2 and 3 of the table below.

Year	Ranked Return of X_i(Assigned same year)	Ranked Return of Y_i	Rank of X_i	Rank of Y_i
3	−4%	6%	1	3
2	1%	5%	2	2
5	2%	9%	3	5
1	3%	8%	4	4
4	5%	2%	5	1

Then, derive the ranks of X; and Y;. This is done in columns 4 and 5.
We have five observation pairs and therefore 5 x (5 - 1)/ 2 = 10 combinations of pairs to evaluate.
These are:
[(1,3),(2,2)], [(1,3),(3,5)], [(1,3),(4,4)], [(1,3),(5,1)]
[(2,2),(3,5)], [(2,2),(4,4)], [(2,2),(5,1)]
[(3,5),(4,4)], [(3,5),(5,1)]
and [(4,4),(5,1)]

A pair of observations (X_i, Y_i) and (X_j, Y_j) is considered concordant if both $X_i < X_j$ and $Y_i < Y_j$ or both $X_i > X_j$ and $Y_i > Y_j$. It is considered discordant if $X_i < X_j$ and $Y_i > Y_j$ or $X_i > X_j$ and $Y_i < Y_j$.

Concordant pairs

[(1,3),(3,5)], [(1,3),(4,4)], [(2,2),(3,5)], and [(2,2),(4,4)]

Discordant pairs
[(1,3),(2,2)], [(1,3),(5,1)], [(2,2),(5,1)], [(3,5),(4,4)], [(3,5),(5,1)], and [(4,4),(5,1)]

$$\text{The Kendall} \quad \tau = \frac{n_c - n_d}{\frac{n(n-1)}{2}}$$

Where:
n_c = number of concordant pairs
n_d = number of discordant pairs
n = number of time periods

$$\tau = \frac{4-6}{\frac{5(5-1)}{2}} = \frac{-2}{10} = -0.2$$

Q.4041 An investor is analyzing the data of two assets X and Y for a period of 7 years. He applied all three statistical models to measure the correlation coefficient. The results were as follows:
Pearson correlation coefficient = -0.8501
Spearman correlation coefficient = -0.9
Kendall’s t = -0.4

He again analyzed the same data but changed two values of asset X without affecting its ranking. What would be the impact of this change on the results?

A. The Spearman results would change but the results of Pearson and Kendall approaches would remain unchanged.

B. The Pearson and the Kendall results would change but the Spearman results would remain unchanged.

C. The Kendall results would change but the results of the Spearman and Pearson approaches would remain unchanged.

D. The Pearson results would change but the results of the Spearman and the Kendall approaches would remain unchanged.

The correct answer is **D.**

The Pearson correlation coefficient measures the linear correlation between two variables, assuming both variables are normally distributed. It is sensitive to the exact values in the dataset, so if you change the values for asset X, the Pearson correlation coefficient would likely change.

On the other hand, the Spearman correlation coefficient and Kendall's tau are rank-based measures of correlation. They measure the degree of association between the rankings of the data points in each variable, rather than the raw data values themselves. Therefore, if you change the values of asset X but not the relative rankings of the values, the Spearman correlation coefficient and Kendall's tau would remain unchanged.

Options A, B, and C are incorrect because they suggest changes in Spearman and Kendall results. Both these correlation measures are based on ranks rather than raw data, so as long as the ranking of values in Asset X isn't changed, these results will not change.

Things to Remember

- Pearson correlation coefficient measures the linear correlation between two variables assuming normal distribution of data.
- Spearman correlation coefficient and Kendall's tau are rank-based measures of correlation.
- Pearson correlation coefficient is sensitive to the exact values in the dataset.

- Spearman correlation coefficient and Kendall's tau measure the degree of association between the rankings of the data points.
- If values in a dataset are changed but the rankings remain the same, Spearman and Kendall results will not change.

Q.4042 Consider the following data.

Time	Price
0	100
1	98.65
2	98.50
3	97.50
4	95.67
5	96.54

What is the value of the simple return at time 4?

A. 1.88%

B. -1.97%

C. -1.88%

D. 1.97%

The correct answer is **C.**

The simple return is given by:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

So that:

$$R_4 = \frac{P_4 - P_3}{P_3} = \frac{95.67 - 97.50}{97.50} = -0.0188 = -1.88\%$$

Q.4043 Consider the following data.

Time	Price
0	100
1	98
2	98
3	97
4	99
5	P

If the simple return for the 5th period is 1.75%, what is the value of p?

A. 99.56

B. 100.5

C. 100.73

D. 99.98

The correct answer is **C.**

Using the analogy:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Then

$$0.0175 = \frac{p - 99}{99} \Rightarrow p = (0.0175 \times 99) + 99 = 100.7325$$

Q.4044 The prices of a portfolio at different times are as shown in the table below.

Time	Price
0	105.62
1	104.10
2	105.54
3	103.68
4	103.56

What is the value of continuously compounded return at time 4?

A. 0.81%

B. -0.0081

C. 0.0091

D. -0.0012

The correct answer is **D.**

The continuously compounded returns is given by:

$$r_t = \ln P_t - \ln P_{t-1}$$

So that:

$$r_4 = \ln P_4 - \ln P_3 = \ln 103.56 - \ln 103.68 = -0.0012\%$$

Q.4045 A simple return for an investment in a particular holding period is 15%. What is the equivalent continuously compounded return over the same holding period?

A. 0.1256

B. 0.1578

C. 0.1398

D. 0.1278

The correct answer is **C.**

Using the relationship:

$$1 + R_t = e^{r_t}$$
$$\Rightarrow r_t = \ln(1 + R_t) = \ln 1.15 = 0.1398 = 13.98\%$$

Q.4046 If the daily volatility of the price of gold is 0.3% in a given year. What is the annualized volatility of the gold price assuming 252 trading days?

A. 4.67%

B. 3.56%

C. 2.56%

D. 4.76%

The correct answer is **D.**

Using the scaling analogy, the corresponding annualized volatility is given by:

$$\sigma_{\text{annual}} = \sqrt{252 \times \sigma^2_{\text{daily}}} = \sqrt{252 \times 0.003^2} = 0.047623 = 4.7623\%.$$

Q.4047 A financial analyst wishes to model the returns from investment using the normal distribution. The analyst approximates the skewness of the data to 0.35 and kurtosis of 3.04. The analyst performs the JB test at a 95% confidence level. What is the value of the test statistic as per the analyst's results if the sample size is 100? You might need to use the following chi-square table:

df	0.1	0.05	0.025	0.01	0.005
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589
10	15.987	18.307	20.483	23.209	25.188
11	17.275	19.675	21.920	24.725	26.757
12	18.549	21.026	23.337	26.217	28.300

A. 2.028

B. 5.991

C. 1.0256

D. 6.484

The correct answer is **A.**

According to the Jarque-Bera (JB) test, the test statistic is given by:

$$JB = (T-1)\left(\frac{\hat{S}^2}{6} + \frac{(\hat{k}-3)^2}{24}\right)$$

So,

$$JB = (100-1)\left(\frac{0.35^2}{6} + \frac{(3.04-3)^2}{24}\right) = 2.028$$

Recall that $JB \sim \chi^2_2$ so that critical value at 5% with 2 degrees of freedom (df) is 5.991 (which can be seen from a chi-square table).

Since the test statistic is less than the critical value, we fail to reject the null hypothesis that the returns are normal, and hence the analyst can use the normal distributions in analyzing the returns.

Reading 24: Simulation and Bootstrapping

Q.572 In which of the following situations would bootstrapping be ineffective?

A. Use of non-independent data.

B. Sampling with replacement.

C. If there are no outliers in the data.

D. Re-sampling from regression residuals.

The correct answer is **A.**

Bootstrapping is ineffective in the use of non-independent data. The bootstrapping method is based on the assumption that the observations in the data are independent and identically distributed (i.i.d). This means that each observation in the dataset is independent of all other observations and they all follow the same probability distribution. If the data are not independent, for example, if there are correlations in the data, then the bootstrapping method would be ineffective. This is because the resampling process in bootstrapping, which involves sampling with replacement, would not accurately reflect the original data's structure and dependencies. Therefore, the resulting bootstrap samples would not be representative of the original data, leading to inaccurate estimates of the sampling distribution.

Choice B is incorrect. Bootstrapping involves sampling with replacement. This means that after a data point is selected for the sample, it's put back in the data set for potential re-selection, which is a fundamental aspect of bootstrapping.

Choice C is incorrect. The presence or absence of outliers does not affect the effectiveness of bootstrapping. Bootstrapping can handle datasets with outliers as it does not make any assumptions about the underlying distribution.

Choice D is incorrect. Resampling from regression residuals can be an effective use of bootstrapping, especially when testing hypotheses about regression coefficients or constructing confidence intervals around predictions.

Things to Remember

- Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by resampling with replacement from the original data.
- Bootstrapping is a non-parametric method and does not rely on assumptions about the underlying distribution of the data.
- Bootstrapping can be used for various statistical purposes such as estimating confidence intervals, hypothesis testing, and model validation.

- Bootstrapping is particularly useful when the underlying distribution of the data is unknown or when the sample size is small.
- Bootstrapping can handle datasets with missing values or outliers effectively.

Q.574 Construct a 95% confidence interval for the ending mutual fund capital amount where the number of simulations is 100, the mean ending capital is $200,000, and the standard deviation is $34,456.

A. [$193,247, $206,753]

B. [$193,177.7, $200,000]

C. [$193, $206]

D. [$180,000, $220,000]

The correct answer is **A.**

We need to find the 2.5th percentile and the 97.5th percentile for the z-distribution with 100 observations. The formula to apply is:

$$\bar{X} - 1.96 * \left(\frac{s}{\sqrt{N}}\right), \bar{X} + 1.96 * \left(\frac{s}{\sqrt{N}}\right)$$

Where N = 100

$$= \$200,000 - 1.96\left(\frac{\$34,456}{\sqrt{100}}\right), \$200,000 + 1.96\left(\frac{\$34,456}{\sqrt{100}}\right)$$
$$= [\$193,246.624, \$206,753.376]$$

Q.577 Under which of the following situations would you prefer bootstrapping to pure simulation?

A. If you have a very small sample of actual data.

B. If the distribution of the actual data is unknown.

C. If the distribution of the data is known exactly.

D. If you have a large sample and the data follows a normal distribution.

The correct answer is **B.**

Bootstrapping is a resampling technique that is used when the distribution of the data is unknown. It involves repeatedly sampling from the observed data with replacement and then estimating the parameter of interest on each sample. This process is repeated many times to generate a distribution of the parameter estimates. The advantage of bootstrapping is that it makes no assumptions about the underlying distribution of the data. This is particularly useful when the distribution of the data is unknown, as it allows for the estimation of the parameter of interest without the need for assumptions about the distribution. In contrast, pure simulation requires knowledge of the distribution from which to draw random samples. Therefore, if the distribution of the actual data is unknown, bootstrapping would be preferred over pure simulation.

Choice A is incorrect. Bootstrapping may not be advantageous with a very small sample of actual data. This is because bootstrapping relies on resampling from the original dataset to generate new samples, and a small dataset may not provide enough diversity for this process, leading to biased estimates.

Choice C is incorrect. If the distribution of the data is known exactly, pure simulation would be more advantageous than bootstrapping. In such cases, we can directly simulate from the known distribution rather than relying on resampling techniques like bootstrapping.

Choice D is incorrect. If you have a large sample and the data follows a normal distribution, pure simulation would generally be more advantageous. With a known distribution, direct simulation is more efficient and accurate compared to bootstrapping, which is more useful when the distribution is unknown or when making fewer assumptions about the data's distribution.

Things to Remember

- Bootstrapping is a resampling technique that involves repeatedly sampling from the observed data with replacement.
- Bootstrapping is useful when the distribution of the data is unknown.
- Bootstrapping makes no assumptions about the underlying distribution of the data.
- Pure simulation requires knowledge of the distribution from which to draw random samples.
- Bootstrapping may not be advantageous with a very small sample of actual data as it can lead to biased estimates.
- If the distribution of the data is known exactly, pure simulation would be more

advantageous than bootstrapping.

- With a large sample and data following a normal distribution, pure simulation is generally more advantageous than bootstrapping.

Q.579 Which of the following statements best explains why Monte Carlo simulations are considered exceptionally effective compared to probability trees when appraising a capital project?

A. Monte Carlo simulations incorporate scenarios that span the entire probability space.

B. Monte Carlo simulations consume less time.

C. Monte Carlo simulations are embedded within most modern computer programs.

D. Monte Carlo simulations allow for the comparison of only the net present values that are positive.

The correct answer is **A.**

Monte Carlo simulations are a computational algorithm that relies on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems that might be deterministic in principle. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other approaches. Monte Carlo simulations are used to model the probability of different outcomes in a process that cannot easily be predicted due to the intervention of random variables. It is a technique used to understand the impact of risk and uncertainty in prediction and forecasting models. Monte Carlo simulations can be used to tackle a range of problems in virtually every field such as finance, engineering, supply chain, science and, more. In the context of appraising a capital project, Monte Carlo simulations are considered exceptionally effective because they incorporate scenarios that span the entire probability space. This means that they take into account all possible outcomes and their likelihoods, which provides a comprehensive view of the risks and potential returns of the project. This is not the case with probability trees, which only consider a limited set of outcomes and their probabilities.

Choice B is incorrect. While Monte Carlo simulations can be computationally intensive and may require more time than simpler methods like probability trees, their advantage lies in their ability to incorporate scenarios that span the entire probability space, not in their speed.

Choice C is incorrect. The prevalence of Monte Carlo simulations in modern computer programs does not inherently make them superior for evaluating capital projects. Their effectiveness comes from their comprehensive approach to risk assessment, as they consider all possible outcomes and probabilities.

Choice D is incorrect. The assertion that Monte Carlo simulations only allow for the comparison of positive net present values is false. In fact, one of the strengths of Monte Carlo simulations is its ability to model a wide range of outcomes, including both positive and negative net present values.

Things to Remember

- Monte Carlo simulations rely on repeated random sampling to obtain numerical results.
- They are used to model the probability of different outcomes in processes with random variables.
- Monte Carlo simulations are effective in understanding the impact of risk and uncertainty in prediction models.
- They are widely used in various fields such as finance, engineering, and science.
- Monte Carlo simulations provide a comprehensive view of risks and potential returns by considering all possible outcomes and their likelihoods.
- Probability trees only consider a limited set of outcomes and their probabilities, unlike Monte Carlo simulations.

Q.580 John Neur, FRM, runs a Monte Carlo simulation to estimate the ending amount of capital in 25 years using monthly returns for three investments as the basis. Investments A and B are highly correlated while C has zero correlation with both A and B. In order to compute the output of the Monte Carlo simulation, John:

A. Cannot measure the correlations between the three investments.

B. Must accurately determine the probability distribution of the output.

C. Can easily examine effects on output variables when changing scenarios.

D. Must assume that the output is normally distributed.

The correct answer is **C.**

John can easily examine the effects on output variables when changing scenarios. Monte Carlo simulations are a powerful tool for risk management and financial forecasting. They allow for the modeling of complex systems and the examination of the effects of different scenarios on the output variables. This is particularly useful in situations where the relationships between variables are complex and non-linear, such as in the case of John's three investments. By changing the scenarios, John can see how the output variables (in this case, the ending amount of capital) are affected. This can provide valuable insights into the potential risks and returns of the investments, and can help John make informed decisions about how to manage his portfolio. Furthermore, the ability to examine the effects of changing scenarios does not require the identification of the probability distribution of the output variable, which can be a complex and time-consuming task.

Choice A is incorrect. John can indeed measure the correlations between the three investments. In fact, understanding these correlations is crucial for accurately conducting a Monte Carlo simulation. The correlation between investments A and B, as well as the lack of correlation with investment C, will significantly impact the results of his simulation.

Choice B is incorrect. While determining the probability distribution of output can be helpful in some cases, it's not a necessity for conducting a Monte Carlo simulation. The primary requirement for this type of simulation is to have a good understanding of the input variables (in this case, monthly returns), rather than their output distribution.

Choice D is incorrect. Assuming that output is normally distributed may lead to inaccurate results in many cases because financial data often exhibit skewness and kurtosis that deviate from normality assumptions. Therefore, it's not necessary to assume normal distribution when conducting a Monte Carlo simulation.

Things to Remember

- Monte Carlo simulation is a computational technique that uses random sampling and probability distributions to model the impact of risk and uncertainty in forecasting and prediction.
- Correlation measures the relationship between two variables and is crucial in portfolio management and risk assessment.
- Understanding the probability distribution of input variables is important for accurate simulations, but it is not always necessary to know the distribution of the output variable.
- Changing scenarios in a Monte Carlo simulation allows for sensitivity analysis and helps in understanding the impact of different factors on the output variables.
- Normal distribution assumptions may not always hold in financial data, and it is important to consider alternative distributions that better fit the data.

Q.582 An analyst runs a simulation to estimate the future value of an investment of $10,000 today over a 40-year period. He uses random monthly returns that are normally distributed. How does the analyst's situation create a discretization error bias?

A. By using normally distributed returns.

B. By using a simulation period that's too long (40 years).

C. By assuming that returns are random.

D. By assuming that returns are generated on a monthly basis instead of a continuous basis.

The correct answer is **D.**

Discretization error bias arises when a continuous variable is incorrectly assumed to take on only discrete values. In this case, the analyst is assuming that returns are generated on a monthly basis, which is a discrete time interval. However, in reality, returns are generated continuously. This assumption of discrete returns can lead to a bias in the simulation results because it does not accurately reflect the continuous nature of returns. The error is termed as 'discretization error bias'. This bias can significantly affect the accuracy of the simulation, leading to potential misestimations of the future value of the investment.

Choice A is incorrect. Using normally distributed returns does not lead to discretization error bias. In financial analysis, it is common to assume that returns follow a normal distribution due to the Central Limit Theorem, which states that the sum of a large number of independent and identically distributed variables will be approximately normally distributed.

Choice B is incorrect. The length of the simulation period (40 years in this case) does not contribute to discretization error bias. Discretization error arises from approximating a continuous process with discrete steps, not from the overall length of time being simulated.

Choice C is incorrect. Assuming that returns are random does not cause discretization error bias either. In fact, randomness is an inherent characteristic of financial markets and many financial models incorporate randomness in their assumptions.

Things to Remember

- Continuous vs. discrete variables in financial modeling
- Central Limit Theorem and its implications for financial analysis
- Importance of accurately modeling the nature of returns in simulations
- Effects of different time intervals (monthly, daily, continuous) on simulation results
- Considerations for handling biases and errors in financial simulations

Q.3390 Tim Yang, FRM, is working on building a model using a Monte Carlo simulation. However, he is concerned about the accuracy of the simulation which is measured by its standard error. Tim initially runs a model with 81 simulations and the standard deviation was found to be 27%. He then runs the model with 144 simulations and the standard deviation is still 27%.
What are the standard errors for the simulations?

A. Standard error for the first simulation: 0.33%; Standard error for the second simulation: 0.19%

B. Standard error for the first simulation: 27%; Standard error for the second simulation: 27.00%

C. Standard error for the first simulation: 2.25%; Standard error for the second simulation: 3%

D. Standard error for the first simulation: 3%; Standard error for the second simulation: 2.25%

The correct answer is **D.**

$$\text{Standard error estimate} = \frac{\text{Standard deviation}}{\sqrt{n}}$$

$$\text{Standard error estimate} = \frac{27}{\sqrt{81}} = 3\% \text{ for the first simulation;}$$

And $\frac{27}{\sqrt{144}} = 2.25\%$ for the second simulation

Things to Remember

- Standard error is a measure of the statistical accuracy of an estimate, representing the standard deviation of the sampling distribution of a statistic.
- As the number of simulations increases, the standard error typically decreases, indicating a more precise estimate.
- Monte Carlo simulation is a technique used to understand the impact of risk and uncertainty in prediction and forecasting models.
- Standard deviation measures the dispersion of a dataset relative to its mean, while standard error measures the precision of a statistic.
- Increasing the number of simulations in a Monte Carlo simulation can help improve the accuracy and reliability of the model's results.

Q.4196 Which of the following statements correctly describes the difference between Monte Carlo Simulation and bootstrapping?

A. Monte Carlo Simulation generates variables and shocks from a particular distribution while bootstrapping generates the variables from observed data through random sampling.

B. Monte Carlo Simulation generates variables and shocks from observed data while bootstrapping generates the variables from a particular distribution.

C. In Monte Carlo Simulation, random samples are used to draw indices when selecting data to be included in the simulation sample.

D. Bootstrapping is used to simulate future outcomes by altering the underlying distribution, while Monte Carlo Simulation strictly uses historical data for its projections.

The correct answer is **A.**

The simulation uses the draws from a particular distribution to simulate the model quantities while a bootstrap uses random sampling from the observed data to draw a new dataset that has similar features.

Option B is incorrect because it contradicts Option A.

Option C is incorrect because it is only bootstrapping that draws the indices when selecting the data to be included in the bootstrap sample.

Option D is incorrect because bootstrapping does not alter the underlying distribution; it uses random sampling from the observed data to create new datasets with similar features. Monte Carlo Simulation can involve generating variables from theoretical distributions, not strictly historical data.

Things to Remember

- Monte Carlo Simulation is a technique used to understand the impact of risk and uncertainty in prediction and forecasting models by running multiple simulations using random variables.
- Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by resampling with replacement from the observed data.
- Monte Carlo Simulation can involve generating variables and shocks from theoretical distributions, not just observed data.

- Bootstrapping does not alter the underlying distribution of the data; it simply resamples from the observed data to create new datasets.
- Both Monte Carlo Simulation and bootstrapping are widely used in finance for risk management, option pricing, and other quantitative analysis.

Q.4197 Which of the following is the most significant limitation of bootstrapping?

A. The "Black Swan" problem.

B. Bootstrapping can potentially construct samples that are significantly larger than historically observed datasets if the assumed distribution possesses the same feature.

C. Bootstrapping is suitable where the data being bootstrapped has a time dependence feature.

D. Bootstraps requires the use of more antithetic variables.

The correct answer is **A.**

The 'Black Swan' problem is indeed the most significant limitation of bootstrapping. The term 'Black Swan' is used in the context of events that are beyond the realm of normal expectations and have potentially severe consequences. These events are characterized by their extreme rarity, severe impact, and the widespread insistence they were obvious in hindsight. In the context of bootstrapping, the 'Black Swan' problem refers to the inability of the bootstrap method to generate data that has not occurred in the sample. This is because bootstrapping is a resampling method that generates new samples by drawing observations from the original data. Therefore, it cannot create or predict 'Black Swan' events that are not represented in the original data. This limitation can lead to underestimation of risk or uncertainty in various applications, including financial risk management and predictive modeling.

Choice B is incorrect. Bootstrapping does not inherently construct samples that are significantly larger than historically observed datasets. The size of the bootstrapped samples is determined by the analyst and can be set to match the size of the original dataset. Therefore, this statement does not represent a limitation of bootstrapping.

Choice C is incorrect. While it's true that bootstrapping can be applied to data with time dependence features, it's not always suitable or accurate in these cases due to its assumption of independence between observations. However, this doesn't represent a significant limitation as there are variations of bootstrap methods designed specifically for time series data.

Choice D is incorrect. The use of antithetic variables isn't a requirement for bootstrapping; rather, it's an optional technique used in Monte Carlo simulations to reduce variance and improve efficiency. This choice seems to confuse two different statistical techniques and thus doesn't accurately reflect a limitation of bootstrapping.

Things to Remember

- Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by resampling with replacement from the original data.
- It is a non-parametric method that does not rely on assumptions about the underlying distribution of the data.
- Bootstrapping is commonly used in finance for risk management, portfolio optimization, and model validation.
- One variation of bootstrapping is the parametric bootstrap, where the resampling is done based on a specific parametric model.
- Bootstrapping can be computationally intensive, especially for large datasets, but it provides a robust way to estimate uncertainty and variability.

Q.4198 A financial analyst is using Monte Carlo simulation to price a complex derivative. They are concerned about the computational cost associated with achieving a high level of accuracy, as a large number of simulations are required to reduce sampling error. The analyst is considering using variance reduction techniques to improve the efficiency of the Monte Carlo simulation. Which of the following statements best describes how antithetic and control variates reduce Monte Carlo sampling error?

A. Antithetic variates reduce sampling error by increasing the number of independent random samples used in the simulation, while control variates reduce error by replacing complex calculations with simpler approximations.

B. Antithetic variates reduce sampling error by creating pairs of negatively correlated random variables, while control variates reduce error by exploiting the known relationship between the derivative price and a related, analytically tractable variable.

C. Antithetic variates reduce sampling error by smoothing out the probability distribution of the underlying asset, while control variates reduce error by introducing additional randomness into the simulation to explore a wider range of possible outcomes.

D. Antithetic variates reduce sampling error by generating random numbers that are closer to the mean of the distribution, while control variates reduce error by using stratified sampling techniques to ensure better coverage of the sample space.

The correct answer is **B.**

Antithetic variates work by generating pairs of random variables that are negatively correlated. If Z is a standard normal random variable used in the simulation, then -Z is its antithetic counterpart. By using both Z and -Z in the simulation, the variance of the resulting estimate is reduced because the negative correlation between the two variables offsets some of the random fluctuations. Control variates, on the other hand, exploit the known relationship between the derivative being priced and a related, simpler variable (the control variate) for which an analytical solution is known. By comparing the Monte Carlo estimate of the control variate with its known analytical value, an adjustment can be made to the Monte Carlo estimate of the target derivative, thereby reducing the variance and improving accuracy.

A is incorrect. Antithetic variates do not increase the number of independent samples; they create dependent pairs. Control variates do not replace complex calculations with simpler approximations of the target derivative; they use a separate, simpler variable (the control variate) with a known analytical solution.
C is incorrect. Antithetic variates do not smooth the probability distribution; they use the existing distribution to generate negatively correlated pairs. Control variates do not introduce additional randomness; they use a related variable with a known analytical solution to improve the accuracy of the simulation.
D is incorrect. Antithetic variates do not force random numbers closer to the mean; they generate negatively correlated pairs that explore both sides of the distribution. Control variates are not related to stratified sampling; they use a related variable with a known analytical solution.

Things to Remember

- Antithetic variates and control variates are variance reduction techniques used to improve the efficiency of Monte Carlo simulations.
- Antithetic variates exploit the property that if Z is a standard normal random variable, then -Z is also a standard normal random variable and they are perfectly negatively correlated.
- By using both Z and -Z, the variance of the average of the function evaluated at these points will be lower than the variance obtained by using two independent draws.
- Control variates use a related variable for which an analytical solution is known. The difference between the simulated and analytical values of the control variate is used to adjust the simulated value of the target variable, reducing variance.
- A common example of a control variate is using the price of a European call option (which has a closed-form solution) as a control variate when pricing an American call option (which typically requires numerical methods).

- These techniques allow for more accurate results with fewer simulations, saving computational time and resources.

Q.4199 Assume that you want to generate random variables from U(-1,5) using random variables from U(0,1). What is the corresponding random variable of $0.10 \sim U(0,1)$?

A. -0.8

B. 0.5

C. -0.4

D. 0.7

The correct answer is **C.**

The distribution function for $X \sim U(-1,5)$ is given by:

$$F(x) = \frac{x+1}{6}$$

this is due to the fact that the CDF of a uniform distribution is given by:

$$\frac{x-a}{b-a}, \quad \text{for} \quad a \leq x \leq b$$

Now, define the random variable U, so that:

$$U = \frac{x+1}{6} \Rightarrow X = 6U - 1$$

So for, U=0.10, the corresponding random variable is

$$X = 6 \times 0.10 - 1 = -0.40$$

Q.4200 Which of the following is the first step in a typical iid bootstrapping?

A. Generating data according to an assumed distribution.

B. Constructing a bootstrap sample.

C. Selecting an appropriate block size.

D. Generating a set of a random collection of m integers (1,2,3…n) with replacement.

The correct answer is **D.**

The first step in the independent and identically distributed (iid) bootstrapping method is to generate a set of m integers (1,2,3…n) with replacement. This step is crucial as it forms the basis for the construction of the bootstrap sample. The integers generated are used as indices to select observations from the original data set. This process is done with replacement, meaning that the same integer can be selected more than once, allowing for the creation of a bootstrap sample that is the same size as the original data set but with some observations repeated and others possibly omitted.

Choice A is incorrect. Generating data according to an assumed distribution is not the initial step in an iid bootstrapping procedure. This step might be part of other statistical methods, but in bootstrapping, we work with the original dataset and generate bootstrap samples from it.

Choice B is incorrect. Constructing a bootstrap sample is indeed a part of the bootstrapping process, but it's not the initial step. The first step involves generating a set of random integers which will then be used to construct the bootstrap sample.

Choice C is incorrect. Selecting an appropriate block size pertains more to block bootstrapping methods where data may have time-dependent structure or when observations are not independent and identically distributed (iid). In iid bootstrapping, this concept does not apply as all observations are treated as independent.

Things to Remember

- Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by resampling with replacement from the original data.
- The key assumption in bootstrapping is that the data is independent and identically distributed (iid).
- Bootstrap samples are created by randomly selecting observations from the original dataset with replacement.
- Bootstrapping is often used to estimate the sampling distribution of a statistic, such as the mean, variance, or regression coefficients.
- Bootstrap methods can provide valuable insights into the variability of a statistic and can be particularly useful when the underlying distribution of the data is unknown or

complex.

Q.5610 A financial analyst at a technology company is evaluating the risk profile of a new investment product using historical data. The analyst is considering applying a resampling technique to estimate the distribution of potential returns. The analyst debates whether to use Monte Carlo simulation or bootstrapping for the analysis. Which of the following statements about bootstrapping is correct?

A. Bootstrapping requires that the data be independent and identically distributed (i.i.d).

B. In bootstrapping, data must be resampled without replacement to preserve statistical integrity.

C. Bootstrapping can only be applied to data that is normally distributed.

D. Bootstrapping involves resampling with replacement from the original dataset.

The correct answer is **D.**

Bootstrapping is a statistical technique that involves repeatedly resampling with replacement from an original dataset to estimate the distribution of a statistic. This method allows analysts to assess the variability of a statistic without making strict assumptions about the underlying data distribution. By resampling with replacement, each bootstrap sample can be the same size as the original dataset, and each observation in the original dataset has the same probability of appearing in each bootstrap sample. This process helps in creating a sampling distribution of the statistic, which can be used to estimate measures like confidence intervals.

A is incorrect. While having i.i.d. data is ideal for many statistical methods, bootstrapping can be applied even when data does not strictly meet these criteria. However, it's important to be cautious when applying bootstrapping to dependent data, as it can affect the validity of the results.

B is incorrect. Bootstrapping specifically involves resampling with replacement. Resampling without replacement would not be bootstrapping but rather a different type of resampling method.

C is incorrect. One of the main advantages of bootstrapping is that it does not assume any specific distribution of the data, including normal distribution. It can be applied to data with any distribution, making it a flexible tool in statistical analysis.

Things to Remember

- Bootstrapping can be used to estimate the bias and variance of a statistic. By comparing the bootstrap estimates to the original sample estimate, analysts can understand how much bias and variability are present in their estimation process.

- In some cases, bootstrapping provides a more accurate or robust alternative to traditional parametric methods, especially when the sample size is small or when the underlying distribution of the data is unknown or complex.
- Bootstrapping is used in machine learning algorithms, particularly in ensemble methods like bagging (bootstrap aggregating). It helps in reducing overfitting and improving model accuracy by aggregating multiple models trained on different bootstrap samples.
- Bootstrapping may not be suitable for data with strong serial correlations, like time series data, where the assumption of independent resamples is violated. Specialized bootstrapping techniques, such as block bootstrapping, are sometimes used to handle such data.

Q.5197 Which of the following best describes the difference between machine learning and classical econometrics?

A. Machine learning mainly relies on strong assumptions about data, while classical econometrics is flexible and more data driven.

B. Machine learning is better suited for analyzing larger and more complex data sets, while classical econometrics is better suited for analyzing smaller data sets.

C. Machine learning has a solid foundation in mathematical statistics and probability, and economic theory, while classical econometrics automatically learns from data on the best features that should be included in the model.

D. Machine learning algorithms can only be used for binary classification tasks, while classical econometrics can be used for a wide range of statistical tasks.

The correct answer is **B.**

Machine learning is indeed better suited for analyzing larger and more complex data sets. This is because machine learning algorithms are designed to automatically identify patterns and relationships within data, without the need for explicit programming. This makes them particularly effective when dealing with large volumes of data, where manual analysis would be impractical or impossible. Furthermore, machine learning algorithms can handle unstructured data, such as text or images, which are often found in large data sets.

On the other hand, classical econometrics is better suited for analyzing smaller data sets. Econometric models are typically built on strong theoretical foundations and require a clear understanding of the underlying relationships between variables. This makes them less flexible than machine learning algorithms, but more interpretable. Because of these characteristics, econometric models are often used when the data set is small and the relationships between variables are well understood.

Choice A is incorrect. This statement is not accurate because machine learning does not necessarily rely on strong assumptions about data. In fact, it's the opposite; machine learning algorithms are designed to be flexible and can handle a wide variety of data types and structures, while classical econometrics often relies on certain assumptions about the underlying data.

Choice C is incorrect. While it's true that machine learning has a solid foundation in mathematical statistics and probability, it does not typically incorporate economic theory into its models. On the other hand, classical econometrics does not automatically learn from data on the best features to include in the model; this process usually requires human intervention and expertise.

Choice D is incorrect. Machine learning algorithms are capable of handling more than just binary classification tasks. They can also perform regression, clustering, anomaly detection, among others tasks depending upon their design and purpose.

Things to Remember

- Machine learning algorithms are designed to automatically identify patterns and relationships within data, without the need for explicit programming.
- Classical econometrics models are typically built on strong theoretical foundations and require a clear understanding of the underlying relationships between variables.
- Machine learning is better suited for analyzing larger and more complex data sets, while classical econometrics is better suited for analyzing smaller data sets.
- Machine learning algorithms can handle unstructured data, such as text or images, which are often found in large data sets.
- Machine learning algorithms can perform regression, clustering, anomaly detection, among others tasks depending upon their design and purpose.

Q.5198 Which of the following *best* describes the main difference between unsupervised learning and supervised learning?

A. Unsupervised learning involves training a model on labeled data, while supervised learning involves training a model on unlabeled data.

B. Unsupervised learning involves training a model to make predictions, while supervised learning involves training a model to classify data.

C. Unsupervised learning involves training a model to learn from the data itself, while supervised learning involves training a model using explicit guidance from a human.

D. Unsupervised learning involves training a model to maximize a reward signal, while supervised learning involves training a model to minimize a loss function.

The correct answer is **C.**

Unsupervised learning involves training a model to discover data patterns without explicit guidance, while supervised learning involves training a model to make predictions or classify data using labeled examples.

A is incorrect. Unsupervised learning involves training a model on unlabeled data, while supervised learning involves training a model on labeled data.

B is incorrect. Unsupervised learning recognizes data patterns without an explicit/predefined target. It is aimed at identifying patterns rather than making predictions.

D is incorrect. Reinforcement learning is the one that trains a model to maximize a reward signal.

Things to Remember

- Unsupervised learning: Involves training a model to discover patterns in data without explicit guidance or labeled examples.
- Supervised learning: Involves training a model to make predictions or classify data using labeled examples provided by a human.
- Reinforcement learning: Involves training a model to maximize a reward signal by interacting with an environment.
- Clustering: A common task in unsupervised learning where data points are grouped based on similarities.
- Classification: A common task in supervised learning where data points are categorized into classes based on labeled examples.

Q.5201 How does the K-means algorithm separate a sample into clusters?

A. By minimizing the distance between points within the same cluster.

B. By maximizing the distance between points in different clusters.

C. By minimizing the variance within each cluster.

D. By maximizing the variance between clusters.

The correct answer is **A.**

The K-means algorithm separates a sample into clusters by minimizing the sum of squared distances between data points and the centroid of the cluster to which they belong. The algorithm begins by randomly selecting K initial cluster centers and then assigns each point in the sample to the cluster with the nearest center. The algorithm then recalculates the cluster centers as the mean of all the points in the cluster and repeats the process until the cluster assignments do not change or a maximum number of iterations is reached. The goal of the algorithm is to minimize the sum of the squared distances between each point and its assigned cluster center, which results in clusters with points that are more similar to each other.

Choice B is incorrect. The K-means algorithm does not work by maximizing the distance between points in different clusters. Instead, it focuses on minimizing the distance within each cluster to ensure that all points in a single cluster are as similar as possible.

Choice C is incorrect. While minimizing variance within each cluster can be a result of the K-means algorithm, it's not the primary objective of this method. The main goal of K-means is to minimize the distance between points within each cluster, which may indirectly lead to minimized variance.

Choice D is incorrect. Maximizing variance between clusters isn't how K-means operates either. This algorithm aims at creating distinct clusters by reducing intra-cluster distances rather than increasing inter-cluster variances.

Things to Remember

- K-means algorithm is an iterative clustering algorithm that aims to partition n observations into k clusters.
- The algorithm requires the number of clusters (k) to be specified in advance.
- K-means is sensitive to the initial selection of cluster centers, as it can converge to different solutions based on the initial random assignment.
- The algorithm may not always converge to the global optimum and can get stuck in local optima.
- K-means is computationally efficient and works well with large datasets.

Q.5202 Which of the following *best* describes the main difference between underfitting and overfitting?

A. Underfitting occurs when a model is too simple and cannot capture the underlying patterns in the data, while overfitting occurs when a model is too complex and fits the noise in the data.

B. Underfitting occurs when a model is too complex and cannot generalize to new data, while overfitting occurs when a model is too simple and cannot capture the underlying patterns in the data.

C. Underfitting occurs when a model is too complex and overfits the data, while overfitting occurs when a model is too simple and underfits the data.

D. Underfitting occurs when a model is trained on too little data, while overfitting occurs when a model is trained on too much data.

The correct answer is **A.**

Underfitting and overfitting are two common problems in machine learning and data modeling. Underfitting occurs when a model is too simple to capture the underlying patterns in the data. This could be due to the model not having enough parameters or the model not being complex enough. As a result, the model performs poorly on both the training data and new, unseen data because it cannot accurately represent the data's complexity.

On the other hand, overfitting occurs when a model is too complex and starts to fit the noise in the data. This typically happens when the model has too many parameters relative to the number of observations. The model performs well on the training data because it can fit the data perfectly, including the noise. However, it performs poorly on new, unseen data because the noise it learned from the training data does not generalize to new data. Therefore, the model's ability to predict future observations is compromised.

Choice B is incorrect. Underfitting does not occur when a model is too complex; rather, it happens when a model is too simple and fails to capture the underlying patterns in the data. Overfitting, on the other hand, does not happen when a model is too simple; instead, it occurs when a model is overly complex and fits even the noise in the data.

Choice C is incorrect. This choice incorrectly swaps the definitions of underfitting and overfitting. Underfitting refers to situations where a model fails to capture underlying patterns due to its simplicity, while overfitting refers to situations where an overly complex model fits even noise in the data.

Choice D is incorrect. While it's true that insufficient or excessive training data can contribute to underfitting or overfitting respectively, these are not their primary causes. The main distinction between underfitting and overfitting lies in their relation with complexity of models: underfitted models are typically too simple while overfitted models are typically too complex.

Things to Remember

- Underfitting occurs when a model is too simple and cannot capture the underlying patterns in the data.
- Overfitting occurs when a model is too complex and fits the noise in the data.
- Model complexity plays a crucial role in determining whether a model is underfitting or overfitting.
- Regularization techniques can help prevent overfitting by penalizing overly complex models.
- Cross-validation is a common method used to detect and prevent overfitting in machine learning models.

Q.5203 Which of the following *best* describes the differences among the training, validation, and test data sub-samples, and how each is used?

A. Training data is used to evaluate the model’s performance during the training process, validation data is used to build and finetune the model, and test data is used to identify issues or biases in the model.

B. Training data is used to build and finetune the model, validation data is used to evaluate the model’s performance during the training process, and test data is used to evaluate the final performance of a ML model.

C. Training data is used to evaluate the final performance of a ML model, validation data is used to build and finetune the model, and test data is used to compare the model’s performance to other models.

D. Training data is used to identify issues or biases in the model, validation data is used to evaluate the model’s performance during the training process, and test data is used to build and finetune the model.

The correct answer is **B.**

In the process of developing a machine learning model, the data is typically divided into three distinct sub-samples: training, validation, and test data. The training data is used to build and fine-tune the model. This is the initial phase where the model learns to make predictions by adjusting its parameters based on the input and output data. The validation data is used to evaluate the model’s performance during the training process. This helps in tuning the model's hyperparameters and in deciding when to stop training to avoid overfitting. Finally, the test data is used to evaluate the final performance of the machine learning model. This is the ultimate test of the model's ability to generalize to unseen data. It provides an unbiased estimate of the model's performance as it is data that the model has not seen during the training or validation phases.

Choice A is incorrect. Training data is not used to evaluate the model’s performance during the training process, but rather it is used to build and finetune the model. Validation data, on the other hand, is used to evaluate the model's performance during training and not for building or fine-tuning.

Choice C is incorrect. Training data does not serve to evaluate the final performance of a ML model; this role belongs to test data. Also, validation data helps in tuning parameters and adjusting weights but it doesn't serve as a basis for comparing models' performances.

Choice D is incorrect. The purpose of training data isn't identifying issues or biases in a model; instead it's utilized for building and refining models. Test data isn't employed for building or fine-tuning models but rather assessing their final performance.

Things to Remember

- Training data: Used to build and fine-tune the model by adjusting parameters based on input and output data.

- Validation data: Used to evaluate the model's performance during the training process and tune hyperparameters to avoid overfitting.
- Test data: Used to evaluate the final performance of the machine learning model and provide an unbiased estimate of its generalization ability.
- Overfitting: When a model performs well on the training data but fails to generalize to unseen data.
- Hyperparameters: Parameters that are set before the learning process begins and control the learning process of the model.

Q.5204 Which of the following *best* describes the Monte Carlo method for reinforcement learning?

A. It looks only one decision ahead when updating strategies.

B. It requires a lot of computational resources to simulate the entire environment.

C. The agent initially knows only the possible states and actions that it can take in an environment.

D. It requires a priori knowledge of the environment in order to learn a policy.

The correct answer is **B.**

Monte Carlo simulation involves simulating the entire environment to completion in order to generate sufficient samples for estimating the strategy. This makes it computationally intensive.

A is incorrect as it only applies to temporal difference learning. In the Monte Carlo method, the strategy is updated based on the total future rewards that are obtained through interaction with the environment rather than just looking at one decision ahead.

C is incorrect and only applies to temporal difference learning. In the Monte Carlo method, the agent has to wait until the completion of an episode to update the strategy.

D is incorrect. Monte Carlo method does not require any a priori knowledge of the environment and can learn from the outcomes of actions taken in the environment rather than by explicitly learning a policy.

Things to Remember

- Monte Carlo methods are a class of computational algorithms that rely on repeated random sampling to obtain numerical results.
- In reinforcement learning, Monte Carlo methods are used to estimate the value of actions based on sample returns obtained by running simulations in the environment.
- Monte Carlo methods are model-free, meaning they do not require knowledge of the underlying dynamics of the environment.
- One key advantage of Monte Carlo methods is that they can handle episodic tasks where the agent interacts with the environment over a sequence of episodes.
- Monte Carlo methods are particularly useful in scenarios where the dynamics of the environment are stochastic and not fully known.

Q.5206 Which of the following statements is the *most accurate?*

A. In reinforcement learning, the algorithm is trained on a smaller dataset and is expected to generalize to new situations, while in supervised learning the algorithm is trained on a larger dataset and is expected to perform well on the training data.

B. In reinforcement learning, the algorithm is given correct output values, similar to how supervised learning algorithms are presented with labeled data.

C. Reinforcement learning is similar to supervised learning in that the algorithm must learn through trial and error but is not presented with the correct output values for the training set.

D. Reinforcement learning is not based on labeled training data, but rather it uses feedback in the form of rewards or punishments to learn how to take actions in an environment to maximize a reward

The correct answer is **D.**

Reinforcement learning is a type of machine learning where an agent learns to make decisions by taking actions in an environment to maximize some type of reward. Unlike supervised learning, reinforcement learning does not rely on labeled training data. Instead, the agent learns from the consequences of its actions, which are presented as rewards or punishments. The agent's goal is to learn a policy, which is a mapping from states to actions, that maximizes the cumulative reward over time. This process involves exploring the environment, taking actions, and learning from the feedback. The agent uses this feedback to update its policy and improve its future actions. This iterative process continues until the agent's policy converges to an optimal policy, which is the one that maximizes the expected cumulative reward. Therefore, choice D accurately encapsulates the essence of reinforcement learning.

Choice A is incorrect. The size of the dataset used for training does not fundamentally distinguish reinforcement learning from supervised learning. Both types of algorithms can be trained on large or small datasets, depending on the specific application and available data.

Choice B is incorrect. Unlike supervised learning, reinforcement learning does not involve presenting the algorithm with correct output values or labeled data. Instead, it uses feedback in the form of rewards or punishments to learn how to take actions in an environment to maximize a reward.

Choice C is incorrect. While it's true that both reinforcement and supervised learning involve some form of trial and error, this statement incorrectly suggests that reinforcement learning does not receive correct output values for its training set. In fact, reinforcement learning doesn't use a traditional "training set" at all; instead, it learns through interaction with its environment.

Things to Remember

- Reinforcement learning is a type of machine learning where an agent learns to make decisions by taking actions in an environment to maximize some type of reward.

- Unlike supervised learning, reinforcement learning does not rely on labeled training data.
- The agent learns from the consequences of its actions, which are presented as rewards or punishments.
- The goal of the agent is to learn a policy that maximizes the cumulative reward over time.
- Reinforcement learning involves exploring the environment, taking actions, and learning from the feedback received.
- The agent updates its policy based on the feedback to improve its future actions.
- The iterative process continues until the agent's policy converges to an optimal policy that maximizes the expected cumulative reward.

Q.5207 Which of the following would be the *most appropriate* machine-learning algorithm for predicting the value of a house price index in ten years.

A. Supervised learning

B. Unsupervised learning

C. Reinforcement learning

D. All of the above

The correct answer is **A.**

Supervised learning is the most appropriate machine learning algorithm for predicting the value of a house price index in ten years. In supervised learning, the algorithm learns from a labeled dataset to make predictions or decisions without being explicitly programmed to perform the task. The dataset is divided into an input (feature) set and an output (target) set. The algorithm learns the mapping function from the input to the output. The aim is to approximate the mapping function so well that when new input data is given, the algorithm can predict the output for that data. In this case, the target variable is the value of the house price index in ten years, and the goal is to make a prediction for this value based on the given data. Therefore, a supervised learning algorithm would be appropriate for this problem.

Choice B is incorrect. Unsupervised learning algorithms are typically used for tasks such as clustering and dimensionality reduction, where the goal is to discover hidden patterns or structures in data. In this case, we have a specific target variable (house price index) that we want to predict, which is not the kind of task unsupervised learning algorithms are designed for.

Choice C is incorrect. Reinforcement learning involves an agent that learns how to behave in an environment by performing certain actions and observing the results or rewards of those actions. This type of algorithm would not be suitable for predicting a house price index as it requires a clear reward system and interaction with an environment, which does not apply in this scenario.

Choice D is incorrect. While all types of machine learning algorithms can be useful depending on the task at hand, they are not all equally suited for every problem. In this case, supervised learning would be the most appropriate choice because it involves using labeled data (i.e., data where the target variable is known) to train a model that can make predictions about unseen data.

Things to Remember

- Supervised learning is a type of machine learning where the algorithm learns from labeled data to make predictions.
- Unsupervised learning is a type of machine learning where the algorithm learns patterns and structures from unlabeled data.
- Reinforcement learning is a type of machine learning where an agent learns to make decisions by interacting with an environment.
- Supervised learning is commonly used for regression and classification tasks.
- Feature engineering is an important step in supervised learning to extract relevant features from the data.

Q.5209 Which of the following *best* describes the relationship between model size and the bias-variance trade-off in machine learning?

A. Large models with many features have low bias and low variance, while smaller models with fewer features have high bias and high variance.

B. Large models with many features have high bias and high variance, while smaller models with fewer features have low bias and low variance.

C. Large models with many features have low bias and high variance, while smaller models with fewer features have high bias and low variance.

D. Large models with many features have high bias and low variance, while smaller models with fewer features have low bias and high variance

The correct answer is **C.**

In machine learning, a model's complexity is often determined by the number of features it incorporates. Large models with many features are typically characterized by low bias and high variance. Low bias indicates that the model makes fewer assumptions about the form of the underlying truth. In other words, it is flexible enough to model complex patterns in the data. However, this flexibility can lead to high variance, meaning the model is sensitive to fluctuations in the training data and may overfit, performing poorly on new, unseen data.

Conversely, smaller models with fewer features tend to exhibit high bias and low variance. High bias means the model makes more assumptions about the form of the underlying truth, potentially oversimplifying the problem. This can lead to underfitting, where the model fails to capture important patterns in the data. However, these models typically have low variance, meaning they are less sensitive to fluctuations in the training data and may perform better on new, unseen data. This relationship between bias and variance is a fundamental aspect of the trade-off in machine learning model design.

Choice A is incorrect. Large models with many features do not have low bias and low variance. In fact, they tend to have low bias because they can fit the training data very well, but high variance because they are likely to overfit the data and perform poorly on new, unseen data.

Choice B is incorrect. Large models with many features do not have high bias and high variance. High bias would mean that the model is oversimplified and cannot capture the complexity of the data, which is unlikely for large models with many features.

Choice D is incorrect. Large models with many features do not have high bias and low variance. As explained above, large models are more likely to have low bias due to their ability to fit complex patterns in the training data but may suffer from high variance due to overfitting.

Things to Remember

- Bias-Variance Trade-off: The bias-variance trade-off is a key concept in machine learning that refers to the balance between a model's ability to capture the true underlying patterns in the data (bias) and its sensitivity to fluctuations in the training data (variance).

- Overfitting: Overfitting occurs when a model learns the noise in the training data rather than the underlying patterns, leading to poor generalization to new, unseen data.

- Underfitting: Underfitting happens when a model is too simple to capture the underlying patterns in the data, resulting in poor performance on both the training and test data.

- Model Complexity: The complexity of a model is often determined by the number of features it incorporates, with larger models typically having more features and higher complexity.
- Flexibility vs. Generalization: Large, complex models are more flexible and can capture intricate patterns in the data, but they may struggle with generalization to new data. Smaller, simpler models may generalize better but could miss important patterns in the data.

Q.5210 Which of the following *best* describes the primary importance of feature scaling in clustering?

A. Feature scaling helps to improve the accuracy of the clustering algorithm by reducing the variance of the features.

B. Feature scaling is necessary to ensure that all features are on the same scale so that the distance measure used by the clustering algorithm is meaningful.

C. Feature scaling is used to reduce the computational complexity of the clustering algorithm by reducing the number of features.

D. Feature scaling helps improve the clusters' interpretability by making it easier to understand the characteristics of each cluster.

The correct answer is **B.**

Feature scaling is a crucial step in the data preprocessing stage, especially when dealing with clustering algorithms. The primary reason for this is to ensure that all features are on the same scale, making the distance measure used by the clustering algorithm meaningful. Clustering algorithms, such as K-means, use distance measures like Euclidean or Manhattan distance to determine the similarity between data points. If the features are on different scales, the distance measure can be dominated by the feature with the largest scale, leading to inaccurate clustering results. By scaling all features to the same scale, the distance measure becomes more meaningful, and the clustering algorithm can accurately group similar data points together. This process improves the accuracy and interpretability of the clustering results, making it easier for data scientists to draw meaningful insights from the data.

Choice A is incorrect. Feature scaling does not primarily aim to improve the accuracy of the clustering algorithm by reducing the variance of the features. While it can help in normalizing data and dealing with outliers, its main purpose is to ensure that all features are on a similar scale so that distance measures used by clustering algorithms are meaningful.

Choice C is incorrect. Feature scaling does not reduce computational complexity by reducing the number of features. It only transforms the values of numerical variables in your dataset into a common scale, without distorting differences in ranges or losing information.

Choice D is incorrect. Although feature scaling may aid in interpreting clusters by standardizing their scales, this isn't its primary purpose within clustering algorithms. The main reason for using feature scaling in clustering is to ensure that all features contribute equally to the final result and no particular feature dominates others due to its scale.

Things to Remember

- Feature scaling is essential for distance-based algorithms like K-means clustering to ensure that all features contribute equally to the clustering process.
- Common feature scaling techniques include Min-Max scaling, Standardization (Z-score normalization), and Robust scaling.
- Feature scaling can help improve the convergence speed of clustering algorithms and prevent numerical instability.
- Outliers in the data can significantly impact clustering results, and feature scaling can help mitigate their effects.
- Feature scaling is also important in other machine learning algorithms like Support Vector Machines (SVM) and Principal Component Analysis (PCA).

Q.5211 Why is it *mostly uncommon* for econometric models to be evaluated using validation samples?

A. Validation samples are not necessary because econometric models are always accurate.

B. Validation samples are not typically used because researchers are primarily interested in understanding the underlying relationships in the data, rather than in making predictions.

C. Validation samples are not available for the specific research context.

D. Validation samples are not used because they are too expensive to obtain

The correct answer is **B.**

Econometric models are primarily tools for understanding the relationships between different variables. They are used to analyze how changes in one variable might affect another. The primary goal of these models is not to make predictions, but to understand these underlying relationships. Therefore, validation samples, which are typically used to test the predictive accuracy of a model, are not commonly used in econometrics. Instead, researchers focus on the model's ability to accurately represent the relationships in the data. This focus on understanding rather than prediction is what sets econometrics apart from other fields that use statistical models.

Choice A is incorrect. It is not accurate to say that econometric models are always accurate. In fact, all models are simplifications of reality and therefore inherently contain some degree of error. Furthermore, the accuracy of a model can vary depending on the specific context in which it is used.

Choice C is incorrect. While it may be true that validation samples are not always available for every research context, this does not explain why they are typically not used in the evaluation of econometric models. Even when validation samples are available, researchers often choose not to use them for reasons other than their availability.

Choice D is incorrect. The cost of obtaining validation samples may be a consideration in some cases, but it does not represent the primary reason why these samples are typically not used in econometrics. The main reason has more to do with the objectives of econometric analysis - understanding relationships rather than making predictions - as explained above.

Things to Remember

- Validation samples are typically used to test the predictive accuracy of a model.
- Econometric models focus on understanding relationships between variables rather than making predictions.
- All models, including econometric models, are simplifications of reality and contain some degree of error.
- The accuracy of a model can vary depending on the specific context in which it is used.
- Validation samples may not always be available for every research context, but their absence does not explain why they are not commonly used in econometrics.
- The cost of obtaining validation samples may be a consideration in some cases, but it is not the primary reason why they are typically not used in econometrics.

Q.5213 Which machine learning model is most suitable for a data scientist who wants to split 2,000 customers into 20 distinct groups?

A. Clustering

B. Regression

C. Support vector machines (SVMs)

D. K-nearest neighbors

The correct answer is **A.**

Clustering is the most suitable machine learning model for this task. Clustering is a type of unsupervised learning algorithm that is used to group similar instances on the basis of features into clusters. In this case, the data scientist wants to divide 2,000 customers into 20 distinct groups. Since there is no labeled training data provided, an unsupervised learning algorithm like clustering is the best choice. Clustering algorithms can identify and learn patterns and similarities from the data and group them accordingly. This makes it an ideal choice for tasks like customer segmentation, where the goal is to identify distinct groups or segments within a larger dataset.

Choice B is incorrect. Regression is a supervised learning model used for predicting a continuous outcome variable (also called dependent variable) based on one or more predictor variables (also called independent variables). It's not suitable for categorizing customers into distinct groups without pre-labeled data.

Choice C is incorrect. Support Vector Machines (SVMs) are also supervised learning models that are typically used for classification and regression analysis. They require labeled data to train the model, which the data scientist does not have in this case.

Choice D is incorrect. K-nearest neighbors (KNN) is a type of instance-based learning or non-generalizing learning where the function is only approximated locally and all computation is deferred until function evaluation. It requires labeled data to classify new instances based on their similarity to existing instances, which isn't applicable here as there's no pre-labeled data available.

Things to Remember

- Clustering is a type of unsupervised learning algorithm that groups similar instances into clusters based on features.
- Supervised learning models like regression and SVMs require labeled data for training.
- K-nearest neighbors (KNN) is an instance-based learning algorithm that classifies new instances based on similarity to existing instances.
- Customer segmentation is a common application of clustering in marketing and business analytics.

Reading 26: Machine Learning and Prediction

Q.5166 Which of the following best describes the role of linear regression and logistic regression in machine learning?

A. Linear regression is used to predict continuous variables, while logistic regression is used to predict categorical variables.

B. Linear regression is used for unsupervised learning, while logistic regression is used for supervised learning.

C. Linear regression is used for classification, while logistic regression is used for regression.

D. Both linear regression and logistic regression can be used for both classification and regression.

The correct answer is **A.**

Linear regression is used to predict continuous variables, while logistic regression is used to predict categorical variables. Linear regression is a type of regression analysis that is used to predict a continuous target variable based on one or more input features. Logistic regression, on the other hand, is a classification algorithm that is used to predict a binary outcome (such as "yes" or "no") based on one or more input features.

B is incorrect. Both linear and logistic regressions are supervised machine learning techniques.

C is incorrect. Linear regression is instead used for regression, while logistic regression is used for classification.

D is incorrect. Linear regression cannot be used for classification. When using a linear regression model for binary classification, where the dependent variable Y can only be 0 or 1, it is possible for the model to predict probabilities outside of the range of 0 to 1. This occurs because the model is attempting to fit a straight line to the data, and the predicted values may not be restricted to the valid range of probabilities. As a result, the model may produce predictions that are less than zero or greater than one.

Things to Remember

- Linear regression is a type of linear approach to modeling the relationship between a dependent variable and one or more independent variables.
- Logistic regression is a type of regression analysis used for predicting the probability of a binary outcome.
- Linear regression assumes a linear relationship between the dependent and

independent variables, while logistic regression uses the logistic function to model the probability of a binary outcome.

- Linear regression is sensitive to outliers, while logistic regression is more robust to outliers.

- Both linear and logistic regressions require the assumption of independence of observations.

Q.5167 Which of the following *best describes* the primary importance of encoding categorical variables before using them as input features in a machine learning model?

A. Most machine learning models can only work with numerical input data

B. Encoding categorical variables improves the prediction accuracy of the model

C. Encoding categorical variables helps to reduce the size of the dataset

D. Encoding categorical variables eliminates the need for feature scaling.

The correct answer is **A.**

The primary reason for encoding categorical variables is that most machine learning models are designed to work with numerical input data. Machine learning models, in general, are mathematical models that pcomputations on the input data to make predictions or decisions. Since mathematical operations are performed on numbers, these models require numerical input data. Categorical data, which can be in the form of text or labels, cannot be directly used in these mathematical computations. Therefore, it is necessary to convert or encode these categorical variables into numerical form before they can be used as input features in a machine learning model. This process of encoding transforms the categorical data into a format that the machine learning model can understand and work with.

Choice B is incorrect. While encoding categorical variables can sometimes improve the prediction accuracy of a model, it is not the main reason for encoding. The primary reason for encoding categorical variables is to convert them into a format that machine learning algorithms can understand, as most of these algorithms require numerical input data.

Choice C is incorrect. Encoding categorical variables does not necessarily reduce the size of the dataset. In fact, certain types of encoding techniques such as one-hot encoding can increase the dimensionality of the dataset because they create additional binary features to represent each category.

Choice D is incorrect. Encoding categorical variables does not eliminate the need for feature scaling. Feature scaling is a separate preprocessing step that is applied to numerical data to ensure that all features contribute equally to the model's predictions.

Things to Remember

- One-hot encoding is a common technique used to encode categorical variables by creating binary dummy variables for each category.
- Label encoding is another method of encoding categorical variables where each category is assigned a unique numerical value.
- Ordinal encoding is used when the categorical variables have an inherent order or ranking.
- Encoding categorical variables incorrectly can introduce bias or noise into the model, so it is important to choose the appropriate encoding method based on the data and the machine learning algorithm being used.
- Feature scaling is a preprocessing step that standardizes the range of independent variables or features of data, which is important for certain machine learning algorithms like support vector machines or k-nearest neighbors.

Q.5168 A financial institution is building a machine learning model to predict the likelihood of default for a portfolio of loans. The model includes several categorical variables, such as loan purpose and borrower credit score, which are transformed into dummy variables. If an intercept term and correlated dummy variables are included in the model, which of the following is a potential issue that may arise?

A. The model will have a single solution

B. The model will not be able to find a unique best-fit solution

C. The model will have a high bias

D. The model will have a high variance

The correct answer is **B.**

The model will not be able to find a unique best-fit solution. This issue arises due to a phenomenon known as the 'dummy variable trap'. The dummy variable trap is a scenario in which the inclusion of dummy variables in a model creates perfect multicollinearity. Perfect multicollinearity is a statistical phenomenon in which two or more predictor variables in a multiple regression model are highly correlated. In this case, one variable can be linearly predicted from the others with a substantial degree of accuracy. When this happens, it becomes impossible for the model to identify a unique best-fit solution because there are multiple sets of coefficients that could explain the data with equal accuracy. This is a significant problem because it undermines the reliability of the model's predictions and the statistical significance of the coefficients. Therefore, when building a model, it is crucial to avoid the dummy variable trap to ensure the model's accuracy and reliability.

Choice A is incorrect. The inclusion of an intercept term along with correlated dummy variables does not guarantee that the model will have a single solution. In fact, it can lead to a problem known as multicollinearity, where the predictor variables are highly correlated with each other. This can make it difficult for the model to estimate the individual effect of each predictor variable on the outcome, leading to unstable and unreliable estimates.

Choice C is incorrect. High bias in a machine learning model refers to a situation where the model consistently makes errors by making assumptions about the data that do not hold true. Including an intercept term along with correlated dummy variables does not necessarily lead to high bias in this context; instead, it may cause multicollinearity which affects variance inflation factor and standard errors but doesn't directly relate to bias.

Choice D is incorrect. High variance in a machine learning model refers to overfitting where models perform well on training data but poorly on unseen data due its complexity or noise within training data set itself . However, including an intercept term along with correlated dummy variables doesn't necessarily increase variance or overfitting risk; rather it leads towards multicollinearity which impacts reliability of coefficient estimates but doesn't directly contribute towards increasing variance.

Things to Remember

- Dummy variables are used in regression analysis to represent categorical data as numerical data.
- Perfect multicollinearity occurs when two or more independent variables in a regression model are perfectly correlated.
- Multicollinearity can lead to unstable coefficient estimates and unreliable statistical significance tests.
- Variance inflation factor (VIF) is a measure of multicollinearity in a regression model.
- High bias in a model refers to systematic errors in predictions, while high variance refers to overly sensitive models that do not generalize well to new data.

Q.5170 Which of the following statements about regression techniques and regularization is most accurate?

A. Ridge regression sets some coefficient estimates to 0

B. LASSO regression adds a penalty term, which is equal to sum of the absolute values of the coefficients, to the sum of squared residuals (RSS)

C. Elastic net is a hybrid of LASSO regression and ordinary least squares (OLS) regression in a single loss function

D. L2 regularization works by adding a penalty term, which is equal to sum of the absolute values of the coefficients, to the sum of squared residuals (RSS)

The correct answer is **B.**

LASSO (Least Absolute Shrinkage and Selection Operator) regression is a type of linear regression that uses shrinkage. Shrinkage is where data values are shrunk towards a central point, like the mean. The lasso procedure encourages simple, sparse models (i.e. models with fewer parameters). This particular type of regression is well-suited for models showing high levels of multicollinearity or when you want to automate certain parts of model selection, like variable selection/parameter elimination. The acronym "LASSO" stands for Least Absolute Shrinkage and Selection Operator.

Lasso regression performs L1 regularization, which adds a penalty equal to the absolute value of the magnitude of coefficients. This type of regularization can result in sparse models with few coefficients; Some coefficients can become zero and eliminated from the model. Larger penalties result in coefficient values closer to zero, which is the ideal for producing simpler models. On the other hand, L2 regularization (Ridge regression) doesn't result in elimination of coefficients or sparse models. This makes the Lasso far easier to interpret than the Ridge.

Choice A is incorrect. Ridge regression does not set some coefficient estimates to 0. Instead, it shrinks the coefficients towards zero but does not make them exactly zero. This is a key difference between Ridge regression and LASSO regression, which can indeed set some coefficients to zero.

Choice C is incorrect. Elastic net is not a hybrid of LASSO regression and ordinary least squares (OLS) regression in a single loss function. Rather, Elastic net combines the properties of both Ridge and LASSO regressions by adding both types of penalty terms to the loss function.

Choice D is incorrect. L2 regularization does not add a penalty term equal to the sum of absolute values of coefficients to RSS; instead, it adds a penalty term equal to the square of magnitude (L2 norm) of coefficient vector.

Things to Remember

- Ridge regression and LASSO regression are both types of linear regression models that use regularization techniques to prevent overfitting.

- Ridge regression adds a penalty term equal to the square of the magnitude of coefficients (L2 norm), while LASSO regression adds a penalty term equal to the absolute value of the magnitude of coefficients (L1 norm).

- Elastic net regression combines the penalties of both Ridge and LASSO regression, offering a balance between variable selection and coefficient shrinkage.

- Regularization techniques like Ridge, LASSO, and Elastic net are commonly used in machine learning and statistical modeling to improve model performance and interpretability.

Q.5171 Which of the following statements best describe the difference between hyperparameters and model parameters?

A. Hyperparameters are learned from data during training, while model parameters are fixed before training

B. Hyperparameters are fixed before training, while model parameters are learned from data during training

C. Hyperparameters are fixed before training, while model parameters are learned from data during testing

D. Hyperparameters are used to make predictions, while model parameters are used to specify the structure of the model

The correct answer is **B.**

Hyperparameters are values that are set before training a machine learning model, while model parameters are learned from data during training and are used to make predictions. Hyperparameters specify the structure of the model, such as the learning rate or the regularization coefficient, and have a significant impact on the performance of the model. Model parameters, on the other hand, are the coefficients or weights that are learned from the data and are used to make predictions based on the structure specified by the hyperparameters.

A is incorrect. Hyperparameters are values that are set before training a machine learning model, while model parameters are learned from data *during* training.

C is incorrect. Model parameters are learned from data during training, not during testing.

D is incorrect. Hyperparameters specify the structure of the model, while parameters are used to make predictions.

Things to Remember

- Hyperparameters are external configuration settings that are not learned from the data, such as learning rate, number of hidden layers, etc.
- Model parameters are internal variables whose values are learned from the training data, such as weights and biases in a neural network.
- Hyperparameters are set by the data scientist before the training process begins, while model parameters are updated during the training process.
- Choosing the right hyperparameters is crucial for the performance of the model, as they directly impact how the model learns and generalizes from the data.
- Hyperparameter tuning is the process of finding the best hyperparameters for a given model and dataset, often done through techniques like grid search or random search.

Q.5172 When constructing a decision tree, the feature that is *most likely* considered at each node is:

A. The feature with the highest information gain

B. The feature with the lowest information gain

C. The first feature in the dataset

D. The last feature in the dataset

The correct answer is **A.**

The feature with the highest information gain is considered at each node when constructing a decision tree. Information gain is a measure of the reduction in entropy or impurity in the data after a dataset is split on a feature. It is used to decide which feature to split on at each step in building the tree. The feature with the highest information gain is considered to be the most effective for reducing uncertainty and splitting the sample. Therefore, it is selected at each node during the construction of the decision tree. This approach helps to improve the accuracy of the decision tree by ensuring that the most informative features are used to split the data.

Choice B is incorrect. The feature with the lowest information gain is not typically selected at each node during the construction of a decision tree. This is because low information gain implies that the feature does not contribute significantly to our understanding of the target variable, and therefore would not be an effective choice for splitting at a node.

Choice C is incorrect. The order of features in the dataset has no bearing on their selection for nodes in a decision tree. Therefore, it's not necessarily true that the first feature in the dataset will be considered at each node during construction of a decision tree.

Choice D is incorrect. Similar to Choice C, it's also incorrect to assume that just because a feature appears last in a dataset, it will be considered at each node during construction of a decision tree. The selection of features for nodes depends on their ability to improve our understanding or prediction of the target variable, rather than their position within the dataset.

Things to Remember

- Entropy: Entropy is a measure of impurity in a dataset. It is used in decision tree algorithms to determine the homogeneity of a sample.
- Information Gain: Information gain is the measure of the effectiveness of a feature in splitting the data. It helps in deciding the best feature to split on at each node.
- Gini Impurity: Gini impurity is another measure of impurity in a dataset. It is commonly used in decision tree algorithms as an alternative to entropy.
- Decision Tree Pruning: Decision tree pruning is a technique used to reduce the size of a decision tree by removing nodes that do not provide much information.
- Overfitting: Overfitting occurs when a model learns the noise in the training data rather than the underlying pattern, leading to poor generalization to new data.

Q.5176 Which of the following *best* describes a neural network as a technique for machine learning?

A. A neural network is a precise replica of the human nervous system

B. A neural network is based on various linear regression models

C. A neural network is made up of interconnected nodes and links, and it is particularly effective at modeling non-linear relationships between the input data

D. A neural network is structured like a tree, with nodes representing the features and the relationships between them being linear

The correct answer is **C.**

A neural network is indeed made up of interconnected nodes and links, and it is particularly effective at modeling non-linear relationships between the input data. This machine learning model is inspired by the structure and function of the human brain, which is made up of interconnected neurons that transmit and process information. In a neural network model, the nodes represent artificial neurons, and the links represent the connections between them. The model is designed to learn patterns and relationships in the input data and make predictions based on those patterns. This ability to model non-linear relationships is a key strength of neural networks, making them particularly useful in complex tasks such as image recognition, natural language processing, and more.

Choice A is incorrect. While neural networks are inspired by the human nervous system, they are not precise replicas of it. They do not replicate all the complexities and functionalities of the human brain.

Choice B is incorrect. Neural networks are not based on linear regression models. They can model both linear and non-linear relationships between inputs, making them more versatile than simple linear regression models.

Choice D is incorrect. The structure described in this option corresponds to a decision tree, not a neural network. Neural networks consist of interconnected nodes or "neurons", but their relationships aren't necessarily linear and their structure isn't tree-like.

Things to Remember

- Neural networks are a type of machine learning model inspired by the structure and function of the human brain.
- Nodes in a neural network represent artificial neurons, and links represent connections between them.
- Neural networks are effective at modeling non-linear relationships in data, making them versatile for various tasks.
- Neural networks are used in complex tasks such as image recognition, natural language processing, and more.
- Neural networks can have different architectures like feedforward, recurrent, convolutional, etc.

Q.5194 Consider the following confusion matrices. **Model A**

	Predicted: No default	Predicted: Default
Actual: No default	TN = 100	FP = 50
Actual: Default	FN = 50	TP = 900

Model B

	Predicted: No default	Predicted: Default
Actual: No default	TN = 120	FP = 80
Actual: Default	FN = 30	TP = 870

The model that is *most likely* to have a higher accuracy and higher precision, respectively, is:

A. Model A, Model B

B. Model B, Model A

C. Model A, Model A

D. Model B, Model B

The correct answer is **C.**

Model accuracy is calculated as:

$$\text{Model accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$$\text{Model A accuracy} = \frac{(900 + 100)}{(900 + 100 + 50 + 50)} = 0.909$$

$$\text{Model B accuracy} = \frac{(870 + 120)}{(870 + 120 + 80 + 30)} = 0.900$$

Model A has a slightly higher accuracy than model B.

Model precision is calculated as:

$$\text{Model precision} = \frac{(TP)}{(TP + FP)}$$

$$\text{Model precision of A} = \frac{(900)}{(900 + 50)} = 0.9474$$

$$\text{Model precision of B} = \frac{(870)}{(870 + 80)} = 0.9158$$

Model A has a higher precision relative to B.

Q.5196 A financial institution is using a logistic regression model to predict the likelihood of a loan default based on various borrower characteristics. To make predictions using the model, a threshold value Z is chosen, and the predicted probability p_i is compared to the threshold. Which of the following statements is *most likely* accurate? If p_i is:

A. Greater than or equal to Z, the model predicts that the loan will default

B. Less than or equal to Z, the model predicts that the loan will default

C. Greater than Z, the model predicts that the loan will default

D. Less than Z, the model predicts that the loan will default

The correct answer is **A.**

If p_i is greater than or equal to Z, the model predicts that the loan will default. In this scenario, the predicted probability p_i is compared to the threshold value Z to make predictions.

B is incorrect. if p_i is less than the threshold Z, the model predicts that the loan will not default.

C and D are incorrect.

This can be expressed as:

$y_j = \{ (1 \ if \ p_i \geq Z) \ (0 \ if \ p_i < Z)$

Where 1 is the positive outcome (in this case, default), and 0 is the negative outcome (no default).

Things to Remember

- Logistic regression is a statistical model that is commonly used for binary classification problems, where the outcome variable is categorical and has two possible values.
- The logistic regression model predicts the probability of the outcome variable belonging to a particular category based on the input features.
- The threshold value Z in logistic regression is used to classify the predicted probabilities into the different categories.
- Receiver Operating Characteristic (ROC) curve is often used to evaluate the

performance of a logistic regression model by plotting the true positive rate against the false positive rate at various threshold values.

- Area Under the Curve (AUC) of the ROC curve is a common metric used to quantify the overall performance of a logistic regression model.

Q.5256 Which measure is best for evaluating the quality of a split in a decision tree model?

A. Mean squared error

B. Root mean squared error

C. Gini impurity

D. Information gain

The correct answer is **D.**

Information gain is the most suitable measure for evaluating the quality of a split in a decision tree model. It quantifies the reduction in entropy or uncertainty in the target variable after making a split. Entropy is a measure of the impurity or disorder in a set of instances. When a decision tree is built, the aim is to split the nodes in a way that decreases entropy and increases information gain. The higher the information gain, the more effective the split is in segregating the data. Therefore, information gain is a crucial measure in decision tree models as it helps in creating an effective and accurate model.

Choice A is incorrect. Mean squared error (MSE) is a measure of the average of the squares of the errors or deviations. It is not suitable for assessing the quality of a split in a decision tree model because it does not provide information about how well the split segregates data into pure subsets.

Choice B is incorrect. Root mean squared error (RMSE) is also an inappropriate measure for evaluating splits in decision trees as it, like MSE, focuses on quantifying average prediction errors rather than assessing data segregation.

Choice C is incorrect. Gini impurity measures the degree or probability of a particular variable being wrongly classified when it's randomly chosen. While Gini impurity can be used to evaluate splits in decision trees, it's not considered as effective as information gain which directly measures improvement in prediction accuracy resulting from each potential split.

Things to Remember

- Entropy is a measure of impurity or disorder in a set of instances, commonly used in decision tree models.

- Information gain quantifies the reduction in entropy or uncertainty in the target variable after making a split in a decision tree.
- Mean squared error (MSE) is a measure of the average of the squares of errors and is not suitable for evaluating splits in decision trees.
- Root mean squared error (RMSE) is the square root of the average of the squares of errors and is also not appropriate for assessing the quality of splits in decision trees.
- Gini impurity measures the probability of a particular variable being wrongly classified when randomly chosen, but it is not as effective as information gain for evaluating splits in decision trees.